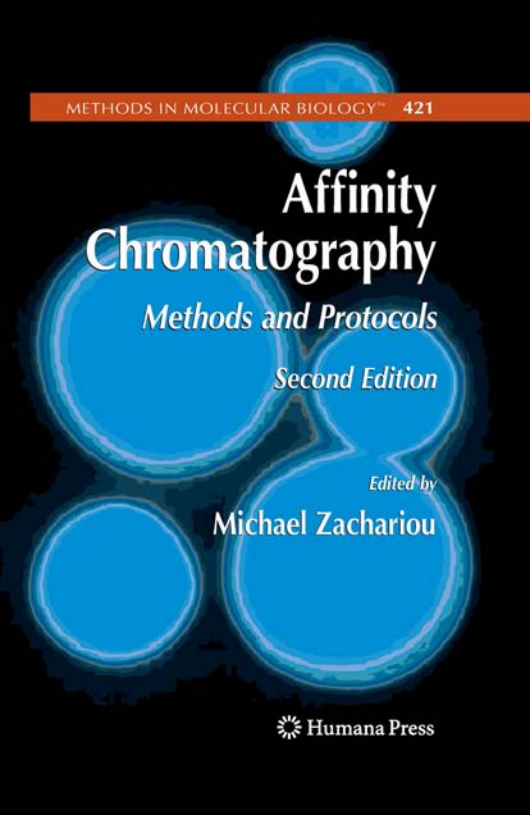




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Affinity Chromatography

Methods and Protocols

Second Edition

Edited by

Michael Zachariou

 Humana Press

Affinity Chromatography

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Michael Zachariou

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Cover illustration: Fig. 4, Chapter 7, "Rationally Designed Ligands for use in Affinity Chromatography: An Artificial Protein L," by Ana Cecilia A. Roque and Christopher R. Lowe

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To Tina, Emmanuella, Natalie, and Ashez

Preface

Forty years after the term “affinity chromatography” was introduced, this mode of chromatography remains a key tool in the armory of separation techniques that are available to separation and interaction scientists. Affinity chromatography is favored because of its high selectivity, speed, and ease of use. The rapid and selective isolation of molecules using affinity chromatography has allowed a better understanding of biological processes, accelerated the identification of target molecules, and spawned new process areas such as immobilized enzyme reactors. It has had ubiquitous application in most areas of science ranging from small molecule isolation to biopolymers such as DNA, proteins, polysaccharides, and even whole cells. The number of applications of affinity chromatography continues to expand at a rapid rate. For example, more than 60% of purification protocols include some sort of affinity chromatography step, while a database search of PubMed reveals more than 36,000 publications making use of the term “affinity chromatography,” more than 3000 of which refer to it in their title. The US patent office reports more than 16,000 references to the term “affinity chromatography”, while there are more than 270 references to the same term in the patent title.

The aim of this edition of *Methods in Molecular Biology, Affinity Chromatography: Methods and Protocols, Second Edition* is to provide the beginner with the practical knowledge to develop affinity separations suitable for various applications relevant to the post-genomic era. This second edition expands on the first edition by introducing more state-of-the-art protocols used in affinity chromatography. This current edition also describes protocols that demonstrate the concept of affinity chromatography being applied to meet the modern high throughput screening demands of researchers and development scientists, while expanding on some more traditional affinity chromatography approaches that have become of greater interest to separation scientists. This volume begins with an overview of affinity chromatography authored by one of the pioneers of affinity chromatography, Professor Christopher Lowe. Part I expands on affinity chromatography techniques that currently enjoy frequent citation in the literature from those purifying biomolecules. These affinity chromatography techniques include immobilized metal affinity chromatography, immunoaffinity chromatography and dye-ligand chromatography.

Affinity tags for purification of proteins have become useful and common tools in academic and industrial research laboratories for rapid protein isolation. The sequencing of the human genome along with a multitude of prokaryotic genomes has forced research laboratories and biotechnology companies to find rapid and high-yielding approaches to screen for protein targets. Affinity chromatography techniques allow for high-yielding, rapid approaches to target identification. Part II presents a number of protocols describing the use of various fusion tags as well as how to cleave them, so as to allow the scientists to study the native phenotype of the protein. This section also discusses methods for selecting ligands through rational combinatorial design and phage display for use in affinity chromatography. Part III ventures into diverse applications of affinity chromatography such as its use in catalytic reactions, DNA purification, whole cell separations, and for the isolation of phosphorylated proteins. Protocols are also presented on analytical applications of affinity chromatography, such as in capillary electrophoresis and quantitative affinity chromatography.

Affinity Chromatography: Methods and Protocols, Second Edition is aimed at those interested in separation sciences, particularly in the pharmaceutical and biological research sectors that have an interest in isolating macromolecules rapidly, quantitatively, and with high purity.

Michael Zachariou

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Affinity Chromatography

History, Perspectives, Limitations and Prospects

Ana Cecília A. Roque and Christopher R. Lowe

Summary

Biomolecule separation and purification has until very recently steadfastly remained one of the more empirical aspects of modern biotechnology. Affinity chromatography, one of several types of adsorption chromatography, is particularly suited for the efficient isolation of biomolecules. This technique relies on the adsorbent bed material that has biological affinity for the substance to be isolated. This review is intended to place affinity chromatography in historical perspective and describe the current status, limitations and future prospects for the technique in modern biotechnology.

Key Words: Affinity; chromatography; biomimetic; ligands; synthetic; proteins; purification; design; combinatorial synthesis.

1. Introduction

Traditional techniques for biomolecule separation based on precipitation with pH, ionic strength, temperature, salts, solvents or polymers, ion exchange or hydrophobic chromatography are slowly being replaced by sophisticated chromatographic protocols based on biological specificity. Affinity techniques exploit highly specific biorecognition phenomena and are ideally suited to the purification of biomolecules. In affinity chromatography, the specific adsorption properties of the bed material are realized by covalently attaching the ligand complementary to the target biomolecule onto an insoluble matrix. If a crude

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cell extract containing the biologically active target is passed through a column of such an immobilized ligand, then all compounds displaying affinity under the given experimental conditions will be retained by the column, whereas compounds showing no affinity will pass through unretarded. The retained target is then released from the complex with the immobilized ligand by changing operational parameters such as pH, ionic strength, buffer composition or temperature. Conceptually, the technique represents chromatographic nirvana: Exquisite selectivity combined with high yields and the unparalleled simplicity of a 'load, wash, elute' philosophy. However, experience over the last 3–4 decades has shown that there is a very high penalty to pay for the implicit specificity and simplicity of affinity chromatography, which has important ramifications for commercial use and process development.

2. Historical Perspective

Affinity chromatography is a particular variant of chromatography in which the unique biological specificity and reversibility of the target analyte and ligand interaction is utilized for the separation (*1*). It is possible to distinguish four phases in the development of the technique (*see Fig. 1*) starting from the early

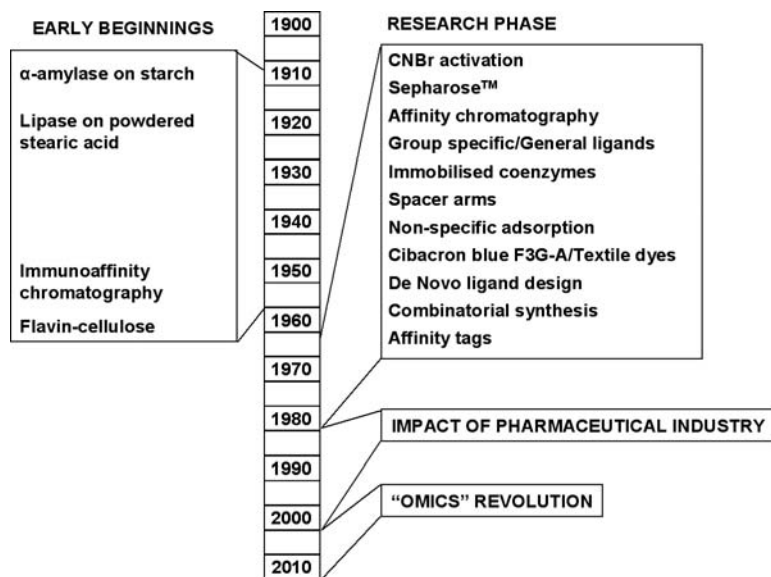


Fig. 1. Development of affinity chromatography as a technique: (i) Early beginning; (ii) Research phase; (iii) Impact of pharmaceutical industry and (iv) 'Omics' revolution.

realization of the technique, through the research phase, the impact of the nascent biopharmaceutical industry to the likely effect of the 'omics' revolution.

2.1. Early Beginnings

The concept of resolving complex macromolecules by means of biospecific interactions with immobilized substrates has its antecedents reaching back to the beginning of the 20th century. The German pharmacologist Emil Starkenstein (1884–1942) in a paper published in 1910 (2) on the influence of chloride on the enzymatic activity of liver α -amylase was generally considered to be responsible for the first experimental demonstration of the biospecific adsorption of an enzyme onto a solid substrate, in this case, starch. Not long after, Willstätter et al. (3) appreciably enriched lipase by selective adsorption onto powdered stearic acid. It was not until 1951, however, that Campbell and co-workers (4) first used the affinity principle to isolate rabbit anti-bovine serum albumin antibodies on a specific immunoabsorbent column comprising bovine serum albumin coupled to diazotised *p*-aminobenzyl-cellulose. This technique, now called immunoaffinity chromatography, became established before the development of small-ligand selective chromatography, where Lerman (5) isolated mushroom tyrosinase on various *p*-azophenol-substituted cellulose columns, and Arsenis and McCormick (6,7) purified liver flavokinase and several other FMN-dependent enzymes on flavin-substituted celluloses. Insoluble polymeric materials, especially the derivatives of cellulose, also found use in the purification of nucleotides (8), complementary strands of nucleic acids (9) and certain species of transfer RNA (10).

2.2. Research Phase

The general notion of exploiting strong reversible associations with highly specific substrates or inhibitors to effect enzyme purification was evident in the literature in the mid-1960s (11), although the immense power of biospecificity as a purification tool was not generally appreciated until 1968 when the term 'affinity chromatography' was coined (12). It was recognized that the key development required for wider application of the technique was that the solid-phase adsorbent should have a number of desirable characteristics:

... the unsubstituted matrix or gel should show minimal interactions with proteins in general, both before and after coupling to the specific binding group. It must form a loose, porous network that permits easy entry and exit of macromolecules and which retains favourable flow properties during use. The chemical structure of the supporting material must permit the convenient and extensive attachment of the specific ligand under relatively mild conditions, and through chemical bonds that are stable to the conditions of adsorption and elution. Finally, the inhibitor

groups critical in the interaction must be sufficiently distant from the solid matrix to minimise steric interference with the binding process (12).

In this seminal paper, the general principles and potential application of affinity chromatography were enunciated and have largely remained unchanged until the present date. The paper contained several important contributions. First, it generalized the technique to all potential enzyme purifications via immobilized substrates and inhibitors and exemplified the approach by application to staphylococcal nuclease, α -chymotrypsin and carboxypeptidase A. Second, it introduced for the first time a new highly porous commercially available 'beaded' matrix of agarose, Sepharose, which displayed virtually all of the desirable features listed above (13) and circumvented many of the issues associated with conventional cellulosic matrices available at that time. Agarose is a linear polysaccharide consisting of alternating 1,3-linked β -D-galactose and 1,4-linked 3,6-anhydro- α -L-galactose units (13). Third, the report exploited the activation of Sepharose by treatment with cyanogen bromide (CNBr) to result in a derivative that could be readily coupled to unprotonated amino groups of an inhibitory analogue to generate a highly stable Sepharose-inhibitor gel with nearly ideal properties for selective column chromatography (14,15). The use of CNBr activation chemistry was a milestone in the development of the technique, because the complex organic chemistry required for the synthesis of reliable immobilized ligand matrices had previously prevented this technique from becoming generally established in biological laboratories. Fourth, the report introduces the notion of spacer arms to alleviate steric interference and exemplifies the concept by showing the dramatically stronger adsorption of α -chymotrypsin to the immobilized inhibitor D-tryptophan methyl ester when a 6-carbon chain, ϵ -amino caproic acid, was interposed between the Sepharose matrix and the inhibitor. When the inhibitor was coupled directly to the matrix, incomplete and unsatisfactory resolution of the enzyme was observed. Fifth, the report emphasizes the importance of selective affinity for the immobilized inhibitor by demonstrating the absence of adsorption of chemically inhibited enzymes such as DFP-treated α -chymotrypsin or CNBr-treated nuclease to their respective adsorbents (12). Finally, this paper emphasizes the efficacy of relatively low-affinity inhibitors and suggests that unusually strong affinity constants are not an essential requirement for utilization of these techniques for the rapid single-step purification of proteins.

Affinity chromatography caught the eye of many researchers worldwide and there followed a spate of publications purporting to purify proteins and other biomolecules by every conceivable class of immobilized ligand. However, troubling issues relating to the chemistry of the ligand attachment still remained. For example, there was much debate on how adsorbents should be synthesized (16); the 'solid-phase assembly' approach was more facile and advocated the

attachment of ligands to spacer arms already present on the pre-activated affinity matrix, whereas the 'pre-assembly' approach uses conventional organic chemistry to modify the ligand with a suitably derivatized spacer arm, after which the whole assembly is coupled to the matrix. The solid-phase assembly approach lead to inhomogeneity problems where there were multiple sites on the target ligand or the coupling chemistries were incomplete, whereas the pre-assembled ligand spacer arm unit could be pre-characterized by conventional chemical techniques and studies in solution to yield useful advance information on binding specificity and kinetic constants. The present authors believe that a combination of both strategies represents an effective means of developing new and well-characterized affinity adsorbents for the purification of target proteins.

A further key development introduced in the early 1970s was that of 'group-specific' (17) or 'general ligand' (18) adsorbents. An important advantage of ligands with a broad bioaffinity spectrum, such as the coenzymes, lectins, nucleic acids, metal chelates, Protein A, gelatine and heparin, is that it was not obligatory to devise a new organic synthetic strategy for every projected biospecific purification. However, a possible disadvantage of the group-specific approach is that the broad specificity of the adsorption stage required a compensatory specific elution step to restore the overall biospecificity of the chromatographic system. Nevertheless, of the thousands of enzymes that have been assigned a specific Enzyme Commission number, approximately one-third involve one of the four adenine coenzymes (NAD^+ , NADP^+ , CoA and ATP), and not surprisingly, these classes of enzymes were the first to be targeted by this approach (17–19) and subsequently extensively exploited in the purification of oxido-reductases by affinity chromatography and in enzyme technology (20–22).

Until this point in time, most of the studies had generated rules-of-thumb on how to apply the technique of affinity chromatography to selected purifications. However, it became apparent on even a rudimentary examination of the theoretical basis of the technique (23) that the implicit assumption that the observed chromatographic adsorption of the target protein to the immobilized ligand was due exclusively to biospecific enzyme–ligand interactions was misguided. The large discrepancies observed between what was anticipated on the basis of the biological affinity for the immobilized ligand and what was observed experimentally to be the case were found to be due to the largely unsuspected interference by non-biospecific adsorption, which, in many cases, completely eclipsed the biospecific adsorption (24–25). O'Carra and co-workers (24–25) demonstrated that spacer arms do not always act simply as passive links between biospecific ligands and the polymer matrix and described methods for the control of interfering non-specific adsorption effects and for the optimization of affinity chromatography performance by a logical and systematic appraisal

of reinforcement effects and, where applicable, kinetic and mechanistic factors. Whilst the necessity for spacer arms interposed between the ligand and matrix was recognized very early in order to alleviate steric interference (12,26,27), it was not until later that it was realized that the aliphatic hydrocarbons commonly employed as spacers could act as hydrophobic ligands in their own right. In a study with pre-assembled AMP ligands containing spacer arms of varying degrees of hydrophilicity and hydrophobicity, it was found that enzymes bound preferentially to ligands tethered via hydrophobic spacer arms and that the notion of constructing adsorbents comprising a ligand attached to a matrix via a hydrophilic arm in order to ameliorate non-specific hydrophobic interactions may not be a viable proposition (28). Alternative strategies of combating these undesirable effects, such as inclusion of low concentrations of water-miscible organic solvents in the buffers (e.g., ethylene glycol, glycerol or dioxane), were adopted as they resulted in dramatically improved recoveries of the released enzyme (29).

Several other advances in ligand selection also had a dramatic effect on the development of the technique of affinity chromatography. Originally, selective adsorbents were fabricated with natural biological ligands as the exquisite selectivity of enzymes, antibodies, receptor and binding proteins and oligonucleotides for their complementary ligands was rational and easily justified on economic grounds. However, experience has shown that the majority of biological ligands are difficult to immobilize with retention of activity and often lead to prohibitively expensive adsorbents that have limited stability in a multi-cycle sterile environment. Paradoxically, the key feature of affinity chromatography, exquisite selectivity, is also its biggest weakness, because off-the-shelf adsorbents other than those with group specificity are often commercially unavailable. Ideal adsorbents for large-scale applications should combine features of selective and non-selective adsorbents, be inexpensive, have general applicability and be stable to a variety of adsorption, elution and sterilization conditions, with specially synthesized quasi-biological ligands offering the best hope of finding general purpose, inexpensive and stable adsorbents.

2.2.1. Synthetic Ligands

The reactive textile dyes are a group of synthetic ligands that have been widely exploited to purify an astounding array of individual proteins (30,31). The archetypal dye, Cibacron blue F3G-A, contains a triazine scaffold substituted with polyaromatic ring systems solubilized with sulphonate or carboxylate functions and decorated with electron withdrawing or donating groups. It has been the subject of intensive research (30) ever since it was found serendipitously to bind to yeast pyruvate kinase when co-chromatographed with blue

dextran on a Sephadex G-200 gel filtration column (32). Subsequent studies demonstrated that it was the reactive chromophore of blue dextran, Cibacron blue F3G-A, that was responsible for binding and not the dextran carrier itself (33,34). Sepharose-immobilized Cibacron blue F3G-A (35) is advantageous for large-scale affinity chromatography as it is low cost, generally available, easily coupled to a matrix and exhibits protein-binding capacities that exceed those of natural ligand media by factors of 10–100 (30). Furthermore, synthetic dyes are almost completely resistant to chemical and enzymatic attack and are hence readily cleaned and sterilized in situ, are less prone to leakage than other ligands and yield high capacity, easily identified adsorbents.

It is believed that these dyes mimic the binding of natural anionic heterocyclic substrates such as nucleic acids, nucleotides, coenzymes and vitamins (36,37). However, concerns over the selectivity, purity, leakage and toxicity of the commercial dyes limited their use and led to the search for improved “biomimetic” dyes and the adoption of rational molecular design techniques (38). For example, inspection of the interaction of Cibacron blue F3G-A with horse liver alcohol dehydrogenase provided a sound basis for rational ligand design. X-ray crystallography and affinity labelling studies showed that the dye binds to the coenzyme-binding domain of the enzyme with the anthraquinone, diaminobenzene sulphonate and triazine rings adopting similar positions as the adenine, adenosine ribose and pyrophosphate groups respectively of NAD^+ (39). It appeared that the terminal aminobenzene sulphonate ring of the dye was bound to the side of the main NAD^+ -binding site in a crevice bounded by the side chains of cationic (Arg/His) residues. Thus, the synthesis, characterization and assessment of a number of terminal ring analogues of the dye confirmed the preference for a small, anionic o- or m-substituted group and substantially improved the affinity and selectivity of the dye for the protein (39). These conclusions have been confirmed with more recent studies with a range of new analogues and demonstrate how the use of modern design techniques can greatly improve the selectivity of biomimetic ligands.

2.2.2. De Novo Ligand Design

The recently acquired ability to combine knowledge of X-ray crystallographic, nuclear magnetic resonance (NMR) or homology structures with defined or combinatorial chemical synthesis and advanced computational tools has made the rational design of affinity ligands even more feasible, powerful, logical and faster (40). The target site on the protein may be a known active site, a solvent-exposed region or motif on the protein surface or a site involved in binding a natural or complementary ligand. However, the design of a complementary affinity ligand is at best only a semi-rational process, as numerous

unknown factors are introduced during immobilization of the ligand. The affinity of the immobilized ligand for the complementary protein is determined partly by the characteristics of the ligand *per se* and partly by the matrix, activation, spacer and coupling chemistry. Studies in free solution with soluble ligands do not fairly reflect the chemical, geometrical and steric constraints imposed by the complex three-dimensional matrix environment. Nevertheless, three distinct approaches to ligand design can be distinguished: first, investigation of the structure of a natural protein–ligand interaction and the use of the partner as a template on which to model a biomimetic ligand (40); second, construction of a molecule which displays complementarity to exposed residues in the target site (41–44); and third, direct mimicking of natural biological recognition interactions (45).

Peptidal templates comprising two or three amino acids have been used to design highly selective affinity ligands for IgG (40–42), kallikrein (46) and elastase (44) and were synthesized by combinatorial substitution of a triazine scaffold with appropriate analogues of the amino acids.

2.2.3. Combinatorial Ligand Synthesis

However, in many cases, there is inadequate or insufficient structural data on the formation of complexes between the target protein and a substrate, inhibitor or binding protein, to design molecules *de novo* to interact with the exposed residues of a specified site and ensure that the ligand has complementary functionality to the target residues. A good example of this approach is the design, synthesis and evaluation of an affinity ligand for a recombinant insulin precursor (MI3) expressed in *Saccharomyces cerevisiae* (43). Preliminary molecular modelling showed that a lead ligand comprising a triazine scaffold substituted with aniline and tyramine, showed significant π - π overlap with the aromatic side chains of B:16-Tyr and B:24-Phe from the biomolecule, and was thus used as a guide to the type of directed solid-phase combinatorial library that might be synthesized. A library of 64 members was synthesized from 26 amino derivatives of bicyclic, tricyclic and heterocyclic aromatics, aliphatic alcohols, fluorenes and acridines substituted with various functionalities. The solid-phase library was screened for MI3 binding and elution, and fractions from each column were analyzed by reversed-phase high performance liquid chromatography by reference to the known elution behaviour of authentic MI3. Under the specified conditions, the most effective ligands appeared to be bisymmetrical ligands substituted with aminonaphthols or aminonaphthoic acids, with very high levels of discrimination being noted with various ring substituents. Modelling studies showed that bisymmetrical bicyclic-ring ligands displayed more complete π - π overlap with the side chain of residues B:16-Tyr and

B:24-Phe, than the single-ring substituents of the original lead compound used to direct library synthesis. However, despite the value of computer modelling in visualizing putative interactions, the complexity of the three-dimensional matrix environment, with largely unknown ligand–matrix, coupling, activation and spacer molecule chemistry interactions, suggests that rational design and combinatorial chemistry together should be evoked to develop effective affinity ligands. Nevertheless, despite these reservations, the symmetrical ligand 23/23 was synthesized *de novo* in solution, characterized and immobilized to agarose beads, whence affinity chromatography of a crude clarified yeast expression system revealed that MI3 was purified on this adsorbent with a purity of >95% and a yield of 90% (43). This study showed that a defined structural template is not required and that a limited combinatorial library of ligands together with the use of parallel screening protocols allows selective affinity ligands to be obtained for target proteins.

One of the most widely used combinatorial technologies is based on biological vehicles as platforms for the presentation of random linear or constrained peptides, gene fragments, cDNA and antibodies. The non-lytic filamentous bacteriophage, M13, and the closely related phages, fd and f1, are the most commonly exploited vectors with random peptides displayed on the surface of the phage by fusion of the desired DNA sequence with the genes encoding coat proteins (47,48). Combinatorial libraries containing up to 10^9 peptides can be generated and selected for the desired activity by ‘biopanning’ of the phage pool on a solid-phase immobilized target receptor. Bound phage particles are eluted, amplified by propagation in *Escherichia coli* and the process repeated several times to enrich iteratively for the peptide with the desired binding properties, and whose sequence is determined from the coding region of the viral DNA. Phage display libraries have been successfully applied to epitope mapping, vaccine development, the identification of protein kinase substrates, bioactive peptides and peptide mimics of non-peptide ligands and are eminently suitable as a source of affinity ligands for chromatography or analysis (49). However, a limitation of the phage display approach is that peptides may only function when the peptide is an integral part of the phage-coat protein and not when isolated in free solution (50). These limitations can be circumvented to some extent by using conformationally constrained peptides (51), although issues relating to retention of their function on optimization, scale-up and use on various solid-phase matrices still remain (52). An alternative approach based on ribosome display for the evolution of very large protein libraries differs from other selection techniques in that the entire procedure is conducted *in vitro* and is particularly appropriate for the screening and selection of folded proteins (53). Other scaffolds exploiting domains from proteins such as fibronectin (“monobodies”), V domains (“minibodies”) or α -helical bacterial

receptor domains (“affibodies”) have been shown to yield specific binders, with usually mM affinities, from libraries of up to 10^7 clones (54).

Recently, it has been shown that peptides of a modest size isolated from a combinatorial library using a simple genetic assay can act as specific receptors for other peptides (55). However, peptide arrays are known to offer advantages, particularly in signal-to-noise ratio and in the chromatographic optimization steps (56). A good example of the use of randomized synthetic peptidomers for the affinity purification of antibodies has been reported (57). The lead peptide mimics *Staphylococcus aureus* protein A in its ability to recognize the Fc fragment of IgG and offers a one-step isolation of 95% pure antibody from crude human serum. Panels of peptides derived from a combinatorial library were also shown to bind human blood coagulation factor VIII (58).

A similar approach to peptide phage display involves the use of oligonucleotide-based combinatorial biochemistry, in which the nucleotides on the DNA polymerase-encoding gene 43 regulatory loop of bacteriophage T4 are randomized (59,60). The so-called systematic evolution of ligands by exponential enrichment (SELEX) technology can yield high affinity/high specificity ligands for virtually any molecular target. Several of the ligands, aptamers, that emerge from this method, where starting libraries may contain up to 10^{14} – 10^{15} sequences, have been shown to have pM–nM affinities for their binding partners. DNA-aptamer affinity chromatography has recently been applied to the purification of human L-selectin from Chinese hamster ovary cell-conditioned medium (61). The aptamer column resulted in a 1500-fold single-step purification of an L-selectin fusion protein with an 83% recovery. **Figure 2** summarizes the various types of affinity ligand and the stages in their development.

2.3. Impact of the Biopharmaceutical Industry

The development of novel therapeutic proteins must rank amongst the most laborious and capital intensive of all industrial activities. The nascent biotechnology industry faces two principal challenges in fulfilling this promise to deliver new therapeutics. The first relates to the production of specified therapeutic proteins at an appropriate price, scale and quality. Many of the potential customers, particularly health service providers, are struggling to contain rising costs and are thus cautious about using high-cost therapies based on biopharmaceuticals. As much as 50–80% of the total cost of biomanufacturing is incurred during downstream processing, purification and polishing. Thus, the need to revise existing production processes to improve efficiency and yields is high on the agenda of many manufacturers. Furthermore, changes in the regulatory climate have shifted the focus of regulation from defining production processes

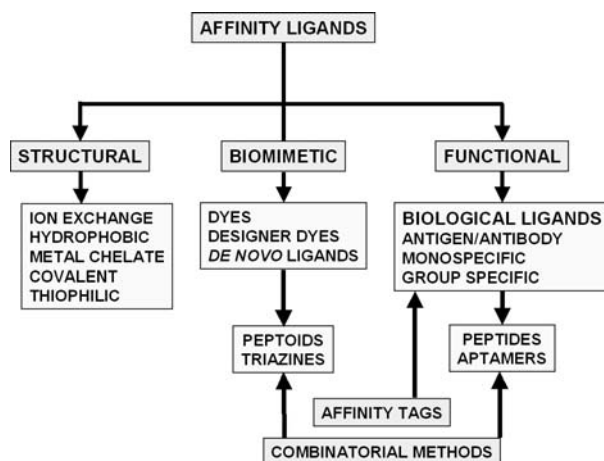


Fig. 2. Types of affinity ligands utilized in the separation of biomolecules.

per se to the concept of the “well-characterized biologic.” Under this regime, the final protein will be required to have defined purity, efficacy, potency, stability, pharmacokinetics, pharmacodynamics, toxicity and immunogenicity. The product should also be analyzed, not only for contaminants such as nucleic acids, viruses, pyrogens, residual host cell proteins, cell culture media, leachates from the separation media and unspecified impurities, but also for the presence of various isoforms, originating from post-translational modifications in the host cell expression system, such as glycosylation, sulphation, oxidation, misfolding, aggregation, misalignment of disulphide bridges and nicking or truncation. A thorough characterization of the potency, purity and safety of proteinaceous drugs using high performance hyphenated techniques is now required.

This new challenge has necessitated a radical re-think of the design and operation of purification processes, with the options being largely dictated by their speed of introduction, effectiveness, robustness and economics. Conventional purification protocols are now being substituted with highly selective and sophisticated strategies based on affinity chromatography (62). This technique provides a rational basis for purification and simulates and exploits natural biological processes such as molecular recognition for the selective purification of the target protein. Affinity chromatography is probably the only technique currently able to address key issues in high-throughput proteomics and scale-up. The principal issue is to devise new techniques to identify highly selective affinity ligands, which bind to the putative target biopharmaceuticals. Not surprisingly, the value of computer-aided design and combinatorial strategies for the design of ultra stable synthetic ligands has been appreciated (43,63).

A further issue of concern to the FDA and relating to both biological and synthetic ligands is that of leakage. The regulatory authorities insist that any biological ligand used in the manufacture of a therapeutic product meet the same requirements as the end product itself. This notion extends even to how the affinity ligand is produced and purified. A good example of this strategy lies in the design, synthesis and chromatographic evaluation of an affinity adsorbent for human recombinant Factor VIIa (**63**). The requirement for a metal ion-dependent immuno-adsorbent step in the purification of the recombinant human clotting factor, FVIIa, and hence scrutiny by the FDA, has been obviated by using X-ray crystallography, computer-aided molecular modelling and directed combinatorial chemistry to design, synthesize and evaluate a stable, sterilizable and inexpensive “biomimetic” affinity adsorbent. The ligand comprises a triazine scaffold bis-substituted with 3-aminobenzoic acid and was shown to bind selectively to FVIIa in a Ca^{2+} -dependent manner. The adsorbent purifies FVIIa to almost identical purity (>99%), yield (99%), activation/degradation profile and impurity content (~ 1000 ppm) as the current immuno-adsorption process, while displaying a 10-fold higher capacity and substantially higher reusability and durability (**63**). A similar philosophy was used to develop synthetic equivalents to Protein A (**40**) and Protein L (**64**).

2.4. The “Omics” Revolution

The “omics” technologies of genomics, proteomics and metabolomics collectively have the capacity to revolutionize the discovery and development of drugs. Genomics specifies the patterns of gene expression associated with particular cellular states, whereas proteomics describes the corresponding protein expression profiles. However, many key aspects of proteomics, such as the concentration, transcriptional alteration, post-translational modification, intermittent or permanent formation of complexes with other proteins or cellular components, compartmentation within the cell, and the modulation of biological activity with a plethora of small effector molecules, are not encoded at the genetic level but influence the function of the protein and can only be clarified by analysis at the protein level. These modulations often play a crucial role in the activity, localization and turnover of individual proteins. The inability of classical genomics to address issues at the protein level in sufficient detail is a crucial shortcoming, as most disease processes develop at this level. Thus, the field of proteomics will require the development of a new toolbox of analytical and preparative techniques that allow the resolution and characterization of complex sets of protein mixtures and the subsequent purification of individual target therapeutic proteins.

Liquid chromatography is regarded as an indispensable tool in proteomics allowing the discrimination of proteins by diverse principles based on reverse-phase, ion exchange, size-exclusion, hydrophobic and affinity interactions (65). The technique is potentially useful not only for the separation of specific groups of proteins, but also for the exploration of post-translational modifications and the study of protein–protein and protein–ligand interactions (66).

Furthermore, the use of affinity chromatography to enrich scarce proteins or deplete over-abundant proteins is a powerful means of enhancing the resolution and sensitivity in two-dimensional electrophoresis (2D-PAGE) or mass spectrometry (MS) analysis. Isotope-encoded affinity tags may represent a new tool for the analysis of complex mixtures of proteins in living systems (67). Alternatively, element-encoded metal chelates may also prove helpful for affinity chromatography, quantification and identification of tagged peptides from complex mixtures by LC-MS/MS (68).

A significant development in affinity techniques for proteomics is the use of fusion tags or proteins for expression and purification (69–71). A large choice of systems is available for expression in bacterial hosts, with a further selection amenable for eukaryotic cells. Amongst the most popular fusion partners for molecular, structural and bioprocessing applications are the polyArg (72), hexaHis-tag (73), glutathione-S-transferase (74) and maltose-binding protein (75). Other less commonly employed expression tags include thioredoxin (76), the Z-domain from Protein A (77), NusA (68), GB1 domain from Protein G (78) and others (79). A recent comparison of the efficiency of eight elutable affinity tags for the purification of proteins from *E. coli*, yeast, *Drosophila* and HeLa extracts shows that none of these tags is universally superior for a particular system because proteins do not naturally lend themselves to high throughput analysis and they display diverse and individualistic physicochemical properties (80). It was found that the His-tag provided good yields of tagged protein from inexpensive, high capacity resins but with only moderate purity from *E. coli* extracts and poor purification from the other extracts. Cellulose-binding protein provided good purification from HeLa extracts. Consequently, affinity tags are invaluable tools for structural and functional proteomics as well as being used extensively in the expression and purification of proteins (81). Affinity tags can have a positive impact on the yield, solubility and folding of their complementary fusion partners. Combinatorial tagging might be the solution to choosing the most appropriate partner in high throughput scenarios (70,81).

2.5. Resolution of Isoforms

Heterogeneity in proteins may arise due to variations in post-translational modifications during the synthesis of a protein in native, recombinant or

transgenic systems. These variations may include altered glycosylation, unnatural or incomplete disulphide bond formation, partial proteolysis, amino- and carboxy-terminal sequence alterations and oxidation or deamidation of amino acids, unnatural phosphorylation or dephosphorylation, myristoylation or sulphation of amino acids. The expressed proteins may then differ in function, kinetics, structure, stability and other properties affecting their biological role. Most proteins produced by recombinant DNA technology for *in vivo* administration are glycosylated and may have glycoform heterogeneity due to variable site occupancy of the sugar moieties on the protein or due to variations in the carbohydrate sequence. Consequently, in the future, it may be important to be able to isolate and purify recombinant glycoforms with defined glycosylation and biological properties prior to administration because mixtures of isoforms could have serious side effects on human health. The concepts of rational design and solid-phase combinatorial chemistry have been used to develop affinity adsorbents for glycoproteins (81,82). The strategy for the resolution of glycoforms involves generation of synthetic ligands that display affinity and selectivity for the sugar moieties on glycoproteins but which have no interaction with the protein *per se*. A detailed assessment of protein-carbohydrate interactions from a number of known X-ray crystallographic structures was used to identify key residues that determine monosaccharide specificity and which were subsequently exploited as the basis for the synthesis of a library of glycoprotein-binding ligands (82,83). The ligands were synthesized using solid-phase combinatorial chemistry and were assessed for their sugar-binding ability with several glycoproteins. Partial and completely deglycosylated proteins were used as controls. A triazine-based ligand, bis-substituted with 5-aminoindan, was identified as a putative glycoprotein-binding ligand, because it displayed particular affinity for mannoside moieties. These findings were substantiated by interaction analysis between the ligand and mannoside moieties through NMR experiments (83). ¹H-NMR studies and molecular modelling suggested involvement of the hydroxyls on the mannoside moiety at C-2, C-3 and C-4 positions. Small peptides selected from a library of 62,000 chemically synthesized peptides have also been shown to display some selectivity for binding monosaccharides, although their application in the chromatographic resolution of glycoproteins was not established (84).

3. Conclusions

This review has looked at the history, current status and prospects for affinity chromatography and identified techniques that are able to rationalize the design and selection of affinity ligands for the purification of pharmaceutical proteins.

Two strategies are evident: first, screening for target binding to large combinatorial libraries of peptides, oligonucleotides, antibodies, various natural binding motifs and synthetic ligands and, secondly, the introduction of a design step to reduce the size of the directed libraries. The approach adopted depends to a large extent on what information is available at the outset; if structural data is at hand, the design approach is possible, whilst in the absence of such information, which may be the case in many proteomics applications, a combinatorial screen would be the only route available. The present author prefers the 'intelligent' approach, because it drastically reduces the chemistry and screening necessary to identify a lead ligand. Nevertheless, combinatorial screening is still required to obviate many of the unknowns involved in the interaction of protein with solid-phase immobilized ligands. A key aspect of this system is that the chromatographic adsorption and elution protocols can be in-built into the total package at the screening stage and therefore lead to very rapid conversion of a hit ligand into a working adsorbent. Rapid screening techniques based on fluorescently labelled proteins (85), ELISA (64), surface plasmon resonance (86) and the quartz crystal microbalance (87) are now available.

The use of synthetic ligands offers a number of advantages for the purification of pharmaceutical proteins. First, the adsorbents are inexpensive, scaleable, durable and reusable over multiple cycles. Secondly, the provision of a ligand with defined chemistry and toxicity satisfies the regulatory authorities. Finally, the exceptional stability of synthetic adsorbents allows harsh elution and cleaning-in-place and sterilization-in-place protocols to be used. These considerations remove the potential risk of prion or virus contamination, which may arise when immuno-adsorbents originating from animal sources are used. Other types of affinity ligand based on peptide, oligonucleotide or small protein libraries are likely to be less durable under operating conditions, which employ harsh sterilization, and cleaning protocols.

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I

**VARIOUS MODES OF AFFINITY
CHROMATOGRAPHY**

Immobilized Metal Ion Affinity Chromatography of Native Proteins

Adam Charlton and Michael Zachariou

Summary

Immobilized metal affinity chromatography (IMAC) is a common place technique in modern protein purification. IMAC is distinct from most other affinity chromatography technologies in that it can operate on a native, unmodified protein without the need for a specialized affinity “tag” to facilitate binding. This can be particularly important where a protein of interest is to be separated from a complex mixture such as serum or an environmental isolate. Relying on the interaction of specific surface amino acids of the target protein and chelated metal ions, IMAC can provide powerful discrimination between small differences in protein sequence and structure. Additionally, IMAC supports have been demonstrated to function effectively as cation exchangers, allowing for two modes of purification with a single column. This chapter provides methodologies to perform IMAC in its most fundamental form, that of the interaction between histidine and immobilized metal ions, those that enable purification of proteins that lack surface histidines and the operation of IMAC supports in cation exchange mode.

Key Words: IMAC; protein purification; native protein; cation exchange.

1. Introduction

Immobilized metal affinity chromatography (IMAC) of proteins is a high resolution liquid chromatography technique. It has the ability to differentiate a single histidine residue on the surface of a protein (**1**), it can bind proteins with dissociation constants of 10^{-5} – 10^{-7} (**2**) and has had wide application in the field of molecular biology for the rapid purification of recombinant proteins.

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Since the first set of work was published describing the immobilization of metal ions using a chelating agent covalently attached to a stationary support to purify proteins (3,4), there have been several modifications and adaptations of this technique over the years. The fundamental approach remains to use immobilized metal ions, and, in particular borderline Lewis metal ions such as Cu^{2+} , Ni^{2+} and Zn^{2+} , to purify proteins on the basis of their histidine content (3).

In 1985, there were indications that electrostatic interactions were also occurring between proteins and immobilized Fe^{3+} -iminodiacetic acid (IDA) stationary phases (5), and in 1996, it was demonstrated that IMAC adsorbents in general could also be used in pseudo-cation exchange mode, independently of histidine interaction (6). Yet another mode of interaction involved in the IMAC of proteins was the mixed mode interactions involving aspartate and/or glutamate surface residues on proteins along with electrostatic interactions, again independent of histidine interactions (7). It is the purpose of this work to describe the methodologies involved in the traditional histidine-based IMAC interactions, the mixed mode interactions involving aspartate, glutamate and electrostatic interactions and then the purely electrostatic interactions. The reader is referred to reviews of IMAC of proteins for a more detailed perspective (8,9,10,11).

The traditional use of IMAC for proteins has been to select proteins on the basis of their histidine content. The approach uses a chelating agent immobilized on a stationary surface to capture a metal ion and form an immobilized metal chelate complex (IMCC). The chelating agent has usually been the tridentate IDA, despite a plethora of chelating stationary supports available for such work (12). Generally, Cu^{2+} , Ni^{2+} and Zn^{2+} have been used in this mode, but other metal ions such as Co^{2+} , Cd^{2+} , Fe^{2+} and Mn^{2+} have also been examined as the metal ions of choice. Histidine selection by the IMCC exploits the preference of borderline Lewis metals (*see* **ref. 13** for a review of the concept of hard and soft acids and bases and their preferred interactions) to accept electrons from borderline Lewis bases such as histidine. With a pK_a of 6, histidine will be able to donate electrons effectively at $\text{pH} > 6.5$ and thus bind to the IMCC, although this may vary depending on the microenvironment the histidine finds itself in. Once the protein has bound, a specific elution can be deployed by using imidazole, which is the functional moiety of histidine. Alternatively, the pH may be decreased to < 6.5 to prevent histidine from donating electrons thereby inducing elution of the bound protein.

An alternative use of IMAC can be to exploit the interaction between aspartate and glutamic acid residues on the surface of proteins with IMCCs. This was first demonstrated in 1992 when a protein with no surface accessible histidines bound to a group of hard Lewis metal ions immobilized on

8-hydroxyquinoline (7,14). In this context, at $\text{pH} > 4$, the carboxyl groups of aspartate and glutamate are fully deprotonated and able to donate electrons. By including imidazole and $\geq 0.5 \text{ M NaCl}$ in the binding buffer, any histidine or electrostatic interactions will be quenched, leaving aspartate and glutamate as the only amino acids able to donate electrons and interact with the IMCC. This type of interaction can be further enhanced by using hard Lewis metal ions as part of the IMCC so as to exploit the preference of hard Lewis metal ions for hard bases such as those found in oxygen-rich compounds like the carboxyl groups of aspartate and glutamate. This type of interaction has been observed to occur predominantly in the pH region of 5.5–6.5 and may involve some electrostatic component. Above this pH range, electrostatic influence becomes more pronounced, and the IMCCs exhibit pseudo-cation exchange behaviour.

The traditional use of IMAC has involved the inclusion of 0.5–1 M NaCl in the binding buffer to prevent the protein from interacting with the IMCC on the basis of non-specific electrostatic interactions. The contribution of such interactions comes from charges presented to the protein by unoccupied chelate sites, a variety of hydrolytic species that exist on the IMCC, as well as the metal ion itself (6,15). The overall contribution results in a net negative charge on the IMCC, which becomes increasingly negative as the pH becomes more alkaline. This phenomenon occurs with any IMCC and will vary depending on the metal ion and immobilized chelator involved. By encouraging this phenomenon instead of quenching it, IMAC can be used in cation exchange mode. In this mode, the binding buffers are of low ionic strength ($< 0.1 \text{ M}$) and a pH of between 5 and 6.5, and elution can be afforded either by increasing the ionic strength and/or by increasing pH . Furthermore, by including imidazole in the binding buffer, histidine interactions can be minimized if desired. The use of IMCCs in cation exchange mode has been demonstrated previously using model proteins as well as crude mixtures (6,16). With the large number of possible combination of metal ion and chelators, use of IMAC in pseudo-cation exchange mode provides a number of options during purification beyond the traditional use of IMAC or the use of traditional cation exchangers.

2. Materials

2.1. Purification of Proteins Using IMAC Based on Histidine Selection

1. Stationary support: Chelating Sepharose FF (Amersham-Pharmacia Biotech, UK).
2. Charge solution: 0.1 M CuNO_3 .
3. Metal rinsing solution: 0.2 M acetic acid pH 4.
4. Pre-equilibration buffer: $0.2 \text{ M K}_2\text{HPO}_4/\text{KH}_2\text{PO}_4$ + 0.5 M NaCl pH 7.4.

5. Equilibration buffer: 0.02 M $\text{K}_2\text{HPO}_4/\text{KH}_2\text{PO}_4$ + 0.5 M NaCl pH 7.4.
6. Elution buffer: 0.05 M imidazole + 0.5 M NaCl pH 7.
7. Regeneration buffer: 0.2 M EDTA + 0.5 M NaCl pH 8.

2.2. Purification of Proteins Using IMAC Based on Non-Histidine Selection and High Ionic Strength

1. Stationary support: Chelating Sepharose FF (Amersham-Pharmacia Biotech).
2. Charge solution: 0.05 M metal salts.
3. Metal rinsing solution: 0.05 M acetic acid + 0.1 M KNO_3 .
4. Pre-equilibration buffer: none.
5. Equilibration buffer: 0.03 M morpholinoethane sulphonic acid (MES) + 0.03 M imidazole + 0.5 M NaCl pH 5.5/pH 6.
6. Elution buffer: 0.03 M MES + 0.03 M imidazole + 0.1 M K_2HPO_4 + 0.14 M NaCl pH 5.5/pH 6.
7. Regeneration buffer: 0.2 M EDTA + 0.5 M NaCl pH 8.
8. Storage solution: 0.01 M NaOH.

2.3. Purification of Proteins Using IMAC in Pseudo-Cation Exchange Mode

1. Stationary support: Chelating Sepharose FF (Amersham-Pharmacia Biotech, UK).
2. Charge solution: 0.05 M metal salts.
3. Metal rinsing solution: 0.05 M acetic acid + 0.1 M KNO_3 .
4. Pre-equilibration buffer: none.
5. Equilibration buffer: 0.03 M MES + 0.03 M imidazole + 0.05 M NaCl pH 5.5/pH 6.
6. Elution buffer: 0.03 M HEPES + 0.03 M imidazole + 0.5 M NaCl pH 8.
7. Regeneration buffer: 0.2 M EDTA + 0.5 M NaCl pH 8.
8. Storage solution: 0.01 M NaOH.

3. Method

3.1. Purification of Proteins Using IMAC Based on Histidine Selection

1. Wash packed Cu-IDA column with 2 column volumes (CV) of metal rinsing solution, 0.2 M acetic acid pH 4 (*see Note 1*).
2. Wash column with 5 CV of Milli Q water.
3. Pre-wash packed Cu-IDA column with 10 CV of 0.2 M $\text{K}_2\text{HPO}_4/\text{KH}_2\text{PO}_4$ + 0.5 M NaCl, pH 7.4.
4. Equilibrate the column with 10 CV of 20 mM $\text{K}_2\text{HPO}_4/\text{KH}_2\text{PO}_4$ + 0.5 M NaCl, pH 7.4.
5. Confirm equilibration by measuring pH and conductivity. Continue equilibration until pH and conductivity of effluent matches equilibration buffer.
6. Load sample containing target molecule ensuring the sample pH is between pH 7 and 7.2. As a general rule, loading linear velocities should be between 10 and

33% the maximum operating linear velocity allowed by the stationary support (see **Note 2**), that is, 70–235 cm/h for the stated support. Assume a loading of no more than 1 mg target protein per ml of stationary support (see **Note 3**). However, target proteins in ratio volumes of 300:1 cell culture per support have been successfully loaded by the author (see **Note 4**).

- 7. Wash stationary support with 10 CV of equilibration buffer at the loading linear velocity or until the $A_{280\text{ nm}}$ reading is at baseline (see **Note 5**).
- 8. Subsequent wash steps can be carried out if deemed necessary (see **Table 1**). If a wash step is required follow **step 7** with the appropriate wash buffer.

Table 1

Wash type	Effect	Comment
Glycine, Arginine, ~0.5 M NH_4Cl and pH 7	Mild eluents that compete for Ni with histidine	These are mild eluents that will not elute the His-tag protein but may displace weaker bound proteins
Non-amine salts, e.g., ~0.5 M – 1 M NaCl; in 20 mM Imidazole + 50 mM NaCl pH 7	Will disrupt any non-specific electrostatic interactions	Such interactions are common in IMAC particularly if the equilibration and wash steps had <0.2 M salt present
Non-ionic detergents, e.g., Triton, Tween No more than 1% v/v Chaotropic agents, e.g., 4 M Urea, e.g., 4 M Guanidine-HCl	Disrupts hydrophobic interactions	In particular will disrupt any interactions between the spacer arm and proteins as well as any protein–protein hydrophobic interactions that may be occurring with the target protein. This is more effective when applied as part of the equilibration conditions so as to prevent such interactions from taking place. Inclusion of detergent will also assist in removing lipids or DNA (20)
Decreasing pH (<7) and/or increasing imidazole concentration (>20 mM)	Disrupts the histidine bond to the IMCC	This step can also be used to elute the target protein so care must be taken to select a condition that ensures good differentiation between contaminants and target protein

IMCC, Immobilized Metal Chelate Complex.

9. Elute protein with up to 5 CV of 50 mM imidazole + 0.5 M NaCl pH 7 at 33% of the recommended maximum linear velocity of the stationary support, 235 cm/h for Chelating Sepharose FF. If this is insufficient to effect elution, imidazole should be taken up to 0.5 M. If the target molecule is still bound then elution with 0.5 M imidazole + 0.5 M NaCl at pH 5.5 should be tried (*see Note 6*). Samples should be examined on sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) for purity (*17*).
10. After elution of the target protein, the column should be regenerated using 3 CV of 0.2 M EDTA + 0.5 M NaCl pH 8. Washing linear velocity is not critical as long as it does not exceed the maximum linear velocity of the stationary support (*see Note 7*).
11. Wash with 10 CV of Milli Q water.
12. Load column with 2 CV of 0.1 M CuNO₃ (*see Notes 8 and 9*).
13. Wash with 10 CV of Milli Q water.
14. Store column at 4°C.

3.2. Purification of Proteins Using IMAC Based on Non-Histidine Selection and High Ionic Strength (*see Note 8*)

1. Load column with 2 CV of 50 mM metal salt.
2. Wash packed Mⁿ⁺-IDA column with 2 CV of metal rinsing solution, 50 mM acetic acid + 0.1 M NaCl pH 4 (*see Note 1*).
3. Wash column with 5 CV of Milli Q water.
4. Equilibrate packed Mⁿ⁺-IDA column with 10 CV of 30 mM MES + 30 mM imidazole + 0.5 M NaCl pH 5.5 or 6 (*see Note 10*). Confirm equilibration by measuring pH and conductivity. Continue equilibration until pH and conductivity of effluent matches equilibration buffer.
5. Load sample containing target molecule that has been pre-equilibrated in equilibration buffer. As a general rule, loading linear velocities should be between 10 and 33% the maximum operating linear velocity allowed by the stationary support (*see Note 2*), that is, 70–235 cm/h for the stated support. Assume a loading of no more than 1 mg target protein per ml of stationary support (*see Note 3*). However, target proteins in ratio volumes of 300:1 cell culture per support have been successfully loaded by the author (*see Note 4*).
6. Wash stationary support with 10 CV of equilibration buffer at the loading linear velocity or until the A_{280 nm} reading is at baseline (*see Note 5*).
7. Subsequent wash steps can be carried out if deemed necessary (*see Table 2*). If a wash step is required follow **step 6** with the appropriate wash buffer.
8. Elute protein with up to 5 CV of 30 mM MES + 30 mM imidazole + 0.1 M K₂HPO₄ + 0.14 M NaCl pH 5.5 or 6 at 33% of the recommended maximum linear velocity of the stationary support, 235 cm/h for Chelating Sepharose FF. If this is insufficient to effect elution, phosphate should be taken up to 0.2 M. If the target molecule still remains bound, then elute with 30 mM HEPES + 30 mM

Table 2

Wash type	Effect	Comment
Oxygen-rich buffers such as phosphate, glutamate, aspartate, acetate; at 0.1 M strength	Eluents competing with metal ion for aspartate and glutamate surface residues	This step can also be used to elute the target protein, so care must be taken to select a condition that ensures good differentiation between contaminants and target protein. Acetate is the mildest and phosphate is the strongest eluent from this set
Non-ionic detergents, e.g., Triton, Tween No more than 1% v/v Chaotropic agents, e.g., 4 M Urea, e.g., 4 M Guanidine-HCl	Disrupts hydrophobic interactions	In particular will disrupt any interactions between the spacer arm and proteins as well as protein-protein hydrophobic interactions that may be occurring with the target protein. This is more effective when applied as part of the equilibration conditions so as to prevent such interactions from taking place. Inclusion of detergent will also assist in removing lipids or DNA (20)
Increasing pH (>6) and/or increasing phosphate concentration (>0.1 M)	Disrupts the aspartate and glutamate bonds to the IMCC as well as disrupting electrostatic interactions if protein has bound in mixed mode	This step can also be used to elute the target protein, so care must be taken to select a condition that ensures good differentiation between contaminants and target protein

IMCC, Immobilized Metal Chelate Complex.

imidazole + 0.1 M K_2HPO_4 + 0.14 M NaCl pH 8. Samples should be examined on SDS-PAGE for purity (17).

9. After elution of the target protein, the column should be regenerated using 3 CV of 0.2 M EDTA + 0.5 M NaCl pH 8. Washing linear velocity is not critical as long as it does not exceed the maximum linear velocity of the stationary support (see **Note 7**).
10. Wash with 10 CV of Milli Q water.
11. Wash with 5 CV of storage solution, 0.01 M NaOH, as a preservative.
12. Store column at 4°C.

3.3. Purification of Proteins Using IMAC in Pseudo-Cation Exchange Mode (see Note 11)

1. Carry out **steps 1–3, Subheading 3.2.**
2. Equilibrate packed M^{n+} -IDA column with 10 CV of 30 mM MES + 30 mM imidazole + 0.05 M NaCl pH 5.5 or 6 (see **Note 10**). Confirm equilibration by measuring pH and conductivity. Continue equilibration until pH and conductivity of effluent matches equilibration buffer.
3. Carry out **steps 5–7, Subheading 3.2.**
4. Subsequent wash steps can be carried out if deemed necessary (see **Table 3**). If a wash step is required follow **step 6, Subheading 3.2** with the appropriate wash buffer.
5. Elute protein with up to 5 CV of 30 mM MES + 30 mM imidazole + 0.5 M NaCl pH 5.5 or 6 at 33% of the recommended maximum linear velocity of the stationary support, 235 cm/h for Chelating Sepharose FF. If this is insufficient to effect elution, NaCl should be taken up to 1 M. If the target molecule still remains

Table 3

Wash type	Effect	Comment
Non-ionic detergents, e.g., Triton, Tween No more than 1% v/v	Disrupts hydrophobic interactions	In particular will disrupt any interactions between the spacer arm and proteins as well as protein–protein hydrophobic interactions that may be occurring with the target protein. This is more effective when applied as part of the equilibration conditions so as to prevent such interactions from taking place. Inclusion of detergent will also assist in removing lipids or DNA (20)
Increasing pH (>6)	Adjusting the pH to beyond the isoelectric point of the protein will make it more negative and interfere with the interactions on the adsorbent	This step can also be used to elute the target protein, so care must be taken to select a condition that ensures good differentiation between contaminants and target protein
Increasing ionic strength to between 0.5 M and 1 M	Disrupts electrostatic interactions	NaCl is used traditionally as an eluent, however, other similar salts could also be used

bound, then elute with 30 mM HEPES + 30 mM imidazole + 1 M NaCl pH 8. Samples should be examined on SDS-PAGE for purity (17).

6. Follow **steps 10–12, Subheading 3.2.**

4. Notes

1. All columns pre-charged with metal should be washed with acid to release any loosely bound metal ions.
2. A slow loading velocity improves the diffusion of proteins (particularly, large proteins) through pores and onto the IMCC and hence improves yields. The stated linear velocities have been derived from the author's personal experience and will vary depending on the stationary support. For example, Poros supports can have linear dynamic capacities, in some cases up to 7000 cm/h, before decreases in capacities are observed. The maximum linear velocity of the support stated for these methods, Chelating Sepharose FF, is 700 cm/h (18). Care must also be taken to ensure that if prolonged loading times are chosen, the target protein is not subject to destabilizing factors such as proteolysis or any intrinsic instability such as deamidation or oxidation and should be monitored during the process. In these instances, the molecule stability needs to take precedence over slow loading velocities.
3. This amount is conservative relative to the manufacturer's claims of 5 mg of protein per ml Chelating Sepharose FF resin (18). However, capacities of <5 mg/ml have been observed often, particularly when cell culture is used where proteins with a higher His content are present as well as the presence of a large amount of free amino acids used as media components which can bind through their α -amino groups (19). Hence, allowing for excess stationary support will reduce the possibility of your target molecule not binding because of capacity issues. Furthermore, any non-specific interactions that may occur because of excess stationary support not interacting with the target molecule is addressed through the proposed stringent pre-equilibration, equilibration and washing regimens.
4. In these instances, significant metal leaching may occur during loading, reducing the capacity of the IMAC support but not below 1 mg of protein per ml of support.
5. If monitoring $A_{280\text{ nm}}$ note that imidazole absorbs at this wavelength and so achieving baseline should only be relative to the absorbance of the equilibration buffer at $A_{280\text{ nm}}$.
6. If this does not elute your protein, then it is highly probable that harsher specific elution conditions are required or that it is non-specifically bound to the backbone matrix. In the former case, elution with regeneration buffer can be tried so as to strip the metal of the chelator. In the non-specific case, chaotropes (e.g., 4 M Urea) or detergents (e.g., 1% Triton X-100) can be tried.
7. This step should also be considered as a last resort for eluting the target molecule in the event that all other elution conditions failed.
8. Metal ions that could be used for this work are preferably the hard Lewis metal ions such as Fe^{3+} and any of the lanthanides. Hard Lewis metal ions such as Ca^{2+}

could also be used; however, a good chelating stationary phase to use this metal ion in IMAC for the purification of proteins does not exist commercially. Al^{3+} is also another example, however, the commercially available 8-hydroxyquinoline support would be more useful over IDA stationary phases for this metal ion. Borderline Lewis metal ions like Cu^{2+} can also be used in this mode (7,14).

9. Not all supports should be stored charged with metal ions. Silica-based supports should be stored free of metal ion and only charged when required. The charged metal ion causes a localized low pH microenvironment that can damage these supports over time, decreasing the life expectancy of the column.
10. Under these conditions, histidine interaction with the IMCC should be quenched (7). Furthermore, the use of oxygen-rich buffers such as phosphate, acetate, carbonate and so on should be avoided whilst equilibrating hard Lewis IMCCs. Sulphonic acid-based buffers such as MES and other Good's buffers used at ≤ 20 mM have minimal interference and can be used.
11. Any metal ion that can be hydrolyzed can be employed with any commercially available chelating stationary support for this section of work.

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Affinity Precipitation of Proteins Using Metal Chelates

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Summary

Metal affinity precipitation has been successfully developed as a simple purification process for the proteins that have affinity for the metal ions. The copolymers of vinylimidazole with *N*-isopropylacrylamide are easily synthesized by radical polymerization. When loaded with Cu(II) and Ni(II) ions, these copolymers are capable of selectively precipitating proteins with natural metal-binding groups or histidine-tagged recombinant proteins.

Key Words: Metal chelate affinity precipitation; thermoresponsive copolymers; affinity macroligands; thermoprecipitation; bioseparation; recombinant histidine-tagged proteins.

1. Introduction

Development of efficient and fast purification protocols in bioseparation has always been a challenging task. With the rapid advancement of gene technology, it has been possible to get any desired protein product, but the recovery of such products still poses a major problem. Affinity techniques for protein purification provide means to purify a specific protein from a complex mixture. Many affinity-based systems have been developed in recent years for the rapid purification of recombinant proteins. The methods utilize specific interactions between an affinity tag (usually a short peptide with specific molecular recognition properties, such as polyhistidines (**1–3**), STREP tag (**4**), maltose-binding

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protein (5), cellulose-binding domain (6), glutathione-S transferase (7), and thioredoxin (8)) and an immobilized ligand.

The concept of using metal chelating in affinity techniques, like immobilized metal-affinity chromatography (IMAC), was a breakthrough introduction (9). IMAC technique has a wide application in protein purification particularly when dealing with recombinant proteins (10,11). This offers a number of important advantages over other “biospecific” affinity techniques for protein purification particularly with respect to ligand stability, protein loading, and recovery (10). The technique is generally based on the selective interaction between metal ions like Cu(II) or Ni(II) that are immobilized on the solid support and electron donor groups on the proteins. The amino acids histidine, cysteine, tryptophan, and arginine have strong electron donor groups in their side chains, and the presence of such exposed residues is an important factor for IMA-binding properties (12). In the recombinant proteins, polyhistidine tag (His-tag) fused to either the N- or C-terminal end of the protein has become the selective and efficient separation tool for applying in IMAC separation. Proteins containing a polyhistidine tag are selectively bound to the matrix, whereas other cellular proteins are washed out. IMAC has also been utilized for the separation of nucleic acids through the interactions of aromatic nitrogens in exposed purines in single-stranded nucleic acids (13,14). At present, it is one of the most popular and successful methods used in molecular biology for the purification of recombinant proteins. The widespread application of metal affinity concept has also recently gained usefulness by adopting the technique in a non-chromatographic format like “metal chelating affinity precipitation” (2,15–17). Such separation strategy makes metal affinity methods more simple and cost-effective when the intended applications are for large-scale processes.

This chapter discusses affinity precipitation method using metal chelating polymers for selective separation of proteins. Affinity precipitation is a relatively new technique, which allows protein separation from crude homogenates with rather high yields compared to conventional chromatography (18). By combining the versatile properties of metal affinity with affinity precipitation, the technique presents enormous potential as a simple and selective separation strategy. Affinity precipitation methods have two main approaches that have been described in the literature (18), namely, precipitation with homo- or hetero-bifunctional ligands. Previously, there have been a few attempts to utilize the metal affinity concept in affinity precipitation methods in homo-bifunctional format. The addition of a bis-ligand at an optimum concentration creates a cross-linked network with the target protein provided the latter has two or more metal-binding sites. The cross-linked protein–bis–ligand network precipitates from the solution eventually. The first such application was reported by Van Dam et al. (19) when human hemoglobin and sperm whale hemoglobin

were quantitatively precipitated in model experiments with bis-copper chelates. In another study, Lilius et al. (20) described the purification of genetically engineered galactose dehydrogenase with polyhistidine tail by metal affinity precipitation. The histidines functioned as the affinity tail and the enzyme could be precipitated when the bis-zinc complex with ethylene glycol-bis-(β -aminoethyl ether)*N,N,N',N'*-tetraacetic acid, EGTA (Zn)₂, was added to the protein solution. However, in general, the application of affinity precipitation with homo-bifunctional ligands has been quite limited (21). The requirement of a multi-binding functionality of the target protein and slow precipitation rate restricts the use of this type of affinity precipitation process (19,22,23). The concentration dependence and the risk of terminal aggregate formation further complicates its use (22).

On the other hand, hetero-bifunctional format of affinity precipitation is a more general approach, wherein affinity ligands are covalently coupled to soluble–insoluble polymers. The ligand selectively binds the target protein from the crude extract. The protein–polymer complex is precipitated from the solution by a simple change of the environment property (pH, temperature, or ionic strength). Finally, the desired protein is dissociated from the polymer, and the latter can be recovered and reused for another cycle (18).

In metal chelating affinity precipitation, metal ligands are covalently coupled to the reversible soluble–insoluble polymers (mainly thermoresponsive polymers) by radical copolymerization. The copolymers carrying metal chelating ligands are charged with metal ions and the target protein binds the metal-loaded copolymer in solution via the interaction between the histidine on the protein and the metal ion. The complex of the target protein with copolymer is precipitated from the solution by increasing the temperature in the presence of NaCl, whereas impurities remain in the supernatant and are discarded after the separation of precipitate. The precipitated complex is solubilized by reversing the precipitation conditions, and the target protein is dissociated from the precipitated polymer by using imidazole or EDTA as eluting agent. The protein is recovered from the copolymer by precipitating the latter at elevated temperature in presence of NaCl. The metal chelating affinity precipitation technique is presented schematically in **Fig. 1**. The technique uses mainly the thermoresponsive polymers, and these polymers constitute a major group of reversibly soluble–insoluble polymers. Among these, poly(*N*-isopropylacrylamide), poly(vinyl methyl ether), and poly(*N*-vinylcaprolactam) have been widely studied and used for various applications (24). Copolymers of *N*-isopropylacrylamide (NIPAM) were mostly used in affinity precipitation methods. Poly(NIPAM) has a critical temperature of precipitation at about 32°C in water and changes reversibly from hydrophilic below this temperature to hydrophobic above it (25). This transition occurs rather abruptly at what

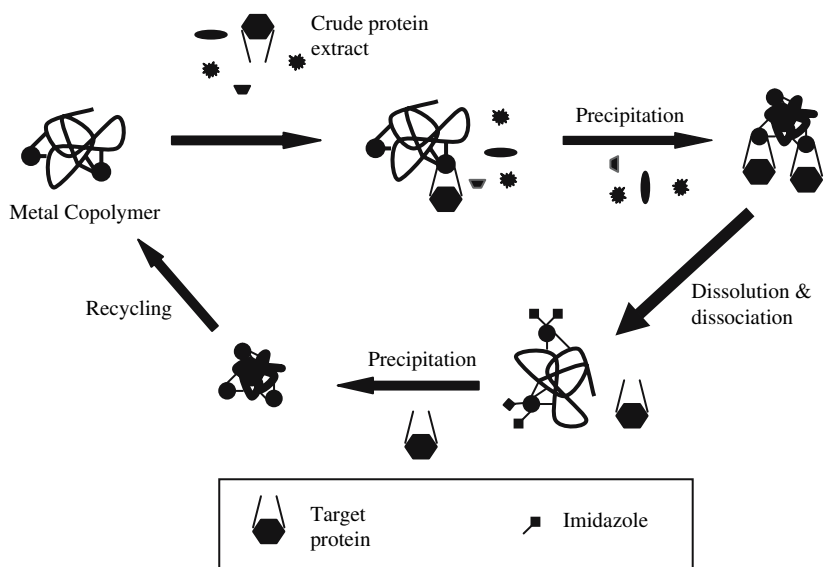


Fig. 1. Scheme of metal chelate affinity precipitation of proteins (reproduced from ref. 37).

is known as cloud point. The lowest cloud point on the composition cloud point diagram is designated as the lower critical solution temperature (LCST). Poly(NIPAM) has no reactive groups to be used directly for coupling of affinity ligand, thus, NIPAM copolymers were used as macroligands.

Traditionally, polydentate carboxy-containing ligands like iminodiacetic acid (IDA) or nitrilotriacetic acid (NTA) have been quite successful in IMAC for metal chelating-mediated purification of proteins (26). The ligands co-ordinate well with the metal ion and still leave coordinating sites on the metal ion available for binding the target protein. Such ligands, however, show some limitations in metal chelating affinity precipitation when copolymerized with NIPAM (27). The introduction of highly charged comonomers (at neutral conditions) such as IDA or NTA into the polymer results in a drastic decrease in the efficiency of precipitation with temperature compared with the behavior of NIPAM homopolymer. Negatively charged moieties render the macromolecule more hydrophilic and hinder the aggregation and precipitation of the polymer. The phase transitions of such copolymers after metal loading have been above 35°C, which makes its application limited to thermostable proteins (27).

The breakthrough in this direction came when a new ligand, imidazole, was successfully incorporated into NIPAM, and the copolymer achieved efficient precipitation (17). Copolymers of vinylimidazole (VI) with NIPAM, poly(VI-NIPAM), can be synthesized by radical polymerization in aqueous solution where

VI concentrations up to 25 mol% can be incorporated in the copolymer. Imidazole is a monodentate ligand in Cu complexes. Up to four imidazoles bind to one Cu(II) ion, the $\log K$ (where K is association constant, M^{-1}) for each imidazole ligand is decreasing from $\log K_1 = 3.76$ for binding the first imidazole ligand to $\log K_4 = 2.66$ for binding the fourth imidazole ligand (28). The binding of single imidazole ligand to the Cu(II) ion in solution is much weaker compared to the binding of tridentate IDA [$\log K = 11$, (29)]. On the other hand, when Cu(II) ion forms a complex with four imidazole ligands, the combined binding constant $\log K = \log K_1 + \log K_2 + \log K_3 + \log K_4 = 12.6\text{--}12.7$. The strength of this complex is close to that of Cu(II) ion complex with poly(1-vinylimidazole), $\log K = 10.64\text{--}14.21$ (28–30) and comparable with the binding of tridentate IDA ligand, $\log K_4 = 5.5\text{--}6$. When coupled to solid matrices, imidazole ligands are spatially separated, and the proper orientation of the ligands to form a complex with the same Cu(II) ion is unlikely and the imidazole ligands are not used for IMA chromatography (17). In solution, the flexible polymer like poly(VI-NIPAM) can adopt a solution-phase conformation where two to three imidazole ligands are close enough to form a complex with the same Cu(II) ion providing significant strength of interaction (*see Fig. 2*). It is clear that not all available

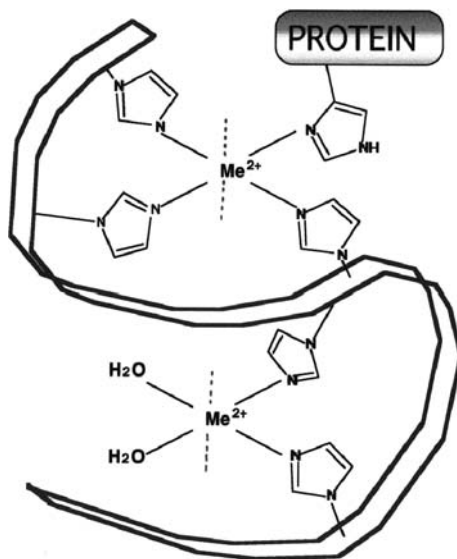


Fig. 2. Imidazole–metal complex formation of flexible poly[vinylimidazole-*N*-isopropylacrylamide (VI-NIPAM)] copolymer with surface His-containing protein. Each metal ion coordinate with two or three imidazole groups in the poly(VI-NIPAM) copolymer (reproduced from **ref. 15**)

coordination sites of the metal ion are occupied by imidazole ligands of the polymer. The unoccupied coordination sites of the metal ion could be used for complex formation with the protein molecule via histidine residues on its surface.

A Cu(II) charged copolymer of poly(VI-NIPAM) can also be applied for the separation of single-stranded nucleic acids such as RNA from double-stranded linear and plasmid DNA by affinity precipitation (31). The separation method utilizes the interaction of metal ions to the aromatic nitrogens in exposed purines in single-stranded nucleic acids (13–14).

Very recently, a metal affinity purification method for His-tagged proteins based on temperature-triggered precipitation of the chemically modified elastin-like proteins (ELPs) biopolymers have been demonstrated (16). ELPs are biopolymers consisting of the repeating penta-peptide, VPGVG. They behave very similar to poly(NIPAM) polymers and have been shown to undergo reversible-phase transitions within a wide range of conditions (32,33). By replacing the valine residue at the 4th position with a lysine in a controlled fashion, metal-binding ligands such as imidazole can be specifically coupled to the free amine group on the lysine residues, creating the required metal coordination chemistry for metal affinity precipitation. ELPs with repeating sequences of $[(VPGVG)_2(VPGKG)(VPGVG)_2]_{21}$ were synthesized, and the free amino groups on the lysine residues were modified by reacting with imidazole-2-carboxyaldehyde to incorporate the metal-binding ligands into the ELP biopolymers. Biopolymers charged with Ni(II) were able to interact with a His-tag on the target proteins based on metal coordination chemistry. Purifications of two His-tagged enzymes, β -D-galactosidase and chloramphenicol acetyltransferase, were used to demonstrate the application of metal affinity precipitation using this new type of affinity reagent. The bound enzymes were easily released by addition of either EDTA or imidazole. The recovered ELPs were reused with no observable decrease in the purification performance.

Other types of metal chelating polymers for affinity precipitation of proteins were reported recently by synthesizing highly branched copolymers of NIPAM and 1,2-propandiol-3-methacrylate (GMA), poly(NIPAM-co-GMA) using the technique of reversible addition fragmentation chain transfer polymerization using a chain transfer agent that allows the incorporation of imidazole functionality in the polymer chain ends. The LCST of the copolymers can be controlled by the amount of hydrophobic and GMA comonomers incorporated during copolymerization procedures. The copolymers demonstrated LCST below 18°C and were successfully used to purify a His-tagged BRCA-1 protein fragment by affinity precipitation (34,35).

It is important to mention here that metal chelating copolymers as discussed above for metal chelating affinity precipitation for proteins are not yet available commercially. Thus for carrying out affinity precipitation of proteins using

metal interaction, the copolymers of poly(VI-NIPAM) need to be synthesized and this is further discussed in **Subheading 2**.

2. Materials

2.1. Chemicals

1. NIPAM (Aldrich, Steinheim, Germany).
2. 1-Vinylimidazole (Aldrich).
3. Ammonium persulfate (Bio-Rad, Solna, Sweden).
4. Tetraethylene methylenediamine (TEMED) (Bio-Rad).
5. Bicinchoninic acid (BCA) protein assay reagent (Sigma, St Louis, MO, USA)

All other reagents were of analytical grade.

2.2. Reagents

1. Metal affinity macroligands:
 - i) Cu(II)-poly(*N*-vinylimidazole-*co*-isopropylacrylamide).
 - ii) Ni(II)-poly(*N*-vinylimidazole-*co*-isopropylacrylamide).
2. Metal charging solution: 0.1 M CuSO₄ or 0.1 M NiSO₄.
3. Precipitating solution: 2 M NaCl.
4. Washing buffer: 10 mM phosphate buffer, pH 7.4.
5. Elution solution: 200 mM imidazole, pH 7.4 or 50 mM EDTA, pH 8.

2.3. Synthesis of Poly(Vinylimidazole-Isopropylacrylamide) Copolymer

1. Copolymerization of VI to NIPAM is carried out by radical polymerization. Add 0.5 ml of VI to 3 g NIPAM (copolymer solution I) and 1 ml of VI to 3 g NIPAM (copolymer solution II) in 30 ml each of degassed water separately and flush with nitrogen gas for 5 min. This gave total polymer concentration of 10% and incorporated 15 and 25 mol% of VI in the synthesized copolymers respectively (*see* **Notes 1** and **2**).
2. Initiate the polymerization by adding 100 µl of a freshly prepared solution of ammonium persulfate (10% w/v), followed by adding 10 µl TEMED in above mixture (*see* **step 1**) and incubate at room temperature overnight.
3. Precipitate the synthesized copolymers of poly(VI-NIPAM) by adding 2 M NaCl to the final concentration of 0.5 M and heat at 60°C for 5 min. Collect the precipitate by decanting the supernatant (*see* **Note 3**).
4. Dissolve the copolymer precipitate collected above (*see* **step 3**) in 60 ml water by stirring at 4°C, till all the precipitate is completely solubilized. Again precipitate the polymer as above (*see* **step 3**) and collect the precipitate by centrifugation at 13,000 g for 5 min.

5. Repeat the precipitation and dissolution of the copolymer as above (*see step 4*). Measure the dry weight of the copolymer solution after drying the copolymer solution at 80°C overnight.
6. Finally, dissolve the copolymer solution (*see step 5*) in water to give final 2% (w/v, dry weight) copolymer solution.

2.4. Metal Loading to the Poly(VI-NIPAM) Copolymer

1. The Cu(II) and Ni(II) loading to the above copolymer solutions of poly(VI-NIPAM) is carried out separately by adding an excess of copper or nickel sulfate solutions as follows. Add 10 ml each of 0.1 M CuSO₄ or 0.1 M NiSO₄ solutions to 20 mL of 2% poly(VI-co-NIPAM) solutions I and II, respectively, slowly while stirring. Stir the metal ion-loaded copolymers for 1 h at room temperature (*see Notes 4 and 5*).
2. Precipitate the metal-loaded copolymers by adding 2 M NaCl to a final concentration of 0.4 M and heat at 40°C for 5 min during continuous mixing using a glass rod (*see Note 6*).
3. Decant the supernatants and dissolve the precipitates of metal ion–copolymer complex in 15 ml of water by stirring at 4°C till the copolymer is completely solubilized (*see Note 7*).
4. Repeat the precipitation and dissolution step of the metal copolymer three times as above (*see Subheading 2.3, step 4*) to completely wash out the unbound metal ions (*see Note 6*). Determine the dry weight of the metal–copolymer solution after drying the copolymer solutions at 80°C overnight.
5. Finally, the metal ion-loaded copolymers are dissolved in water to give a 2% (w/v, dry weight) solution.

3. Methods

3.1. Purification of His-Tag Proteins or Proteins with Natural Metal-Binding Groups

3.1.1. Binding Stage

1. Add protein extract (1–5 ml; depending upon the concentration of the target protein) to 5 ml of the 2% metal ion–copolymer solution and make the total volume up to 10 ml by adding the required volumes of distilled water (*see Notes 8–10*).
2. Keep the samples for a short period on ice (to prevent polymer precipitation) before the pH is adjusted to 7 (for Cu(II) copolymers) and 7.5 (for Ni(II) copolymers) (*see Notes 11 and 12*).
3. Incubate the polymer–protein mixture at 4°C for 30 min with constant mixing on a rotating shaker.
4. Precipitate the protein–copolymer complex by adding 2.5 ml 2 M NaCl to give final concentration of 0.4 M NaCl.

5. Incubate at 30°C for 10 min. The precipitated protein–polymer complex is centrifuged at 14,000 g for 5 min at room temperature (*see* **Notes 13** and **14**).

3.1.2. Washing Stage

1. Collect the supernatant and solubilize the protein–copolymer precipitate in 5 ml of washing buffer containing 0.15 M NaCl.
2. Precipitate again by adding 1.5 ml 2 M NaCl and incubate at 30°C for 10 min. The precipitate is collected by centrifugation at 14,000 g for 5 min at room temperature.

3.1.3. Recovery Stage

1. Dissociate the target protein from the protein–polymer complex by dissolving the protein–polymer pellet in 5 ml of elution solution (50 mM EDTA buffer, pH 8 for His-tag proteins, or 200 mM imidazole buffer, pH 7.4 for proteins with natural metal-binding groups), while the mixture is kept on ice (*see* **Note 15**).
2. Precipitate the polymer by adding 1.5 ml 2 M NaCl and incubate at 30°C for 10 min leaving free target protein in the solution.
3. Collect the dissociated protein in the supernatant by centrifugation of the polymer precipitate at 14,000 g for 5 min (*see* **Note 16**).
4. If required repeat the protein dissociation (*see* **steps 1–3**), to achieve complete recovery of the bound protein (*see* **Note 17**).
5. Dialyze the recovered protein in 1 l of 10 mM phosphate buffer, pH 7.4, or any other buffer suitable for the protein to remove the metal ions leached out by EDTA elution or imidazole buffer (*see* **Note 18**).
6. Determine the protein concentration by BCA method (**36**), using bovine serum albumin as standard (*see* **Note 19**).

3.1.4. Recycling of the Metal Copolymer

1. Recover the metal–copolymer pellet and wash by dissolving and reprecipitating again (*see* **Subheading 3.1.3, steps 2** and **3**).
2. Recycle the metal copolymer by dissolving the recovered polymer in 5 ml of distilled water, pH 7, and use for further cycles of affinity precipitation of the proteins (*see* **Note 20**).

3.1.5. Reloading with Metal

1. To increase the protein-binding capacity of the recycled metal copolymer to its original capacity, reload the metal copolymers with fresh portions of metal ions. Add 5 ml 0.01 M CuSO₄ or NiSO₄ to 10 ml of the recycled metal–copolymer solutions, slowly while stirring. Stir and incubate metal-reloaded copolymers for 1 h at room temperature (*see* **Note 20**).
2. Wash the metal-reloaded copolymers by following the **steps 2–5** as described in **Subheading 2.4**.

4. Notes

1. In synthesizing the poly(VI-*co*-NIPAM) copolymers, it is important to optimize the concentration of VI comonomer. Very high concentrations of VI affect drastically the precipitation behavior of the copolymer. Poly(VI-*co*-NIPAM) copolymers incorporated with about 30 mol% of VI can be precipitated from the solution by heating, but above that concentration, precipitation of the copolymer becomes difficult (27). High concentrations of VI are useful in providing high metal-binding capacity for the copolymer. However, this can lead to poor precipitation behavior, which makes it difficult to recover the synthesized copolymer and thus gives low yields. VI concentrations in the range of 15–25 mol% are considered to be optimal, which provide efficient precipitation properties for the copolymers and also give sufficient binding capacity for the metal ions.
2. Separating the precipitated poly(VI-NIPAM) copolymer from the liquid phase will mainly depend upon the type of precipitate aggregates formed. In such cases, making the copolymers with about 6–12% (w/v, total comonomer concentrations in the starting reaction mixture) ensures easily aggregated precipitate formation. The precipitate can simply be collected by decanting the liquid, which also allows the precipitate to resolubilize fast by adding excess water. Too low concentrations of comonomers (<5% w/v) in the reaction mixture produce dispersed precipitates, which can only be recovered by centrifugation, and long times may be needed to resolubilize the polymer pellet.

This is the main reason that in subsequent steps of copolymer washings, recovery of the polymer precipitate by centrifugation is required as the polymer concentrations in the solution decreases after re-solubilizing the precipitates in excess of water. On the other hand, too high initial concentrations (>20% w/v) of comonomers in the reaction mixture will produce gel type copolymers, which can be rather difficult to handle for subsequent precipitation and solubilization.

3. Washing and recovery of the synthesized poly(VI-*co*-NIPAM) copolymer is carried out by precipitating the copolymer by heating in the presence of 0.4–0.6 M NaCl. Keeping temperatures as high as possible up to 60°C and incubating for about 5–10 min at this temperature will ensure better aggregation of the copolymer. The pH of the copolymer solution is an important factor for the better precipitation of the copolymer and should be kept in the range of 7–8. The incorporation of relatively hydrophilic imidazole moieties hinder the hydrophobic interactions of the native poly(NIPAM) and results in substantial increase in the precipitation temperature (27). The effect is more pronounced at lower pH values (<6) where imidazole moieties are protonated and hence render more hydrophilicity to the copolymer molecule.
4. On loading poly(VI-*co*-NIPAM) copolymer with metal salts such as CuSO₄ or NiSO₄, the hydrophilicity of the copolymer is even more increased (metal ions induce more positive charges) (15). The copolymers are not precipitated even upon increasing the temperature up to 70°C. For complete precipitation of metal-loaded copolymers, conditions promoting hydrophobic interactions and reducing mutual repulsion caused by similar charges should be employed. The

increase in ionic strength, and hence decrease in charge repulsion, by adding NaCl facilitates precipitation of metal-bound copolymers. At relatively moderate salt concentrations of 0.4 M NaCl, the metal-bound copolymers are precipitated quantitatively below 25°C (see Fig. 3).

5. The poly(VI-co-NIPAM) showed good chelating capacity of metal ions. Cu(II) and Ni(II) ion binding to poly(VI/NIPAM) increases initially during the first 45–60 min and then levels off toward the equilibrium level (37). The capacities of poly(VI-NIPAM) (at same VI concentrations in the copolymer) for chelating Cu(II) ions are slightly more than chelating Ni(II) ions, which can further lead to different capacities for binding the target protein (15). Our studies have shown that about two and three imidazole groups co-ordinate with each Cu(II) and Ni(II) ion, respectively (15). With about two to three imidazole ligands bound to the metal ion, one could expect binding strength of $\log K = 6-9$ (28,30), providing a significant strength of interaction. At 15 and 25 mol% VI copolymers (i.e., 1.35 and 2.16 μmol /mg copolymer, respectively), the Cu(II) and Ni(II) ion content bound to the copolymers was the same (about 0.6–0.7 μmol /mg copolymer). So the precipitation efficiency of Cu(II)- and Ni(II)-loaded copolymer at 15 and 25 mol% of VI, respectively for target protein, can be almost the same.
6. Washing of the metal-loaded copolymers three to four times with water (pH 6–7) and by adding 0.4–0.6 M NaCl ensures complete removal of unbound or

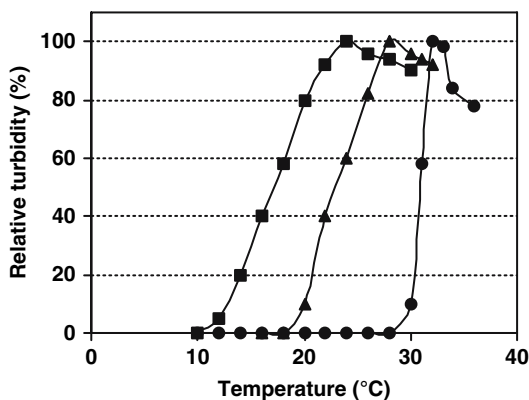


Fig. 3. Thermoprecipitation of poly[*N*-isopropylacrylamide (NIPAM)] and metal-loaded copolymers of poly[vinylimidazole (VI)-NIPAM] from aqueous solution monitored as turbidity at 470 nm. Maximum turbidity was taken as 100%, and relative turbidities were calculated from that. Polymer concentration 1 mg/ml. Poly(NIPAM) precipitation (●-●-); Ni(II)-poly(VI-NIPAM) precipitation (▲-▲-); Cu(II)-poly(VI-NIPAM) precipitation (■-■-), at different temperatures in presence of 0.4 M NaCl. VI concentration was 15 and 25 mol% in case of Cu(II) and Ni(II) copolymers, respectively (reproduced from ref. 2)

loosely bound metal ions. No pre-washing is needed with buffers containing low amounts of imidazole, like 10–50 mM imidazole buffer ideally used in traditional IMAC for pre-washing. Plain poly(NIPAM) shows almost negligible non-specific interactions toward metal ions, and there is no entrapment of the metal ions within the soluble polymer.

7. If precipitated pellets of copolymer take a long time for solubilization, incubate it on ice and use glass rod to mechanically promote the dissolution of the polymer pellet. Ensure that the NaCl concentrations entrapped inside the pellet is very low, otherwise dilute by adding more water.
8. The capacity of the metal copolymer for binding the target protein needs to be optimized by adding different amounts of the protein to the metal–copolymer solution, and precipitation of the target protein above 90% is generally achieved.
9. Protein extracts preserved in azide or anti-proteases, such as benzamidine, should be dialyzed to remove these compounds before applying to the metal copolymers, as these can lead to poor precipitation/binding efficiency of the target protein.
10. Cell supernatants containing large amount of small peptides should also be dialyzed to remove the peptides, which generally compete for the metal binding and hence decrease the precipitation efficiency for the target protein (2).
11. Optimum precipitation of His-tagged proteins or proteins containing natural metal-binding residues (especially histidines on the surface) with Cu(II) copolymer can occur in the pH range of 6–7, whereas for Ni(II) copolymer, this range can be slightly higher (pH 7–8) (2). Quantitative precipitation of the target protein (above 90%) can be achieved in these pH ranges. On either side of this optimum pH range, there can be decreases in the efficiency of precipitation of the target molecules. Under acidic conditions below pH 6, the imidazole groups in histidines are partially unprotonated (38) and hence show a low propensity to coordinate metal ions. In alkaline conditions above pH 8, the decrease in the selective binding of proteins is probably caused by the binding of other proteins through increased competition for hydroxyl ions or coordination with partially deprotonated α -amino groups (10).
12. Cu(II) copolymers show higher capacity for protein precipitation than Ni(II) copolymers, while the latter show slightly higher selectivity for the target protein than Cu(II) copolymers (2).
13. Do not use refrigeration during centrifugation of the copolymer precipitates, as it can solubilize the copolymer.
14. If the polymer precipitate is not sufficiently recovered during centrifugation, make an empty run of the centrifuge (which can slightly increase the temperature inside the centrifuge) before the precipitate is centrifuged.
15. The bound protein can be dissociated directly by dissolving the precipitate of protein–metal–copolymer complex in elution buffer. His-tag proteins bind strongly to the metal-loaded copolymers and are eluted only by using EDTA buffer (2). The elution with imidazole buffer shows very low efficiency for dissociating the His-tag proteins (2). On the other hand, imidazole buffer can

completely recover the proteins bound through natural metal-binding residues (39,40).

16. To ensure that the polymer precipitate is efficiently precipitated and completely recovered by centrifugation, warm the recovered supernatant in the presence of 0.4 M NaCl. That no visual turbidity changes occur in the supernatant means the polymer is precipitated completely. If the supernatant turns cloudy, it means that the polymer was not precipitated completely. In such cases, recover the precipitate from the supernatant by slightly increasing both temperature and NaCl concentration.
17. The metal affinity precipitation technique optimized in the present format using the set of copolymers as discussed here can be essentially used for purifying proteins that are relatively thermostable. However, it is possible to establish copolymers with more hydrophobic side chains that can be utilized to carry out precipitation at low temperatures as well.
18. The metal ions leached out with the recovered protein after EDTA or imidazole elutions can be removed by dialysis. Determine the protein amounts or enzyme activity of the recovered protein after the dialysis.
19. Protein measurements using BCA reagent show no interferences with high salt concentrations or traces of polymers if present in the samples.
20. The metal poly(VI-co-NIPAM) copolymers recovered after the first use of affinity precipitation of the protein can be reused for the precipitation of the same amount of protein in the subsequent cycles, provided the copolymer is re-charged with fresh portions of metal ions.

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Immunoaffinity Chromatography

Stuart R. Gallant, Vish Koppaka, and Nick Zecherle

Summary

Immunonaffinity chromatography is a powerful technique for rapid purification of proteins. In a single-step purification, it is possible to purify proteins for testing in model systems and for conducting enzyme kinetic studies. Because the immunoaffinity-purified proteins are typically >90–95% pure, depending on the starting material, interference from remaining contaminants is rare. This method describes an immunoaffinity chromatography technique for purifying proteins from over-expression in mammalian cell culture. The immobilization of the monoclonal antibody or polyclonal antiserum is presented. Conditions for purifying up to milligram quantities of protein are given, including a representative chromatogram.

Key Words: Immunoaffinity chromatography; antibody; IgG; mammalian cell culture; purification.

1. Introduction

Human immunoglobulins are capable of a high degree of diversity, on the order of 5×10^{13} distinct antibodies maybe expressed by B cells (*1*). As a biological reagent, antibodies provide the backbone of many analytical and preparative laboratory methods, including ELISA, Western blot, and immunoaffinity chromatography.

Immunoaffinity chromatography offers a rapid method of obtaining purified protein that is relatively insensitive to the composition of feedstream. Either

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monoclonal antibodies or purified polyclonal antisera maybe used (2). Because elution from a polyclonal column often requires extremely low pH, some consideration should be given to the stability of the protein of interest below pH 3 if that strategy is pursued. In contrast to the use of a polyclonal column, use of a monoclonal column usually allows for milder elution conditions; this is achieved by screening for a monoclonal antibody of intermediate affinity.

A number of companies (*see Note 1*) offer economical production of monoclonal antibodies or purified polyclonal antisera (3). These antibody sources can be purified using standard protocols with Protein A and Protein G affinity chromatography (2). Subsequently, the antibodies may be covalently attached to activated resin using a range of chemistries (4). Two of the most common approaches are attachment through surface lysines and site-directed attachment through the carbohydrate chains (*see Note 2*).

In the following method, a purified polyclonal antiserum is coupled to Amersham Biosciences NHS-activated Sepharose 4 HP (5,6). The resin is a 34- μ m average particle size agarose resin appropriate for protein purification using low pressure chromatography equipment. The resulting immunoaffinity resin is used to purify a recombinant protein from mammalian cell culture.

2. Materials

2.1. Coupling

1. NHS-activated Sepharose HP column, 5 ml (Amersham/GE Healthcare P/N 17-0717-01) (*see Note 3*).
2. Protein G purified antiserum (2). A concentration of >10 mg/ml is convenient; lower concentrations may be used with recirculation during immobilization.
3. Vivascience Vivaspin 15R Hydrosart 30k spin filters (VS15RH21) or equivalent.
4. Phosphate-buffered saline (PBS).
5. Slide-A-Lyzer Dialysis Cassettes (Pierce, <http://www.piercenet.com>) or equivalent dialysis tubing. The 3–12 ml size will be convenient for 5 ml of antiserum.
6. Pierce Coomassie Plus protein assay kit (23236) or equivalent.
7. Solution of hydrochloric acid, 1 mM, on ice.
8. Disposable syringes, 10 ml and 60 ml. Alternatively, a peristaltic pump may be used to pass solutions over the column. The use of syringes can make the immobilization more convenient; however, a flow rate of 2 ml/min should not be exceeded in order to insure that the resin is not damaged by high pressure.
9. Coupling buffer: 0.2 M ammonium bicarbonate, 0.5 M NaCl, pH 8.3.
10. Blocking buffer: 0.2 M Tris-HCl, 0.5 M NaCl, pH 8.3 (we have also used 0.2 M glycine, 0.5 M NaCl, pH 8.3 for this step).
11. Coupling wash buffer: 0.1 M sodium acetate, 0.5 M NaCl, pH 4.
12. Storage buffer: PBS with 0.05% sodium azide or 0.2 M imidazole, 0.5 M NaCl, pH 7.

2.2. Immunoaffinity Chromatography

1. Equilibration buffer: PBS.
2. Elution buffer: 0.1 glycine-HCl, pH 2.25.
3. Cleaning: 100 mM sodium phosphate, 1.5 M NaCl, pH 7.4.
4. Neutralizing buffer: 2 M Tris-HCl, pH 8.6.
5. Fraction collection tubes, 10 ml; screw cap conical centrifuge tubes are convenient.
6. GE Amersham AKTAExplorer or equivalent, including fraction collector.

3. Method

3.1. Coupling

1. Thaw purified antiserum. For a 5-ml pre-packed NHS Sepharose column, the antiserum should be approximately 5 ml at a concentration of 10 mg/ml. If the antibody is more dilute, it can be concentrated (*see step 2*). Affinity-purified antiserum is typically exchanged (by dialysis or diafiltration) into PBS for storage.
2. If the antiserum is substantially more dilute than 10 mg/ml, use Vivaspin 15R Hydrosart 30k spin filters to concentrate the antiserum. Add up to 15 ml of antiserum solution to concentrator and centrifuge at a maximum centrifugal force of 3000 *g*. Stop the centrifuge periodically in order to observe the remaining volume. Do not overconcentrate the antiserum, as this will result in precipitation.
3. Dialyze the antiserum into coupling buffer using dialysis cassettes or dialysis tubing. Wet the dialysis membrane prior to beginning dialysis. Add the sample and dialyze while slowly stirring with a magnetic stir bar. A ratio of at least 100:1 should be maintained for the dialysis (1 l of dialysis buffer for each 10 ml of sample to be dialyzed).
4. Measure total protein concentration using Bradford protein assay. This value will be used later to calculate coupling efficiency.
5. Wash NHS-activated Sepharose HP column with ice-cold 1 mM HCl. Use 5–10 column volumes, 25–50 ml, at a flow rate of 2 ml/min (60 cm/h). The solution may be passed using a peristaltic pump, or using a disposable syringe.
6. Inject antiserum solution into column. The antiserum will remain in contact with the resin for 2–4 h at room temperature or overnight at 2–8°C. If the entire antiserum solution is greater than the column volume, recycle the excess through the column during the immobilization at a flow rate of 2 ml/min (60 cm/h) for the time period specified above.
7. After immobilization, collect the uncoupled antibody for a Bradford protein assay to verify coupling efficiency (*see Note 4*).
8. Remove the uncoupled antibody by passing at least 3 column volumes of blocking buffer at 2 ml/min (60 cm/h).
9. Replace the blocking buffer with 3 column volumes of coupling wash buffer, then wash with a further 3 column volumes of blocking buffer. This solution remains in the column for 30 min at room temperature to block unreacted NHS sites.

10. Flush the column successively with 3 column volumes each of coupling wash buffer, blocking buffer and coupling wash buffer at 2 ml/min (60 cm/h).
11. Store at 2–8°C in PBS with 0.05% sodium azide (or in 0.2 M imidazole, 0.5 M NaCl, pH 7) to prevent microbial growth.
12. Using the initial and final antisera titers measured by Bradford protein assay, calculate the coupling efficiency as:

$$\text{Coupling Efficiency} = \frac{(\text{Final Antisera Titer in Coupling Solution} \times \text{Final Volume})}{(\text{Initial Antisera Titer} \times \text{Initial Volume})}$$

Coupling efficiency should exceed 90%.

3.2. Immunoaffinity Chromatography

1. Remove storage buffer and replace with equilibration buffer at a flow rate of 1 ml/min (30 cm/h, *see Note 5*).
2. Load preparation (*see Note 6*).
 - a. Cell culture supernatant is clarified to 0.2- μ m filtration using a combination of centrifugation and filtration (*see Note 7*). Verify quantitative product yield during clarification using activity assay.
 - b. Clarified cell culture fluid should be held at 2–8°C until loading on the antibody column. For extended storage, sterility of the load should be maintained. Sodium azide, 0.05%, can be added to the harvest (as a protection against microbial growth) provided that there is no loss of target protein activity.
3. Pass the load of 0.03 mg of target protein per ml of resin (*see Note 8*) over the antibody column at a flow rate of 1 ml/min (30 cm/h, *see Note 5*). Collect the flow-through as fractions (20% of the load per fraction is convenient). The flow-through fractions may be analyzed for activity to verify that the capacity of the column has not been exceeded.
4. Wash the column with equilibration buffer until the $A_{280 \text{ nm}}$ trace returns to baseline (~20–30 column volumes). Retain the wash fraction for activity analysis.
5. Add 0.12 ml of neutralization buffer to each of 20 fraction collection tubes; these will be required in the next step. It is important that the fractions be neutralized as they emerge from the column to maximize the preservation of protein activity.
6. Elute using 48 ml of elution buffer at 1 ml/min. Collect the column eluate in fractions of 2.4 ml (0.5 column volume) using the fraction collection tubes from the previous step. The final volume of each fraction (including 0.12 ml of neutralization buffer) will be approximately 2.5 ml.
7. Pass 25 ml (3–5 column volumes) of cleaning buffer at 1 ml/min.
8. Pass 25 ml (3–5 column volumes) of storage buffer at 1 ml/min and store column at 2–8°C until the next use of the column.

A sample chromatogram for an immunoaffinity purification is shown in **Fig. 1**. In this figure, the large flow-through peak can be seen. This flow-through peak is comprised of contaminating proteins, DNA, and other molecules cleared by the immunoaffinity chromatography. The quality of the eluate can be judged from the sodium dodecyl sulfate–polyacrylamide gel electrophoresis

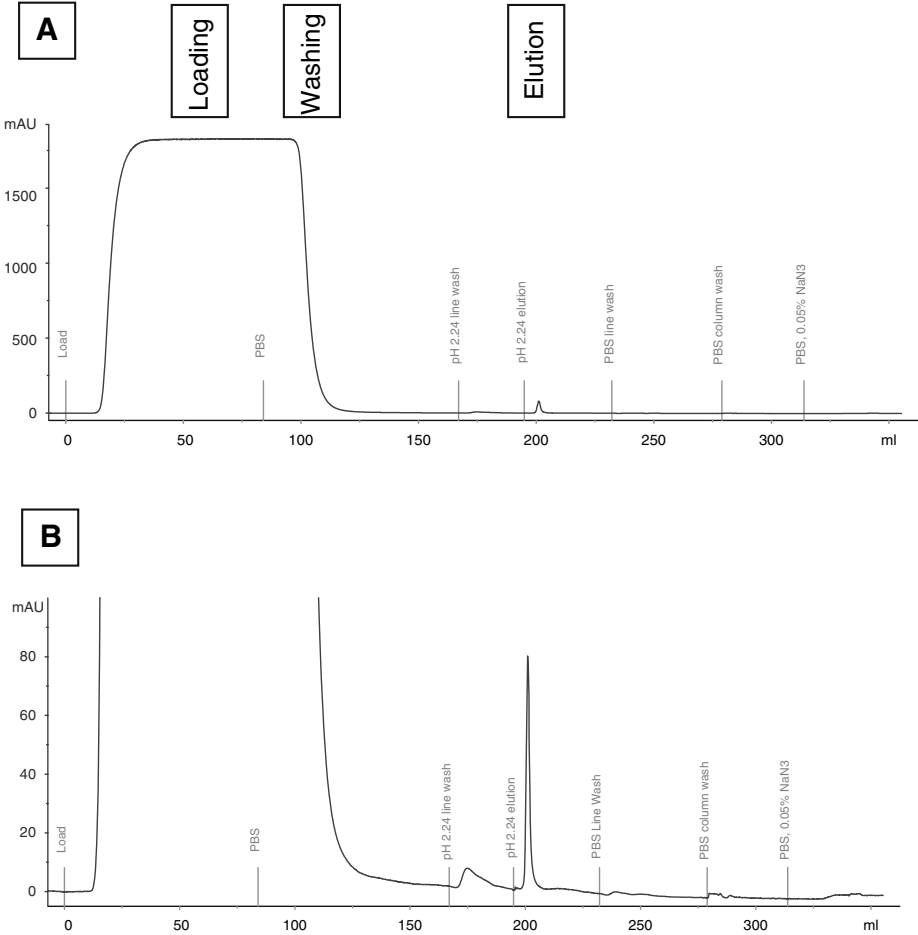


Fig. 1. Representative $A_{280\text{ nm}}$ chromatograms of immunoaffinity chromatography at two magnifications. **(A)** At lower magnification, the relative clearance of various contaminants can be seen by comparing the flow-through peak to the elution peak. **(B)** At higher magnification, the elution peak can be seen to be quite sharp. It is preceded by a low flat peak that has no significance; this section of the chromatogram was generated during line flushing of the chromatography system.

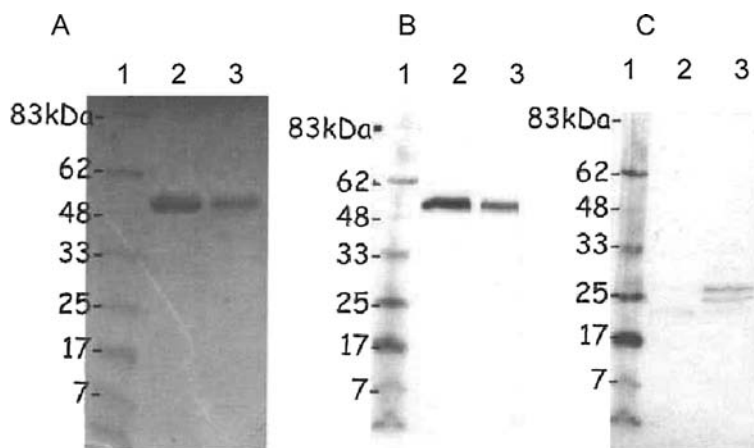


Fig. 2. Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) and Western blot of immunoaffinity eluate with MW standards (lane 1), reference protein (lane 2), immunoaffinity eluate (lane 3). (A) SDS–PAGE gel stained with Coomassie Blue, (B) anti-target Western blot, and (C) anti-host cell Western blot.

and Western blots seen in **Fig. 2**. Only a small amount of host cell protein remains as a contaminant of the immunoaffinity-purified protein.

4. Notes

1. A large number of companies provide these services. Two that the authors have found to be reliable and economical are Covance (<http://www.covance.com>) and Sierra Biosource (now Celliance, <http://www.celliancecorp.com>). Some approximate times for production for polyclonal sera is 3 months and for monoclonal hybridoma development is 6 months.
2. Some considerations when selecting the activated support include that the pore size should be adequate for the antibody and the protein of interest. Because a layer of bound antibody extends approximately 10 nm away from the pore wall, antibody immobilization can significantly narrow a 50-nm pore. Relatively, large pore-activated supports (in the range of 100–300 nm) are available through Millipore in their Prosep line of activated supports; Millipore has aldehyde activated controlled pore glass, as well as glyceryl CPG that must be oxidized to the aldehyde form. Coupling through the lysines with an aldehyde-activated support is a particularly effective means of covalently attaching antibodies. Evaluation of several different methods of coupling for coupling efficiency is useful at the beginning of a project. Measuring coupling efficiency through UV absorbance is only appropriate if the coupling reaction does not release a UV-absorbent product. For example, NHS chemistry releases a UV-absorbent NHS group for each covalent bond formed during coupling, so another method of protein concentration determination than UV

absorbance must be used during coupling with NHS-activated supports. Vendors who offer activated supports include GE (<http://www.amershambiosciences.com>), Millipore (<http://www.millipore.com>), BioRad (<http://www.biorad.com>), JT Baker (<http://www.jtbaker.com>), and Tosoh Bioscience (<http://www.tosohbiosep.com>).

3. NHS-activated Sepharose HP is only available in prepacked HiTrap columns; however, NHS-activated Sepharose 4 Fast Flow is available as unpacked gel in a range of resin volumes. The unpacked gel has the advantage of allowing packing of a broad range of column sizes.
4. As NHS interferes with absorbance measurements near 280 nm, $A_{280\text{ nm}}$ will not be effective for determination of coupling efficiency.
5. A flow rate of between 1 and 5 ml/min (30–150 cm/h) may be applicable for a 2.5 cm (long) by 1.6 cm (in diameter) column of NHS-activated Sepharose HP. This resin has an average diameter of 34 μm and should provide relatively low pressure drop. However, excessive pressure should be avoided as this can damage the packing of the column (normally the resin is not damaged as it is compressible).
6. If the load has previously been partially purified by some other means (ion exchange chromatography, precipitation, and so on), the purification will be enhanced by putting the load into a good loading buffer such as 10 mM HEPES, 500 mM NaCl, 0.01% Tween 80, pH 7. This can be accomplished by dialysis (described above) of the load into the buffer or by dilution of the protein solution with the buffer until the desired pH is reached (dilution should not be less than 1 part load to 2 parts buffer). This prepared load should be 0.2 μm filtered and loaded onto the column as described above.
7. Clarification to a final 0.2- μm filtration can be accomplished at small scale by centrifugation followed by vacuum filtration with a glass fiber prefilter (provided that the protein of interest does not bind to the glass prefilter). Centrifugation at $G_t = 10^6$ sec will remove cells and large cell debris. Small insoluble particles that can foul the column will be removed by 0.2- μm filtration. If only filtration is to be employed, the following filter train works quite well for most mammalian cell culture supernatants: Sartorius Sartopure PP2 1.2- μm filter followed by a Sartopore 2 0.45/0.2- μm filter.
8. The amount of product to load depends on a number of factors, including protein size, load composition, and chromatography resin. In this example, 0.03 mg of target protein per ml of resin was loaded. Loading more than the capacity of the column will result in loss of the product in the flow-through and wash.

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Dye Ligand Chromatography

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Summary

Dye affinity chromatography is a purification technique offering unique selectivities and high purification factors. Dye ligands may act as substrate analogs, offering affinity interactions with their corresponding enzymes. This chapter describes a dye ligand chromatography technique for purifying proteins from overexpression, in mammalian cell culture. The method begins with batch binding in order to rapidly select binding and elution conditions. Subsequently, gradient elution is employed to maximize the selectivity of the final packed bed chromatography method. Conditions for purification of a protein from mammalian cell culture on Cibacron blue are given with an accompanying sample chromatogram.

Key Words: Dye ligand chromatography; Cibacron blue; mammalian cell culture; affinity chromatography; purification.

1. Introduction

Dye ligand chromatography offers the convenience and high capacity of ion-exchange chromatography in combination with unique selectivities that can allow purification of some proteins difficult to purify by any other means (1–3). Today, many of the major chromatography resin suppliers (GE, Tosoh Bioscience, Prometic, and others) manufacture dye ligand chromatography supports. Frequently, these resins now have linkages stable to 1 M NaOH sanitization, making reuse over many cycles possible.

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Binding and elution conditions on dye ligand chromatography are a function of several variables (*see Table 1*). In some cases, for some proteins and dye ligands, binding through charge–charge interactions may dominate. In those cases, binding and elution will most effectively be controlled by varying the conductivity of the mobile phase (i.e., salt concentration) and by varying the pH (4). In other cases, the hydrophobic component of the interaction may be quite strong. In those cases, addition of a solvent or a detergent to the elution buffer may be required in order to elute the product.

Table 1
Control of Protein Binding and Elution in Dye Ligand Chromatography

Physical variable	Effect
pH	pH exerts a strong affect on binding and elution. pH for binding may range from 4 to 8 and may be controlled using sodium acetate (pKa 4.8), MES (pKa 6.3), phosphate (pKa 7.2), HEPES (pKa 7.6) or other appropriate buffers. Choosing an alternate buffer system can sometimes resolve solubility problems and is worth considering during batch screening (5).
Salts	Low ionic strength (typically below 100 mM) enhances binding to the charged dye ligands. Extremely low ionic strengths (below 20 mM) can enhance protein solubility problems. One convenient means of elution is to increase ionic strength (100–1000 mM). If 1 M salt is insufficient to elute the protein, then either the pH must be modified during elution or solvent or detergent must be added during elution (<i>see Note 6</i>).
Solvents	Hydrophobic interactions enhance the affinity of dye ligands for proteins. To increase protein yield, the elution buffer strength may be enhanced by addition of non-denaturing solvents such as ethylene glycol or glycerol. Up to 50% maybe used.
Chaotropic agents and detergents	Chaotropic agents, such as urea and guanidine, may be employed to enhance either the elution effect or the washing affect of a buffer. Non-ionic detergents, such as Tween 80 and Triton X100, may also be employed (6). These components modulate the hydrophobic interactions of proteins with the dye ligand. Care should be taken to insure that the selected concentration is compatible with protein activity.

Development of a dye affinity chromatography step requires optimization of binding, washing and elution conditions. This chapter describes both batch chromatography (for scouting binding and elution conditions) and column chromatography for purification optimization. The most efficient means of establishing the binding conditions for a dye affinity purification is to use batch chromatography. Use of this screening method can save substantial time and expense by focusing the chromatographer's efforts on the stationary phase chemistries and the mobile phase conditions most likely to succeed (*see Note 1*).

Having selected the appropriate dye affinity support and established the possibility of quantitatively recovering protein activity, the chromatographer can move onto column chromatography. This chapter describes a dye affinity technique successfully used to purify a recombinant protein. Provided that the reader takes the time to carry out the batch development successfully, the adaptation of the column chromatographic technique should follow quickly.

2. Materials

2.1. Batch Binding

During buffer preparation, adjust pH to specified values using concentrated hydrochloric acid or concentrated sodium hydroxide as appropriate.

1. Batch binding/wash buffers:

- a. 20 mM sodium acetate, 50 mM NaCl, pH 4.
- b. 20 mM sodium acetate, 50 mM NaCl, pH 5.
- c. 20 mM MES, 50 mM NaCl, pH 6.
- d. 20 mM HEPES, 50 mM NaCl, pH 7.
- e. 20 mM HEPES, 50 mM NaCl, pH 8.

2. Batch elution buffers:

- a. 20 mM sodium acetate, 1 M NaCl, pH 4.
- b. 20 mM sodium acetate, 1 M NaCl, pH 5.
- c. 20 mM MES, 1 M NaCl, pH 6.
- d. 20 mM HEPES, 1 M NaCl, pH 7.
- e. 20 mM HEPES, 1 M NaCl, pH 8.

3. Slide-A-Lyzer Dialysis Cassettes (Pierce, <http://www.piercenet.com>). Choose the largest molecular weight cutoff that will not pass the protein of interest. The 3–12 ml size will allow one cassette per batch binding condition. A ratio of at least 100 to 1 should be maintained for the dialysis (1 l of dialysis buffer for each 10 ml of sample to be dialyzed).

4. Dye ligand resins to be screened (see **Subheading 1** for vendors).

5. Mixing device: A rotator, shaking platform or rocking platform may be used. The protein/resin mixtures should mix gently without allowing the resin to settle to one point in the test tubes. Having the test tubes on their sides can be helpful.
6. Miscellaneous: 15-ml polypropylene test tubes with screw caps, 10-ml serological pipettes and autopipettor, 1-ml pipettor and tips, razor blade and stir plate(s) for dialysis.

2.2. Dye Ligand Chromatography

1. Load: Cell culture supernatant with appropriate sample preparation (pH and salt concentration adjustment); should be filtered to 0.2 μm prior to loading on the chromatography column (*see Note 2*).
2. Dye ligand chromatography support(s) selected above in the batch binding experiments. For the example, GE Amersham Blue Sepharose 6 FF was used.
3. Binding and elution buffers selected above in batch screening:
 - a. Binding Buffer: Selected in the batch binding experiment to be a pH that gives good product binding. Capacity of the resin for the product should be greater than 1 mg/ml in the presence of impurities. Ideally, capacity would be greater than 5 mg/ml. In the example below, 50 mM sodium acetate, pH 5, is the binding buffer.
 - b. Gradient Buffer A: The elution gradient starts at 100% Gradient Buffer A and goes to 100% Gradient Buffer B. This is convenient to arrange using a GE Amersham AKTA Explorer or equivalent. Highest purity will be obtained by eluting using only a single variable (only pH or only salt). Transition from binding condition to Gradient Buffer A allows this to be possible. In the example below, binding occurs at pH 5, washing using Gradient Buffer A occurs at pH 6.5. Then a salt gradient to 100% Gradient Buffer B allows product elution based on increasing sodium chloride. In the example below, Gradient Buffer A is 10 mM sodium phosphate, pH 6.5.
 - c. Gradient Buffer B: This buffer should elute the product with good efficiency. In dye affinity chromatography, product recoveries in the range from 80 to 90% are typical. In the example below, Gradient Buffer B is 10 mM sodium phosphate, 1 M NaCl, pH 6.5.
4. Other buffers:
 - a. 0.1 M NaOH for column sanitization (Blue Sepharose 6 FF will not tolerate to 1 M NaOH).
 - b. 20% ethanol for column storage.
5. Chromatography system:
 - a. GE Amersham AKTA Explorer or equivalent. The automated gradient formation, fraction collection and data logging of this type of chromatography equipment will save substantial amounts of time and effort (*see Note 3*).

6. Miscellaneous: Fraction tubes, chromatography columns (from GE, Millipore, Omnifit or equivalent).

3. Method

3.1. Batch Binding

1. Obtain 50 ml of cell culture supernatant per dye resin to be tested. This should be clarified down to $0.2\mu\text{m}$ by a combination of centrifugation, capsule filtration and vacuum filtration (*see Notes 2 and 4*).
2. Ten milliliters of the cell culture supernatant is dialyzed against each binding buffer (*see "Instructions: Slyde-A-Lyzer Dialysis Products"*). A stir plate will be required to mix each of the dialysis containers at room temperature overnight (*see Note 5*).
3. To prepare the dye resin for use, aliquot 10 ml of each binding buffer into a new set of five labeled test tubes (one for each separate pH). Aliquot 50 ml of the dye ligand resin into the appropriate test tube (taking into account the slurry factor; for a 50% slurry transfer 100 ml). The slurry may be in the shipping solution because the first step below will rinse the gel. This is an important step, so care should be taken to aliquot the correct amount of resin.
 - a. Use the razor to cut the tip off of the micropipette to be used. This will insure that the gel is not prevented from freely entering the micropipette.
 - b. Calculate the amount of slurry to be added: $50\text{ ml} \times \frac{\text{total slurry volume}}{\text{settled resin volume}}$. Pipette this amount into each polypropylene tube.
 - c. Cap the test tubes and vortex.
4. Spin the resin down in a centrifuge ($G_t = 10^6\text{ s}$) and remove the buffer using a serological pipette, without disturbing the gel. Do not decant; gel will inevitably be lost. Use a serological pipette.
5. Add the dialyzed protein solution to the appropriate test tubes (pH 4 dialysate with pH 4 binding buffer, etc.). If dialysis has resulted in a change of volume, add the equivalent of 10 ml of starting cell culture supernatant (i.e., if the Slyde-A-Lyzer contents swell from 10 to 13 ml, add the entire 13 ml). Mix overnight at room temperature. Note that prior knowledge of protein stability may dictate specific incubation conditions (temperature, incubation time, etc.) at this point.
6. Spin the resin down in a centrifuge ($G_t = 10^6\text{ s}$) and remove the supernatants without disturbing the gel. Transfer the supernatants to labeled tubes. The supernatants, containing unbound protein, should be stored prior to assay under conditions favorable to target protein stability.
7. Add 5 ml of the appropriate elution buffer to each of the resin pellets. (The batch binding experiments are carried out at constant pH, that is, use the same pH binding and elution buffer conditions.). Mix for 10 min.
8. Spin the resin down in a centrifuge ($G_t = 10^6\text{ s}$) and remove the eluents using a serological pipette as described above, label and store.
9. Assay the binding supernatants and the eluents for protein activity.

10. Data interpretation:

- a. Binding: The primary event to be looked for is binding. Provided reasonable protein capacity is achieved, the protein can usually be eluted with one of the strategies mentioned in **Table 1**. Look for samples in which little activity remains in the binding supernatant.
- b. Elution: In the initial screen described above, both pH and sodium chloride are examined as possible eluents. Look for pH conditions and sodium chloride conditions at which the protein is eluted and is found in the supernatant.

3.2. Dye Ligand Chromatography

For the protein purification by dye ligand chromatography described below, the following conditions are used:

- Binding condition: Cell culture supernatant titrated to pH 5 with 10% acetic acid; binding to GE Amersham Blue Sepharose 6 FF.

After binding, the pH is increased to 6.5 without eluting the protein, while the sodium chloride concentration remains low. The wash condition and the elution condition were the following:

- a. Wash condition: 10 mM sodium phosphate, pH 6.5 (Gradient Buffer A).
- b. Elution condition: A linear gradient between Gradient Buffer A and Gradient Buffer B (10 mM sodium phosphate, 1 M NaCl, pH 6.5).

The method employed in the chromatography is as follows:

1. A loading of 0.78 mg target protein/ml of packed resin is used. Seventy milligrams of the protein of interest is loaded on a 90-ml column (17 × 2.6 cm).
2. A flow rate of 13.3 ml/min is used. This flow rate is quite conservative and the manufacturer would allow up to five times the flow rate based on this column's cross-sectional area. See individual manufacturer's resin specifications.
3. Consult the manufacturer's instruction to pack the column.
4. Sanitize the column by passing 3 column volumes of cleaning buffer (0.1 M NaOH) at 13.3 ml/min.
5. Equilibrate the column at 13.3 ml/min with 3 column volumes of binding buffer and check the eluent pH. Repeat until pH is correct.
6. Load the sample at 13.3 ml/min.
7. Wash with 10 column volumes of Gradient Buffer A or until detector baseline (typically $A_{280\text{ nm}}$) is reached.
8. Run a linear gradient at 13.3 ml/min from 0 to 100% B in 20 column volumes. Collect fractions of 0.5 column volume.
9. Repeat sanitization and store in 20% ethanol or equivalent bacteriostatic solution.
10. Analyze fractions for activity.
11. Data analysis: In the example chromatogram (*see Fig. 1*), 85% of the activity was recovered in main $A_{280\text{ nm}}$ peak (fractions 6–14).

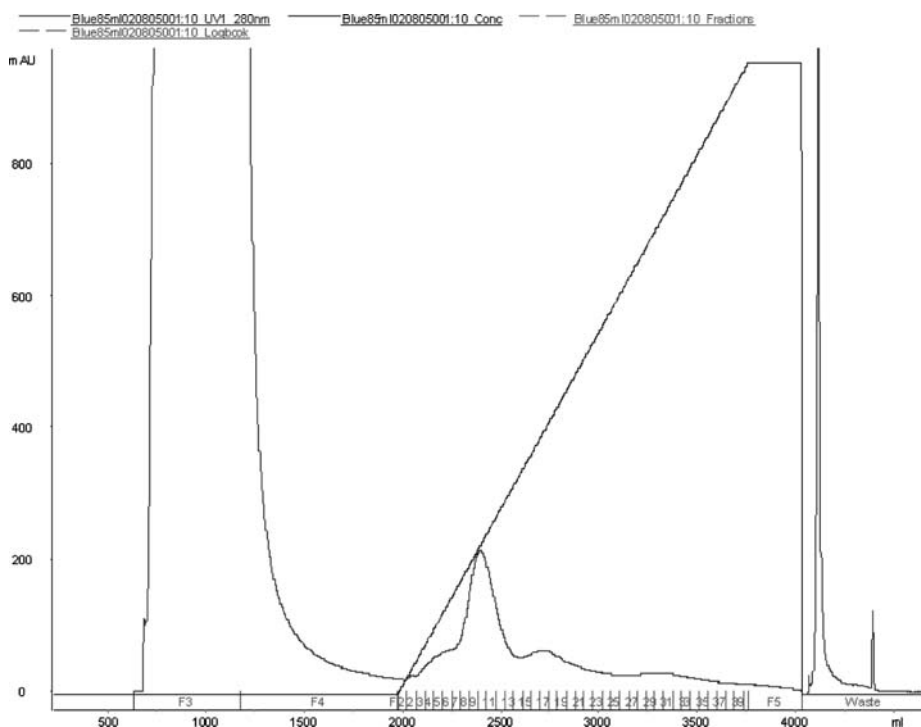


Fig. 1. Dye ligand chromatogram.

4. Notes

1. Prior to initiation of the resin screening, the protein of interest may be screened for compatibility with planned binding and elution conditions. Understanding the protein's stability can be critically important to interpreting chromatographic data during purification optimization. Dialysis of the protein against a range of buffers followed by activity assays of each condition will define the borders of the optimization space of the chromatography. Basic screening conditions for protein activity include the following:
 - a. pH: 50 mM buffer + 100 mM NaCl, where the buffers are sodium acetate (pH 4 and 5), MES (pH 6) and HEPES (pH 7 and 8).
 - b. Salt/Detergent/Chaotrope/Solvent: Begin with 50 mM buffer + 100 mM NaCl, where the buffer is chosen to give good protein activity. Add 0.01% Tween, 0.02% Triton X-100, 20% glycerol, 30% ethylene glycol, 1 M NaCl, 1 M guanidine or 1 M urea to separate aliquots of the basic buffer. Verify protein activity after exposure to each buffer. Use this information in selecting the wash and elution conditions to be tested during batch screening.

2. Adjustment of the load proceeds in two steps: pH and conductivity adjustment, followed by clarification. The pH should be adjusted using a dilute acid (10% acetic acid) or base solution (1 M Tris base), depending on whether the pH is to be decreased or increased. Conductivity should be adjusted by adding a concentrated sodium chloride solution (4 M) or deionized water, depending on whether conductivity is to be increased or decreased. Clarification to a final 0.2- μ m filtration can be accomplished at small scale by centrifugation followed by vacuum filtration with a glass fiber prefilter (provided that the protein of interest does not bind to the glass prefilter). A glass fiber free filter train which works quite well for most mammalian cell culture supernatants consists of the Sartorius Sartopure PP2 1.2- μ m filter followed by a Sartopore 2 0.45/0.2- μ m filter. Some thought should be given to the filtration method because product losses can be quite large if an inefficient method of filtration is selected.
3. In some cases, a chromatographic system may be unavailable or undesirable. The later case occurs with a feedstock which may be inappropriate to contact with a system which is used repeatedly (e.g., a load sample containing virus). In that case, a peristaltic pump with disposable tubing is used to pump the solutions. Small discrete steps in buffer concentration can be substituted for a linear gradient. Three column volumes of 10% B, then 3 column volumes of 20% B and so on. Column fractions are analyzed by UV spectrophotometer.
4. Selection of the appropriate amount of resin per batch binding experiment is important to the success of the experiment:
 - a. A typical expression level of a recombinant protein in mammalian cell culture is 0.02 mg/ml of supernatant (although some titers may be 10-fold above or below this). For low titer cell culture fluid, a concentration step (ultrafiltration) may be desirable in order to reduce the volumes of feedstock needed in the batch binding experiments.
 - b. A desirable binding capacity for capture of a protein from cell culture is 4 mg/ml of gel (although 5-fold above this is possible for high affinity proteins).
 - c. In order to load 50 ml of resin to 4 mg/ml with a 0.02 mg/ml cell culture supernatant, 10 ml of cell culture supernatant will be required for each condition to be tested.

Loading of the gel in batch binding experiments should be substantial or it will be difficult to interpret the results. Overloading the resin is not generally a problem. Conversely, underloading the resin provides only limited data regarding the binding capacity for the target protein and may lead to poor recovery (accountability).

5. If necessary, dialysis at 5°C may be used to preserve protein activity. Disposable desalting columns may be used to reduce processing time. If precipitation is observed during dialysis, do not terminate that experimental condition. After dialysis is complete, clear the precipitate by centrifugation and continue with the binding experiment using the clarified supernatant. Frequently, the protein of interest remains in solution; a specific assay (activity in the case of an enzyme) is required to verify loss of the protein of interest.

6. Secondary screening: Frequently, secondary screening for elution conditions is useful. Dye affinity resins may bind the protein of interest with high affinity and prevent good product recovery using the initial elution conditions. In the secondary screen, focus on the resins that have generated reasonable capacities for the protein of interest during primary screening. Carry out binding on these resins using the optimal conditions found in the initial binding study, then vary the elution condition either by variation in pH or conductivity or through the addition of other elution modulators: 0.01% Tween, 0.02% Triton X-100, 20% glycerol, 30% ethylene glycol, 1 M guanidine and 1 M urea, provided that each of these is compatible with protein activity/stability. The goal is to find an elution condition capable of eluting the protein of interest quantitatively while preserving any relevant biological characteristics (full activity, native structure, etc.). In some cases, extremely high affinity between the ligand and the target protein may preclude the identification of suitable elution conditions. Usually, a moderate affinity dye ligand can be found which will release active protein. The chances of finding a moderate affinity dye ligand are enhanced by screening a large number of resins initially.

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Purification of Proteins Using Displacement Chromatography

Nihal Tugcu

Summary

Displacement chromatography has several advantages over the nonlinear elution technique, as well as the linear elution mode, such as the recovery of purified components at high concentrations, less tailing during elution, high throughput and high resolution. Displacer affinity and its utilization are the critical components of displacement chromatography. Particularly, the nonspecific interactions between the displacer and the stationary phase can be exploited to generate high affinity displacers. This chapter will discuss the design and execution of displacer selection and implementation in a separation specifically focusing on its utilization in ion exchange chromatography.

Key Words: Displacement; chromatography; protein purification; steric mass action isotherm.

1. Introduction

Even though the vast majority of chromatographic bioseparations are performed in the elution mode, displacement chromatography is rapidly emerging as a powerful preparative bioseparations tool because of the high throughput and purity associated with it. These characteristics make displacement chromatography an attractive alternative to elution chromatography. Displacement is a mode of chromatography as are isocratic and gradient elutions. However, because of the nonlinear adsorption inherent to displacement

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chromatography, the sorptive capacity of the stationary phase is fully utilized. Displacement chromatography is fundamentally different from any other modes of chromatography in that the solutes are not desorbed in the mobile phase modifier and separated by differences in migration rates. In displacement, molecules are forced to migrate down the chromatographic column by an advancing shock wave of a displacer molecule that has a higher affinity for the stationary phase than any feed solute. It is this forced migration that results in higher product concentrations and purities compared to other modes of operation. Displacement, invented by Tiselius in 1943 (*1*), was first used for the separation of amino acids and peptides using activated carbon adsorbents. In the 40 years that followed, displacement chromatography was primarily used for the isolation of transuranic elements (*2,3*), rare earth metals (*4–6*) and simple biochemicals (*7,8*). The technique also found application for the enrichment of trace components (*9,10*). Displacement chromatography had limited success until the 1980s due to the unavailability of efficient stationary phases. In parallel with progress in high performance liquid chromatography (HPLC), along with advances in the manufacture of stationary phases with increased capacity, mechanical strength and stability, rapid kinetics and mass transfer, the technique was revived by Horvath and his co-workers (*11*) and has since found many applications, especially for the purification of biomolecules.

In displacement chromatography, the column is subjected to sequential step changes in the inlet conditions—in a manner very similar to step-gradient chromatography. The column is initially equilibrated with a buffer that would provide relatively strong binding conditions for the feed components (such as low ionic strength buffers for ion exchange chromatography). The feed is then loaded onto the column under conditions of pronounced overloading and followed by a constant infusion of the displacer solution. The displacer molecule is selected such that it has the highest affinity for the stationary phase compared to any of the feed components. This enables the displacer front to stay behind, displace and separate the feed components into adjacent zones in the order of increasing affinity for the stationary phase (*see Fig. 1*). It is important to note that the displacer enables the feed components to develop into “square-wave” zones that have high capacity and purity, forming the “displacement train.” After the breakthrough of the displacer from the column effluent, the column is regenerated and re-equilibrated with the carrier buffer allowing the process to be repeated.

Displacement chromatography, an intrinsically nonlinear mode, has several advantages over the nonlinear elution technique, as well as the linear elution mode. In displacement chromatography, the components are resolved into consecutive zones of pure substances rather than “peaks.” Because the process takes advantage of the nonlinearity of the isotherms, a larger feed can be

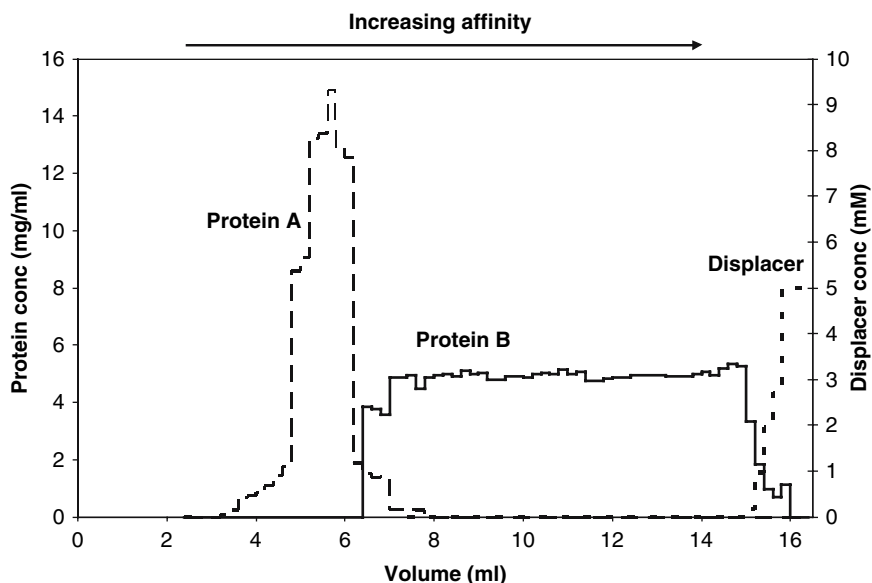


Fig. 1. Sample chromatogram from displacement chromatography.

separated on a given column with the purified components recovered at significantly higher concentrations. In addition, the tailing observed in nonlinear elution chromatography is greatly reduced in displacement chromatography due to the self-sharpening boundaries formed in the process. In displacement chromatography, the displacer suppresses the adsorption of feed components in the displacer zone and thus prevents tailing of the most strongly retained feed component. In a fully developed displacement train, each of the components displaces the component ahead of it, leading to a suppression of tailing in all solute zones. This makes displacement chromatography less sensitive to feed loads resulting in high throughputs without sacrificing resolution and purity. Displacement chromatography exploits the nonlinear, multi-component competition amongst the components to be separated, resulting in higher resolution, particularly among closely related species. In addition, product recovery is possible under relatively low mobile phase modifier concentrations (e.g., salt). This combination of high throughput and high resolution in a single process makes displacement chromatography an attractive mode of operation for preparative separations.

Displacer affinity and its utilization are the critical components of displacement chromatography. It has been accepted that retention in ion exchange systems is not purely based on electrostatic interactions (12–15), and there are a few reports in the literature concerning the relative importance of

non-specific interactions, such as hydrogen bonding and hydrophobic interaction, in governing affinity in ion exchange materials (16). Most of the time, it is those nonspecific interactions that can be exploited to generate the unique displacer–stationary phase interaction leading to high affinity displacers. Studies done using a homologous set of displacers (17–19) have shed light into the structural components that would increase the affinity of a displacer on a particular stationary phase. For example, increased displacer affinity with increasing flexibility and number of aromatic rings was observed on polymethacrylate-based Waters strong cation exchange resin (17,18). Similarly, on hydrophilic resins, such as agarose-based SP Sepharose XL (from GE healthcare), displacer affinity was shown to be dominated by the electrostatic interactions (charge of the displacer), whereas hydrophobicity was the key component for displacer affinity on polystyrene-divinylbenzene-based supports (19).

In the early years (1978–1995), high molecular weight displacers were utilized for displacement chromatography for purification of many proteins, such as the use of carboxymethyl dextrans for purification of β -lactoglobulins, ovalbumin, α -lactalbumin and soy-bean trypsin (20–23). Other examples of such displacers are chondroitin sulfate (24) and Nalcolyte 7105 (25). Nalcolyte 7105 was utilized as a displacer for the purification of a four-component protein mixture composed of ribonuclease, α -chymotrypsinogen, cytochrome A and lysozyme resulting in successful purification at preparative scale on a cation exchange support (26). Nontoxic displacers, such as protamine and heparin sulfate, were reported by Gerstner et al. (27–29) for use in anion exchange systems. Protamine sulfate was later utilized by Barnthouse et al. (30) for purification of recombinant human brain-derived neurotrophic factor, rHuBDNF, using cation exchange displacement chromatography.

One of the advances in displacement chromatography came with the introduction of low molecular weight displacers (<2 kDa), such as dendritic polymers in 1995, by Jayaraman et al. (31). This work was particularly important because it represented the first evidence that very low molecular weight displacers could be effective for protein separations. Low molecular weight displacers have significant operational advantages compared to large polyelectrolyte displacers. The most important advantage is if there is any overlap between the displacer and the protein of interest, these low molecular weight materials can be readily separated from the purified protein during post-displacement downstream processing involving size-based purification methods. Many other examples of low molecular weight displacers followed protected amino acids such as N- α -Benzoyl-L-arginine-ethyl ester and N- ϵ -carbobenzoxy-L-lysine-methyl ester and antibiotics including streptomycin and neomycin sulfate (32,33) used for cation exchange displacement chromatography. The sulfonic acid salts of aromatic compounds (34) and sulfonated sugars such as sucrose octasulfate

(SOS) (35) have been employed as displacers for anion exchange systems. While most of these separations were carried out on ion exchange resins, the use of displacement chromatography on reversed phase and hydroxyapatite (HA) resins was also demonstrated. Viscomi et al. (36) used the combination of reversed-phase and ion exchange displacement chromatography for the purification of a synthetic peptide, the fragment 163-171 of human interleukin-B. In the reversed-phase displacement chromatography step, the displacer was benzyltributyl ammonium chloride, whereas in the ion exchange displacement step, the displacer was an ammonium citrate solution. The use of displacement to separate proteins in immobilized metal affinity chromatography (IMAC) has also been reported (37,38). Freitag et al. (39) presented the application of displacement chromatography on HA stationary phases.

The remainder of this chapter aims to provide the reader with all of the tools necessary to determine the best operating conditions for a successful displacement experiment for ion exchange systems. However, knowing that sometimes material and/or time requirements may not allow the reader to go through all of the steps described in this chapter, where appropriate an abbreviated version of methods development will be described.

2. Methods

2.1. Identification of Stationary Phase and Operating Conditions for Selectivity

Linear elution chromatography can be employed to select an appropriate stationary phase with sufficient selectivity as well as an operating condition (buffer pH, mobile phase additives such as salt type and concentration) that provides a sufficient resolution between the feed components.

2.2. Constructing the Adsorption Isotherms

If pure feed components are available, the next step will be obtaining the adsorption isotherms for the feed components. If pure feed components are not available, proceed to the steps in **Subheading 2.3** for the ranking and selection of displacers. This chapter will focus on the use of the steric mass action (SMA) isotherm as a tool to define operating conditions for a successful ion exchange displacement chromatography (see **Note 1**). The SMA model (40) has been shown to be a convenient methodology for examining the chromatographic behavior of proteins in ion exchange systems. In this model, adsorption has been described using three (SMA) parameters: characteristic charge (ν), which is the number of interaction sites each molecule has with the stationary phase material; the equilibrium constant (K) of the reaction between the solute and

the salt counter ions on the surface; and the steric factor (σ) which is the number of adsorption sites sterically shielded by the adsorbed molecule. The single component SMA isotherm (40) is

$$C = \left(\frac{Q}{K} \right) \left(\frac{C_{\text{salt}}}{\Lambda - (\nu + \sigma)Q} \right)^\nu, \quad (1)$$

where Q and C are the solute concentrations on the stationary and mobile phases, respectively. C_{salt} is the mobile phase salt concentration and Λ (see **Note 2**) is the total ionic capacity of the stationary phase represented. Two approaches are available to determine ν and K .

In the first method, isocratic experiments at different mobile phase salt concentrations are carried out, and the retention times of the proteins or displacers at these different salt concentrations are recorded. The following **Eq. 2 (41)** can then be solved to obtain the linear SMA parameters:

$$\log k' = \log(\beta K \Lambda^\nu) - \nu \log C_{\text{salt}} \quad (2)$$

where k' is the capacity factor and β is the phase ratio. Thus, a plot of $\log k'$ versus $\log C_{\text{salt}}$ yields a straight line with a slope of ν and an intercept of $\log(\beta K \Lambda^\nu)$.

Alternatively, in the second method, gradient experiments may be used to obtain the linear parameters. This approach can enable the simultaneous determination of linear SMA parameters for all components of the feed mixture. In addition, this technique is more suitable for displacers, as they will have a high affinity for the stationary phase. Once the retention volumes are obtained, using at least two different gradient conditions, the values are substituted into the following equation to solve for the linear parameters (42):

$$V_g = \left[\left(x_i^{\nu+1} + \frac{V_0 K \varepsilon \Lambda^\nu (\nu + 1) (x_f - x_i)}{V_G} \right)^{\left(\frac{1}{\nu+1} \right)} - x_i \right] \frac{V_G}{x_f - x_i}, \quad (3)$$

where V_g is the solute retention volume, x_i and x_f are the initial and final salt concentrations respectively, V_G is the total gradient volume, V_0 is the dead volume and ε ($\varepsilon = 1/(\beta + 1)$) is the column porosity.

The non-linear parameter, σ , for displacers or proteins can be determined by non-linear frontal experiments with the displacer or protein at very low mobile phase salt concentrations. These experiments can also provide an independent measure for the characteristic charge in addition to the steric factor (41). The ratio of the magnitudes of the induced salt gradient to the concentration of the displacer (d)/protein (p) in the front gives the value of the characteristic charge.

$$\nu = \frac{\Delta C_{\text{salt}}}{C_{\text{displacer/protein}}} \quad (4)$$

The breakthrough volume of the displacer/protein front (with known concentrations of C_d or C_p at a salt concentration of C_{salt}) can be used to calculate the capacity of the stationary phase for displacer/protein (Q_d or Q_p) as

$$Q = \frac{C \left(\frac{V_{\text{br}}}{V_0} - 1 \right)}{\beta}, \quad (5)$$

where V_{br} is the breakthrough volume for the displacer or the protein. Using this value along with knowledge of the SMA parameters, K and ν , the steric factor (σ) can then be determined from **Eq. 1**.

2.3. Dynamic Affinity and Affinity Ranking Plots

Once the SMA isotherm parameters are obtained, the next steps will be predicting the elution order, selecting the right displacer and the operating conditions for conducting the displacement chromatography. This can be done one of two ways: via a dynamic affinity plot or an affinity ranking plot as described below. For non-chromatographic methods of selecting high affinity displacers, *see Note 3*.

It has been shown that a stability analysis can be carried out to determine the elution order of feed components in a displacement train from the following expression (**43**):

$$\left(\frac{K_a}{\Delta} \right)^{1/v_a} < \left(\frac{K_i}{\Delta} \right)^{1/v_i} \quad (6)$$

where,

$$\Delta = Q_d / C_d, \quad (7)$$

where Δ is the partition ratio of the displacer and Q_d and C_d are the concentrations of the displacer on the stationary and mobile phases, respectively. The left-hand side of **Eq. 6** can be written as the dynamic affinity (λ) of component “a”:

$$\lambda_a = \left(\frac{K}{\Delta} \right)^{1/v_a}, \quad (8)$$

which is dependent on the value of Δ that represents the operating conditions of the displacement experiment, such as the mobile phase salt concentration and the displacer concentration. Taking the logarithm of both sides of **Eq. 8**:

$$\log K = \log(\Delta) + \nu \log(\lambda) \quad (9)$$

On a plot of $\log K$ versus ν (dynamic affinity plot, *see Fig. 2A*), **Eq. 9** defines two regions separated by a line with a slope of $\log(\lambda)$ and an intercept of $\log(\Delta)$. The line intercepts the point $\log(\Delta)$ on the y-axis and passes through the point defined by the parameters K_a and ν_a of component “a” (determined as described previously). Because this plot gives information regarding the elution order of the components in a displacement train (increasing dynamic affinity in the upwards direction), it may also be used to compare displacer efficacies for separating a protein mixture under specific salt and displacer concentrations.

Figure 2A illustrates a dynamic affinity plot for three solutes “A,” “B” and “C” at a Δ value of 10. Under the experimental conditions specified by the

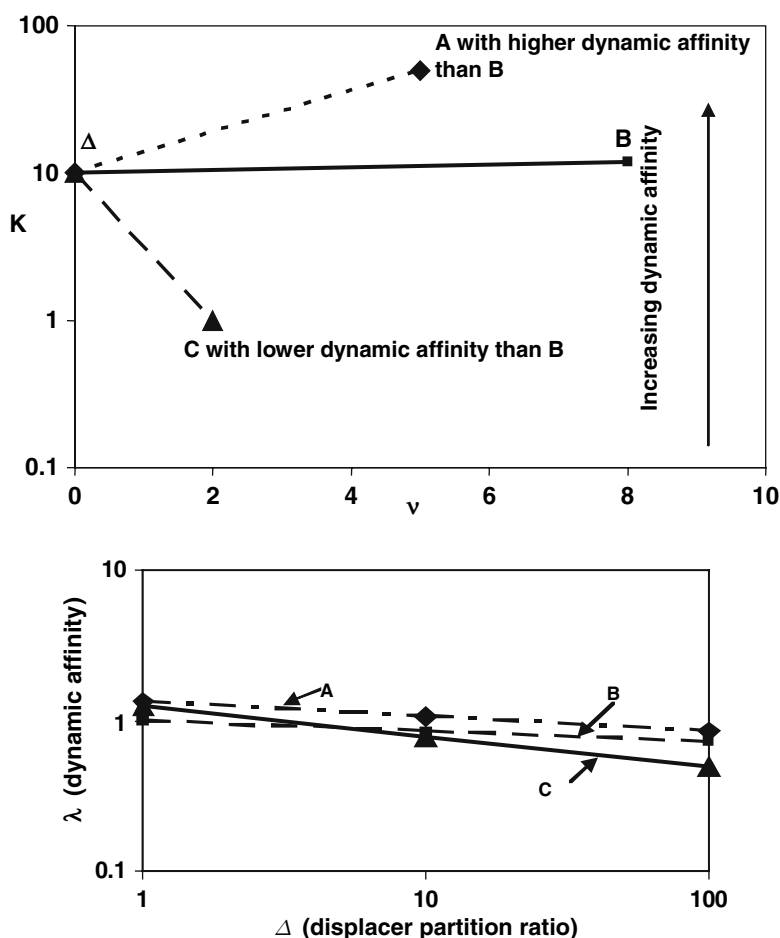


Fig. 2. (Continued).

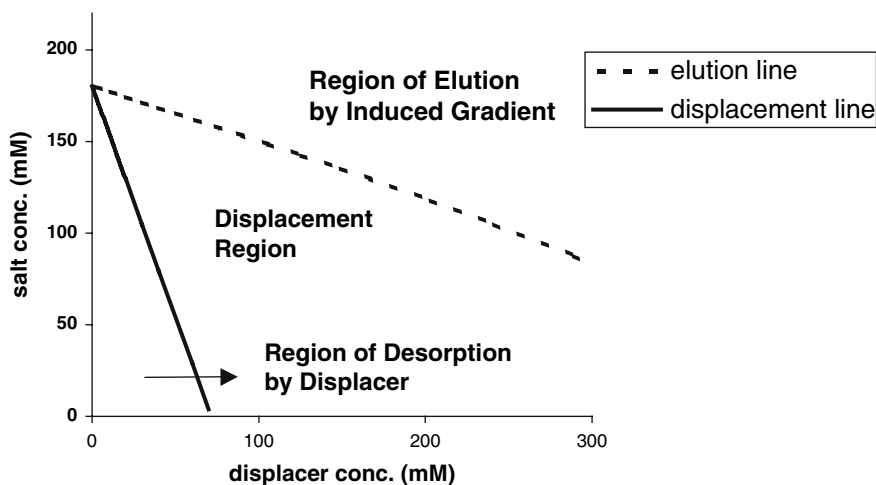


Fig. 2. (A) Dynamic affinity plot for “A,” “B” and “C.” Parameters: A ($\nu = 5$, $K = 50$), B ($\nu = 8$, $K = 12$) and C ($\nu = 2$, $K = 1$), $\Delta = 10$. (B) Affinity ranking plot for “A,” “B” and “C.” Parameters: A ($\nu = 10$, $K = 20$), B ($\nu = 15$, $K = 1$) and C ($\nu = 5$, $K = 3$). (C) Operating regime plot.

value of Δ , “A” has a greater dynamic affinity than both “B” and “C” while “C” has a lower dynamic affinity than “B.” Therefore, in a displacement train, the order of elution will be “C” followed by “B” and then by “A.”

Displacer affinity ranking plots (42), on the other hand, serve a different purpose. These plots enable the ranking of the relative efficacies of displacers. In contrast to the dynamic affinity plot which is constructed for a specific Δ (operating condition), this type of ranking plot can show the variation of the dynamic affinity of a molecule over a range of Δ values. Thus, these plots provide a means of comparing the affinity of various displacers over a range of operating conditions and give a realistic understanding of the efficacy of a molecule as a displacer. Displacer affinity ranking plots originate from the rearrangement of Eq. 8 as follows:

$$\log(\lambda) = \frac{1}{\nu} \log(K) - \frac{1}{\nu} \log(\Delta). \quad (10)$$

Thus, a plot of $\log(\lambda)$ versus $\log(\Delta)$ (see Fig. 2B) can be constructed using the linear SMA parameters, K and ν . On these plots, higher values of Δ correspond to lower values of displacer or salt concentrations. Thus, lower values of Δ result in higher values of the dynamic affinity (λ).

Figure 2B illustrates a typical affinity ranking plot for three displacers, “A,” “B” and “C.” For this range of Δ , “A” has a greater dynamic affinity than “B” and “C.” However, the relative efficacies of “B” and “C” can change as indicated by the plot. At Δ values less than about 10, “B” has a higher dynamic affinity than “C.” However, the order changes for Δ values greater than 10. A range of Δ values could be picked, once the dynamic affinity lines for feed components and displacer candidates are plotted using an affinity ranking plot.

If time or material is not available for the detailed screening described above, then evaluating displacer candidates via linear elution chromatography could be a replacement. In that case, the suggestion will be to pick the displacer with the highest affinity (longest retention time) while making sure that a regeneration protocol for this displacer on the specific resin is available (*see Note 4*).

2.4. Operating Regime Plots

Once a displacer has been selected and its corresponding Δ was determined based on its affinity to displace feed components as described above, the next step would be calculating the corresponding displacer concentration at a given mobile phase salt concentration. A detailed analysis of the displacer concentration necessary for displacement of feed components as a function of mobile phase salt concentration is done via use of operating regime plots described later in this section. However, if the reader has already established a salt concentration leading to relatively strong binding of the feed components and the displacer, then once the Δ is determined, the SMA isotherm (**Eq. 1**) can simply be used to calculate the displacer concentration.

As mentioned previously, Δ is a function of displacer and mobile phase salt concentrations. Therefore, having an operating regime plot that shows Δ as a function of salt concentration would be invaluable. To create these plots, a displacement line that separates the displacement and desorption regions should be determined. It has been shown that low molecular weight displacers will generally have a critical partition ratio (Δ) at which they cease to act as a displacer and begin to act as a desorbent (**44**). D and P in these equations refer to displacer and protein, respectively. The equation for the displacement line is given by

$$C_{\text{salt}} = \left(\frac{K_D}{\Delta} \right)^{1/\nu_D} \{ \Lambda - [(\nu_D + \sigma_D) C_D \Delta] \}, \quad (11)$$

where the critical displacer partition ration Δ is calculated as

$$\Delta = \frac{K_P^{\nu_D/(\nu_D-\nu_P)}}{K_D^{\nu_P/(\nu_D-\nu_P)}} \quad (12)$$

By selecting values of C_D and substituting them into **Eq. 11**, the boundary between displacement and desorption may be mapped onto a plot of salt concentration versus displacer concentration (solid line, *see Fig. 2C*).

To draw the boundary between displacement and elution, the following equations are solved sequentially

$$C_D = \frac{\Delta \left[1 - \left(\frac{K_D}{\Delta} \right)^{1/\nu_D} \left(\frac{\Delta}{K_P} \right)^{1/\nu_P} \right]}{\left(\frac{\Delta}{K_P} \right)^{1/\nu_P} \left\{ \nu_D - \left[\left(\frac{K_D}{\Delta} \right)^{1/\nu_D} (\nu_D + \sigma_D) \Delta \right] \right\}} \quad (13)$$

$$C_{\text{salt}} = \left(\frac{K_{ID}}{\Delta} \right)^{1/\nu_D} \{ \Delta - [(\nu_D + \sigma_D) C_D \Delta] \} \quad (14)$$

By selecting values of the displacer partition ratio, Δ , and substituting into **Eq. 13** and **Eq. 14**, the boundary between displacement and elution may also be mapped onto a plot of salt concentration versus displacer concentration. In **Fig. 2C**, the boundary between displacement and elution is shown as a dashed line. To the left of the line, displacement occurs; to the right, elution occurs.

This type of plot is, by definition, specific to a particular protein and a particular displacer. However, by overlaying several plots for a particular displacer paired with each of the major components in a feed mixture to be purified, it is possible to gain significant insight into the effect of displacer concentration and salt concentration on a given separation.

The next step would be running the displacement experiment under the conditions established based on the methods described in this section in order to test and optimize the operating conditions if necessary. Fractions should be collected for the regions where the feed components elute and the displacer desorbs. A practical approach would be analyzing displacer containing fractions via size exclusion chromatography due to the differences between the molecular weights of proteins and the displacers. There must also be an analytical technique to differentiate between the feed components. With these assays in place, purity and yield calculations can be made. It should be noted that if abbreviated methods have been used due to insufficient time and/or material, it may take longer to identify optimized operating conditions for the displacement.

2.5. Running Displacement Chromatography for Purification of Proteins

In this section, displacement of a model protein mixture consisting of β -lactoglobulin A and B using two different displacers will be explained. For cases 1 and 2, operating conditions for the use of saccharin or SOS as the displacer will be summarized, respectively. These two displacers differ from each other in terms of the characteristic charge they carry. Although both of these displacers are low molecular weight (<2 kDa), saccharin interacts with the stationary phase with one charge, while SOS interacts with six.

2.5.1. Materials

2.5.1.1. CASE 1 (45)

1. Stationary phase: Source 15Q (GE Healthcare formerly known as Amersham Biosciences) strong anion exchange.
2. Column dimensions: 0.46 cm ID $\times$ 10 cm L (15 μ m particle size).
3. Carrier/equilibration buffer: 100 mM Tris (consists of Trizma base and Tris-HCl), pH 7.5.
4. Displacer: 100 mM saccharin prepared in the carrier buffer (*see Note 5*).
5. Model proteins: 10.8 mg each of β -lactoglobulin A and B prepared in 2 mL carrier buffer.
6. Flow rate: 0.2 mL/min.
7. Regeneration buffer: 2 M NaCl solution.

2.5.1.2. CASE 2 (35)

1. Stationary phase: Protein-Pak Q-40 HR strong anion exchange.
2. Column dimensions: 1 cm ID $\times$ 10 cm L (40 μ m particle size).
3. Carrier/equilibration buffer: 50 mM Tris (consists of Trizma base and Tris-HCl), pH 7.5.
4. Displacer: 10 mM SOS prepared in the carrier buffer (*see Note 5*).
5. Model proteins: Mixture of β -lactoglobulin A and B prepared at a total concentration of 0.56 mM in 9.6 mL carrier buffer.
6. Flow rate: 0.8 mL/min.
7. Regeneration buffer: 1.5 M NaCl solution, at a pH of 2.5.

2.5.2. Equipment

The equipment required is common for both cases.

If available, chromatography systems such as AKTAExplorer 100 or AKTAPurifier can be used for the application. In these systems, two different sample lines can be used for the sequential perfusion of the column with

proteins (feed) and displacer, respectively. Otherwise, a system that will include an HPLC pump, an injection valve (a valve that can accommodate two different injection loops such as a Model C10W 10 port valve (Valco) will be preferable), UV-Vis detector, fraction collector and data recorder can be assembled.

2.5.3. Methods

The methods described here are common for both cases.

1. Equilibrate the column with 5–10 column volume (CV) of the equilibration buffer. If preferred, completion of equilibration can be checked via an in-line conductivity meter or pH meter (AKTAExplorer 100) or by simply collecting the effluent and checking the pH and conductivity with stand alone detectors.
2. Prepare the injection valve to first load the protein mixture and then the displacer solution on to the column. Make sure the lines from the protein mixture and displacer solution are primed with the corresponding solutions.
3. Start loading the protein mixture onto the column by switching the valve position to the loop (line) that contains the protein mixture. Start monitoring the column effluent at 280 nm. If an increase in absorbance is detected, start the fraction collector to collect the effluent (200–400 μ L fractions can be collected).
4. As soon as loading of the protein mixture is over, switch the injector valve position to the loop (line) that contains the displacer solution. Monitor the effluent absorbance. When the absorbance at 280 nm starts increasing (indicating the elution of proteins), start to collect fractions.
5. Once it is established that displacer breakthrough has occurred (*see Note 6*), regenerate the column with 10–20 CV of regeneration buffer. Collect fractions during regeneration for analysis (large fractions such as 5–10 mL will be sufficient) for further analysis. Re-equilibrate the column as described in step 1.
6. Analyze the fractions collected during the displacement experiments. The protein mixture β -lactoglobulin A and B can be analyzed using anion exchange chromatography (Source 15Q) at isocratic conditions at a flow rate of 1 mL/min. The mobile phase used is 50 mM Tris-HCl + 130 mM NaCl buffer at a pH of 7.5. The fractions are diluted threefold to fivefold and 5- μ L samples were injected. Column effluent is monitored at 235 nm. Saccharin can be assayed using size exclusion chromatography. The fractions are diluted threefold, and 5- μ L samples are injected. The column effluent is monitored at 254 nm. For analysis of SOS, a phenol-sulfuric acid assay can be used (*see Note 6*) (35).
7. Construct the displacement chromatogram based on the fraction analysis and determine the purity and yield of the protein components of interest. If the resolution and purity of the protein components are sufficient, pool the fractions based on the fraction analysis. If separation and/or yield are not satisfactory,

then the operating conditions should be reevaluated (see **Notes 7** and **8**). After re-evaluation, repeat steps 1–7 for the displacement experiment with new conditions.

8. If there is a concern about any displacer present in the product pool, simply carry out ultrafiltration or dialysis to remove any displacer in order to increase the purity of the product pool.

3. Notes

1. If the SMA formalism is not preferred, then the adsorption isotherm can be measured experimentally and used to predict displacement. An alternative way of predicting a displacement operating condition is by using the isotherm with the operating line. The order of the isotherms will predict the increasing affinity of components for the stationary phase (the higher the curve, the higher the affinity) and predicts the elution order of the displacer and the feed components. **Figure 3** shows the use of isotherms to predict operating conditions for displacement. The only necessary condition for displacement to occur is the presence of concave downward isotherms. If it is found out that the isotherms are not concave downward or cross each other, then the stationary phase and mobile phase conditions should be re-evaluated to satisfy this condition.

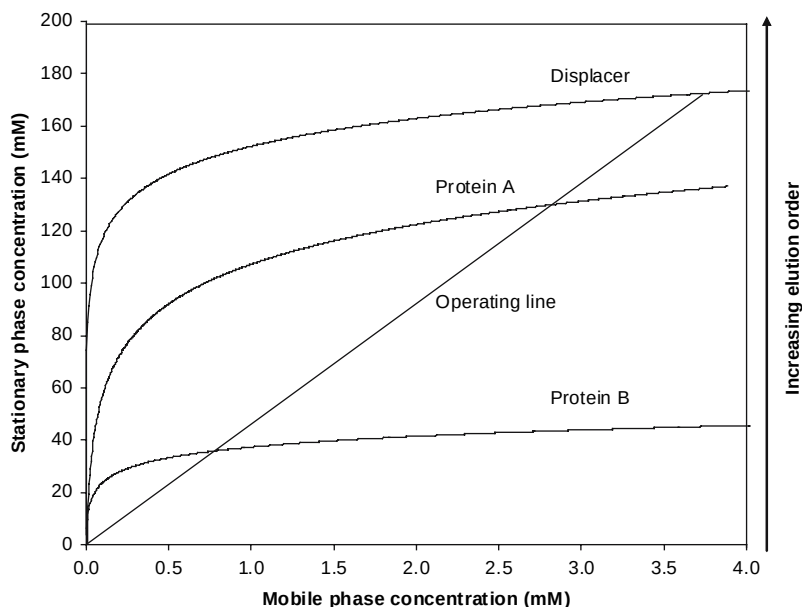


Fig. 3. Use of adsorption isotherms and operating line for determining elution order and concentration.

2. The column capacities for a monovalent salt counterion (Λ) of cation exchange stationary phases were determined using a titration method. Ten to twenty column volumes of acetic acid at either pH 3.5 or 2.5 (depending on the stability of the stationary phase) is passed through the column. This treatment is followed with 10 CV of deionized water. Then, 50–60 CV of 1 M KNO_3 is passed through the column, and the column effluent is collected. The column effluent is finally titrated against 0.01 M NaOH using phenolphthalein as an indicator. From this volume, the capacity is obtained. The ionic capacities for anion exchange resins are determined using a frontal method. The column is perfused with at least two different concentrations of sodium nitrate solutions in the equilibration buffer (such as 50 mM Tris–HCl, 30 mM NaCl, pH 7.5), and breakthrough of sodium nitrate can be determined by measuring the effluent absorbance at 310 nm. After each frontal, the stationary phase is regenerated using 2 M NaCl. The breakthrough volumes of the sodium nitrate are used to calculate the ionic capacity of the stationary phases.
3. Batch adsorption techniques (46) can also be employed for displacer screening. Computational methods have also been used to identify and predict high affinity displacers according to their structural components (47).
4. Commonly used solutions for removing displacers from stationary phases are up to 2.5 M NaCl solution, 1 N NaOH, acetonitrile or ethanol solutions and/or a combination of NaOH and solvents (such as 1 N NaOH with 25% acetonitrile or ethanol). The pH may need to be adjusted based on the type of the ion exchangers. For example, high pH will work better on cation exchangers, and a low pH will work better on anion exchangers.
5. The displacer concentration necessary for displacement of proteins will be a function of the salt concentration and its affinity. Saccharin, being a relatively low affinity displacer, requires a higher concentration whereas the opposite is true for SOS.
6. While some displacers have a chromophore that enables the use of UV-Vis detection, there are other cases where the displacer solution needs to be detected via refractive index or specific chemical assays. The breakthrough volume of the displacer can be determined separately with a frontal experiment (using the same displacer concentration that will be used for the displacement experiment) before the displacement experiment is performed with the proteins. If chemical assays are to be used, effluent fractions can be collected and assayed in order to determine the breakthrough volume.
7. If protein concentrations in the displacement zone need to be increased, increase the displacer concentration. This will increase the protein concentration and narrow the zone that the protein eluted in. However, if the protein concentration is too high, precipitation may occur and lead to high pressure drops and low recoveries. Solubility limits for feed components should be established prior to any displacement experiment. If wider protein displacement zones are preferred, decrease the displacer concentration.

Table 1
Troubleshooting for Displacement Chromatography

Problem	Reason	Solution
Displacement zones are not well developed	Total protein mass is too high	Increase the column length and/or decrease the total protein mass
Diffuse boundaries in between the displacement zones	High linear velocity, large particle size	Decrease the linear velocity and/or use a smaller particle size stationary phase
Displacement zones are too narrow, purity of proteins are low	Displacer concentration is too high, protein load is too low	Decrease the displacer concentration or increase the protein load (both will help widen the displacement zones)
Complete mixing of displacement zones	Crossing or not concave downward isotherms	Establish conditions where a concave downward isotherm condition is achieved and isotherms do not cross
Feed components are detached from the displacer	Operating line does not intersect the isotherm of feed components	Increase the displacer concentration or establish a higher retention (better adsorption) condition for the feed components

8. If a displacement experiment does not give satisfactory results, refer to **Table 1** for troubleshooting and possible solutions.

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II

AFFINITY CHROMATOGRAPHY USING PURIFICATION TAGS

Rationally Designed Ligands for Use in Affinity Chromatography

An Artificial Protein L

Ana Cecília A. Roque and Christopher R. Lowe

Summary

Synthetic affinity ligands can circumvent the drawbacks of natural immunoglobulin (Ig)-binding proteins by imparting resistance to chemical and biochemical degradation and to in situ sterilization, as well as ease and low cost of production. Protein L (PpL), isolated from *Peptostreptococcus magnus* strains, interacts with the Fab (antigen-binding fragment) portion of Igs, specifically with kappa light chains, and represents an almost universal ligand for the purification of antibodies. The concepts of rational design and solid-phase combinatorial chemistry were used for the discovery of a synthetic PpL mimic affinity ligand. The procedure presented in this chapter represents a general approach with the potential to be applied to different systems and target proteins.

Key Words: Affinity; biomimetic; ligands; synthetic; proteins; purification; design; combinatorial synthesis; screening; Protein L.

1. Introduction

The manufacturing process of a biotherapeutic must follow Good Manufacturing Practice guidelines, such that the final product is a “well characterized biologic” complying with the exigencies from regulatory bodies, such as the Food and Drug Administration (FDA) (**1**). Antibodies represent an important and growing class of biotherapeutics, with a multibillion dollar

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market, with 14 FDA-approved monoclonal antibodies, 70 in late stage clinical (Phase II+) trials, and more than 1000 in preclinical development in 2003 (2). Engineering the downstream processing of antibodies has been a principal task in research and industry, by exploring different types of interactions and separation techniques. Affinity chromatography is undoubtedly the most widespread technique in use for the purification of antibodies. It has seen improvements in classical chromatographic techniques (such as the expanded bed adsorption mode) and in non-chromatographic techniques, namely, affinity precipitation and aqueous two-phase systems. Biospecific affinity ligands, mainly immunoglobulin (Ig)-binding proteins isolated from the surface of bacteria (proteins A, G, and L), have been the most popular ligands for antibody purification. The “traditional” pseudobiospecific affinity matrices include, for example, thiophilic, hydrophobic, and mixed-mode adsorbents, and are also well liked for antibody purification purposes although they lack in specificity (3). Combinatorial approaches applied to affinity chromatography identified a new class of pseudobiospecific ligands, termed as biomimetics, as an improved version of the natural affinity ligands. Low-molecular-weight substances, able to bind Igs in the same fashion as protein A, have been developed (4). These include the multimeric peptide TG19318 (5) and the artificial protein A (ApA) (ligand 22/8), a triazine-based fully synthetic ligand (6). The latter belongs to a class of *de novo* designed non-peptidic ligands developed by Lowe and co-workers and represents an appealing concept for the generation of highly resistant, specifically tailor-made affinity ligands.

Protein L (PpL) has received special attention since its discovery in 1985, mainly for being an Ig light chain-binding protein and, as a consequence, being particularly suitable for the purification of scFv (single-chain variable fragment), Fab and F(ab')₂ biomolecules (7). PpL binds with high affinity (K_d of 1 nM) to a large number of Igs with $\kappa 1$, $\kappa 3$, and $\kappa 4$ light chains (but not to $\kappa 2$ and λ subgroups) and thus recognizes 50% of human and more than 75% of murine Igs (8). Although displaying high selectivity, PpL adsorbents suffer from high costs of production and purification, low binding capacities, limited life cycles, and low scale-up potential, which is attributable to the biological nature of the ligand. Biomimetic ligands, as the ApA, are fully synthetic in nature and can circumvent problems associated with biological ligands, while maintaining the affinity and specificity for the target proteins. In this chapter, we describe the process followed for the design and development of an Ig-binding ligand, mimicking the interaction of PpL with the light chains (named as artificial PpL), following the concept of *de novo* designed biomimetics (9) (see Fig. 1).

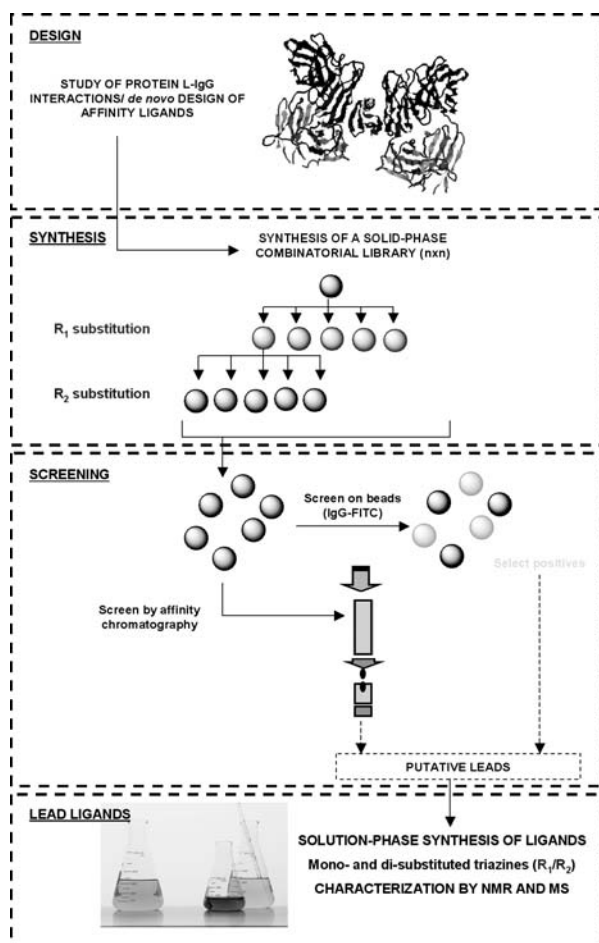


Fig. 1. Strategy followed for the development of synthetic affinity ligands mimicking the interaction of Protein L with the Fab fragment of immunoglobulins.

2. Materials

2.1. Study of the PpL-Fab Binding Site and De Novo Design of Affinity Ligands

1. Computer-aided molecular modelling: Different software packages are commercially available to perform molecular modelling, including Quanta2000 and InsightII from Accelrys, which can run on a IRIX®6.5 Silicon Graphics®Octane® workstation from Silicon Graphics Inc. Some molecular modelling studies were also carried out in a microsoft windows environment using WebLabViewerLite (<http://www.msi.com>), SwissPDBViewer3.7 (<http://www.expasy.ch/spdbv>),

and RasMol V2.7.1.1 (<http://www.umass.edu/microbio/rasmol/>). Protein X-ray and nuclear magnetic resonance (NMR) crystallographic structures are available from the Brookhaven database (<http://www.rcsb.org/pdb/>), which possesses over 33,000 entries. For the development of a PpL mimic, we have utilized the crystal structure of the complex between a single PpL domain and a human antibody Fab fragment (Fab 2A2; human V_L - $\kappa 1$) refined to 2.7 Å (PDB code: 1HEZ) (10).

2.2. Synthesis of Bis-Substituted-Triazine Ligands

1. Sepharose® CL-6B: Product available from Amersham Biosciences-GE Healthcare (Piscataway, NJ), which can be obtained as a suspension of beads in a 20% (v/v) aqueous ethanol solution. Must be stored at 4°C, avoiding periods of dryness. Agitation of gel suspensions, when required, should be made with an orbital shaker and not using a magnetic stirrer.
2. Epichlorohydrin (1-chloro-2,3-epoxypropane): Widely available chemical (a high purity (+99%) or equivalent should be used) which is utilized to epoxy-activate the Sepharose® CL-6B beads or other surfaces. It is a very unstable compound and must be stored in an anhydrous environment at 0–4°C. The extent of epoxy activation of beads can be determined (*see Note 1*). Hazards: Flammable, poison, toxic by inhalation, and in contact with skin and if swallowed may cause cancer. Toxicity data: LD50 90 mg/kg oral, rat. Note: Should be handled in a fume hood with safety glasses and gloves and treated as a possible cancer hazard.
3. Ammonia aqueous solution (35% (v/v)): Widely available chemical, which is used to introduce free amino groups in the epoxy-activated beads and can be quantified by the 2,4,6-trinitrobenzenesulfonic acid (TNBS) test (*see Note 2*). Hazards: Poison, corrosive alkaline solution, causes burns, harmful if swallowed, inhaled, or absorbed through skin. Toxicity data: LD50 3500 mg/kg oral, rat. Note: Should be handled in a fume hood with safety glasses and gloves.
4. Ninhydrin (1,2,3-triketohydrindene monohydrate): Widely available chemical that is light sensitive. Ninhydrin reacts with free amines (2:1 molar ratio) giving a purple product (Ruhemann's purple resonance structure). Used as a 0.2% (w/v) solution in ethanol for the qualitative determination of aliphatic amines on the agarose beads (*see Note 3*). Hazards: Harmful if swallowed; skin, eye, and respiratory irritant. Toxicity data: LD50 78 mg/kg intraperitoneal, mouse. Note: Should be handled in a fume hood with safety glasses and gloves.
5. Cyanuric chloride (2,4,6-Trichloro-*sym*-1,3,5-triazine; Chloro-triazine; Trichloro-cyanidine): This is widely available. A high purity (99%) compound should be used. It is a very reactive compound and must be stored at 2–8°C in an anhydrous environment. It is recommended to recrystallize in petroleum ether (*see Note 4*). Hazards: Poison, lachrymator, and irritant to eyes, skin, and respiratory system. May be harmful if swallowed. Toxicity data: LD50 485 mg/kg oral, rat. Note: Should be handled in a fume hood with safety glasses and gloves and treated as a possible cancer hazard.
6. Amines: For the development of the artificial PpL, the compounds utilized to sequentially substitute the chlorines of the triazine molecule were: L-alanine

(1), 1,5-diaminopentane (2), tyramine (3), *m*-xylylenediamine (4), phenethylamine (5), isoamylamine (6), 4-aminobutyric acid (7), 4-aminobenzamide (8), 1-amino-propan-2-ol (9), β -alanine (10), 2-methylbutylamine (11), 4-aminobutyramide (12), whereas ammonia was considered as a control (0). Apart from compound 12 (synthesized according to procedure described by Boeijen and Liskamp (*II*)), all the amines were commercially available, and hazards and toxicity data were considered individually for each compound according to suppliers' recommendations. The compounds must be dissolved in an appropriate buffer, either an aqueous solution (usually for hydrophilic amines) or an organic solvent such as a 50% (v/v) aqueous solution in dimethylformamide (DMF). In any case, usually 1 molar equivalent of NaHCO_3 is added in order to neutralize the HCl released during nucleophilic substitution. Caution: DMF is harmful and considered a potential carcinogen. Should be handled in a fume hood.

2.3. Assessing the Affinity of Ligands for the Target Protein

1. Proteins tested: The human proteins utilized in the search of a PpL mimetic ligand are widely available from various suppliers and included IgG, Fab, F(ab')_2 and Fc (crystallizable fragment). Reagent grade proteins with $\geq 95\%$ purity must be used. Caution: Human proteins are considered biohazardous, handle as if capable of transmitting infectious agents.
2. Buffers: The buffers used for the screening of the ligands vary from case to case, being dependent on the type of protein studied, the standard conditions recommended for its use and the type of interactions exploited in the affinity purification. The regeneration buffer usually utilized is 0.1 M NaOH in 30% (v/v) isopropanol. The regeneration buffer is used to remove any physically adsorbed ligand prior to screening and after the screening procedure to remove retained protein. Special care should be taken when using iso-propanol (Hazards: flammable, irritant to eyes, respiratory system, and skin). Toxicity data: LD50 10g/kg oral, human (Should be handled with gloves, safety glasses and avoid vapors). The equilibration/binding and elution buffers were selected for according to the usual operational conditions used in PpL affinity chromatographic assays (12). The former consisted of phosphate-buffered saline (PBS) (10 mM sodium phosphate, 150 mM NaCl, pH 7.4) and the latter contained 0.1 M glycine-HCl pH 2 (1 M Tris-HCl, pH 9, was then added to the elution samples to neutralize the pH).

2.3.1. Screening Techniques

1. Fluorescein isothiocyanate (FITC)-based screening: The requirements are as follows.
 - a. Target protein: Must be conjugated with FITC-isomer I (F), and the conjugation occurs through free amino groups of proteins or peptides, forming a stable thiourea bond. Conjugated proteins can be bought from most suppliers of biochemical products, but the protein (P) can also be chemically modified in

house using, for example, the FluoroTag™ FITC-conjugation kit (Sigma), with which different conjugation ratios can be obtained (molar F/P of 2 is recommended (**13**)). The conjugates may then be purified using pre-packed PD-10 columns (Amersham Biosciences-GE Healthcare) and characterized in terms of the F/P ratio:

$$\text{Molar } \frac{F}{P} = \frac{\text{MW}_{\text{protein}}}{389} \times \frac{A_{495}/195}{A_{280} - [(0.35 \times A_{495})]/\epsilon_{280}^{0.1\%}}$$

where $\epsilon_{280}^{0.1\%}$ is the absorption at 280 nm of a protein at 1 mg/ml; $A_{280 \text{ nm}}$ is the absorbance measured at 280 nm.

- b. Glass slides.
 - c. A fluorescence microscope with appropriate filters for the fluorophore used.
2. Affinity chromatography: When performing preparative small-scale assays, disposable empty columns, for example, Bond Elut TCA® (4-ml propylene columns with 20- μm frits) from Varian Inc. can be used. Alternatively, if choosing an automatic system of sample/buffer loading and sample collection, for example, the FPLC system from Amersham Biosciences-GE Healthcare, the affinity resins must be properly packed in columns recommended by the supplier. The determination of bound/washed and eluted protein can be performed with different techniques, such as measurement of absorbance at 280 nm (using a conventional spectrophotometer), quantitation of protein with colorimetric assays (such as the Pierce BCA Protein Assay Reagent Kit from Pierce Biotechnology), or quantitative ELISA (when utilizing small amounts of protein (**14**)), among others.

2.3.2. Characterization of the Affinity Interactions Ligand-Protein

1. Partition equilibrium studies: Requires several Eppendorf tubes containing solutions of the target proteins [usually 5–0.1 mg/ml in equilibration buffer] and the agarose-immobilized ligand.
2. Competitive ELISA: Requires 96-well microtiter plates and ELISA plate reader equipment. Proteins utilized were human IgG and human Fab (unconjugated and conjugated to EZ-Link™ Activated Peroxidase (HRP) according to the supplier instructions; Pierce Biotechnology) and PpL. Solutions needed included coating buffer (0.05 M sodium carbonate-bicarbonate, pH 9.6); PBS-Tween (PBST 20; 0.05% (v/v)), ligand 8/7 solution (82 μM in 50% DMF : PBS); freshly prepared substrate solution (5 mM Na_2HPO_4 , 2 mM citric acid, 1.85 mM *o*-phenylenediamine dihydrochloride (OPD; Merck) and 0.04% (v/v) H_2O_2); stopping solution (50 μl of H_2SO_4 , 2 M). Caution: OPD is harmful and considered a potential carcinogenic; hydrogen peroxide and sulphuric acid are harmful and corrosive—all these chemicals should be handled in a fume hood with safety glasses and gloves.

3. Methods

3.1. Design of PpL Mimic Ligands(15)

1. Study of the complex between PpL and Fab: The complex structure is asymmetric because a single PpL domain contacts similar V_L regions of two Fab molecules via independent interfaces; the PpL domain is, in effect, sandwiched between two antibody Fab molecules (*see Fig. 2*). In the first interface, there are six hydrogen bonds joining the β -sheets of the PpL domain and the V_L domain into a unique sheet, through a β -zipper type of interaction. In total, there are 13 residues from the Fab involved in the interaction with the C* PpL domain. There are 12 residues from the PpL domain (strand $\beta 2$ and α helix) involved in the interaction: Lys24, Ile34, **Gln35**, **Thr36**, **Ala37**, **Glu38**, **Phe39**, **Lys40**, Glu49, Arg52, **Tyr53** and Leu56. Residues in bold are critical residues in the interaction with the light chains, not only by being conserved in different PpL domains, but also by being largely buried upon complex formation. The second binding interface involves 15 residues from the V_L domain, 10 of them in common with the first binding interface. None of the PpL domain residues that contribute significantly for the second binding interface are involved in the first one. Arg52 is a common residue to both interfaces, although this position is not conserved amongst different Ig-binding domains from PpL (it is replaced by an Ala). The 14 PpL domain residues involved in the second interaction are located in strand $\beta 3$ and α helix (Phe43, Glu44, Thr47, Ala48, Tyr51, Arg52, Asp55, Tyr64, Thr65, Ala66, Asp67, Leu68,

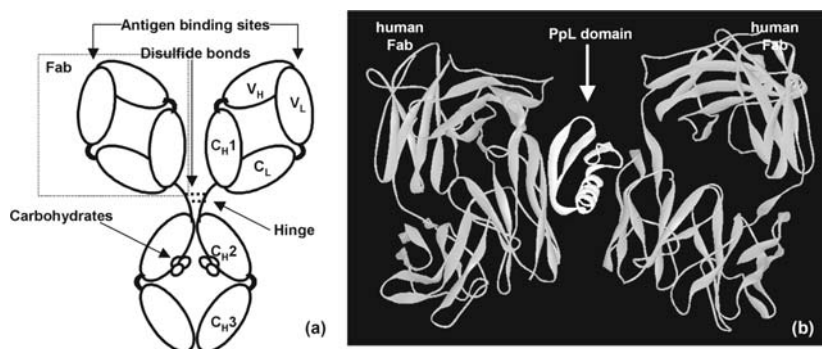


Fig. 2. Basic structure of immunoglobulins (Ig) (a) showing the main components of IgG: the Fab fragments contain the antigen-binding sites of the molecule whereas the Fc fragment comprise of the C_H2 and C_H3 domains as well as the carbohydrate portion. The hinge region is responsible for the flexibility of the Ig molecules, particularly conferring a wide range of movements to the Fab portions. The X-ray crystallographic structure of the complex formed between two human Fab fragments and one Protein L (PpL) domain is shown on part (b) of the figure (1HEZ.pdb). The structural information inferred from this biological interaction was used as the basis for the de novo design of PpL biomimetic ligands.

Gly71 and Gly72). Six hydrogen bonds and two salt bridges mediate the interaction between Fab and the second PpL binding interface.

2. Selection of compounds to be included in the solid-phase combinatorial library: There is a total of 11 different amino acid residues of the PpL domain (including interfaces 1 and 2) involved in the interaction with the light chains, which are generally exposed to the solvent, promoting hydrogen bonds or salt bridges or being largely buried upon complex formation. These amino acid residues—Ala, Asp, Gln, Glu, Gly, Ile, Leu, Lys, Phe, Thr and Tyr—were used as the basis for the design of analog compounds. The analogue compounds all possess an amine-terminal group to react with cyanuric chloride, and their structures are equivalent to the side chains of the amino acid residues they mimic. Amine 4 (*m*-xylylenediamine) resembles a lysine side chain by possessing a $-\text{CH}_2\text{NH}_2$ terminal group but with the addition of an aromatic ring. Similarly, compound 8 (4-amino-benzamide) bears a resemblance to glutamine and asparagine residues by having a terminal amide group.

3.2. Synthesis of Bis-Substituted-Triazine Ligands

3.2.1. Solid-Phase Combinatorial Synthesis of a Ligand Library

1. Epoxy activation of agarose beads: The required amount of Sepharose® CL-6B is washed with 40 ml of distilled water/g of gel on a sinter funnel. The washed agarose is transferred to a 1-l conical flask and 1 ml of distilled water/g of gel added. To this moist gel, 0.8 ml of 1 M NaOH/ml of gel and 1 ml of epichlorohydrin/ml of gel are added. The slurry is incubated for 10–12 h, at 30°C on a rotary shaker. The epoxy-activated gel is washed with 40 ml of distilled water/g of gel on a sinter funnel and used directly for amination. The epoxy content is determined according to **Note 1**.
2. Amination of agarose beads: The washed epoxy-activated gel is suspended in 1 ml of distilled water/g of gel in a 1-l conical flask. About 1.5 ml of ammonia/g of gel is added, and the gel is incubated for 12 h at 30°C in a rotary shaker. The aminated gel is washed with 40 ml of distilled water/g of gel on a sinter funnel and stored in 20% (v/v) ethanol at 0–4°C. The extent of amination is determined as described in **Note 2**. Aminated beads can also be purchased from Amersham Biosciences-GE Healthcare.
3. Cyanuric chloride activation: Aminated agarose is suspended in a 1-l conical flask, in a solution of acetone/water 50% (v/v), using 1 ml of solution/g of gel. This mixture is maintained at 0°C in an ice bath on a shaker. Recrystallized cyanuric chloride (5 molar excess to aminated gel) is dissolved in acetone (8.6 ml/g cyanuric chloride) and divided into 4 aliquots. The aliquots are added to the aminated gel, with constant shaking at 0°C and the pH maintained neutral by the addition of 1 M NaOH. Each aliquot is added with an interval of about 30 min, and samples of gel are taken in order to evaluate the presence of free amines (*see Note 3*). When the four aliquots are added, the gel is washed, with 1 l of each of the following

- mixtures acetone : water (v/v)—1:1, 1:3, 0:1, 1:1, 3:1, 1:0, 0:1. Cyanuric chloride activated gel is not stored but used immediately for R₁ substitution.
4. Nucleophilic substitution of R₁: Cyanuric chloride activated gel is divided into n aliquots, where n is the number of different amines used to synthesize the combinatorial library. A twofold molar excess (relative to the amount of amination of the gel) of each amine is dissolved in the appropriate solvent (1 ml/g gel). The n aliquots are suspended in the previous mixture and incubated at 30°C in a rotary shaker (200 rpm) for 24 h. After this period, each R₁ substituted gel is thoroughly washed on a sintered funnel with the appropriate buffer for each amine. The resulting gel is stored in 20% (v/v) ethanol at 0–4°C or used immediately for R₂ substitution.
 5. Nucleophilic substitution of R₂: The n amines selected are dissolved in 15 ml of appropriate solvent. Each amine is in 5 molar excess to the amount of amination of the gel. Each aliquot of R₁ substituted gel is divided into 5 ml fractions, suspended in the previous mixture and incubated at 85°C for 72 h. At the end of the synthesis, the gels are washed with appropriate solvent, weighed and stored at 0–4°C in 20% (v/v) ethanol.

3.2.2. Solution-Phase Synthesis of Lead Ligands

The conditions vary from case to case and need to be optimized accordingly. Solution-phase synthesized ligands are characterized by ¹H-NMR, ¹³C-NMR and mass spectroscopy and further immobilized on a solid support (*see Note 5*). The synthesis of the PpL-mimic lead ligand, ligand 8/7 was done as shown in **Fig. 3**.

3.2.2.1. SYNTHESIS OF 4-(4,6-DICHLORO-[1,3,5]TRIAZIN-2-YLAMINO) BENZAMIDE

Cyanuric chloride (3.68 g, 20 mmol) was dissolved in acetone (90 ml) and ice water (20 ml) at 0°C. To this, a mixture of 4-aminobenzamide (2.72 g, 20 mmol)

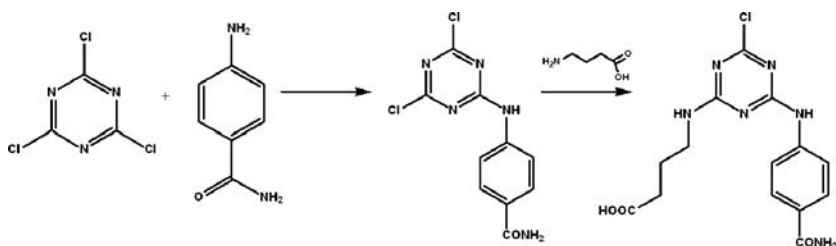


Fig. 3. Basic steps followed on the solution-phase synthesis of the lead ligand (ligand 8/7). Details of the synthesis are given in **Subheading 3.2**.

dissolved in acetone (30 ml) and water (60 ml) and NaHCO_3 (1.68 g, 20 mmol) in water (30 ml) were added dropwise. The reaction mixture was stirred for 2 h at 0°C . The reaction was monitored by TLC (solvent system: ethyl acetate/methanol 95:5, v/v) and stopped when no cyanuric chloride was detected. The resultant yellowish solid product was filtered off, washed with hot water and heptane and dried in vacuo over solid P_2O_5 . Yield: 90% (5.15 g, 18.1 mmol). R_f 0.6 (EtOAc/MeOH 95:5, v/v). $^1\text{H-NMR}$ (400 MHz, $[\text{D}_6]$ DMSO, 25°C): δ 7.34 (s, 1H, NH), δ 7.65, 7.67 (d, 2H, ArH), δ 7.87, 7.89 (d, 2H, ArH), δ 7.92 (s, 1H, NH), δ 11.32 (s, 1H, NH). $^{13}\text{C-NMR}$ (500 MHz, $[\text{D}_6]$ DMSO, 25°C): δ 120.16, 128.42 (ArC), δ 129.63, 140.11 (ArC, quaternary), δ 166.88 (CONH_2), δ 166.18, 167.69, 169.41 (C triazine). MS (EI, CONCEPT) calculated for $\text{C}_{10}\text{H}_7\text{Cl}_2\text{N}_5\text{O}$: 283.00, found 283.00. MS (ESI, Q-tof) calculated for $\text{C}_{10}\text{H}_7\text{Cl}_2\text{N}_5\text{O}$ ($\text{M}+\text{H}$) $^+$: 284.0, found 284.0. Melting point $>250^\circ\text{C}$.

3.2.2.2. SYNTHESIS OF 4-[4-(4-CARBAMOYL-PHENYLAMINO)-6-CHLORO-[1,3,5]TRIAZIN-2-YLAMINO]-BUTYRIC ACID

To a solution of 4-(4,6-dichloro-[1,3,5]triazin-2-ylamino)benzamide (1.98 g, 7 mmol) in DMF (100 ml) and water (15 ml), a mixture of 4-aminobutyric acid (0.72 g, 7 mmol) in water (30 ml) and NaHCO_3 (0.58 g, 7 mmol) in water (30 ml) was added. The reaction was carried out at $45\text{--}50^\circ\text{C}$ with constant stirring for 24 h, monitored by TLC (solvent system: ethyl acetate/methanol 95:5, v/v) and stopped when no 4-aminobutyric acid was detected by the ninhydrin coloration test. The white precipitate formed was filtered off, washed with water and dried in vacuo over solid P_2O_5 . The white solid was dissolved in an aqueous solution of K_2CO_3 5% (w/v) and washed four times with ethylacetate. The aqueous phase was neutralized with HCl (5 M) and the resultant white precipitate filtered, washed with water and dried in vacuo over solid P_2O_5 . Yield: 36% (0.87 g, 2.5 mmol). R_f 0.4 (EtOAc/MeOH 95:5, v/v). $^1\text{H-NMR}$ (400 MHz, $[\text{D}_6]$ DMSO, 25°C): δ 1.71–1.82 (m, 2H, $\text{NHCH}_2\text{CH}_2\text{CH}_2\text{COOH}$), δ 2.25–2.31 (m, 2H, $\text{NHCH}_2\text{CH}_2\text{CH}_2\text{COOH}$), δ 3.26–3.29 (t, 2H, $\text{NHCH}_2\text{CH}_2\text{CH}_2\text{COOH}$), δ 7.20 (s, 1H, NH), δ 7.76–7.84 (m, 4H, ArH and 1H, NH), δ 8.16, 8.24 (s, 2H, $-\text{CONH}_2$), δ 10.11, 10.23 (s, 1H, $-\text{COOH}$). $^{13}\text{C-NMR}$ (400 MHz, $[\text{D}_6]$ DMSO, 25°C): δ 24.31, 24.55, 31.38 (aliphatic CH_2), δ 119.47, 128.32 (ArC), δ 128.61, 128.67 (ArC quaternary), δ 142.04 (CONH_2), δ 165.80 (COOH), δ 168.30, 168.36, 174.65 (C triazine). MS (LSIMS, CONCEPT) calculated for $\text{C}_{14}\text{H}_{15}\text{ClN}_6\text{O}_3$ ($\text{M}+\text{H}$) $^+$: 351.09, found 351.0. MS (ESI, CONCEPT) calculated for $\text{C}_{14}\text{H}_{15}\text{ClN}_6\text{O}_3$ ($\text{M}+\text{Na}$) $^+$: 373.09, found 373.1. Melting point: $218\text{--}219^\circ\text{C}$.

3.3. Screening of Affinity Ligands and Characterization of Affinity Interactions

3.3.1. Screening with the Conjugate FITC-Protein

Each synthesized affinity matrix (50 μ l) is mixed with 100 μ l of distilled water in an Eppendorf tube, centrifuged for 2 min at $1430 \times g$, the supernatant discarded and $2 \times 100 \mu$ l regeneration buffer added to the resin (*see Fig. 4*). The components are gently mixed and centrifuged for 2 min at $1430 \times g$, the supernatant discarded and $2 \times 100 \mu$ l of distilled water added to the resin. The components are mixed and centrifuged for 2 min at $1430 \times g$, the supernatant discarded and $2 \times 100 \mu$ l of equilibration buffer added. The components are again mixed and centrifuged for 2 min at $1430 \times g$. A conjugate FITC-target protein (50 μ l; 1 mg/ml in equilibration buffer) is added to the resin, and the mixture incubated in the absence of light for 15 min with orbital agitation. After this period, the resin is washed in the dark with 3×1 ml equilibration buffer (centrifuging the incubated resin with buffer at $1430 \times g$ and then discarding the supernatant). Each immobilized ligand matrix (1.5 μ l) is placed on a microscope slide and observed under a fluorescence microscope (FITC, $\lambda_{\text{exc}} = 495$ nm; $\lambda_{\text{em}} = 525$ nm). The control experiments consist of repeating the procedure described above using Sepharose® CL-6B, aminated agarose and control ligand 0/0. The results obtained by this screening system were compared with the data resultant from the affinity chromatography test (*13*).

3.3.2. Screening of Affinity Ligands by Affinity Chromatography (Performed at Room Temperature)

The affinity ligands (1 g of moist gel) are packed into 4-sml columns (0.8×6 cm). Each matrix is washed with 2×3 ml regeneration buffer and then with distilled water to bring the pH to neutral. The resins are equilibrated with 10 ml of equilibration buffer. Protein to be tested is reconstituted to 1 mg/ml in

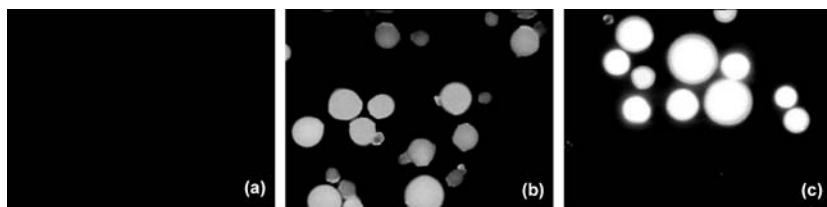


Fig. 4. Typical results obtained with the fluorescein isothiocyanate-based screening system, showing examples of non-binding ligands (a), binding ligands (b) and strongly binding ligands (c).

equilibration buffer and the absorbance at 280 nm measured. Protein solution (1 ml) is loaded onto each column. The columns are washed with equilibration buffer until the absorbance of the samples at 280 nm reaches ≤ 0.005 . Bound protein is eluted with the elution buffer (1 ml fractions collected). After elution, the columns are regenerated with regeneration buffer, followed by distilled water and equilibration buffer, and stored at 0–4°C in 20%(v/v) ethanol.

3.3.3. Characterization of Affinity Interactions by Partition Equilibrium Experiments

The immobilized ligand in study is treated with regeneration buffer and then equilibrated in equilibration buffer. A series of Eppendorf tubes are prepared with 1 ml of standard protein solutions in equilibration buffer (5 tubes at 0.1 mg/ml; confirm concentration by $A_{280\text{ nm}}$ measurement). Immobilized ligand (0.1 g of moist weight gel previously dried under vacuum in a sintered funnel) is added to each Eppendorf tube and incubated for 24 h, at room temperature and under orbital agitation. After this period, the Eppendorf tubes are centrifuged (1 min; 1430 g) to settle the matrix, and the supernatant is taken to measure the $A_{280\text{ nm}}$. The control experiment comprised of incubating the partitioning solute with unmodified Sepharose® CL-6B. The data collected from these experiments are then utilized to calculate the affinity constants for the interaction of the ligand with the target protein (*see Note 6*).

3.3.4. Competitive ELISA (15)

The wells of an ELISA (*see Fig. 5*) microplate were coated with 100 μ l of PpL (10 μ g/ml) in coating buffer overnight at 0–4°C. After three washing steps with PBST, the plate was blocked with PBST (200 μ l/well) and incubated for 1 h at room temperature. The plate was extensively washed with PBST and 100 μ l of PBST added to each well except the first row. For the determination of the inhibition of ligand 8/7 in the interaction between PpL with IgG and Fab, 200 μ l of ligand 8/7 solution was added to the first row and diluted (1:2) by transferring 100 μ l from well to well along the plate. Protein conjugated to HRP (hIgG-HRP, 1:1,000; hFab-HRP, 1:500 in PBST) (100 μ l) was added to all wells and the plate incubated for 2 h at room temperature. After incubation, the plates were carefully and extensively washed with PBST. Substrate solution (100 μ l) was added to the wells. The plates were incubated at room temperature in the dark (10 min: hIgG-HRP; 30 min: hFab-HRP). After the incubation period, 50 μ l of stopping solution was added to each well and the absorbance read at 490 nm. The control wells contained (i) no protein-HRP, (ii) no protein-HRP and

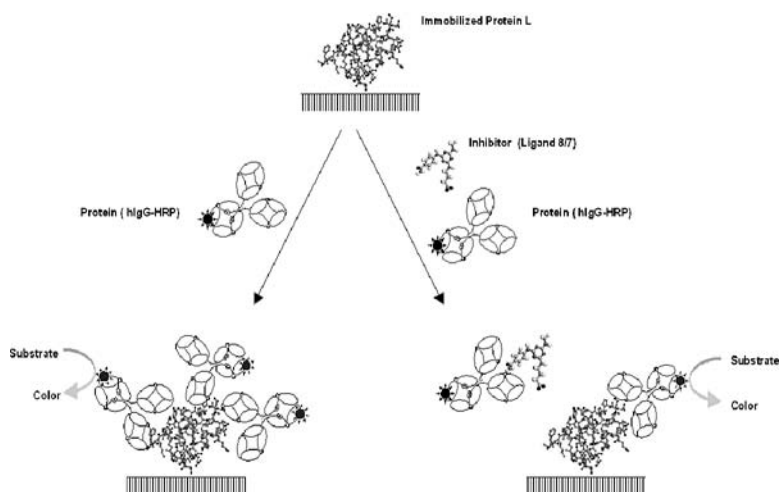


Fig. 5. Schematic representation of the competitive ELISA assay.

no ligand, (iii) protein-HRP and no ligand (corresponding to 100% binding—inhibition data were calculated relative to this value). For the determination of the affinity constants, *see* **Note 7**.

4. Notes

1. Extent of epoxy activation of agarose beads: Sodium thiosulphate (1.3 M) (3 ml) is added to 1 g of epoxy-activated gel and incubated at room temperature for 20 min. This mixture is neutralized with 0.1 M HCl and the amount of HCl used registered. The volume of 0.1 M HCl added corresponds to the number of OH⁻ moles released (10 μ moles per each 100 μ l added), which equals to μ mole epoxy groups/g gel. Therefore, the extent of epoxy activation is expressed as μ l HCl used/10 (μ mol/g gel). The protocol usually results in 25- μ mol epoxy groups/g moist weight gel.
2. Extent of amination on agarose beads with the TNBS test (**16**): Aminated gel (0.1 g) is hydrolyzed with 500 μ l of 5 M HCl at 50°C for 10 min. Upon cooling, the hydrolyzed sample is neutralized with 5 M NaOH and added to 1 ml of 0.1 M sodium tetraborate buffer (pH 9.3) and 25 μ l of 0.03 M TNBS. Samples are incubated at room temperature for 30 min prior to measuring their absorbance at 420 nm. The negative control is 1 ml of distilled water to which sodium tetraborate buffer and TNBS solution (amounts cited above) are added. Calibration curves are constructed with 6-aminocaproic acid (0–2 μ mol/ml). Usual values obtained are 20–25 μ mol amine groups/g moist weight gel.
3. Qualitative test for aliphatic amines: A small amount of moist gel ($\sim$ 1 ml) is placed on a filter paper and ninhydrin in ethanol (0.2%, (w/v)) sprayed on it. The filter paper is heated with a hairdryer (very carefully to avoid burning), until development

of color. Purple or brown coloration indicates, respectively, the presence or absence of free aliphatic amines. Alternatively, the sample of moist gel was placed in a test tube, 2–3 drops of ninhydrin solution added and the test tube heated until development of color (adapted from **ref. 17**).

4. Cyanuric chloride recrystallization: Cyanuric chloride (30 g, 0.16 mol) is dissolved in hot petroleum ether (500 ml) with constant stirring in an oil bath. Heated petroleum ether is poured over a fluted filter paper and the solution of cyanuric chloride filtered into a 1-l conical flask. The saturated solution of cyanuric chloride is left overnight, covered, to allow formation of crystals. The crystals are filtered and dried under reduced pressure. The dried crystals are stable at room temperature in an airtight container. The yield is about 95%.
5. Coupling disubstituted-triazinyl ligands to aminated agarose: To 1 g of moist aminated agarose (24 $\mu\text{mol/g}$) is added a solution containing 5 molar equivalent of the disubstituted-triazinyl and 5 molar equivalent of NaHCO_3 in an appropriate solvent (usually 50%(v/v) DMF : H_2O). The coupling reaction is carried out at 85°C (30 rpm) for 72 h. Agarose beads are then sequentially washed with DMF : water (1:1; 1:0; 1:1; 0:1, v/v) and stored in a solution of ethanol 20% (v/v) at $0\text{--}4^\circ\text{C}$. The ligand concentration on density of immobilized ligands can be determined. Immobilized ligands are washed with regeneration buffer and then neutralized by washing with distilled water. Moist gel (30 mg) containing the immobilized ligand is hydrolyzed in 5 M HCl (0.3 ml) at 60°C for 10 min. On cooling, ethanol (3.7 ml) is added to the hydrolyzed ligand and its absorbance read at the characteristic wavelength estimated for each ligand, against a solution of unmodified agarose submitted to the same treatment. The determination of the extinction coefficient, ϵ_{λ} , for each ligand is made by constructing a standard curve with the measurements of the absorbance read at the characteristic wavelength for different free ligand concentration solutions. Repeating the above-described procedure with 30 mg of unmodified Sepharose®-CL 6B performed the control experiment.
6. Data obtained from the partition coefficient experiments represent adsorption phenomena that usually follow Langmuir type isotherms and can be therefore represented by,

$$q = \frac{Q_{\max} K_a C}{1 + K_a C},$$

in which q is the bound and C the unbound protein, $Q_{\max}$ corresponds to the maximum concentration of matrix sites available to the partitioning solutes (which can also be defined as the binding capacity of the adsorbent), and K_a the association constant. The adsorption data derived from the isotherms can be rearranged into the form:

$$\frac{q}{C} = K_a Q_{\max} - K_a q$$

that represents the Scatchard plot. Scatchard plots indicate whether the interaction between the protein and ligand is (i) reversible and unimolecular (a 1:1 ratio where the

protein binds to a single ligand population and vice versa), (ii) derived from a positive cooperative binding process between equivalent binding sites or (iii) is due to heterogeneous binding sites/negative cooperativity effects. Accordingly, the shape of the Scatchard plot will be linear, convex or concave. The data may be further transformed to Hill plots that assign numerical values to the degree of cooperativity of the system (18). Therefore, considering the existence of n binding sites in the interaction between the protein and the ligand, taking logarithms to the Scatchard plot equation, and using the estimated $Q_{\max}$, a linear Hill plot equation is obtained,

$$\log \left(\frac{q}{Q_{\max} - q} \right) = \log K_a + n_H \log C,$$

where n_H symbolizes the Hill coefficient. This coefficient is not only an indication of the number of binding sites, but also an index of the degree of positive ($n_H > 1$) or negative ($n_H < 1$) cooperativity of the systems (19).

7. For the determination of the affinity constant between PpL and IgG and its fragments, two strategies were considered: in the first row of wells in the ELISA plate, instead of ligand 8/7 solution, a PpL solution (1 μ M) or human IgG or human Fab solutions (1 μ M) were added and the methodology described in **Subheading 3.3., step 4** was followed. The Cheng–Prusoff equation expressed by

$$\frac{1}{K_2} = \frac{ED_{50}}{1 + p.K_1}$$

relates the affinity constant K_2 (association constant of the interaction inhibitor L_2 and L_1) with the ED_{50} , having as constants p (concentration of labeled ligand L_1) and K_1 (association constant for L_1 receptor) (Cheng and Prusoff (1973) in (ref. 20)). The last parameter may be also determined by the Cheng–Prusoff equation where unlabeled molecule L_1 is considered as the inhibitor L_2 , and therefore it is evaluated by the displacement of labeled L_1 by itself. As an alternative, it is also possible to use the receptor in solution as the inhibitor L_2 .

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Phage Display of Peptides in Ligand Selection for Use in Affinity Chromatography

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Summary

Large repertoires of peptides displayed on bacteriophage have been extensively used to select for ligand-binding molecules. This is a relatively straightforward process involving several cycles of selection against target molecules, and the resulting ligands can be tailored to various applications. In this chapter we describe detailed methods to select peptide ligands for affinity chromatography, with particular focus on selection of peptides that mimic antigen epitopes. The selection process involves screening a phage peptide library against a monoclonal antibody, proving the peptide is an authentic epitope mimic and coupling the peptide mimotope to an affinity resin for purifying antibodies from human serum. There are several other applications of phage peptides that could be used for affinity chromatography; the approaches are outlined, but detailed methods have not been included.

Key Words: Phage display; peptides; mimotopes; peptide ligands.

1. Introduction

Phage display of foreign peptides is an established technique now routinely used in many laboratories since the pioneering work by Smith and colleagues 20 years ago (*1*). The flexibility and versatility of isolating peptides with affinity for virtually any desired target has resulted in the growing use of random peptide libraries for a wide variety of applications. Phage peptide libraries can

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be constructed by fusing DNA containing a degenerate region (typically using NNG/T codons to minimize high frequency of stop codons) to a gene encoding a coat protein usually gene III or gene VIII. This allows the foreign peptide to be expressed as an N- or C-terminal fusion on the surface of the M13 bacteriophage (phage) coat protein. A large number of random peptide libraries displayed on bacteriophage are now available, some are disulfide constrained by inserting two cysteine residues, a typical library size ranges from 6 to 43 amino acid residues (2). We chose to construct a 20-residue library, in order to select for peptides long enough to permit short turns and other three-dimensional structural features yet short enough to permit the production of a large diverse library (3).

Selection of peptides of interest from the library that bind to a target molecule can be performed using a process referred to as "panning." This process allows enrichment in binding affinity to the target molecule, and the resulting phage peptides can easily be sequenced and further characterized. Peptides can be synthesized without the phage framework and can be validated as separate entities.

The advantages of peptide ligands for use in affinity chromatography include the relative low cost of high quality stable peptides. In addition, instead of having to prepare an affinity column using the whole recombinant antigen, peptides that represent, for example, the antibody-binding site can be used. This may also be beneficial as it may allow focus on relevant single specificities and avoid unimportant epitopes if the whole protein was used for affinity purification.

There are several applications of phage peptides that can be used for affinity purification.

1. Peptide mimotopes: Phage mimicking the important epitopes of a given antigen can be selected from a random phage peptide library by panning on antibodies that bind to these epitopes. The peptides in principle could be useful for affinity purification of antibodies specific for these important epitopes. These peptides may be useful for serological monitoring of infectious diseases. Note: Definition of peptide mimotopes: Peptides that bind to antibody-binding sites thereby mimicking the three-dimensional conformational features of linear or conformational epitopes. These peptides are defined as mimotopes as they mimic the essential features of the epitope but do not necessarily bear sequence homology with the primary amino acid sequence of the epitope (4).
2. Antigen-binding peptides: Another application of phage display for use in affinity chromatography is the selection of a peptide that binds directly to an antigen. These peptides could be useful for purification of the antigen itself. For example, peptides of 5–7 residues flanked by two cysteines to form a disulfide bond were selected from a phage display library, were immobilized onto a chromatographic

support and used for affinity purification of factor VIII from a complex mixture of proteins (5).

3. Peptides that bind to a complex target: Peptides could also be selected for binding to the surface of a complex target, for example, a cell surface antigen. These peptides could potentially be useful for purification of this antigen from a cell extract or complex mixture. For example, *in vivo* selection techniques have been used to select for peptides that target various tissues by injecting animals or humans with a phage peptide library, and the selected peptides have been used as affinity ligands to identify cell surface receptors (6,7).

Here, we describe a process to select peptides from a random peptide library displayed on phage that can be useful as affinity ligands. A schematic diagram is shown in **Fig. 1**, outlining the major steps involved in this process. We have chosen to describe in detail, methods to isolate a peptide mimotope from a phage displayed random peptide library by isolation of a peptide that can mimic the shape of the antigen epitope and could be used to select antibodies that bind to this particular region of the antigen. We also describe the process of purifying antibodies from human serum that bind to this peptide mimic. This purification process can be used to emphasize the capacity of a peptide mimotope to mimic the antigen epitope. Furthermore, often the resulting antibodies are functional, for example, if the epitope is protective, use of the mimotope to purify naturally occurring protective antibodies from human serum is indicative of the ability of the peptide to mimic the three-dimensional shape of the epitope. This may have implications for generation of a peptide vaccine or the discovery of new protective epitopes (8,9).

2. Materials

2.1. General Reagents

1. Phage displayed peptide library (for the protocols described here, we generated our own in house library. There are several libraries that are commercially available, for example, the 12 or 7 residue Ph.D. library kit by New England Biolabs, the vector can also be purchased for construction of libraries).
2. Target monoclonal antibody and recombinant antigen.

2.2. Panning a Random Peptide Library for a Peptide Mimotope

1. Coating buffer: 0.1 M sodium carbonate/bicarbonate pH 9.6.
2. Elution buffer: 0.1 M glycine, pH 2.2.
3. Equilibration buffer: 1.5 M Tris-HCl, pH 9.
4. Phosphate-buffered saline/Tween (PBST): PBS, 0.05% Tween 20, pH 7.5.
5. Polyethylene glycol (PEG) solution: 20% PEG 8000, 2.5 M NaCl, to 1 l with dH₂O and autoclave, store at 4°C.

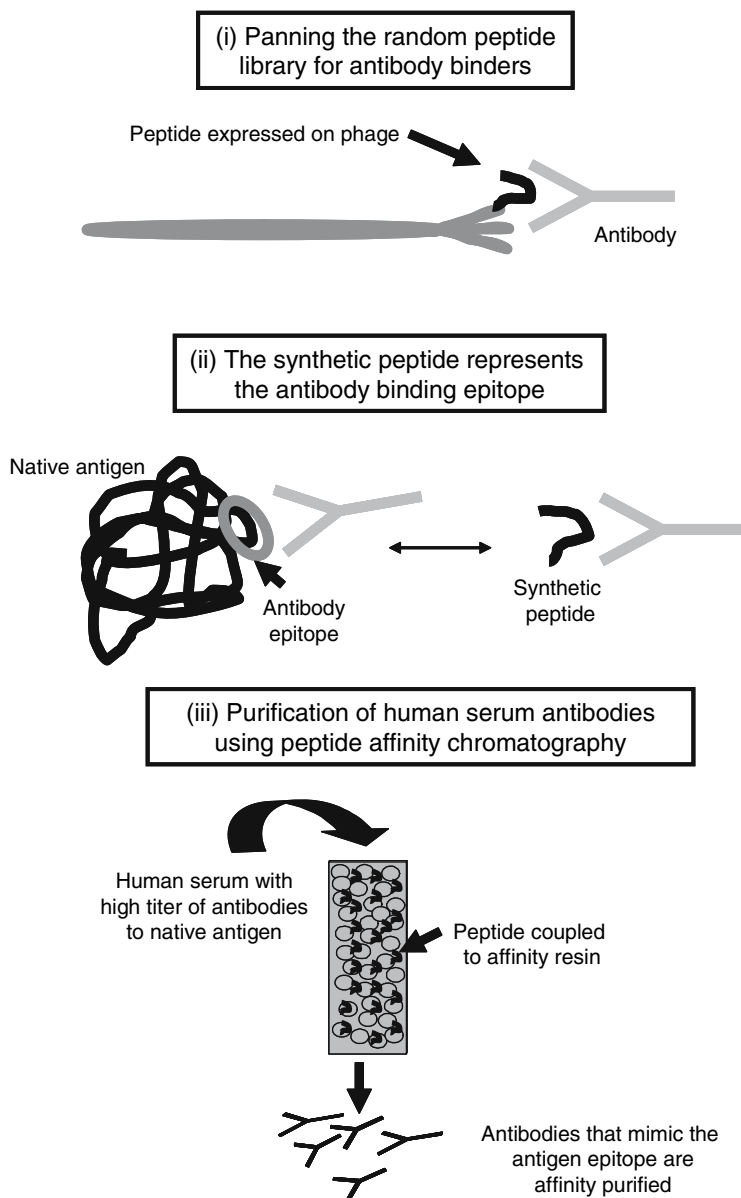


Fig. 1. Schematic diagram of (i) selection of a phage peptide from a random peptide library. (ii) Illustration showing the selected peptide can mimic the shape of the antigen epitope and (iii) peptides can be coupled to an affinity resin and used to selectively purify antibodies specific for this epitope from complex human serum.

6. Blotto: Skim milk powder (any commercial brand) diluted in PBS.
7. Super broth (SB) media: 30 g Tryptone, 20 g Yeast extract, 10 g 3-(N-Morpholino)-propanesulfonic acid (MOPS), to 1 l with dH₂O and autoclave.
8. Yeast tryptone (YT) media: 16 g Tryptone, 10 g Yeast extract, 5 g NaCl, to 1 l with dH₂O and autoclave.
9. YT plates: the same as in step 7 with the addition of 15 g bacto-agar and tetracycline (*see step 10*) when cooled to approximately 50°C. Store plates in the dark as tetracycline is light sensitive.
10. Tetracycline: 40 µg/ml final concentration for plates and liquid media.
11. Minimal media plates: 15 g bacto-agar in 750 ml dH₂O, autoclave and when cooled add 200 ml of 5 × M9 salts (5 × M9 salts: 16.9 g Na₂HPO₄, 7.5 g KH₂PO₄, 1.25 g NaCl, 2.5 g NH₄Cl, 500 ml dH₂O, autoclave), 20 ml of 20% glucose (filter sterilized), 0.5 ml of 1% thiamine-hydrochloride (filter sterilized) and 1 ml of 20% MgCl₂.
12. K91 *Escherichia coli* cells starved culture on minimal media, culture a fresh K91 plate every week.
13. Maxisorp microtiter plates (Nunc), these are recommended for high levels of protein binding.
14. Centrifuge tubes (250 ml clear polypropylene) autoclaved.
15. One-liter flasks autoclaved, baffled flasks are recommended.

2.3. Preparation of a Peptide Affinity Column

1. N-hydroxysuccinimide (NHS)-activated Sepharose 4 fast flow media (Pharmacia Biotech). This resin has been specially developed for coupling of peptides to a solid matrix. It has a highly stable 6-aminohexanoic acid spacer arm which can form an amide linkage with the primary amino group of peptides.
2. Coupling buffer: 0.1 M NaHCO₃, 0.5 M NaCl, pH 7.5
3. In order to maintain the maximum binding capacity of the resin, all solutions should be pre-chilled (0–4°C) and prepared prior to coupling the ligand.

2.4. Affinity Chromatography Using a Peptide Column

1. Wash buffer: 0.1 M boric acid, 0.5 M NaCl, 0.05% Tween 20, pH 8.5.
2. Elution buffer: 0.1 M glycine, pH 2.2.

3. Methods

3.1. Panning a Random Peptide Library for a Peptide Mimotope (See Note 1)

1. Coat 10 wells of an ELISA plate (Nunc Maxisorp) with 100 µl antibody at 5–10 µg/ml diluted in coating buffer overnight at 4°C.
2. Inoculate 10 ml YT media with a colony of K91 cells and grow until log phase (~OD = 0.6 at 600 nm) at 37°C shaking vigorously.

3. Wash the coated plate twice with PBS and block the plate with 200 μ l of 5% blotto for 2–3 hr at room temperature.
4. Take an aliquot of the phage library and dilute to 10^{11} phage/well in 1% blotto. Allow the phage to incubate for 15 min in 1% blotto before adding to the plate to remove the milk binding phage.
5. Wash the plate twice with PBS, then add 100 μ l of the pre-incubated phage to the blocked wells and incubate for 2–3 hr on the bench at room temperature.
6. When the K91 have grown to log phase remove from the shaking incubator and allow to settle. This enables the F-pilus to regenerate.
7. Wash the ELISA plate using increased stringency per round of panning. For example, use the following:
 - a. Round 1: 2 $\times$ PBS washes.
 - b. Round 2: 4 $\times$ PBST, then 2 $\times$ PBS.
 - c. Round 3: 6 $\times$ PBST, then 2 $\times$ PBS.
 - d. Round 4: 8 $\times$ PBST, then 2 $\times$ PBS.
8. Elute the bound phage by adding 100 μ l elution buffer for 10 min, pool the elutions and neutralize with equilibration buffer. Immediately add the pooled phage to the stationary K91 culture and incubate for 1 h at 37°C (mix gently occasionally) to allow re-infection of the eluted phage.
9. Add the re-infected K91 culture to 200 ml SB media (containing 40 μ g/ml tetracycline) and expand the culture overnight at 37°C (vigorous shaking).
10. Centrifuge the culture for 15 min at 4°C at 10,400 g . Prepare glycerol stocks of the pellet (final glycerol concentration 20%). Retain the supernatant and transfer to a centrifuge tube and PEG precipitate overnight, using a 1:5 dilution of PEG solution, shake and incubate on ice overnight in the cold room.
11. Spin the precipitated phage at 16,400 g for 50 min, resuspend in 1.5 ml PBS and re-centrifuge at 15,700 g to remove remaining cell debris. Store phage at –80°C.
12. Repeat **steps 1–11** for subsequent rounds of panning.

3.2. Analyzing Rounds of Panning by ELISA

To ensure the panning process has been successful, an ELISA should be performed. An example of the typical results obtained is shown in **Fig. 2A**.

- 1) Coat a microtiter plate (Nunc Maxisorp) with the antibody that was used for panning using the same conditions (*see Subheading 3.1.*).
- 2) Wash and block as per panning conditions (*see Subheading 3.1.*).
- 3) Prepare phage dilutions of rounds 0–4, usually 10^{10} /ml (*see Subheading 3.3.*) for titration in PBS, apply 100 μ l in duplicate wells and incubate for 1 h on a plate shaker at room temperature. To check for non-specific binding, test for phage binding to the blocking solution only or coat with an isotype control antibody.
- 4) Wash four times with PBST.

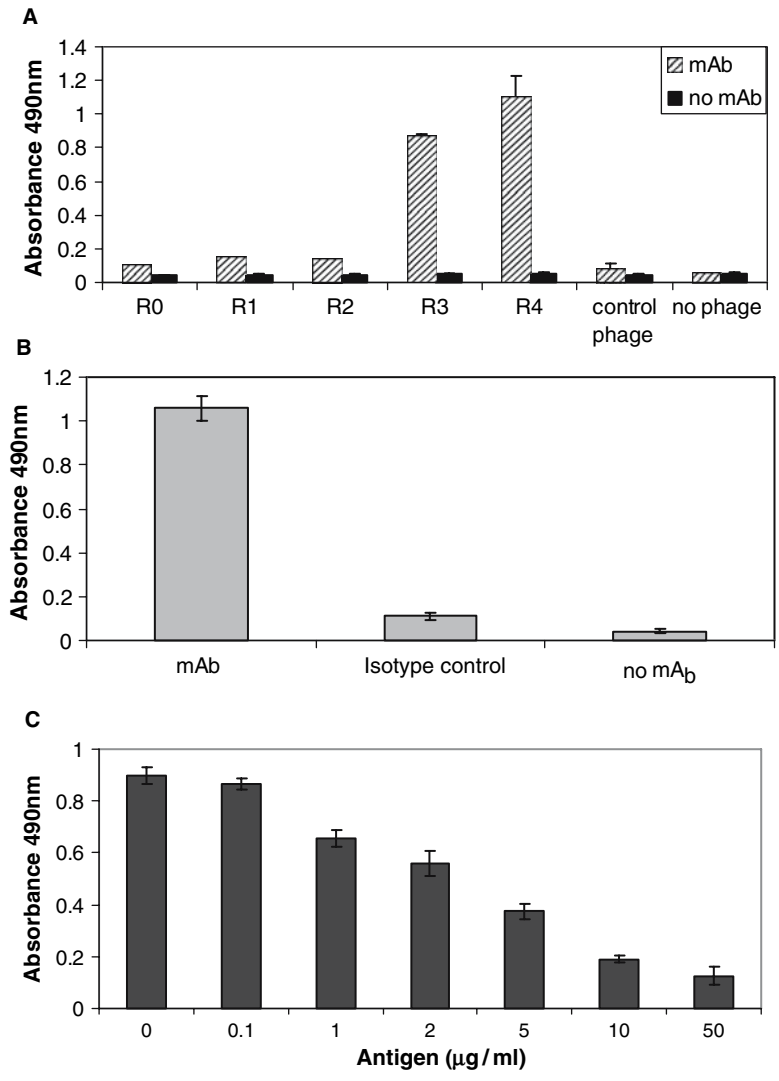


Fig. 2. Selection and characterization of phage clones as mimotopes. (A) Reactivities of selected phages from each round (R) of panning on a monoclonal antibody (mAb) detected by ELISA. (B) Binding of a selected phage clone to the mAb but not to an antibody of the same isotype shown by ELISA. (C) Recombinant antigen is shown to compete with the phage clone for binding to the parent mAb by ELISA.

- 5) Apply anti-M13-horseradish peroxidase (HRP) conjugate (Pharmacia Amersham) diluted in PBST at 1/5000, apply 100 μl /well and incubate with shaking for 1 h at room temperature.

- 6) Wash four times as above, add substrate 100 μ l/well O-phenylenediamine (OPD Sigma P-3804), wait until color develops and stop color reaction with 100 μ l/well 1 M HCl and read plate at OD_{490 nm}.

3.3. Titration of Phage

- 1) The titer of phage should be determined by preparing 10-fold serial dilutions of phage and allowing re-infection of mid-log phase *E. coli* K91 cells for 30 min at room temperature.
- 2) A sample of each dilution should be plated onto Luria broth (LB) agar plates containing 40 μ g/ml tetracycline; the titer can be derived by counting the number of colonies. Phage titers are expressed as colony forming units per ml (CFU/ml).

3.4. Analyzing Individual Clones for Binding by ELISA

- 1) Streak out around four glycerol stocks and pick 10 individual colonies, inoculate 10 ml of SB media containing 40 μ g/ml tetracycline and allow cultures to grow overnight shaking vigorously at 37°C.
- 2) Prepare phage as in **Subheading 3.1., steps 10 and 11**, and prepare glycerol stocks for the individual colonies.
- 3) Perform an ELISA as in **Subheading 3.2** to test whether individual clones bind to the parent antibody.

3.5. Sequencing of Clones Selected from 20-Mer Phage Peptide Library

The sequence of individual phage clones that have been shown to bind (*see Subheading 3.4.*) can easily be obtained. If more than four different sequences are obtained from sequencing of 10 clones, we recommend performing 1–2 additional rounds of panning and increasing the number of washes.

- 1) Streak out round four glycerol stocks, and pick colonies from each plate for sequencing.
- 2) Polymerase chain reaction (PCR) the insert region using forward and reverse primers: Reverse primer, GCCTGTAGCATTCCACAGACAG; Forward primer, GTGTTTTAGTGTATTCTTTTCGCCTCTTTC. PCR (50 μ l): Colony, 1.0 μ l (made to 20 μ l with sterile dH₂O); Forward primer, 0.5 μ l of a 1/5 dilution, (dilute original stock of primer to 1 μ g/ μ l); Reverse primer, 0.5 μ l of a 1/5 dilution (dilute original stock of primer to 1 μ g/ μ l); Taq, 0.25 μ l; 25 mM MgCl₂, 5 μ l; 10 $\times$ buffer, 5 μ l; deoxynucleoside triphosphates (dNTP), 5 μ l (2.5 mM final concentration of each dNTP); sterile dH₂O, 32.75 μ l. PCR conditions: 94°C for 5 min, 30 cycles of 94°C (30 s), 52°C (30 s) and 72°C (30 s) then 72°C for 7 min.
- 3) After the reaction is complete, analyze a sample on a 1% agarose gel, use a PCR clean up kit and send samples for DNA sequencing.

3.6. Characterization of a Phage Peptide as a Mimotope

It is important to characterize the selected phage clones for their ability to mimic the native antigen. Ideally, two ELISAs should be performed involving competition of the phage with the native antigen for binding to the parent antibody and checking the phage does not bind to the constant or framework regions of the antibody using a relevant isotype control antibody. Examples of the typical results are shown in **Fig. 2B and C**. The same ELISA described in **Subheading 3.2** should be used with the following modifications.

3.6.1. Antigen Competition ELISA

Step 3: Various concentrations of antigen competitor (usually 0.1–200 µg/ml) can be mixed with a constant phage dilution (50 µl of each). The optimal phage concentration to be used can be determined first by titrating the phage and taking a dilution at the top of the binding curve where it begins to plateau.

3.6.2. Isotype Control ELISA

Step 1: Wells should be coated with the isotype control antibody and binding compared to the original antibody.

3.7. Characterization of the Synthetic Peptide

Once it has been established the selected phage clone(s) are true mimotopes, it is important to prove the peptides are functional in the absence of the phage framework (data not shown). This can be performed using the ELISA described in **Subheading 3.2** with the following modifications.

Step 3: Various concentrations of peptide (usually 1–500 µg/ml) can be mixed with a constant phage dilution. The optimal phage concentration to be used can be determined by titrating the phage and taking a dilution at the top of the binding curve where it begins to plateau.

3.8. Preparation of the Peptide Affinity Resin (see Note 2)

- 1) For a 2-ml column weigh out 2 mg of peptide. If the peptide requires organic solvent (e.g., dimethyl sulfoxide or dimethyl formamide) keep the volume of solvent to a minimum, approximately 100–200 µl and mix until the peptide is fully dissolved, then make up to 1 ml with coupling buffer. If the peptide is soluble dissolve directly into 1 ml of coupling buffer.
- 2) Mix the NHS-activated Sepharose until an even gel suspension is apparent. Measure 2 ml of the resin and wash with 15 column volumes (CV) of cold 1 mM HCl.

- 3) Mix the washed medium and the peptide in a 15-ml tube, adjust the pH to 6–8, and the volume should be made to 2 ml with coupling buffer. The coupling reaction should be allowed to proceed overnight at 4°C, mixing very slowly end-over end on a rotator.
- 4) After the coupling is completed, excess ligand should be washed away with 10 CV of coupling buffer, and any non-reacted groups on the medium should be blocked by mixing and standing in 1 M Tris–HCl buffer, pH 8, for 2 h.
- 5) To wash the medium after coupling, four alternative washes with high and low pH should be used. Each cycle consists of a wash with 10 ml of 0.1 M Tris, pH 8, containing 0.5 M NaCl, followed by 10 ml of 0.1 M Na-Acetate, pH 4, containing 0.5 M NaCl.
- 6) The coupled affinity resin should be resuspended in PBS, transferred to an empty column and washed with 20 ml PBS. The column should be stored at 4°C; for long-term storage, the column should be stored in 20% ethanol.

3.9. Affinity Purification Using Peptide Column (See Note 3)

In our example, the peptide column is used to purify antibodies having an affinity for the peptide mimotope from human serum. Ethical approval should be obtained for use of human serum. Collect all fractions.

- 1) Filter human serum (2 ml) using a 0.2- μ m filter and dilute 1:10 in PBS. Retain a small sample for analysis.
- 2) Equilibrate the 2-ml peptide column with 10 CV PBS.
- 3) Load diluted serum onto the column, taking care not to disturb the resin, and allow to pass through the column slowly. Note the flow as it may be stopped at any time, and this allows longer contact time for the serum antibodies with the peptide resin.
- 4) Repeat step 3 passing the serum through the column again and collecting the flow through. Steps 3 and 4 should take at least 1 h to ensure sufficient contact time of the serum antibodies with the peptide resin.
- 5) Wash the column with 50 ml PBS.
- 6) Wash the column with 50 ml wash buffer.
- 7) An additional 50 ml PBS wash should be carried out to ensure all the non-specific serum components are washed away.
- 8) To elute the bound antibodies, 10 ml of elution buffer is added and 10 $\times$ 1 ml fractions collected. Fractions are immediately equilibrated with 2 M Tris and stored at 4°C prior to analysis.
- 9) The column is re-equilibrated with 50 ml PBS and stored at 4°C.

3.10. Analysis of Affinity-Purified Antibodies to Ensure Validity of Column

It is important to assess the efficiency of the peptide affinity resin and identify which of the eluted fractions to pool. Sodium dodecyl sulfate–polyacrylamide

gel electrophoresis could be used to analyze each fraction and the wash fractions; however, there will be no distinction between total serum antibodies and antibodies that bind to the peptide and have been eluted from the column. Therefore, we recommend performing an ELISA using the native antigen as the purified antibodies should bind to the same antigen epitope that the peptide mimics. An example of this is shown in **Fig. 3A**. The methods described in **Subheading 3.2** can be used with the following modifications.

Step 1: Coat wells of a microtiter plate with 2–10 $\mu\text{g/ml}$ antigen.

Step 3: The original, wash and eluted fractions should be diluted 1:10–1:50 in PBST.

Step 5: Anti-human IgG conjugated to HRP should be used at the manufacturers suggested concentration (usually 1/5000 dilution for Chemicon AP113P).

The ELISA results should indicate which eluted fractions should be retained; these should be pooled and dialyzed into PBS and concentrated using an

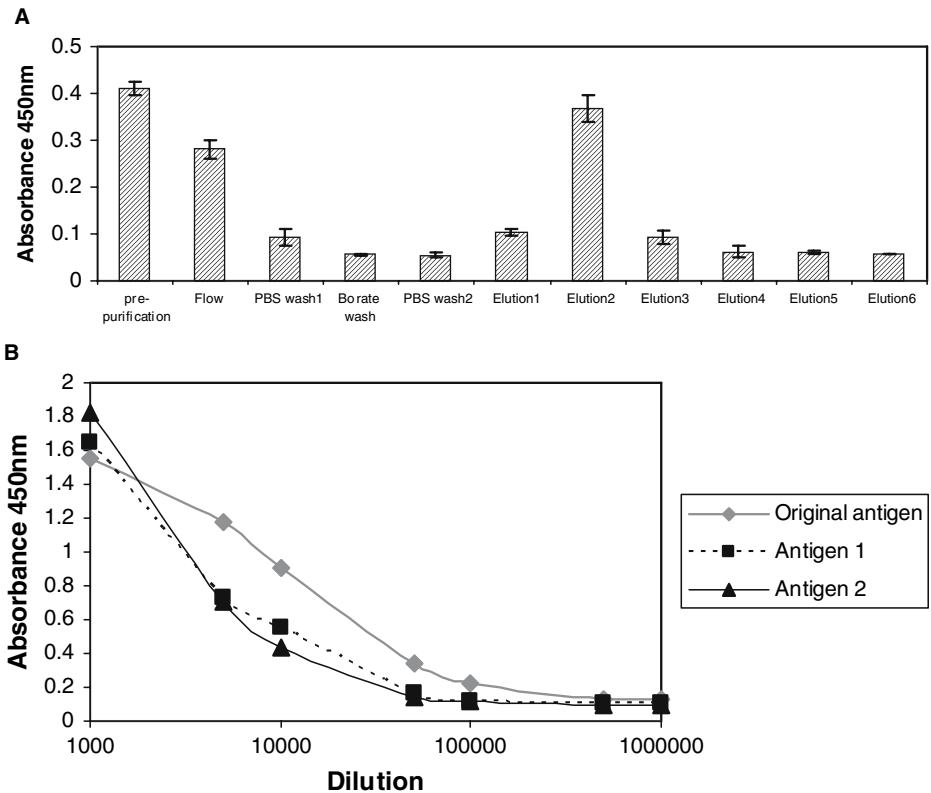


Fig. 3. (Continued)

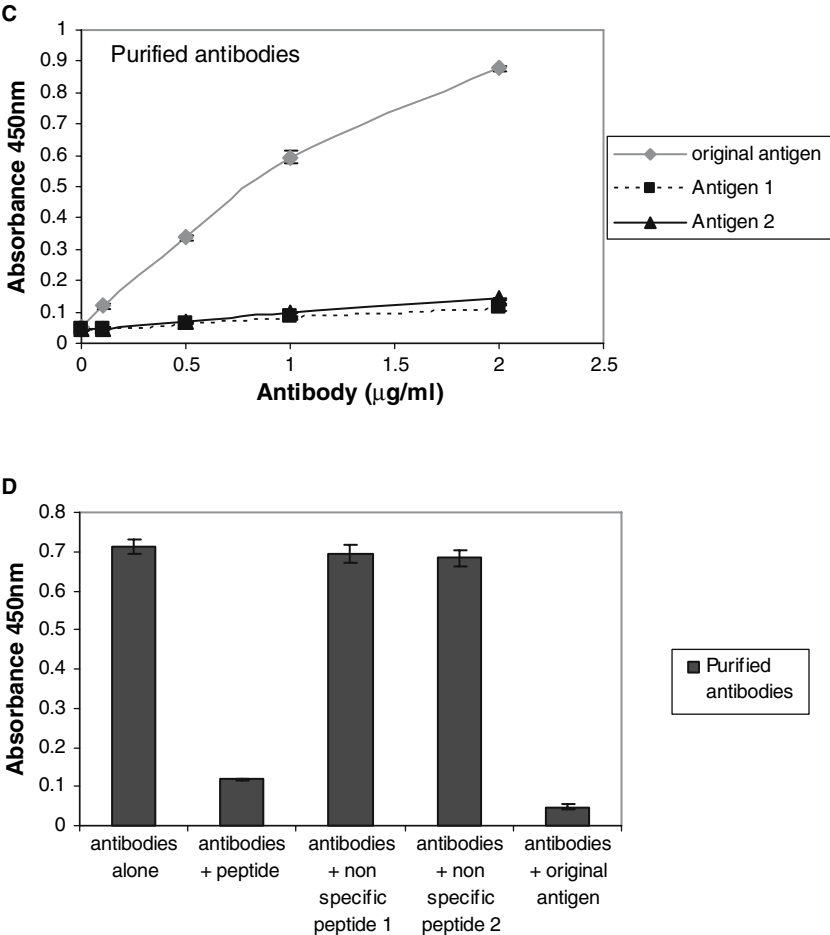


Fig. 3. Purification of human serum antibodies using a peptide affinity chromatography. (A) Reactivity of fractions prior to purification, the flow-through, wash and eluted fractions by ELISA to the parent recombinant antigen. (B) Reactivity of human serum with the parent antigen and two other antigens, prior to peptide affinity purification. (C) Reactivity of the resulting antibodies after peptide affinity purification, demonstrating the purified antibodies specifically bind to the original antigen. (D) The purified antibodies were found to be highly specific for the peptide they were purified against as addition of the peptide or original antigen inhibited binding to the antigen, however, addition of 2 non-specific peptides did not inhibit the binding.

Amicon stirred cell ultrafiltration device (Millipore) if required. The final protein concentration can be determined by measuring the OD_{280 nm} using the extinction coefficient for antibodies of 1.45.

Further ELISA tests can be performed to analyze the efficiency of the peptide affinity resin. Characterization of the serum prior to purification in **Fig. 3B** illustrates binding of the serum antibodies to the original antigen and two other antigens, whereas after purification (*see Fig. 3C*), the eluted antibodies should show higher relative binding to the original antigen than to the two other antigens. This indicates enrichment for antibodies binding to the original antigen via affinity purification using the peptide mimotope. In addition, a competition ELISA could be performed using the resulting peptide-purified antibodies (*see Fig. 3D*). These antibodies should compete with the peptide they were purified against, but should not compete with other non-specific peptides. Furthermore, the peptide-purified antibodies should compete with the original antigen as they share specificity with the peptide mimotope (*see Fig. 3D*). Refer to **Subheading 3.2** for ELISA protocol, **Subheading 3.6** for competition ELISA and the modifications described earlier in this section.

4. Notes

1. Tips for handling bacteriophage: Bacteriophage should be treated with care, and the following points should be considered to prevent possible contamination.
 - a. It is recommended whenever using phage to use filter tips.
 - b. All work surfaces should be cleaned with 2% bleach prior to and after working with phage.
 - c. Pipettes should be cleaned regularly with 2% bleach, certain parts can be autoclaved (check with the manufacturer).
 - d. When performing ELISA washes, use a separate piece of paper towel for blotting. The towel should be placed in a biohazard bag and autoclaved.
 - e. Autoclave all bacteriophage waste.
2. Peptide affinity ligands: This system can be applied to any peptide selected against any potential monoclonal antibodies or polyclonal antibodies.
 - a. The solubility and stability of the peptide will affect the stability of the affinity column and the number of times the column can be used successfully.
 - b. This purification system may result in low yields of protein mainly because antibodies to a single epitope are being selected.
3. Maintenance of the peptide column:
 - a. For long-term storage, the peptide affinity column should be stored in 20% ethanol.
 - b. For sanitation and removal of bacterial contaminants, wash the column with 0.1 M NaOH in 20% ethanol allowing contact for 1 h.
 - c. To prevent clogging of the column, 0.2 μm , filter all buffers and sample prior to loading.

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Preparation, Analysis and Use of an Affinity Adsorbent for the Purification of GST Fusion Protein

Gareth M. Forde

Summary

Methods are presented for the preparation, ligand density analysis and use of an affinity adsorbent for the purification of a glutathione S-transferase (GST) fusion protein in packed and expanded bed chromatographic processes. The protein is composed of GST fused to a zinc finger transcription factor (ZnF). Glutathione, the affinity ligand for GST purification, is covalently immobilized to a solid-phase adsorbent (Streamline™). The GST–ZnF fusion protein displays a dissociation constant of 0.6×10^{-6} M to glutathione immobilized to Streamline™. Ligand density optimization, fusion protein elution conditions (pH and glutathione concentration) and ligand orientation are briefly discussed.

Key Words: Key Words: GST fusion protein; affinity purification; chromatography; expanded bed adsorption.

1. Introduction

Purification based on targeted affinity interactions offers high selectivity and facile purification of biomolecules including the capture of products from complex feed stocks (**1,2,3**). The use of affinity ligands leads to an increased adsorbent selectivity, resulting in higher degrees of purification and potentially higher capacities of adsorbent for the target. Due to its high selectivity, affinity chromatography is a preferred tool in the downstream processing of high-value biomolecules of therapeutic interest (**4**).

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Utilization of the GST-glutathione affinity interaction in a chromatographic process has a number of distinct advantages. It enables a straightforward detection protocol via the use of an enzyme activity assay, a reproducible purification strategy from lysed cell culture via adsorption to immobilized glutathione, high selectivity and a convenient strategy for the regeneration of the affinity adsorbent. An enzyme activity assay facilitates the fast, high-throughput assaying of fractions for the quantitative measurement of GST protein concentration.

Presented is a process for the purification of a GST fusion protein. The protein is composed of GST fused to a zinc finger transcription factor (ZnF). The bi-functional fusion protein displays dual affinity for glutathione, via the GST segment, and a specific DNA sequence, via the zinc-finger motif. The protein was ultimately designed for the affinity purification of plasmid DNA. The zinc finger is also known as the Cys₂His₂ zinc finger and is a transcription factor that regulates the expression of proteins by binding specifically to certain DNA sequences. The production of a zinc finger protein that displayed affinity for a 9-base pair sequence was first reported by Desjarlais and Berg (5).

In this work, glutathione is covalently immobilized to a solid-phase adsorbent (Streamline™). The primary biological function of glutathione is to act as a non-enzymatic reducing agent to help keep cysteine thiol side chains in a reduced state on the surface of proteins, which has led to its use as a medicinal antioxidant. Glutathione prevents oxidative stress in most cells and helps to trap free radicals that can damage DNA and RNA. GST catalyzes the nucleophilic attack of the sulphur atom of the glutathione on electrophilic groups of a variety of hydrophobic substrates, including herbicides, insecticides and carcinogens (6,7). The GST–ZnF fusion protein displayed a dissociation constant of 0.6×10^{-6} M to glutathione immobilized to Streamline™, which is similar to that reported for recombinant GST binding to a glutathione-Sepharose™ affinity adsorbent of 1.15×10^{-6} M (8).

Packed bed and expanded bed operation modes were employed to purify the target GST fusion protein. Expanded bed adsorption (EBA) is a quasi-packed bed unit operation through which large particulates (such as suspended solids in non-clarified feeds) can pass. EBA enables bio-target recovery directly from particulate containing feedstocks like cell homogenates or fermentation broth (for extracellular bio-targets). EBA can complete the functions of clarification, concentration and purification in one stage and thereby increase the total yield and reduce the operation time of a process system by reducing the number of stages.

2. Materials

2.1. Biomolecules

1. pM6: The GST-ZnF Cloning Vector, or pM6, was created by inserting a 319 base pair (bp) segment, coding for the zinc finger gene, into pGEX-2TK (4969 bp, accession number U13851.1) between the BamHI and EcoRI restriction sites [3]. The pM6 plasmid employs antibiotic resistance as a selection marker. The pM6 plasmid encoding for the GST-ZnF was produced by Dr. David Palfrey at the Department of Pharmaceutical Sciences, Aston University (UK) and kindly supplied by Dr. Anna Hine.
2. GST-ZnF: The GST-ZnF molecule, comprised of the GST segment (27.7 kDa) and fused to the zinc finger moiety (10.7 kDa), has a molecular weight of approximately 38.4 kDa.
3. Glutathione: Glutathione (=99%, MW 307) is a tripeptide made up of the amino acids gamma-glutamic acid, cysteine and glycine. In this body of work, the reduced form of glutathione is used as a covalently immobilized affinity ligand and in the elution buffer (*see Note 1*).

2.2. Solid-Phase Adsorbent

1. Glutathione-Streamline: Streamline™, acting as the solid-phase adsorbent, is activated and the glutathione ligand immobilized to create the affinity adsorbent as described in **Subheading 3**.

2.3. Buffers

Where required, adjust buffer pH using 1 M HCl or 1 M NaOH.

1. Phosphate-buffered saline (PBS): PBS is used as the equilibration and running buffer. The buffer can be prepared by dissolving a PBS tablet in 200 ml of deionized (DI) water to yield a buffer containing 10 mM phosphate buffer, 2.7 mM potassium chloride and 137 mM sodium chloride, pH 7.4.
2. Elution buffer: 20 mM reduced glutathione, 100 mM Tris-HCl, pH 9.
3. Phosphate buffer (for GST enzyme activity assay): 1 M KH_2PO_4 with 1 M K_2HPO_4 added until pH 6.5 obtained.
4. GST enzyme activity assay reagent: 22 ml DI water, 2.5 ml phosphate buffer, 0.25 ml 100 mM reduced glutathione, 0.25 ml 100 mM 1-chloro-2,4-dinitrobenzene (CDNB, $\geq 99\%$) (*see Note 2*).
5. Sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) staining solution: 0.12% w/v Coomassie Blue, 48% v/v methanol, 60% v/v DI water and 12% v/v glacial acetic acid.
6. SDS-PAGE de-staining solution: 70% v/v DI water, 20% v/v methanol and 10% v/v glacial acetic acid.

3. Methods

3.1. Production of Glutathione-Streamline Matrix

The following procedure is used to produce a glutathione-Streamline matrix for use in the packed bed and expanded bed chromatography studies. The method uses a bisoxirane to introduce oxirane (epoxy) groups to a hydroxylic polymer adsorbent. An epoxy-activated adsorbent (Streamline) is used to covalently immobilize a ligand (glutathione) containing amine or thiol groups.

1. Wash 30 ml of settled bed volume Streamline particles extensively in DI water (*see Note 3*).
2. Remove excess water and resuspend particles in 23 ml of 0.6 M NaOH containing 45 mg sodium borohydride (*see Note 4*).
3. Add 23 ml of 1,4-butanediol diglycidyl ether (BDGE, MW 202.25, 95%) with constant stirring (*see Note 5*). The matrix is activated by stirring for 24 h at 37°C. Stirring is performed using a rotary incubator (set at 100 rpm) as the use of a magnetic stirrer may result in damage of the adsorbent.
4. The next day, wash the matrix extensively with water to remove excess reagent.
5. Remove excess water and resuspend the particles in 30 ml of 100 mM NaHCO₃, pH 8.5 (*see Note 6*).
6. Remove the resuspension liquid, then add 30 ml of glutathione ligand solution: 0.5 mmol reduced glutathione/ml adsorbent, 100 mM NaHCO₃, pH 8.5 (*see Note 7*).
7. Stir the solution at 37°C for 24 h.
8. Wash the gel three times with PBS buffer to remove unreacted ligand.
9. Block excess active groups on the particles by adding 30 ml of 1 M ethanolamine (pH 9) and stir at room temperature for 6 h.
10. Wash the particles with PBS and store as a 50% slurry at 4°C.

3.2. Ligand Density Measurement: Free Amine Groups

1. Create a ninhydrin reagent by dissolving ninhydrin in DI water to produce a 0.10 M solution (*see Note 8*).
2. Add 1 ml of reagent to 1 ml of a 50% slurry of adsorbent in DI water.
3. Incubate the solution on a rotary stirrer for 1 h at room temperature.
4. Centrifuge the sample briefly (max speed, 1 min) using a microcentrifuge in order to settle out the adsorbent particles.
5. Measure the optical density at 564 nm (OD_{564 nm}) of the supernatant and compare the results to a calibration curve created using a serial dilution of pure reduced glutathione in DI water (0.1 g/ml to 5×10^{-6} g/ml is a good starting point).

3.3. Ligand Density Measurement: Free Thiol Groups

1. Dithiodipyridine is sparingly soluble in water. In order to produce the dithiodipyridine reagent, add a known amount of dithiodipyridine to DI water and

mix for 15 min. Pass the solution through Whatman no. 1 filter paper (approximate pore size of 11 μm) to remove undissolved dithiodipyridine. Perform a mass balance to determine the molar concentration of the reagent (*see Note 9*).

2. Add 1 ml of reagent to 1 ml of a 50% slurry of adsorbent in DI water (*see Note 10*).
3. Incubate the solution on a rotary stirrer for 1 h at room temperature.
4. Measure the optical density of the supernatant at 343 nm ($\text{OD}_{343 \text{ nm}}$) and compare the results against a calibration curve created using a serial dilution of pure reduced glutathione in DI water (0.1 g/ml to 5×10^{-6} g/ml is a good starting point).

3.4. Preparation of Clarified Lysate Containing GST–ZnF Via Freeze/Thaw Lysis

1. BL21 *Escherichia coli* cells containing the expressed fusion protein are harvested from fermentation medium by centrifugation at $5000 \times g$ (5300 rpm in a Beckman JA-10 centrifuge) for 10 min in a room temperature rotor (*see Note 11*).
2. Resuspend the cell pellet in PBS buffer and lyse the cells by six cycles of freeze/thaw lysis (place sample in liquid nitrogen until completely frozen, then place sample in 37°C water bath until completely thawed).
3. Clarify the lysis solution by centrifuging at $15,000 \times g$ (9200 rpm in a Beckman JA-10 centrifuge) for 15 min, then syringe the supernatant through a 0.22- μm filter. The expressed GST–ZnF remains soluble in the liquid fraction, so no further processing or refolding is required. Prepare clarified lysate on the day it is to be used.

3.5. Preparation of Unclarified Lysate Containing GST–ZnF Via Homogenization

1. Load room temperature cell culture into the homogenizer holding unit.
2. Pump the culture through the ceramic homogenizing valve at an operating pressure of 1000 bar (*see Note 12*).
3. Immediately hold the homogenized product on ice until the temperature returns to room temperature. Repeat the procedure a further two times. Prepare homogenized cell lysate on the day it is to be used (*see Note 13*).

3.6. Packed Bed Chromatographic Protein Purification

1. Pack a chromatography column via gravity settling with the adsorbent prepared as given in **Subheading 3.1**.
2. Equilibrate the column with 10 column volumes of PBS buffer.
3. Load GST–ZnF-containing lysate onto the column at an approximate rate of 60 cm/h (*see Note 14*). The binding capacity of GST–ZnF for the affinity adsorbent was approximately 6 mg/ml. Hence, a volume of lysate containing approximately 6 mg of GST–ZnF was loaded per ml of adsorbent.

4. Wash the column with 5 column volumes of PBS buffer at a flow rate of 60 cm/h or until the optical density at 280 nm ($OD_{280\text{ nm}}$) of the column outlet stream returns to base-line levels.
5. Elute the GST–ZnF protein using a solution of 20 mM reduced glutathione (*see Note 15*), pH 9 (*see Note 16*), and 100 mM Tris-HCl for buffering at a flow rate of 60 cm/h.
6. Remove excess glutathione present in the elution fraction by dialysis. Dialysis tubing with a molecular weight cut-off of 12,000 Daltons is suitable. Autoclave the dialysis tubing. Secure one end of the dialysis tubing so that it is water tight (i.e., do not allow the passage of liquids out of one end), load the elution solution into the open end of the dialysis tubing, then secure the second end.
7. Place the loaded tubing into 4°C PBS buffer and stir using a magnetic bar stirrer at 100 rpm. Ensure that the temperature is maintained at 4°C. Approximately 500 ml of PBS buffer should be used per 1 ml of elution fraction. It is recommended that dialysis be performed for a period of at least 24 h. Exchanging the dialysis buffer with fresh PBS buffer enables faster removal of glutathione from the elution solution.

3.7. Expanded Bed Adsorption

1. Load the chromatography column via gravity settling with the adsorbent prepared as given in **Subheading 3.1**.
2. Equilibrate the column with at least 10 settled bed column volumes of PBS buffer using upward flow to expand the column. Expand the bed to twice its settled bed height. In a 1-cm diameter column, use a flow rate of approximately 150 cm/h (*see Note 17*).
3. Using upward flow, pump GST–ZnF-containing lysate into the column at 150 cm/h.
4. Some column expansion is to be expected due to the higher density and viscosity of the feed. To prevent loss of adsorbent through the top of the column, the flow may need to be reduced or the position of the top column frit adjusted.
5. Wash the column with 5 settled column volumes of PBS buffer at 150 cm/h or until the optical density at 280 nm ($OD_{280\text{ nm}}$) of the column outlet stream returns to base-line levels.
6. Reverse the flow of PBS buffer to downward flow and lower the top adaptor in order to operate the column in packed bed mode (*see Note 18*). Continue downward flow at 150 cm/h with PBS buffer until the optical density at 280 nm ($OD_{280\text{ nm}}$) of the column outlet stream is at base-line levels.
7. Elute the GST–ZnF in packed bed mode at 150 cm/h with 20 mM reduced glutathione, pH 9, 100 mM Tris-HCl with approximately 3 settled column volumes. Refer to the optical density at 280 nm ($OD_{280\text{ nm}}$) to monitor the elution peak.

3.8. GST Activity Assay

GST activity assays are performed on the lysates, flow-through and elution fractions in order to determine the concentration of the GST–ZnF molecule and perform a mass balance.

1. Add 10 μ l of sample to 1 ml of GST enzyme activity assay reagent and mix by inverting the sample four times.
2. Perform a rate analysis at OD_{340 nm} to detect the GST-mediated reaction of CDNB with glutathione. Dilute GST–ZnF samples to concentrations less than 2 mg/ml so that the change in activity over time is linear.

3.9. SDS–PAGE

Lysate and protein samples were analyzed by SDS–PAGE using NuPAGE Novex Bis-Tris 4–12% Gels run in an XCell Mini-Cell (Invitrogen, UK).

1. Incubate 10 μ l of protein containing samples with 10 μ l of protein gel loading buffer for 5 min at 95°C.
2. Load the protein sample aliquots into the gel wells. Up to 10 samples can be run simultaneously.
3. Run gels in MOPS SDS running buffer (Invitrogen) for 50 min at 200 V.
4. Remove gels from the cartridge and stain for 2 h using SDS–PAGE staining solution.
5. De-stain overnight in SDS–PAGE de-staining solution. Gels were photographed using the EDAS 290 utilizing a visible light illuminator. When densitometry studies were performed, the images created by the EDAS 290 were analyzed using Kodak Digital Science 1D Image Analysis Software and compared to protein markers of known concentration.

4. Notes

1. Glutathione should be stored at 2–8°C.
2. CDNB IS TOXIC (by inhalation, contact with skin or if swallowed) and should be handled in a fume hood.
3. This first washing stage is to remove ethanol that helps to preserve the adsorbent. Extensive washing usually requires at least three wash/bed settle stages in batch mode or at least 10 column volumes if washed in a chromatography column. In batch mode, the smell and a change in resin morphology indicate that the ethanol has been removed. For a chromatography column, the OD_{260 nm} of the stream exiting the column will be constant. If ethanol is present, the binding capacity of the adsorbent for the target may be affected.
4. Sodium borohydride acts as a reducing agent and assists in stabilizing the bonds between the spacer arm and solid-phase adsorbent.

5. BDGE is toxic (by inhalation, contact with skin or if swallowed) and should be handled in a fume hood. A 2^{3-1} factorial experiment was used to explore the effects of three parameters on the total ligand density: time for glutathione immobilization (24 and 48 h), temperature during immobilization (37 and 45°C) and the length of the spacer arm (BDGE, a 10-carbon spacer arm, and hexane diglycidyl ether, a 12-carbon spacer arm). It was found that all of the parameters have a significant effect on ligand density, and the highest ligand density was obtained for immobilization conditions of 37°C for 24 h using BDGE as the spacer arm. Using a suitable spacer arm is important: binding capacities can be increased by placing the ligand at some distance from the matrix as this helps to reduce the effects of steric hindrance caused by the matrix (9). The ideal spacer arm will have appropriate coupling functionalities on both ends and an overall hydrophilic character (10). The length of the spacer arm is critical. If it is too short, the arm is ineffective and the ligand fails to bind substances in the sample due to the steric interference of the matrix. If it is too long, non-specific effects become pronounced and reduce the selectivity of the separation as very long spacer arms can bind substances via hydrophobic interactions. Non-specific hydrophobic interactions are undesirable in chromatographic systems as contaminants may be co-purified.
6. This stage is required in order to remove water and create an environment that is conducive for glutathione immobilization. NaHCO_3 acts as a buffer for this purpose.
7. The orientation of the glutathione ligand attachment to the base matrix is determined by the pH at which the coupling reaction is conducted. At pH 7.5–8.5, the coupling occurs primarily through the thiol group of glutathione molecule, which leaves the amine group exposed for adsorption of GST protein. This was found to yield significantly higher capture of the GST compared with the opposite case at pH greater than 9 where the ligand coupling was enabled via the amine group of glutathione and GST adsorption was via the thiol group. Analysis of the glutathione-Streamline adsorbent prepared according to the steps described in **Subheading 3** showed that over 95% of free binding groups are amine groups, which indicates that ligand coupling was achieved predominantly via the thiol group as desired.
8. Ninhydrin is toxic (by inhalation, contact with skin or if swallowed) and should be handled in a fume hood. Ninhydrin is used to detect ammonia or primary amines. When reacting with free amines, a deep blue or purple colour is evolved. In order to generate the ninhydrin chromophore, the amine is oxidized to a Schiff base by redox transfer from the ninhydrin moiety.
9. Concentration of dithiodipyridine in reagent (mg/ml) = $(\text{Mass dithiodipyridine added to reagent (mg)} - \text{mass dithiodipyridine collected on filter paper (mg)}) / \text{reagent volume (ml)}$.
10. Dithiodipyridine reacts with thiol groups forming a disulphide bond, which can be monitored by means of the absorbance change at 343 nm. Knowing the ligand density of an adsorbent enables calculations of ligand utilization to be made and what effect the process parameters have on the ligand density. By measuring the

concentration of the amine and thiol groups, the total free ligand concentration can be calculated. Ligand densities as high as 362 $\mu\text{mol/ml}$ were observed for the optimized immobilization protocol.

11. The materials and methods for bacterial cell transformation with the pM6 plasmid and expression of the GST–ZnF are reported elsewhere (3,11).
12. An APV-2000 homogenizer unit (Invensys, Denmark) was used at a nominal pumping rate of 11 l/h for minimum sample sizes of 100 ml.
13. Homogenization requires optimization for different cells and feed cell concentrations. The method described in **Subheading 3.5** was optimized via use of SDS–PAGE gel analysis in order to obtain maximum yield of the GST–ZnF protein (mg/ml) without degradation due to shear and/or an increase in temperature.
14. An Amersham Biosciences 5/5 column, 5 mm inside diameter, containing 1 ml of adsorbent (5.1 cm bed height), was used and operated using an ÄKTA Explorer™ (Amersham Biosciences). For this column, a flow rate of 0.2 ml/min equates to approximately 60 cm/h.
15. Glutathione concentration considerations for GST–ZnF elution: A Biacore CM5 chip with covalently immobilized glutathione was used to determine the effect of reduced glutathione concentration on the elution of GST–ZnF bound to the glutathione ligand. After equilibration with PBS, a 25- μl sample containing 100 $\mu\text{g/ml}$ of pure GST–ZnF was loaded onto the chip followed by washing and then elution. Increasing concentrations of reduced glutathione in a solution of DI water (pH 9) were used to determine the amount eluted, measured by the reduction in response units (RU) from the start to the end of the elution. After each run, the chip was regenerated and equilibrated. The results of the elution study are displayed in **Fig. 1**. Significant increases in elution occurred as the reduced glutathione concentration was increased from 0 up to 20 mM. From 20 mM up to 100 mM, only minimal changes in elution were obtained ($\pm 8\%$). These variations were within the experimental error ($\pm 27\%$). The data indicate that any further increase in reduced glutathione concentration above 20 mM will not necessarily yield a greater amount of eluted GST–ZnF. For an industrial scale operation, economic issues would need to be considered as the ongoing costs of expensive eluting agents (i.e., glutathione) is an important economic consideration and there is the added processing issue of removing the eluting agent from the elution fractions. It is therefore preferable to use the minimum amount of eluting agent whilst maintaining optimal elution yields. The data presented in **Fig. 1** supports the use of 20 mM glutathione in the elution buffer.
16. pH considerations for GST–ZnF elution: Elution of GST–ZnF may be improved by using an elution buffer pH where both the glutathione ligand and GST–ZnF have the same charge (e.g., are both negative). The charge of a protein is determined by its pI and the buffer pH, where the pI of a protein is the pH at which the protein has an equal number of positive and negative charges. The number of net negative charges on a protein increases with increasing pH above the pI (12). The theoretical pI of GST–ZnF determined using the ExPASy ProtParam Tool (13,14) is 8.96. An isoelectric focusing gel confirmed that the theoretical pI of the GST–

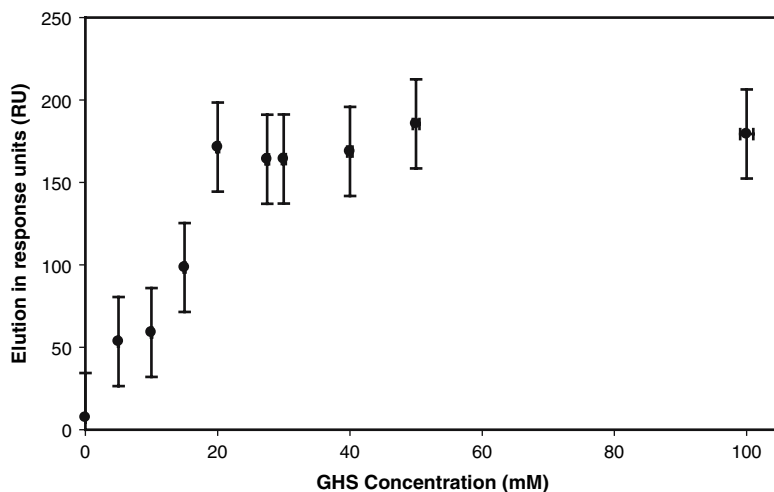


Fig. 1. Elution profile of GST-ZnF eluted from glutathione ligand immobilized onto a Biacore CM5 chip. A 25- μ l sample containing 100 μ g/ml of pure GST-ZnF was loaded onto the chip followed by elution. The amount of GST-ZnF eluted was determined by measuring the change in RU before and after loading of the elution buffer.

ZnF is approximately correct (data not shown). Studies of the effect of pH on the elution of GST-ZnF from an affinity adsorbent were performed. The elution pH was varied whilst maintaining constant glutathione (20 mM) and Tris-HCl (100 mM) concentrations. Clarified lysate containing GST-ZnF (1 ml, 1.34 mg/ml) was bound to 0.1 ml of affinity adsorbent (5-min incubation) and eluted using 1.50 ml of elution buffer (5-min incubation). The amount of total protein eluted was measured by a bicinchoninic acid protein assay and the percentage of total protein that was GST-ZnF determined by a densitometry study of an SDS-PAGE gel, shown in **Fig. 2**.

The amounts of GST-ZnF eluted were 0.41 mg/ml adsorbent for pH 8, 0.73 mg/ml adsorbent for pH 9 and 0.90 mg/ml adsorbent for pH 10. Amersham Biosciences (15) recommends an elution buffer of pH 8 and a maximum elution buffer pH of 9. At pH levels above 9, GST fusion proteins will be denatured (16), jeopardizing the function of the proteins (i.e., ability to bind to affinity ligands or recognition sequences). The pI of recombinant GST expressed by *E. coli* is 6.52 (17). An elution pH of 8 is sufficient for the successful elution of GST from glutathione ligands. However, the pI of the GST-ZnF fusion protein (8.96) is higher than that of GST. Using an elution buffer of pH 9 ensures that the pH is above the pI for both the ligand and target protein whilst the pH is at a level which will not denature the fusion protein.

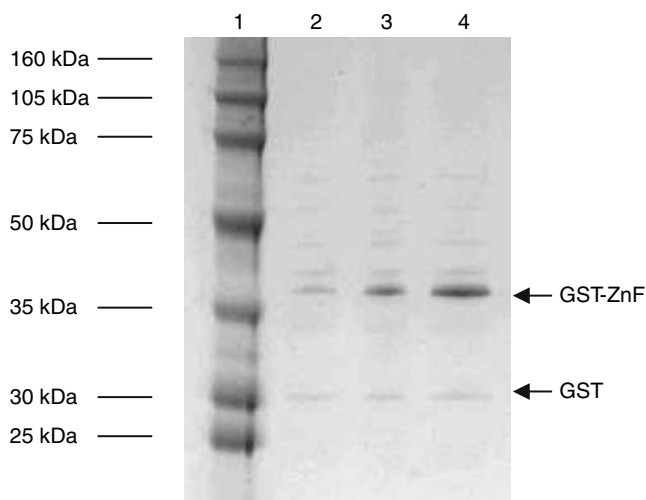


Fig. 2. Sodium dodecyl sulphate–polyacrylamide gel electrophoresis gel showing eluants obtained using three different elution pH of 8, 9 and 10. All elution buffers contained 20 mM reduced glutathione and 100 mM Tris–HCl. Lane 1: Marker. Lane 2: Elution buffer 1, pH 8, 10 μ l. Lane 3: Elution buffer 2, pH 9, 10 μ l. Lane 4: Elution buffer 3, pH 10, 10 μ l.

17. A glass column supplied by Soham Scientific Ltd., UK, with an internal diameter of 1 cm, was used for the EBA work. The bottom of the column incorporated a sintered glass frit as the flow distributor. The nominal pore size of the sintered frit was 100–160 μ m with a thickness of 2 mm. The adjustable top adaptor had no sinter to allow free passage of the solid debris and was fixed in position by a screw connection at the top of the column. A three-way valve was attached to the base of the column in order to ensure that no air bubbles were trapped below the frit. The inlet and outlet tubing was designed to minimize the mixing of liquid entering and exiting the column whilst also minimizing the pressure drop over the system. The column was loaded with 15.7 ml of adsorbent (20 cm settled bed height). A flow rate of 156.6 cm/h (2.06 ml/min) was used in order to expand the bed to twice its settled bed height.
18. Eluting in packed bed mode reduces the volume of the elution fraction and reduces mixing, hence increasing the concentration of the target in the elution fractions.

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Immobilized Metal Ion Affinity Chromatography of Histidine-Tagged Fusion Proteins

Adam Charlton and Michael Zachariou

Summary

Immobilized metal ion affinity chromatography (IMAC) is a ubiquitous technique in modern recombinant production and purification. The wide range of expression vectors for the production of histidine-tagged recombinant proteins as well as the variety of stationary supports for their separation make IMAC an attractive and versatile choice for fast and reliable protein purification. It is not uncommon for IMAC purification to yield near homogenous target protein, with purities over 95%. The small size of the histidine tag means that in many cases it can remain associated with the target protein without interference with its intended function, obviating the need for any potentially complicating tag removal steps. This chapter provides protocols for the routine purification of such histidine-tagged fusion proteins. As with any purification regime, complications with IMAC can arise. Lacking the absolute specificity of a biological ligand/ligate system such as the avidin/biotin interaction or an antibody and its cognate antigen, IMAC can sometimes display non-ideal product purity. The protocols described in this chapter provide strategies for the improvement in the purity of IMAC-purified proteins. Similarly, non-specific binding may reduce product yields and purity in some circumstances. Methodologies for enhancing the yield of the target protein are therefore provided.

Key Words: IMAC; histidine tag; protein purification; affinity chromatography; recombinant protein expression.

1. Introduction

Immobilized metal ion affinity chromatography (IMAC) is a relatively recent protein purification technology that exploits the specific relationship between the side chains of certain amino acids and particular borderline Lewis metal ions (such as Cu^{2+} , Ni^{2+} and Zn^{2+}) (*1,2*), with histidine by far the one of the most common amino acid involved in such binding events. The immobilization of the metal ion is achieved via a chelating agent that is attached to a stationary support, with the capture of the metal ion by said immobilized chelator forming an immobilized metal chelate complex (IMCC). The most commonly employed chelators for such applications are iminodiacetic acid (IDA) or nitrilotriacetic acid (NTA), despite an extensive range of alternatives (*3*). Binding of histidine side chain to the IMCC takes place by donation of electrons from the imidazole moiety of the histidine side chain to the two or three (if tetra- or tri-dentate chelators are used, respectively) available coordination sites of the metal ion (*see Fig. 1*).

The earliest applications of IMAC made use of surface histidines that occur naturally in the target protein (*1*). The concept was extended by the inclusion of a hexahistidine tail or "tag" on the target protein (*4,5*), allowing for more stringent binding conditions and thus a more selective purification. Not inconsequential is the fact that a polypeptide tag on either terminus of a protein is much more likely to be accessible for binding to IMAC resins. The optimum sequence configuration of the histidines in the tag has been shown to be His-(Xaa)₃-His (*6*), hence the canonical hexahistidine provides this motif in two

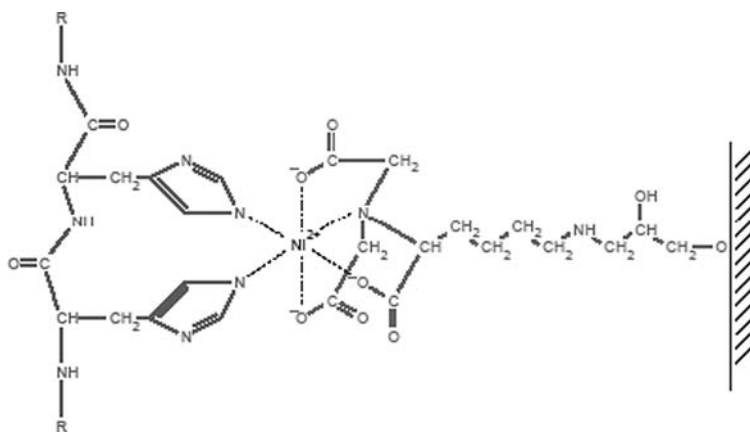


Fig. 1. Coordination binding of the histidine tag to a Ni-nitrilotriacetic acid immobilized metal chelate complex.

binding modes. The reader is referred to recent reviews of IMAC of proteins for a more detailed perspective (7–9).

Histidine tag IMAC has seen widespread adoption in recent years for the purification of fusion proteins. Prior to 1996, only 55 Medline citations returned from a search for “His tag,” but in the last decade, this number has grown to almost 1200. With 1850 citations returned for the more general “affinity tag” search, it suggests that histidine tag IMAC alone accounts for nearly two-thirds of all affinity tag usage in modern recombinant protein expression and purification. IMAC is seeing a similar explosion in commercial application, with the same “His tag” search of the U.S. Patent Office returning no patents prior to 1996, but over 1800 approved in the decade since. Widespread availability of expression vectors designed for producing histidine-tagged fusion proteins is an indication of the pervasiveness of this technology. A subset of commercially available vectors is given in **Table 1**.

The small size of the hexahistidine tag means that it in many cases removal of the histidine tag is not required; it may remain attached to the target protein without interfering with its intended application or biological function. In fact, the literature is replete with examples in which histidine tags have remained attached to multimeric protein subunits without abolishing assembly of the quaternary structure (10–13).

IMAC can be the method of choice for insoluble proteins, because the affinity interaction of IMAC does not rely on biological function, but rather the spatial position of the atoms of the amino acids, it is one of the few affinity chromatography technologies available that can function in denaturing conditions. In fact, due to the equivalent functionality of IMAC in both denaturing

Table 1
Commercial Vectors Bearing Histidine Tags for Immobilized Metal Ion Affinity Chromatography IMAC

Supplier	Vector(s)	Notes
QIAGEN	pQE	pQE-1 designed for use with TAGzyme system
Roche Applied Science Novagen	pIVEX pET	Various configurations available
Promega Invitrogen	FLEXI vector system pBADpTrcHispThioHis	(HQ) ₆ tag Modified thioredoxin binds to IMAC
Stratagene Clontech	pDUAL PRO TM tet vectors	(HN) ₆ tag

and non-denaturing environments, it has been used to refold proteins whilst still bound to the IMCC (14). This approach can allow the user to obtain near homogenous, soluble protein from insoluble input material. Conversely, incorporation of a histidine tag has been shown to improve the soluble yield of some recombinant proteins by its presence alone, presumably by increasing the hydrophilicity of the protein and thus rendering it more compatible with expression in *Escherichia coli* (15).

As a mature affinity chromatographic technology, IMAC has seen application in circumstances outside of its traditional role of protein purification. Significant interest has been in proteomic screening technologies; with chelators immobilized on magnetic beads, IMAC binding is amenable to automation. This allows for rapid expression and purification of large protein libraries (16). IMAC has also functioned as a coupling technique for immobilization of receptors in microarrays (17) and as a tether for membrane proteins in the generation of artificial lipid bilayers (18). Histidine tags have even been incorporated into synthetic oligonucleotides, allowing for their purification by IMAC (19).

The ubiquitous application of histidine-tag IMAC has seen a range of supporting technologies emerge; tools for the specific detection and removal of histidine tags are commercially available. Qiagen's TAGzyme system is a classic example of the latter. The system consists of a series of three enzymes that are specifically tailored to remove N-terminal hexahistidine tags leaving no vector or tag-derived amino acids on the target protein. The system is described in detail elsewhere in this book. Specific detection of histidine-tagged proteins is as readily available as tag-bearing cloning vectors, with detection systems supplied by Pierce, Novagen, Clontech, Qiagen and Invitrogen, among many others. These systems usually rely on variations of an anti-hexahistidine antibody for secondary antibody-reporter enzyme conjugate detection, or a reporter enzyme (horseradish peroxidase and alkaline phosphatase) linked to a chelator for direct metal chelate complex detection. Samples can then be queried by either method for the presence of the histidine tag in a western blot type assay format.

With a wealth of background literature, a wide variety of cloning vectors and stationary supports, IMAC is a popular first choice for many recombinant protein purification applications at any scale, from proteomic screening up to biopharmaceutical production.

2. Materials

2.1. Purification of His-Tagged Proteins Using Ni-NTA

1. Stationary Support: Ni-NTA-Superflow (Qiagen).
2. Charge solution: 0.1 M NiNO_3 .

3. Metal rinsing solution: 0.2 M acetic acid.
4. Pre-equilibration buffer: 0.2 M imidazole + 0.5 M NaCl pH 7.
5. Equilibration buffer: 0.02 M imidazole + 0.05 M NaCl pH 7.
6. Elution buffer: 0.2 M imidazole + 0.5 M NaCl pH 7.
7. Regeneration buffer: 0.2 M EDTA + 0.5 M NaCl pH 8.

2.2. Improving Product Recovery

1. Stationary Support: Ni-NTA-Superflow (Qiagen).
2. Charge solution: 0.1 M NiNO_3 .
3. Metal rinsing solution: 0.2 M acetic acid.
4. Pre-equilibration buffer: 0.2 M imidazole + 0.5 M NaCl pH 7.
5. Equilibration buffer: 0.02 M imidazole + 0.05 M NaCl pH 7.
6. Elution buffer 1: 0.2 M imidazole + 0.5 M NaCl pH 7.
7. Elution buffer 2: 0.5 M imidazole + 0.5 M NaCl pH 7.
8. Elution buffer 3: 0.5 M imidazole + 0.5 M NaCl pH 5.5 (optional).
9. Elution buffer 4: 0.5 M imidazole + 0.5 M NaCl + 0.05 M sodium acetate pH 4 (optional).
10. Regeneration buffer: 0.2 M EDTA + 0.5 M NaCl pH 8.
11. Sanitization solution: 1 M NaOH.

2.3. Improving Product Purity

1. Stationary Support: Ni-NTA-Superflow (Qiagen).
2. Charge solution: 0.1 M NiNO_3 .
3. Metal rinsing solution: 0.2 M acetic acid.
4. Pre-equilibration buffer: 0.2 M imidazole + 0.5 M NaCl pH 7.
5. Basal equilibration buffer: 0.02 M imidazole + 0.05 M NaCl pH 7.
6. Elution buffer 1: 0.2 M imidazole + 0.5 M NaCl pH 7.
7. Regeneration buffer: 0.2 M EDTA + 0.5 M NaCl pH 8.

3. Method

3.1. Purification of His-Tagged Proteins Using Ni-NTA

1. Wash packed Ni-NTA column with 2 CV of metal rinsing solution, 0.2 M acetic acid (*see Note 1*).
2. Wash column with 5 CV of Milli Q water.
3. Pre-wash packed Ni-NTA column with 10 CV of 0.2 M imidazole + 0.5 M NaCl, pH 7 (*see Note 2*).
4. Equilibrate the column with 10 CV of 20 mM imidazole and 50 mM NaCl pH 7 (*see Note 3*). Confirm equilibration by measuring pH and conductivity. Continue equilibration until pH and conductivity of effluent matches equilibration buffer.
5. Load sample containing target molecule ensuring pH is between pH 7 and 7.2. As a general rule, loading linear velocities should be between 10 and 33%

the maximum operating linear velocity allowed by the stationary support (*see Note 4*), that is, 300–1000 cm/h for the stated support. Assume a loading of no more than 1 mg target protein per ml of stationary support (*see Note 5*). However, target proteins in ratio volumes of 300:1 cell culture per Ni-NTA have been successfully loaded by the authors (*see Note 6*).

6. Wash stationary support with 10 CV of equilibration buffer at the loading linear velocity or until the $A_{280\text{ nm}}$ reading is at baseline (*see Note 7*).
7. Elute protein with up to 5 CV of 0.2 M imidazole + 0.5 M NaCl pH 7 at 33% of the recommended maximum linear velocity of the stationary support, 1000 cm/h for Ni-NTA superflow. Samples should be examined on sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS–PAGE) for purity (20). If these conditions have not been able to effect complete elution, follow the steps described in **Subheading 3.2**. If the eluted product is of insufficient purity, follow the steps described in **Subheading 3.3**.
8. After elution of the target protein, the column should be regenerated using 3 CV of 0.2 M EDTA + 0.5 M NaCl pH 8. Washing linear velocity is not critical as long as it does not exceed the maximum linear velocity of the stationary support.
9. Wash with 10 CV of Milli Q water.
10. Load column with 2 CV of 0.1 M NiNO_3 (*see Notes 8 and 9*).
11. Wash with 10 CV of Milli Q water.
12. Store column at 4°C.

3.2. Improving Product Recovery Pilot Investigation

1. Carry out **steps 1–6, Subheading 3.1**.
2. Proceed immediately to resin regeneration (i.e., stripping of Ni^{2+}) with 3 CV of 0.2 M EDTA + 0.5 M NaCl pH 8 (*see Note 10*). Washing linear velocity is not critical as long as it does not exceed the maximum linear velocity of the stationary support.
3. Wash with 10 CV of Milli Q water.
4. Sanitize the column by washing with 5 CV of 1 M NaOH.
5. Wash with 10 CV of Milli Q water.
6. Load column with 2 CV of 0.1 M NiNO_3 (*see Notes 8 and 9*).
7. Wash with 10 CV of Milli Q water.
8. Store column at 4°C.
9. If the protein was recovered in step 2 proceed with the steps described in **Subheading 3.2.1**, if not proceed with the steps described in **Subheading 3.2.2**.

3.2.1. Improving Product Recovery Where Binding is IMCC-Histidine Mediated

1. Carry out **steps 1–3, Subheading 3.1**.
2. Equilibrate the column with 10 CV of 0.1 M imidazole and 0.5 M NaCl pH 7 (*see Note 11*). Confirm equilibration by measuring pH and conductivity. Continue equilibration until pH and conductivity of effluent matches equilibration buffer.

3. To the load sample, containing the target molecule, add imidazole to 0.1 M and NaCl to 0.5 M. Adjust pH to 7. Load the column at 33% of the maximum operating linear velocity allowed by the stationary support (*see Note 4*), that is, 1000 cm/h for the stated support. Assume a loading of no more than 1 mg target protein per ml of stationary support (*see Note 5*).
4. Wash stationary support with 10 CV of equilibration buffer at the loading linear velocity or until the $A_{280\text{ nm}}$ reading is at baseline (*see Note 7*).
5. Attempt to elute the protein with up to 5 CV of 0.5 M imidazole + 0.5 M NaCl pH 7 at 10% of the recommended maximum linear velocity of the stationary support, 300 cm/h for the stated support. Samples should be examined on SDS-PAGE to evaluate elution success (*see Note 7*).
6. If unsuccessful, attempt elution with up to 5 CV of 0.5 M imidazole + 0.5 M NaCl pH 5.5 (*see Note 12*) at 10% of the recommended maximum linear velocity of the stationary support, 300 cm/h for the stated support. Samples should be examined on SDS-PAGE to evaluate elution success (*see Note 7*).
7. If unsuccessful, attempt elution with up to 5 CV of 0.5 M imidazole + 0.5 M NaCl + 50 mM sodium acetate, pH 4 (*see Note 13*) at 10% of the recommended maximum linear velocity of the stationary support. Samples should be examined on SDS-PAGE to evaluate elution success (*see Note 7*).
8. If still unsuccessful, repeat the steps described in **Subheadings 3.1** and **3.2** with a different metal ion (*see Note 14*).
9. Carry out **steps 9–12, Subheading 3.1**. Substitute metal ions where appropriate.

3.2.2. Improving Product Recovery Where Binding is Non-Specific

1. Carry out steps 1–7, **Subheading 3.1**.
2. Select a factor from the **Table 2** and incorporate it into the elution buffer (**step 7, Subheading 3.1**). Repeat **step 7, Subheading 3.1**, iteratively with the factors presented in the Table until protein liberation is detected. Author's recommendation: Commence with the most extreme conditions that the target protein can endure and then work backwards toward milder conditions.
3. Regenerate resin (i.e., strip Ni^{2+}) with 3 CV of 0.2 M EDTA + 0.5 M NaCl pH 8. Washing linear velocity is not critical as long as it does not exceed the maximum linear velocity of the stationary support.
4. Carry out **steps 3–8, Subheading 3.2**.
5. Include the optimum condition determined in **step 2, Subheading 3.2.2**, into the equilibration buffer, load sample and elution buffer and repeat the steps described in **Subheading 3.1** with these modifications (*see Note 15*).

3.3. Improving Product Purity

1. Carry out **steps 1–4, Subheading 3.1**.
2. To the load sample containing the target molecule add imidazole to 20 mM and NaCl to 50 mM. Adjust pH to 7. Load the column at 33% of the maximum

Table 2

Agent	Effect	Comment
Non-ionic detergents, e.g., Triton, Tween; No more than 10% v/v	Disrupts hydrophobic interactions	In particular, will disrupt any interactions between the spacer arm and proteins as well as any protein–protein hydrophobic interactions that may be occurring with the target protein. Inclusion of detergent will also assist in removing lipids or DNA (20)
Ionic detergents, e.g., SDS (anionic), CTAB (cationic) No more than 0.5% w/v		
Chaotropic agents, e.g., 8 M Urea or 6 M Guanidine–HCl		
Organic solvent, e.g., Isopropanol No more than 20% v/v		
pH > 9 pH < 7	Alters ionic status of protein	Introducing localized surface charge into the protein can alter the properties of the non-specific binding. For example, hydrophobic interactions will be disrupted by charging regions of the protein
Decrease temperature	Weakens hydrophobic interactions	Decreasing temperature to <10°C may allow for mild elution of hydrophobically bound proteins. May also enhance product stability

SDS, Sodium diodecyl sulfate.

operating linear velocity allowed by the stationary support (*see Note 4*), that is, 1000 cm/h for the stated support. Assume a loading of no more than 1 mg target protein per ml of stationary support (*see Note 5*).

3. Wash stationary support with 10 CV of basal equilibration buffer at the loading linear velocity or until the $A_{280\text{ nm}}$ reading is at baseline (*see Note 7*).
4. Select a wash condition from the **Table 3** and incorporate it into the basal equilibration buffer and repeat step 3. Author's recommendation: Care should be taken to avoid eluting the target protein, commence with the mildest conditions given in

Table 3

Wash type	Effect	Comment
Glycine, arginine, ~0.5 M NH ₄ Cl and pH 7	Mild eluents that compete for Ni with histidine	These are mild eluents that will not elute the His-tag protein but may displace weaker bound proteins
Non-amine salts, e.g., ~0.5 M–1.0 M NaCl; in 20 mM Imidazole + 50 mM NaCl pH 7	Will disrupt any non-specific electrostatic interactions	Such interactions are common in IMAC particularly if the equilibration and wash steps had <0.2 M salt present (21)
Non-ionic detergents, e.g., Triton, Tween no more than 1% v/v	Disrupts hydrophobic interactions	In particular, will disrupt any interactions between the spacer arm and proteins as well as any protein–protein hydrophobic interactions that may be occurring with the target protein. This is more effective when applied as part of the equilibration conditions so as to prevent such interactions from taking place. Inclusion of detergent will also assist in removing lipids or DNA (20)
Chaotropic agents, e.g., 4 M Urea or 4 M Guanidine–HCl		
Decreasing pH (<7.0) and/or increasing imidazole concentration (>20 mM)	Disrupts the histidine bond to the IMCC	This step can also be used to elute the target protein, so care must be taken to select a condition that ensures good differentiation between contaminants and target protein

IMCC, Immobilized metal chelate complex.

the Table, for example, increasing imidazole concentration, inclusion of amines or other salts, and then increase the stringency of the conditions up to the highest possible levels that do not elute the target protein.

5. Carry out **steps 7–12, Subheading 3.1.**

4. Notes

1. All columns pre-charged with metal should be washed with acid to release any loosely bound metal ions.
2. This step serves to totally quench the immobilized metal ion with imidazole, improving selectivity of the IMCC for proteins. Furthermore, it creates a uniform surface by eluting weakly bound hydroxide species bound to the IMCC surface. Such species have been observed previously and if not controlled can significantly contribute to non-specific electrostatic interactions during IMAC (21). Lower imidazole concentrations are not as effective. In addition, the pre-charge buffer approximates the elution buffer and so can reduce metal ion leakage attributable to such a high imidazole concentration even before elution occurs.
3. The pH of equilibration is varied throughout the literature and can range from 7 to 8. By operating closer to pH 7 than to pH 8 during protein binding, a greater selectivity may be achieved which would ultimately yield greater purity of the final product. Improved capacity may also result because less non-specific interactions will occur. Most His-tagged proteins will bind within pH 7–8 range and should be determined empirically. Other buffers such as 100 mM phosphate are commonly used at pH 7–7.5. In these instances, the Ni-NTA becomes less selective and proteins containing histidine regions are more likely to bind, than if imidazole was used, leading to potential problems downstream of the process.
4. A slow loading velocity improves the diffusion of proteins (particularly large proteins) through pores and onto the IMCC and hence improves yields. The stated linear velocities have been derived from the author's personal experience and will vary depending on the stationary support. For example, Poros and Hyper D supports can have linear dynamic capacities, in some cases up to 7000 cm/h, before decreases in capacities are observed. Care must also be taken to ensure that if prolonged loading times are chosen, the target protein is not subject to destabilizing factors such as proteolysis or any intrinsic instability such as deamidation or oxidation. The status of the protein should therefore be monitored during the process. In these instances, the stability of the molecule needs to take precedence over slow loading velocities.
5. This amount is conservative relative to the manufacturer's claims of 5–10 mg of protein per ml Ni-NTA resin (22); however, capacities of <5 mg/ml have often been observed. This is particularly prevalent when cell culture is used, wherein proteins with a higher His content are present, as well as a large amount of free amino acids (used as media components) which can bind through their α -amino groups (23). Hence, allowing for excess stationary support will reduce the possibility of your target molecule not binding because of capacity issues. Furthermore, any

non-specific interactions that may occur because of excess stationary support not interacting with the target molecule are addressed through the proposed stringent pre-equilibration, equilibration and washing regimens.

6. In these instances, significant metal leaching may occur during loading, reducing the capacity of the Ni-NTA but not below 1 mg of protein per ml of Ni-NTA.
7. If monitoring $A_{280\text{ nm}}$ note that imidazole absorbs at this wavelength and so achieving baseline should only be relative to the absorbance of the equilibration buffer at $A_{280\text{ nm}}$. In wash and elution steps, care should be taken to avoid confusing an increasing $A_{280\text{ nm}}$ signal due to the use of a higher imidazole concentration with that of elution of a protein.
8. Not all supports should be stored charged with metal ions. Silica-based supports should be stored free of metal ion and only charged when required. The charged metal ion causes a localized low pH microenvironment that can damage these supports over time, decreasing the life expectancy of the column.
9. Metal ions that could be used for this work are preferably the hard Lewis metal ions such as Fe^{3+} and any of the lanthanides. Hard Lewis metal ions such as Ca^{2+} could also be used; however, a good chelating stationary phase to use this metal ion in IMAC for the purification of proteins does not exist commercially. Al^{3+} is also another example; however, the commercially available 8-hydroxyquinoline support would be more useful over IDA stationary phases for this metal ion. Borderline Lewis metal ions like Cu^{2+} and Co^{2+} can also be used in this mode (24,25).
10. In this way, insight will be gained as to the mode of binding of the target protein. If the protein is recovered in this step, then the binding is mediated by histidine binding to the IMCC. If not, then the protein is bound in a non-specific manner, such as hydrophobic interaction with the spacer arm of the ligand.
11. It is known from attempting the steps described in **Subheading 3.1** that the target protein remains bound in the presence of 0.2 M imidazole + 0.5 M NaCl. Loading under more stringent conditions may assist later elution by reducing the number of binding modes available to the protein. Higher binding stringency may also improve product purity and column capacity, as less binding sites are occupied by contaminants, this leaves more sites to exclusively bind the target protein.
12. A pH of less than 6.5 can effect elution by protonating the histidine side chain, preventing it from donating electrons to the bond with the IMCC.
13. A localized pH microenvironment may require more extreme shifts in pH to allow elution.
14. Alternative borderline Lewis metal ions will have different affinity for the histidine tag. As a rule of thumb, binding strength is generally in the order $\text{Cu}^{2+} > \text{Ni}^{2+} > \text{Co}^{2+} \approx \text{Zn}^{2+}$ (26), so the use of, for example, Zn^{2+} may allow elution where it was not possible from Ni^{2+} .
15. Incorporation of the altered conditions into the binding and washing phase of the chromatography run. It is often more effective to prevent non-specific interactions from occurring than to disrupt them once established. In these circumstances, it may be possible to achieve elution in the absence of the altered condition, as the causative agent (or its effects) may remain loosely associated with the protein

for some time after it has been washed out of the system (author's personal observations). The exclusion of such agents from the final elution may be beneficial where the agent is refractory to the intended application of the protein target, for example, the inclusion of detergents or chaotropes in the protein preparation. Exclusion of these agents from the elution buffer may obviate the need for a buffer exchange step. Likewise, incorporation of the agent prior to elution may allow for even further reduction in the severity of the conditions determined in **step 2, Subheading 3.2.2.**

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Methods for the Purification of HQ-Tagged Proteins

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Summary

The HQ (H = histidine, Q = glutamine) tag is a small fusion tag that can be isolated using immobilized metal affinity columns. HQ-tagged proteins can be expressed and purified from bacterial cells under native and denaturing conditions, mammalian cells, insect cells, wheat germ and rabbit reticulocyte. Furthermore, HQ-tagged proteins can be purified using magnetic or non-magnetic resins in multiple formats from small to large-scale and manual or automated. In this chapter, we have described various protocols for the purification of HQ-tagged proteins.

Key Words: Protein expression; HQ-tagged proteins; recombinant protein; magnetic resin; non-magnetic resin; protein purification; automated protein purification; high-throughput protein purification.

1. Introduction

Protein fusion tags are essential tools for the isolation and purification of proteins for the study of protein–protein and protein–ligand interactions; and protein structure–function studies (*1–6*). Many fusion tags are available for the expression and purification of recombinant proteins using immobilized affinity metal chromatography (IMAC) (*7–8*). Among these, the polyhistidine tag is most commonly used for several reasons including that the tag is very small, can be used under native or denaturing conditions and is not immunogenic. The HQ tag is a metal affinity tag consisting of 6–10 amino acids (repeating HQs; H = histidine, Q = glutamine) and is similar in function to polyhistidine

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tag. HQ-tagged proteins are not only expressed and purified similarly to a polyhistidine-tagged protein, but can also be purified from bacterial cells under native and denaturing conditions. The characteristics of the HQ tag are (i) small size, (ii) can be purified using IMAC methods, (iii) many HQ-tagged proteins eluted from metal affinity resin at a low imidazole concentration (e.g., 50 mM imidazole) and (iv) the HQ tag can be attached at amino-(N) or carboxy-(C) termini of the proteins.

2. Materials

2.1. Small-Scale Magnetic Nickel Purification for Bacteria

1. MagneHis™ Protein Purification System (cat. no. V8500, Promega)—MagneHis™ Binding/Wash Buffer: 100 mM HEPES + 10 mM imidazole, pH 7.5; MagneHis™ Elution Buffer: 100 mM HEPES (4-2-Hydroxyethyl) piperazine-1-elthanesulfonic acid) + 500 mM imidazole, pH 7.5; MagneHis™ Ni Particles; FastBreak™ Cell Lysis Reagent, 10×; DNase I.
2. Magnetic stand (cat. no. Z5342, Promega).
3. NaCl (5 M).

2.2. Small-Scale Non-Magnetic Nickel Purification for Bacteria

1. HisLink™ Spin Purification System (cat. no. V1320, Promega)—HisLink™ Binding/Wash Buffer: 100 mM HEPES + 10 mM imidazole, pH 7.5; HisLink™ Elution Buffer: 100 mM HEPES + 500 mM imidazole, pH 7.5; HisLink™ Protein Purification Resin; HisLink™ Spin Columns; FastBreak™ Cell Lysis Reagent, 10×; DNase I; collection tubes.
2. Microcentrifuge.
3. Vacuum Manifold (cat. no. A7231, Promega).
4. Vacuum Adapter (cat. no. A1331, Promega).
5. NaCl (5 M).

2.3. Large-Scale Non-Magnetic Nickel Purification for Bacteria

1. HisLink™ Resin (cat. no. V8821, Promega).
2. Columns.
3. Binding buffer: 100 mM HEPES + 10 mM imidazole, pH 7.5.
4. Wash buffer: 100 mM HEPES + 10–20 mM imidazole, pH 7.5.
5. Elution buffer: 100 mM HEPES + 50–1000 mM imidazole, pH 7.5.
6. NaCl (5 M).

2.4. Magnetic Nickel Purification for Mammalian and Insect Cells

1. MagneHis™ Protein Purification System (cat. no. V8500, Promega)—MagneHis™ Binding/Wash Buffer: 100 mM HEPES + 10 mM imidazole, pH 7.5; MagneHis™

Elution Buffer: 100 mM HEPES + 500 mM imidazole, pH 7.5; MagneHis™ Ni Particles; FastBreak™ Cell Lysis Reagent, 10×; DNase I.

2. Magnetic Stand (cat. no. Z5342, Promega).
3. NaCl (5 M).
4. Imidazole (1 M).

2.5. Magnetic Nickel Purification for Cell-Free Expression: Wheat Germ Extract

1. MagneHis™ Protein Purification System (cat. no. V8500, Promega)—MagneHis™ Binding/Wash Buffer: 100 mM HEPES + 10 mM imidazole, pH 7.5; MagneHis™ Elution Buffer: 100 mM HEPES + 500 mM imidazole, pH 7.5; MagneHis™ Ni Particles; FastBreak™ Cell Lysis Reagent, 10×; DNase I.
2. Magnetic Stand (cat. no. Z5342, Promega).
3. TNT® SP6 High-Yield Protein Expression System (cat. no. L3261, Promega)—TNT® SP6 High-Yield Master Mix; Nuclease-Free Water.
4. NaCl (5 M).

2.6. Magnetic Purification Purification for Cell-Free Expression: Rabbit Reticulocyte Lysate

1. MagZ™ Protein Purification System (cat. no. V8830, Promega)—MagZ™ Binding/Wash Buffer: 20 mM sodium phosphate + 500 mM NaCl, pH 7.4; MagZ™ Elution Buffer: 1 M imidazole, pH 7.5; MagZ™ Binding Particles.
2. Magnetic Stand (cat. no. Z5342, Promega).
3. TNT® SP6 Quick Coupled Transcription/Translation System (cat. no. L2080, Promega)—TNT® Quick Master Mix; SP6 Luciferase Control DNA; Methionine (1 mM); Luciferase Assay Reagent; Nuclease-Free Water.

2.7. Magnetic Nickel Purification for Automation

1. Maxwell™16 Instrument (cat. no. AS1000, Promega)—Instrument; Power Cable; RS-232 Cable for Firmware Upgrades; 1.5 mm Hex Wrench; Cartridge Preparation Rack; Magnetic Elution Tube Rack.
2. Maxwell™16 Polyhistidine Protein Purification Kit (cat. no. AS1060, Promega)—Maxwell16 Polyhistidine Protein Purification Sample Cartridges; Elution Buffer: 100 mM HEPES + 500 mM imidazole, pH 7.5.
3. NaCl (5 M).

2.8. Non-Magnetic Nickel Purification for High Throughput

1. Hislink™96 Protein Purification System (cat. no. V3680, Promega)—Binding/Wash Buffer: 100 mM HEPES + 10 mM imidazole, pH 7.5; Elution Buffer: 100 mM HEPES + 500 mM imidazole, pH 7.5; HisLink™ Resin; FastBreak™ Cell Lysis Reagent, 10×; DNase; Filtration Plate; Collection Plate.

2. NaCl (5 M).
3. Vacuum pump.
4. Vacuum holder.

2.9. Mass Spectrometry Elution Conditions from Magnetic Particles

1. Ammonium acetate, 10 mM, pH 7.5.
2. Ethanol, 30%.
3. TFA (trifluoroacetic acid), 0.1%, in 50% acetonitrile.
4. Speed Vac® Concentrator.

2.10. Mass Spectrometry Elution Conditions from Non-Magnetic Particles

1. HEPES, 100 mM, pH 7.5 + 500 mM NaCl.
2. Double-distilled water.
3. TFA, 0.1%, in 50% acetonitrile.
4. Speed Vac® Concentrator.

2.11. Cloning Vectors

The HQ-tag containing Flexi® Vectors are available for the cloning of desired proteins (9). The HQ tag can be appended to any protein-coding region using Flexi® Vectors designed for bacterial or in vitro protein expression. Flexi® Vectors are designed for rapid, high-fidelity transfer of protein-coding regions between vectors containing various expression or peptide tag options (9). These vectors enable expression of native or fusion proteins to facilitate the study of protein structure and function.

3. Methods

3.1. Purification of HQ-Tagged Proteins Expressed in Bacterial Cells

3.1.1. Preparation of Bacterial Cells

Bacterial cultures can be grown in tubes, flasks or 96-well plates. Grow the culture containing the HQ-tagged fusion proteins to an OD_{600 nm} between 0.4 and 0.6 and then induce protein expression. For IPTG (Isopropyl β-D-thiogalactoside) induction, add IPTG to a final concentration of 1 mM and incubate at 37°C for 3 h or 25°C overnight. Determine the OD_{600 nm} of the fresh bacterial culture.

3.1.2. Bacterial Cell Lysis

There are several methods for lysis of bacterial cells such as mechanical disruption (sonication or French press), enzymatic methods (lysozyme) and

detergents (lysis buffers). We have described lysis methods using sonication and lysis reagents (*see* **Notes 1** and **2**).

3.1.3. Purification

HQ-tagged proteins can be purified using different resins and formats from small to large-scale, manual or high-throughput and magnetic and non-magnetic.

3.2. Lysis and Purification from Bacterial Culture Using Magnetic Ni Particles

Lysis of bacterial culture can be done in culture or using pelleted cells; however, the purification protocol is the same for either lysis method.

3.2.1. Lysis of Pelleted Bacterial Cells Using Lysis Buffer

1. Centrifuge 1 ml of bacterial culture at 14,000 rpm for 2 min in a microcentrifuge. Remove the supernatant completely.
2. For every 1 OD_{600 nm} of culture, dilute 10 µl FastBreak™ Cell Lysis Reagent, 10×, to 1× by adding 90 µl NANOpure® or double-distilled water (100 µl total).
3. Resuspend the cell pellet in 100 µl 1× FastBreak™ Cell Lysis Reagent for every 1 OD_{600 nm} (for example, for a 3 OD_{600 nm} culture, use 300 µl 1× FastBreak™ Cell Lysis Reagent).
4. Resuspend lyophilized DNase I as indicated on the vial (*see* **Note 3**) and add 1 µl to the bacterial culture.
5. Incubate with shaking for 10–20 min at room temperature on a rotary mixer or shaking platform to lyse bacteria.

3.2.2. Direct Lysis of Bacterial Cell Cultures Using Lysis Buffer

1. Add 110 µl of FastBreak™ Cell Lysis Reagent, 10×, (1/10 volume) directly to 1 ml of fresh bacterial culture, OD_{600 nm} < 6.
2. Resuspend lyophilized DNase I (*see* **Note 3**) and add 1 µl per ml of original culture volume.
3. Incubate with shaking for 10–20 min at room temperature on a rotary mixer or shaking platform.

3.2.3. Purification Using Magnetic Ni Particles

1. Add 500 mM NaCl to lysed lysate (i.e., 0.03 g NaCl per 1 ml of lysate) for reducing the non-specific binding and increase efficiency of capture of HQ-tagged protein.
2. Vortex the MagneHis™ Ni Particles to a uniform suspension (*see* **Note 4**).
3. Add 30 µl of MagneHis™Ni particles.

4. Invert tube to mix (~10 times) and incubate for 2 min at room temperature. Make sure the MagneHis™ Ni Particles are well mixed.
5. Place the tube in the appropriate magnetic stand for approximately 30 s to capture the MagneHis™ Ni Particles. Using a pipette carefully remove the supernatant.
6. Remove the tube from the magnetic stand. Add 150 µl of MagneHis™ Binding/Wash Buffer to the MagneHis™ Ni Particles and pipette to mix. If NaCl was added for binding, also use the same amount of NaCl during washing. Make sure that particles are resuspended well.
7. Place the tube in the magnetic stand for approximately 30 s to capture the MagneHis™ Ni Particles. Using a pipette, carefully remove the supernatant.
8. Repeat the wash step two times for a total of three washes.
9. Remove the tube from the magnetic stand. Add 100 µl of MagneHis™ Elution Buffer and pipette to mix.
10. Incubate for 1–2 min at room temperature. Place in a magnetic stand to capture the MagneHis™ Ni Particles. Using a pipette, remove the supernatant containing the purified protein.

3.3. Lysis and Purification from Bacterial Culture Using Non-Magnetic Ni Particles (Spin Baskets)

Lysis of bacterial cells and binding of HQ-tagged proteins is done in one step. Purification can be done by centrifugation or using a vacuum manifold.

3.3.1. Direct Lysis of Bacterial Cell Cultures Using Lysis Buffer

Cultures with concentrations up to 8 OD_{600 nm} units/ml have been successfully used with the HisLink™ Spin Protein Purification System. A maximum of 700 µl of bacterial culture can be loaded per HisLink™ Spin Column.

1. Pipette 700 µl of bacterial culture into a 1.5 ml microcentrifuge tube. Add 70 µl of the FastBreak™ Reagent/DNase I solution (*see Note 5*).
2. Resuspend the resin and allow it to settle. Once the resin has settled, use a wide-bore pipette tip to transfer 75 µl of the HisLink™ Resin from the settled resin bed to the 1.5 ml microcentrifuge tube.
3. Adding 200 mM NaCl prior to the addition of the HisLink™ Resin may reduce non-specific binding and improve binding of HQ-tagged proteins. If NaCl is used in binding also use NaCl in the washes.
4. Incubate the sample and resin for 30 min, mixing frequently on a rotating platform or shaker to optimize binding.
5. Continue with either the centrifugation or vacuum spin column protocol.

3.3.2. Centrifugation Protocol for Spin Columns

1. Place a HisLink™ Spin Column onto a collection tube (or a new 1.5 ml microcentrifuge tube). Use a wide-bore pipette tip to transfer the lysate and resin from the original 1.5 ml microcentrifuge tube to the spin column.

2. Centrifuge the spin column with the collection tube for 5 s or until the liquid clears the spin column.
3. To save the flow through, remove the spin column from the collection tube and transfer the flow through from the collection tube to a new 1.5 ml microcentrifuge tube. Otherwise, discard the flow through.
4. Place the spin column back onto the collection tube. Add 500 μ l of HisLink™ Binding/Wash Buffer plus the same amount NaCl used in binding to the spin column, then cap the spin column. Centrifuge for 5 s or until the buffer clears the spin column. Discard the flow through. Repeat for a total of two washes.
5. Take the spin column off the collection tube and wipe the base of the spin column with a clean absorbent paper towel to remove any excess buffer.
6. Place the spin column onto a new 1.5 ml microcentrifuge tube. Add 200 μ l of HisLink™ Elution Buffer. Cap the spin column and tap or flick it several times to resuspend the resin. Wait for 3 min.
7. Centrifuge the HisLink™ Spin Column and microcentrifuge tube at 14,000 rpm for 1 min to collect the eluted protein.

3.3.3. Vacuum Protocol for Spin Columns

1. Place a HisLink™ Spin Column onto a vacuum adapter and then attach the adapter to a vacuum port. Use a wide-bore pipette tip to transfer the lysate and resin to the spin column. Any unused ports on the vacuum manifold must be closed for the manifold to work properly.
2. Apply a vacuum for 5 s or until the lysate clears the spin column.
3. Add 500 μ l of HisLink™ Binding/Wash Buffer plus the same amount of NaCl used in binding to the spin column. Apply a vacuum for 5 s. Repeat for a total of two washes.
4. Take the spin column off the vacuum adapter and wipe the base of the spin column with a clean absorbent paper towel to remove any excess buffer.
5. Place the spin column onto a new 1.5 ml microcentrifuge tube. Add 200 μ l of HisLink™ Elution Buffer. Cap the spin column and tap or flick it several times to resuspend the resin. Wait for 3 min.
6. Centrifuge the spin column with the 1.5 ml microcentrifuge tube at 14,000 rpm for 1 min to collect the eluted protein.

3.4. Large-Scale Column-Based Lysis and Purification of HQ-Tagged Proteins Using HisLink Resin

3.4.1. Lysis of Pelleted Bacterial Cells Using Sonication

1. Centrifuge bacterial culture at $\geq 10,000 \times g$ for 15 min. Remove the supernatant completely.
2. Resuspend pellet in cell lysis reagent or 100 mM HEPES + 10 mM imidazole, pH 7.5, at 10 to 50 fold concentration of the cell culture, depending on the amount of protein expressed in the culture.

3. Sonicate samples on ice. Sonicate with 5s pulse plus 5-s gap until cells are completely lysed.
4. For large-scale column purification, clear the lysate before loading the column by centrifuging at $10,000 \times g$ for 30 min at 4°C and discard pellet.

3.4.2. Column Preparation

1. Determine the column volume required to purify the protein of interest. In most cases, 1 ml of settled resin is sufficient to purify the amount of protein typically found in up to 1 L of culture (cell density of $OD_{600\text{ nm}} < 6$) (see **Note 6**).
2. Once you have determined the volume of settled resin required, precalibrate this amount directly in the column by pipetting the equivalent volume of water into the column and marking the column to indicate the top of the water. This mark indicates the top of the settled resin bed. Remove the water before adding resin to the column.
3. Do not let the resin dry out after you have applied the lysate to the column.

3.4.3. Purification from Cleared Lysate Using Gravity

1. Fill the column with resin to the line marked on the column by transferring the resin with a pipette. Allow the resin to settle and adjust the resin by adding or removing resin as necessary.
2. Allow the column to drain, and equilibrate the resin with 5 column volumes of binding buffer, allowing the buffer to completely enter the resin bed.
3. Gently add the cleared lysate to the resin until the lysate has completely entered the column.
4. Wash unbound proteins from the resin using at least 10–20 column volumes of wash buffer. Divide the total wash buffer used into two or three aliquots and allow each aliquot to completely enter the resin bed before adding the next aliquot.
5. Once the wash buffer has completely entered the resin bed, add elution buffer and begin collecting fractions (0.5 ml fractions). Elution is protein dependent, but HQ-tagged proteins will generally elute in the first 1 ml for a 1 ml resin column. Elution is usually complete after 3–5 ml of buffer has been collected per 1 ml of settled resin, provided the imidazole concentration is high enough to efficiently elute the protein of interest.

3.4.4. Purification from Cleared Lysate on a Vacuum Manifold

1. Place the column onto a vacuum manifold and apply just enough pressure to drain the water from the area above the resin. Equilibrate the resin with 5 column volumes of binding buffer, allowing the entire volume of buffer to enter the resin bed.
2. Add cleared lysate to the resin and apply a vacuum to the column.
3. Wash the resin with at least 10–20 column volumes of wash buffer, applying vacuum to facilitate washing. Divide the total wash buffer used into two or three

aliquots and allow each aliquot to completely enter the column before adding the next aliquot. Care should be taken not to let the resin dry out during this step.

4. Once the final aliquot of wash buffer has completely entered the resin bed, add elution buffer and begin collecting fractions (0.5 ml fractions). Elution may be performed under vacuum if the manifold used allows for the collection of the eluate. Elution is protein dependent, but HQ-tagged proteins will generally elute in the first 1 ml for a 1 ml resin column. Elution is usually complete after 3–5 ml of buffer per 1 ml of settled resin, provided the imidazole concentration is high enough to efficiently elute the protein of interest.

3.4.5. Batch Purification from Cleared or Crude Lysate

1. Batch purification may be performed on either cleared or crude lysate following the same general protocol. To purify in batch mode, first determine the amount of resin required for the amount of cleared or crude lysate. Generally for expression levels on the order of 1–30 mg/l of culture, 2–4 ml of 50% slurry should be sufficient to bind the HQ-tagged protein from 1 L of culture. Add the resin to the cleared or crude lysate and stir with a magnetic stir bar (or other device) for at least 30 min at 4°C, ensuring that the resin is well mixed throughout the lysate solution. Alternatively, the lysate and resin can be added to a conical tube and placed on an orbital shaker for 30 min.
2. Allow the resin to settle for approximately 5 min, then carefully decant the lysate. If necessary, use a pipette to completely remove the lysate leaving the resin behind.
3. To remove non-specifically bound proteins, add wash buffer (10 ml/ml of resin used) to the resin and fully resuspend. Allow the resin to settle for approximately 5 min, then carefully decant the wash solution. If necessary, use a pipette to remove as much of the wash volume as possible without disturbing the resin. Repeat wash step two times for a total of three washes.
4. After the third wash, thoroughly resuspend the resin in a volume of wash buffer sufficient to transfer the resin to a column. Allow the entire amount of buffer to enter the resin bed. Use as much wash buffer as necessary to transfer all of the resin.
5. Add elution buffer and begin collecting fractions (0.5–5 ml fractions). Elution is protein dependent, but HQ-tagged proteins will generally elute in the first 1 ml for a 1 ml resin column. Elution is usually complete after 3–5 ml of buffer per 1 ml of settled resin, provided the imidazole concentration is high enough to efficiently elute the protein of interest.

3.5. Purification Under Denaturing Conditions

Proteins that are expressed as inclusion bodies and have been solubilized with chaotropic agents such as guanidine-HCl or urea can be purified by modifying

the above protocols to include the appropriate amount of denaturant (up to 6 M guanidine-HCl or up to 8 M urea) in binding, wash and elution buffers.

3.6. Purification of HQ-Tagged Proteins Expressed in Insect and Mammalian Cells Using Magnetic Ni Particles

Bacterial expression of recombinant His-tagged proteins is a common technique. However, insect cells and mammalian cells are becoming more widely used expression systems for expression of recombinant proteins. These eukaryotic expression systems may allow more natural processing and modification of recombinant proteins, which are not possible in bacterial expression system. HQ tag can also be used in these expression systems.

3.6.1. Preparation of Insect and Mammalian Cells

Insect or mammalian cells can be cultured under normal conditions. Process cells at a cell density of 2×10^6 cells/ml of culture. Adherent cells may be removed from tissue culture plastic by scraping and resuspending in culture medium to this density. Cells may be processed in culture medium containing up to 10% serum. Processing more than the indicated number of cells per 1 ml sample may result in reduced protein yield and increased non-specific binding (**10**).

3.6.2. Purification of Intracellular Expressed HQ-Tagged Proteins from Cultured Insect or Mammalian Cells

1. Add 110 μ l of FastBreak™ Cell Lysis Reagent, 10x, to 1 ml of insect or mammalian cells in culture medium (*see Note 7*).
2. Add 1 μ l DNase I (*see Note 3*) to the lysed insect or mammalian cell culture.
3. Incubate with shaking for 10–20 min at room temperature on a rotary mixer or shaking platform.
4. Vortex the MagneHis™ Ni Particles to a uniform suspension (*see Note 4*).
5. Add 30 μ l of the MagneHis™ Ni Particles to 1.1 ml of cell lysate.
6. Add 1 M imidazole (pH 8) to a final concentration of 20 mM to decrease non-specific binding of serum proteins (22 μ l of 1 M imidazole per 1.1 ml of sample).
7. Invert tube to mix (~10 times) and incubate for 2 min at room temperature.
8. Place the tube in the appropriate magnetic stand for approximately 30 s to capture the MagneHis™ Ni Particles. Using a pipette, carefully remove the supernatant.
9. Remove the tube from the magnetic stand. Add 500 μ l of MagneHis™ Binding/Wash Buffer containing 500 mM NaCl to the MagneHis Ni™ Particles and pipette to mix. Make sure that the particles are resuspended well.

10. Place the tube in the appropriate magnetic stand for approximately 30 s. Allow the MagneHis™ Ni particles to be captured and carefully remove the supernatant using a pipette.
11. Repeat the wash step two times for a total of three washes.
12. Remove the tube from the magnetic stand. Add 100 µl of MagneHis™ Elution Buffer and pipette to mix.
13. Incubate for 1–2 min at room temperature. Place the tube in a magnetic stand to capture the MagneHis™ Ni Particles with the magnet. Using a pipette, remove the supernatant containing the purified protein.

3.6.3. Purification of Secreted HQ-Tagged Proteins from Insect or Mammalian Cell (See **Note 8**)

1. Vortex the MagneHis™ Ni Particles to a uniform suspension (see **Note 4**).
2. Add 30 µl of MagneHis™ Ni Particles to 1 ml of culture medium after removing cells.
3. Add 1 M imidazole to a final concentration of 20 mM to decrease non-specific binding of serum proteins (20 µl/1 ml sample). Adding 500 mM NaCl may improve HQ-tagged protein binding and decrease non-specific binding.
3. Invert tube to mix (~10 times) and incubate for 2 min at room temperature.
4. Place the tube in the appropriate magnetic stand for approximately 30 s to capture the MagneHis™ Ni Particles with the magnet. Using a pipette, carefully remove the supernatant.
5. Remove the tube from the magnet. Add 500 µl of MagneHis™ Binding/Wash Buffer containing 500 mM NaCl to the MagneHis™ Ni Particles and pipette to mix. Make sure that the particles are resuspended well.
6. Place the tube in the appropriate magnetic stand for approximately 30 s to capture the MagneHis™ Ni Particles with the magnet. Using a pipette, carefully remove the supernatant.
7. Repeat the wash step two times for a total of three washes.
8. Remove the tube from the magnet. Add 100 µl of MagneHis™ Elution Buffer and pipette to mix.
9. Incubate for 1–2 min at room temperature. Place the tube in a magnetic stand to capture the MagneHis™ Ni Particles. Using a pipette, remove the supernatant that contains the purified protein.

3.7. Purification of HQ-Tagged Proteins Expressed in Cell-Free Expression Systems

Cell-free expression systems may be preferred over in vivo or native systems, because they can be used for the expression of toxic, proteolytically sensitive or unstable proteins (**11–13**). In vitro systems provide the ability to incorporate non-natural amino acids containing photoactivatable fluorescent or biotin residues or radioactive amino acids (**14**). The HQ can be utilized in cell-free expression systems.

3.7.1. Purification of HQ-Tagged Proteins Expressed in Wheat Germ

The TNT® SP6 High-Yield Protein Expression System is a single-tube, coupled transcription/translation system which can express up to 100 µg/ml of protein. This cell-free expression system contains all the components (tRNA, ribosomes, amino acids, polymerase and translation initiation, elongation and termination factors) necessary for protein synthesis directly from DNA templates. In general, wheat germ extracts provide some co-translational and post-translational modifications such as phosphorylation (15), farnesylation (16) and myristoylation (17).

1. Add 150 µl of MagneHis™ Bind/Wash buffer + 500 mM NaCl to 50 µl wheat germ reaction.
2. Vortex the MagneHis™ Ni Particles to a uniform suspension.
3. Add 30 µl of MagneHis™ Resin to the reaction. Mix and incubate for 5 min. Mix periodically to keep the particles from settling. Mix by pipetting or flicking tube.
4. Place in magnetic stand and remove supernatant.
5. Add 150 µl of MagneHis™ Bind/Wash buffer + 500 mM NaCl. Mix and place in magnetic stand (see **Note 9**).
6. Repeat step 5 two more times for a total of three washes.
7. Add 100 µl of MagneHis™ Elution buffer and mix. Incubate 1–2 min and then place in magnetic stand. Supernatant will contain purified protein.

3.7.2. Purification of HQ-Tagged Proteins Expressed in Rabbit Reticulocyte Lysate

The TNT® Quick Coupled Transcription/Translation System is a single-tube, coupled transcription/translation reaction that contains RNA polymerase, nucleotides, salts and recombinant Rnasin®, ribonuclease, inhibitor, for eukaryotic in vitro translation. Canine microsomal membranes may be added for post-translational modifications such as signal sequence cleavage and glycosylation.

1. Add 150 µl of MagZ™ Bind/Wash buffer to 50 µl rabbit reticulocyte reaction.
2. Vortex the MagZ™ Particles to a uniform suspension and add 60 µl of MagZ™ Resin to 1.5 ml tube. Place in magnetic stand and remove buffer.
3. Add rabbit reticulocyte reaction diluted in buffer to resin. Mix and incubate for 15 min. Mix periodically to keep the particles from settling. Mix by pipetting or flicking tube.
4. Place in magnetic stand and remove supernatant.
5. Add 150 µl of MagZ™ Bind/Wash buffer. Mix and place in magnetic stand.
6. Repeat step 5 three more times for a total of four washes.
7. Add 100 µl of MagZ™ Elution Buffer and mix. Incubate 1–2 min at room temperature and then place in magnetic stand. Supernatant will contain purified protein.

3.8. Automated and High-Throughput Purification of HQ-Tagged Proteins

3.8.1. Purification Using a Minirobot: Maxwell™16

The Maxwell™16 Purification Instrument is an automated magnetic particle handling device. The instrument is preprogrammed with purification protocols and can process up to 16 samples in a single run of about 40 min. In addition, prefilled reagent cartridges contain the buffers and resin for purification for optimal convenience.

- 1. Buffer configuration in cartridge, predispensed (see Table 1).
- 2. Optimized up to 20 OD_{600 nm} bacterial culture, 2 × 10⁶ mammalian or insect cells, 1 ml culture media or 100–200 µl wheat germ reaction.
- 3. Add 300 µl of elution buffer to the elution tube.
- 4. Follow the Maxwell™16 protocol for protein purification with the necessary modifications for HQ-tagged proteins. Use the manual protocols as guide for the addition of NaCl.
- 5. Increase imidazole concentration (50–100 mM imidazole) in the washes to reduce non-specific binding in wheat germ reactions.

3.8.2. High-Throughput Purification of HQ-Tagged Proteins

Cultures with concentrations up to 8 OD_{600 nm} units/ml have been successfully used with the HisLink™96 Protein Purification System.

- 1. To 1 ml of bacterial culture, add 100 µl of the FastBreak™ Reagent/DNase I solution (see Note 10).
- 2. Resuspend the resin and allow it to settle. Once the resin has settled, use a wide-bore pipette tip to transfer 75 µl of the HisLink™ Resin from the settled resin bed to each well of the plate.

Table 1
Maxwell® 16 polyhistidine/HQ tagged Protein
Purification Kit Reagent Catridge Contents

Well	Buffer
1	1 ml bacterial culture, mammalian cells, insect cells, media or wheat germ (added by customer) 110 µl FastBreak™ Cell Lysis Reagent, 10×
2	750 µl MagneHis™ Ni- Particles
3–6	1 ml MagneHis™ Binding/Wash Buffer
7	Empty

3. Adding 200 mM NaCl prior to the addition of the HisLink™ Resin may reduce non-specific binding and improve binding of HQ-tagged proteins. If NaCl is used in binding also, use NaCl in the washes.
4. Incubate the sample and resin for 30 min, mixing frequently by vortexing or pipetting to optimize binding.
5. Place a filtration plate onto the vacuum manifold base (*see Note 11*).
6. Use a wide-bore pipette to transfer the lysed lysate and resin to the filtration plate.
7. Cover unused filtration plate wells with an adhesive plate sealer.
8. Apply vacuum to the samples for 10 s.
9. If you collected the flow through, remove the filtration plate from the manifold collar and place the filtration plate onto the vacuum manifold base.
10. Add 250 µl of the HisLink™ Binding/Wash Buffer plus the same amount of NaCl used in binding to the wells of the filtration plate. Apply vacuum for 10 s.
11. Repeat step 6 three more times for a total of four washes.
12. Place the filtration plate onto a clean absorbent towel to remove any excess wash buffer from the ports located on the bottom of the plates.
13. Place the collection plate onto the manifold bed.
14. Place the manifold collar on the collection plate, fitting it into the pins of the manifold bed.
15. Place the filtration plate onto the manifold collar. To prevent uneven flow or spattering, remove the vacuum hose from the port on the manifold collar. Reattach the vacuum hose at step 17.
16. Add 200 µl of the elution buffer. Wait for 3 min.
17. Reattach the vacuum hose to the manifold collar.
18. Collect the eluate by applying a vacuum for 1 min.

3.8.3. High-Throughput Purification Using Robotics

Both the MagneHis™ Protein Purification System and the HisLink™96 Protein Purification System are amendable for high-throughput robotics (**18**). The manual protocols can be used as a guide to develop protocols for automated workstations and may need optimization depending on the instrument used. Automated methods have been developed to purify proteins on several workstations such as Beckman and Tecan and are easily scalable to accommodate a variety of sample volumes. These protocols can be downloaded from <http://www.promega.com/automethods>.

3.9. Mass Spectrometry Analysis of HQ-Tagged Proteins

Matrix-assisted laser desorption/ionization-time of flight (MALDI-TOF) and other alternative methods of mass spectrometry (MS) analysis have become essential methods of protein analysis (**19,20**). Using MS methods, we could

identify post-translational modifications, protein profiles, protein–protein interactions and protein–small molecule interactions and study protein structure and function (21–24). HQ-tagged protein purification systems provide large amounts of protein or small amounts of multiple proteins for study. However, the elution buffers used in these systems contain salts (e.g., imidazole) that cannot be used in MS analysis. To be compatible with MALDI-TOF MS analysis, eluted samples need to undergo tedious dialysis methods or size exclusion separation techniques to remove salts. We have developed various methods for the elution of HQ-tagged proteins. These elution conditions allow direct MS analysis and provide clean MS data necessary for high-throughput analysis using MALDI-TOF MS.

3.9.1. Elution from Magnetic Particles

1. After washing the MagneHis™ Ni Particles with MagneHis™ Binding/Wash Buffer, wash the Ni Particles twice with 150 µl of 10 mM ammonium acetate (pH 7.5) or 30% ethanol.
2. Elute with 100 µl of 0.1% TFA in 50% acetonitrile.
3. Dry sample in a Speed Vac® concentrator or air-dry.
4. Resuspend the sample in the solvent or buffer that will be used for MS analysis.

3.9.2. Elution from Non-Magnetic Particles

1. After binding, wash the resin twice with 500 µl of 100 mM HEPES (pH 7.5) plus 0.5 M NaCl to decrease non-specific binding.
2. Wash the HisLink™ Spin Columns four times with 500 µl of double-distilled water to remove the NaCl and buffer from the resin.
3. Elute with 200 µl of 0.1% TFA in 50% acetonitrile.
4. Dry sample in a Speed Vac® concentrator or air-dry.
5. Resuspend the sample in the solvent or buffer that will be used for MS analysis.

4. Notes

1. FastBreak™ Cell Lysis Reagent was designed to lyse cells without the addition of lysozyme. Lysozyme, if added, will co-purify with the HQ-tagged protein unless 500 mM NaCl is added to the wash buffer. These lysis methods have been used successfully with Luria-Bertani (LB) and Terrific Broth (TB) medium. Some bacterial strains may require a freeze-thaw cycle to achieve maximal lysis. This can be achieved by freezing the cell pellet or culture at –20°C for 15 min or –70°C for 5 min.
2. Some proteins purify more efficiently from a cell pellet. Test both direct lysis and lysis of a bacterial culture to determine which is optimal for the target protein.
3. Resuspend lyophilized DNase I (cat. no. Z358B, Promega) with 80 µl double-distilled water.

4. The MagneHis™ Ni Particles are pre-equilibrated and can be added directly to the sample.
5. Resuspend lyophilized DNase I (cat. no. Z358B, Promega) with 80 µl double-distilled water. Mix to dissolve the powder completely. Remove the DNase solution from the vial and add it to 1 ml of double-distilled water. For each reaction, use 5.8 µl DNase solution + 64.2 µl FastBreak™ Cell Lysis Reagent, 10×.
6. In cases of very high expression (e.g., 50 mg/l), up to 2 ml of resin per liter of culture may be needed.
7. We do not recommend adding 500 mM NaCl to the FastBreak™ Cell Lysis Reagent, 10×, as it could result in particle clumping during lysis and binding in this system.
8. Cells should be removed from the medium before protein purification.
9. The amount of imidazole in the washes can be optimized by titrating from 10–100 mM imidazole. The higher the amount of imidazole used for washing, the less background. However, some tagged protein may elute off.
10. Resuspend lyophilized DNase I (cat. no. Z358B, Promega) with 80 µl double-distilled water. Mix to dissolve the powder completely. Remove the DNase solution from the vial and add it to 1 ml of double-distilled water. For each reaction use 8 µl DNase solution + 92 µl FastBreak™ Cell Lysis Reagent, 10×.
11. If you wish to collect the flow through, place an empty deep-well plate on the manifold bed. On top of the deep-well plate place the manifold collar and insert the filtration plate onto the collar before transferring the lysate.

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Amylose Affinity Chromatography of Maltose-Binding Protein

Purification by both Native and Novel Matrix-Assisted Dialysis Refolding Methods

Leonard K. Pattenden and Walter G. Thomas

Summary

Maltose-binding protein (MBP) is a carrier protein for high level recombinant protein and peptide production from either the cytoplasm or periplasm of *Escherichia coli*. The affinity matrix for purifying MBP-passenger proteins utilizes amylose covalently attached to magnetic beads, agarose, or a chemically inert fast protein liquid chromatography (FPLC) matrix – exploiting the natural affinity of MBP for α -(1 $\rightarrow$ 4)-maltodextrins in the stationary phase. A fundamental problem is the expression and purification failure of as much as 30% of all constructs, which is limiting for one of the best solubilizing carrier proteins available for recombinant expression. In this chapter, we have discussed aspects of MBP biology that can impact upon binding to the amylose affinity matrix including cloning considerations, structural complications, hydrophobic buffer additives and the presence of cellular biomolecules that bind or modify the matrix during purification. Chromatography conditions are presented for purification at very small scales of less than 0.5 mL using amylose magnetic beads, a batch and semi-batch method for small to moderate scale purifications up to approximately 35 mg and larger scale FPLC methods. A novel matrix-assisted dialysis refolding method is also described whereby MBP-passenger proteins can be refolded in the presence of amylose matrix in instances where native purification methods fail to bind the amylose matrix.

Key Words: Protein expression; amylose affinity chromatography; maltose-binding protein; maltodextrin-binding protein; maltose regulon; FPLC; protein refolding/chemistry.

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1. Introduction

Affinity chromatography of maltose-binding protein (MBP) (1) exploits the binding of amylose that is functionalized as the stationary phase to magnetic beads, agarose or an inert matrix. As for other forms of affinity chromatography, the successful purification using amylose affinity chromatography is critically linked to an intimate understanding of the biomolecular interactions (and complications) that can occur due to the specific and unique features of MBP and amylose biology. Likewise a limitation to broader applications, greater developments and the reason for misunderstandings that arise with regard to MBP and amylose affinity chromatography are the failures to completely grasp and exploit aspects of this biology. This chapter highlights the facets of MBP and amylose biology and chemistry that are relevant to affinity purification and discusses how these facets negatively impact purification or can be exploited to achieve purification. Methods are presented for native purification in batch, semi-batch and fast protein liquid chromatography (FPLC) modes, and a new matrix-assisted dialysis refolding method is described that is suitable for batch and semi-batch modes.

MBP, also referred to as maltodextrin-binding protein (and sometimes written unhyphenated), can be expressed at arguably the highest levels for any recombinant carrier protein. Developed by New England BioLabs from *Escherichia coli*, there are now six constructs commercially available for periplasmic or cytoplasmic expression with a Factor Xa, Genenase I or Enterokinase protease site engineered into the constructs (2). There is also a series of MBP constructs developed by David S Waugh (National Cancer Institute at Frederick) that can be obtained from the non-profit distributor of biological reagents AddGene (3,4,5) which include Gateway© His6-MBP and non-*E. coli*-sourced MBP, typically using a TEV protease site (tobacco etch virus nuclear inclusion protease site) (4,5). New England BioLabs also provide a range of suitable *E. coli* host cells that are useful for MBP expression free of charge and have very reasonable licensing and royalty terms, making MBP-based recombinant carrier protein expression and purification also one of the most economically achievable affinity chromatography systems available for both research and commercial ventures.

MBP belongs to the bacterial superfamily of periplasmic-binding proteins that are monomeric bilobular proteins with molecular weights in the range of 25–45 kDa, containing a single ligand-binding site with micromolar dissociation constants for diverse ligands ranging from ions (6–9), amino acids (10–13), oligopeptides (14,15) and carbohydrates (16,17) (see Note 1 for MBP biophysical properties). Within Gram negative bacteria, the periplasmic-binding proteins are involved in the chemotaxis and transport of their respective ligands. Unlike Gram positive bacterium that directly sense and responds to specific

ligands in the environment through integral membrane proteins, Gram negative organisms have two membranes separated by the periplasmic space, presenting a challenge to co-ordinate the uptake, movement and translocation across these diverse structural features. In native *E. coli*, MBP mediate processes by acting as a chemoreceptor for α -(1 $\rightarrow$ 4)-D-glucose polysaccharides (maltodextrins) (18); the binding of the ligand induces a conformational change in MBP that allows the selective recognition by specific integral membrane proteins, receptors and porins for the following:

1. Chemotaxis by inner membrane receptors: Maltose chemotaxis is the process by which the bacteria move in response to a maltodextrin concentration gradient through signals that are transmitted to the flagellar.
2. Transport of maltodextrins: Firstly by porins of the outer membrane, raising the periplasmic concentration of the maltodextrins and subsequent energy-dependent active translocation of maltodextrins into the cytoplasm by integral membrane proteins.

The proteins involved in maltodextrin chemotaxis and transport are collectively termed the maltose regulon of *E. coli* (18). In order to mediate the separate processes of chemotaxis and transport, MBP is normally present in a very large ($\sim$ 50 fold) excess compared to the associated membrane protein components of the maltose regulon. Another suggestion for the high levels of MBP is that MBP has molecular chaperone properties that may help in protein folding and renaturation in the periplasm (19,20). There are two ways by which MBP could be involved in protein folding. One is passive – by being a stable and readily ‘foldable’ protein that is attached to the desired recombinant protein (21,22). Alternatively, MBP has been hypothesized to actively refold proteins – through interactions at hydrophobic regions of MBP (19,20), possibly with the hydrophobic surface clusters important for interacting with proteins involved in maltodextrin chemotaxis and transport (23,24).

MBP mediates diverse cellular responses for maltodextrin metabolism in the presence of any α -(1 $\rightarrow$ 4)-D-glucose polysaccharide of up to 8 glucose units in length. Maltose binds to MBP with the glucose ring oxygen atoms all on the same side, and adopting this correct conformation for alignment of hydrogen bonding interactions within MBP is critical for affinity. Amylose is essentially a repeating maltose polymer with flexible polysaccharide chains joined by the α -(1 $\rightarrow$ 4) links. Amylose affinity chromatography exploits the maltodextrin-like affinity of MBP as the basis for purification (see **Note 2** for matrix properties and chromatography conditions).

Structural aspects of MBP are important for amylose affinity chromatography. The polypeptide chain of MBP is present as two globular domains, and the maltodextrin-like ligands bind within a ligand-binding cleft located at an interface formed by the two globular domains (25). Essentially, MBP

functions as a molecular bivalve; the protein adopts two conformations: an unliganded 'open' conformation and a ligand-bound 'closed' conformation that involves a twisting rotation of approximately 8° and bending movement of up to 35° by the N-terminal lobe (26). The residues forming the binding cleft are placed at the surface of the two domains, so upon binding, the ligand induces the conformational changes that allow the two globular domains to enclose the ligand-binding cleft, excluding solvent and forming a stable, bound state. Positioned behind the ligand-binding cleft is a hinge region, which facilitates the opening and closing structural movements that occur with ligand-induced conformational change.

Fusion constructs of MBP are not normally engineered with the passenger protein at the N terminus, as such constructs are not frequently soluble and do not readily purify. The N-terminal region is located external to the ligand-binding cleft but undergoes radical changes upon ligand binding. Therefore, steric or thermodynamic effects may occur with N-terminal constructs to influence the conformational changes in this region of MBP depending on the size and nature of the construct, impinging on the ability of MBP to open and close – precluding binding to the amylose affinity resin (27). Therefore the C terminus is the preferred site of cloning and appears to undergo far less structural changes in response to ligand binding (26).

1.1. Aspects of MBP and Amylose Biology and Chemistry that Impact on Purification

For *E. coli* expression of MBP, the history (including codon usage) and high intrinsic concentration of MBP are features of the biology, making MBP very favourable as a carrier protein for heterologous expression – depending on the nature of the cloning and physico-chemical properties of the passenger moiety. There are two different construct types available from New England BioLabs. The first type allows for typical recombinant *E. coli* expression localized in the cytoplasm at large levels. The second construct-type exploits MBP biology to direct expressed proteins to the periplasmic space at modest levels, but provides a simplified bioprocess (see **Note 3**) from a unique environment where native disulphide bonds may be formed and a proteolytic profile exists which is distinct from the cytoplasm.

New England BioLabs claim purification ranges, as high as 200 mg/L culture have been obtained for MBP fusion proteins with typical yields in the range of 10–40 mg/L culture for cytoplasmic expression (being 20–40% of the total cellular protein), while typical yields from periplasmic expression being ≤ 4 mg/L culture (1–5% of the total cellular protein) (2). It is not uncommon to obtain yields of 100 mg/L from shake-flask cultures (28). With favourable

properties and high expression levels, the question arises as to why MBP is not more actively utilized? Some of the reasons can be attributed to difficulties of the early MBP systems and bioprocessing challenges for unwary users (*see Note 4*). However, subsequent improvements to the systems have removed these issues (2). A fundamental problem that still exists with recombinant protein expression using MBP is that not all MBP fusion constructs work and the failure rate from a screening expression experiment indicates this could be as high 30% (28). This percentage is very high for what is one of the best solubilizing carrier proteins – so why is the percentage so high?

Some of the reasons for failure are common to recombinant protein expression – both cytoplasmic and periplasmic expressions are subject to the standard *E. coli* challenges of inclusion body formation and proteolysis, depending on the growth conditions, host-cell phenotype and physico-chemical properties of the passenger moiety. With MBP, periplasmic localization can create an additional challenge as it requires passage through a membrane and as periplasmic proteins utilize discreet folding machinery, not all MBP fusion proteins are successfully exported or maintained in the periplasm, showing significant folding variations or truncations which may or may not exhibit recombinant protein activity (29,30). Despite these complications, the major causes for failure appear to be particular to MBP and amylose affinity chromatography, especially in cases where the fusion protein is present and soluble but binds inefficiently to the amylose matrix – or even not at all. There are many reasons why this can occur with MBP and amylose affinity chromatography and these will now be discussed.

Factors that negatively impact on amylose affinity chromatography can include buffer additives and cellular biomolecules present in the crude lysis milieu. Specific problems are noted for the non-ionic detergents Triton X-100 and Tween 20; New England BioLabs state there is passenger-specific variability in the ability to bind in the presence of non-ionic detergents (2). However, it is likely that any additive that can perturb hydrophobic interactions will be detrimental to amylose affinity chromatography owing to the importance of aliphatic features of MBP for structure and function and therefore should be avoided during standard purification (*see Note 5* for a further discussions and guidelines to using additives).

Proteins of the maltose regulon are cellular biomolecules present in the crude lysis conditions that can potentially affect amylose affinity chromatography. In the absence of maltodextrins, there is control of protein levels of the maltose regulon to scavenging levels (18). These basal levels can be elevated significantly when using alternate carbohydrate sources such as glycerol (as in terrific broth) or under glucose-limiting growth conditions (as used in a chemostat or potentially certain bioreactor conditions) (18,31,32). The proteins

of the maltose regulon and cellular inducers of the regulon are particularly detrimental as they can bind and/or modify the amylose matrix directly or sequentially, often releasing maltose, maltotriose or analogues as a by-product which can elute MBP fusion proteins from the amylose matrix. These proteins include maltodextrinyl-specific, phosphorylases, transferases, glucosidases, γ -cyclodextrinases, transacetylases, periplasmic and cytoplasmic α -amylases and amylase-like enzymes (18), and these proteins are likely the cause of deterioration of amylose affinity matrices (see Note 6). However, the basal scavenging levels can be maintained with high glucose concentrations that exert strong catabolite repression to the maltose regulon and maltodextrinyl-specific operons (18). A D-glucose concentration of 0.2% is sufficient in Lauria Bertani media to suitably suppress unwanted protein expression including leaky expression of the MBP fusion protein (2), but the concentration of the suppressor will alter depending on media types and growth parameters, such as growth densities and the phase of growth. Leaky expression from pMAL vectors is also controlled by the presence of glucose which ensures the *tac* promoter is not induced in the absence of isopropyl β -D-thiogalactopyranoside.

Another possible cause of purification failure comes from the proposal that certain chaperone-like interactions may be detrimental if constructs form soluble aggregates through physical association, becoming trapped in a folding intermediate state such that the MBP-passenger protein forms a stable globular form termed a sequestered intermediate (19). However, it is important to consider the nature of the cloning at the C terminus, which is close to the ligand-binding cleft and the hinge region. It is possible that excessive removal of nucleotides during cloning will shorten the linker regions introduced with more recent constructs from New England BioLabs, and some constructs may interact with the ligand-binding cleft depending on their physical properties; this may disrupt ligand binding or important hinge movements that are necessary for high affinity binding of the amylose matrix leading to a failure of purification. It is also important to understand detrimental interactions for purification need not occur locally (in *cis* or upon the same MBP-passenger molecule), but could also arise from multiple regions of the MBP-passenger protein interacting in *trans* on neighbouring MBP-passenger molecules or with other biomolecules in the purification milieu. In such a scenario, the involvement of hydrophobic regions in or about the substrate-binding cleft may occlude binding to the amylose matrix or involve the hinge region and thereby impede normal conformational changes necessary for binding to the amylose matrix. Therefore, the purification failure of some MBP-passenger proteins could be exacerbated by molecular crowding as a consequence of high protein expression levels, the growth conditions for expression or protein concentrations when lysis is conducted in small volumes. It should also be noted that detrimental protein–protein

interactions can be independent of size, and the authors have found amylose affinity chromatography can fail even with small peptides of 4 kDa attached to MBP.

Though currently some mechanisms are only hypotheses, approaches to address these problems can result in successful purification following failure, and the overall issue has also been approached using additional accessory tags (33,34). The authors have found it is very speculative to try to consider the three-dimensional topology of the passenger moiety and MBP in stereo and so have developed a novel matrix-assisted dialysis refolding method (*see Subheading 3.6.*) that is useful for troubleshooting purifications that have failed as well as a general means for purification of recombinant MBP-passenger proteins that can refold in light of the failure of conventional methods. The matrix-assisted dialysis refolding method is essentially refolding denatured MBP-passenger protein within a dialysis cassette or membrane in the presence of the amylose resin. Refolding in the presence of the amylose ligand can allow the MBP-passenger protein to refold attached to the matrix (as the binding cleft forms around the ligand) allowing capture. We have found the contaminants in the resin from denatured debris did not carry over as significantly as imagined, and other refolding conditions will no doubt be successful.

1.2. Amylose Affinity Chromatography

As MBP is active over a wide pH and salt range, there are many choices for buffer conditions that can be used, but generally buffers around a neutral slight basic nature (7.5–8) with modest ionic strengths (100–500 mM) are best. Because MBP has an acidic isoelectric point (pI) (*see Note 1*), when concentrated it can affect the pH of the solution and so it is recommended to use an appropriate strength of the buffer (>20 mM). When deciding on the exact buffer composition, it is important to consider the overall bioprocess (including lysis conditions and downstream processes such as proteolytic tag removal and secondary chromatography) and to formulate the buffer to interface with these other processes. For example, if a Factor Xa cleavage is necessary, certain protease inhibitors (*see Note 7*) and ethylene glycol tetraacetic acid (EGTA) (*see Note 8*) are not desired in wash buffers or elution buffers or need to be thoroughly removed before eluting the protein. Protease inhibitors and metal chelating agents are compatible with all matrices used (e.g., Leupeptin, Aprotinin, Pepstatin, phenylmethylsulfonyl fluoride, ethylenediaminetetraacetic acid (EDTA) and EGTA).

It is also important to consider the disulphide context that may be required. In general, the buffers for amylose affinity chromatography can include redox, oxidizing or reducing agents to either maintain or break disulphide bonds as

necessary. If the correct disulphide context is attempted through periplasmic expression, it is important to omit reducing agents from the buffers. The standard reducing conditions use 1 mM dithiothreitol (DTT) or 10 mM β -mercaptoethanol in the equilibration, wash and elution buffer.

For amylose affinity chromatography, there are generally three scales.

1. Very small scales, where MBP is used as an affinity group for a magnetic support for a peptide or protein which acts as a secondary tag (e.g., an antigen) to purify a completely different biomolecule (e.g., an antibody). This batch mode method using magnet beads is for small-scale purifications of MBP-passenger protein for 500- μ L cell culture supernatant (*see Subheading 3.1.*).
2. Small-to-moderate scales for protein/peptide study in the laboratory using amylose agarose or amylose high flow.
3. Larger scales using FPLC apparatus.

Before undertaking purification at moderate or larger scales, a calculation experiment is advised for optimal purification (*see Subheading 3.2.*), which simply approximates the recombinant MBP-passenger protein expression level for purification. The final yield does not always correlate exactly to the calculation due to bioprocess variations forming the cleared lysate but provides a suitable approximation as a starting point. If reliable gel densitometry estimations can be undertaken, the expression level can be approximated in this manner by taking a 1 mL (or smaller) aliquot of cells when harvesting and running standard sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) protocols for in-gel protein estimation and by-passing the steps described in **Subheading 3.2.** When estimating the amount of matrix, it is advised to base volumes on 3 mg/mL binding capacity unless a 1 mL pilot experiment is conducted for further optimization (most appropriate when larger scales are attempted).

The batch and semi-batch method is principally for smaller scale purifications of MBP-passenger protein using cell culture supernatant volumes as low as 500 μ L. The method can be applied at moderate scales with very good success where an FPLC system is not available. The critical parameter limiting the scale is liquid handling associated with the matrix, especially where the MBP-passenger protein has a lower binding capacity (~ 3 mg/mL); more matrix is needed and can often result in clogging and flow restrictions at high protein loads. Flow restrictions can also occur using the agarose matrix and large liquid volumes when a semi-batch column approach is used under gravity. The flow, using columns up to 2 mL volumes can be enhanced during loading and washing steps using a vacuum manifold (e.g., a 'piglet'), especially when using the amylose high flow matrix where an FPLC system is not available and simple PD10 disposable columns (BioRad) work well using such manifolds.

2. Materials

2.1. Chemicals and Reagents

1. Inhibitor cocktail (*see Note 7*).
2. EGTA (*see Note 8*).
3. EDTA (*see Note 8*).
4. DTT.
5. Magnetic Separation Rack (New England BioLabs).
6. SDS.
7. Amylose magnetic beads (New England BioLabs).
8. Amylose agarose resin (New England BioLabs).
9. Amylose high flow resin (New England BioLabs).
10. Snakeskin Dialysis Membrane (10 kDa) (Pierce) or Slide-A-Lyzer 20 kDa Cassette (Pierce).
11. Medical scalpel.

2.2. Amylose Affinity Chromatography at Small Scale Using Magnetic Beads

1. Stationary support: Amylose magnetic beads (New England BioLabs).
2. Column preparation solution: 5% v/v methanol : ddH₂O.
3. Column buffer: 50 mM N-2-Hydroxyethylpiperazine-N'-2-ethanesulfonic acid (HEPES), 150 mM NaCl, 2 mM EDTA, 2 mM EGTA, 1 mM DTT pH 7.4 (*see Note 9*).
4. Elution buffer: 50 mM HEPES, 150 mM NaCl, 1 mM DTT, 100 mM maltose pH 7.4.

2.3. Calculation Experiment

1. Stationary support: Amylose agarose resin or high flow matrix (New England BioLabs).
2. Column preparation solution: 5% v/v methanol : ddH₂O.
3. Pre-equilibration buffer: 50 mM HEPES pH 7.4 (*see Note 9*).
4. Equilibration buffer: 50 mM HEPES, 150 mM NaCl, 2 mM EDTA, 2 mM EGTA, 1 mM DTT pH 7.4.
5. Wash buffer 1: 50 mM HEPES, 150 mM NaCl, 2 mM EDTA, 2 mM EGTA, 1 mM DTT, inhibitor cocktails pH 7.4.
6. Wash buffer 2: 50 mM HEPES, 150 mM NaCl, 1 mM DTT pH 7.4.
7. Elution buffer: 50 mM HEPES, 150 mM NaCl, 1 mM DTT, 100 mM maltose pH 7.4.
8. Regeneration 1: 50 mM HEPES, 4 M Urea, 0.5% w/v SDS pH 7.4.
9. Regeneration 2: 50 mM HEPES, 150 mM (NH₄)₂SO₄, 2 mM EDTA, 2 mM EGTA pH 7.4.
10. Regeneration 3: ddH₂O.
11. Regeneration 4: 20% v/v ethanol : ddH₂O.

2.4. Amylose Affinity Chromatography in Batch and Semi-Batch Modes Using Agarose and High Flow Matrices

1. Stationary support: Amylose agarose resin or high flow matrix (New England BioLabs).
2. Column preparation solution: 5% v/v methanol : ddH₂O.
3. Pre-equilibration buffer: 50 mM HEPES pH 7.4 (*see Note 9*).
4. Equilibration buffer: 50 mM HEPES, 150 mM NaCl, 2 mM EDTA, 2 mM EGTA, 1 mM DTT pH 7.4.
5. Wash buffer 1: 50 mM HEPES, 150 mM NaCl, 2 mM EDTA, 2 mM EGTA, 1 mM DTT, inhibitor cocktails pH 7.4.
6. Wash buffer 2: 50 mM HEPES, 150 mM NaCl, 1 mM DTT pH 7.4.
7. Elution buffer: 50 mM HEPES, 150 mM NaCl, 1 mM DTT, 50 mM maltose pH 7.4.
8. Regeneration 1: 50 mM HEPES, 4 M Urea, 0.5% w/v SDS pH 7.4.
9. Regeneration 2: 50 mM HEPES, 150 mM (NH₄)₂SO₄, 2 mM EDTA, 2 mM EGTA pH 7.4.
10. Regeneration 3: ddH₂O.
11. Regeneration 4: 20% v/v ethanol : ddH₂O.

2.5. FPLC Purification: Amylose High Flow Affinity Chromatography

1. Stationary support: Amylose high flow matrix (New England BioLabs).
2. Column preparation solution: 5% v/v methanol : ddH₂O.
3. Pre-equilibration buffer: 50 mM HEPES pH 7.4 (*see Note 9*).
4. Column buffer: 50 mM HEPES, 150 mM NaCl, 1 mM DTT pH 7.4.
5. Elution buffer: 50 mM HEPES, 150 mM NaCl, 1 mM DTT, 20 mM maltose pH 7.4.
6. Regeneration 1: 50 mM HEPES, 4 M Urea, 0.5% w/v SDS pH 7.4.
7. Regeneration 2: 50 mM HEPES, 150 mM (NH₄)₂SO₄, 2 mM EDTA, 2 mM EGTA pH 7.4.
8. Regeneration 3: ddH₂O.
9. Regeneration 4: 20% v/v ethanol : ddH₂O.

2.6. Matrix-Assisted Dialysis Refolding Methods

1. Stationary support: Amylose agarose resin or high flow matrix (New England BioLabs).
2. Dialysis membrane, 10–30 kDa, or cassette.
3. Denaturation buffer: 50 mM HEPES, 6 M Urea, 1 mM DTT, 5 mM EDTA pH 7.5 (*see Note 9*).
4. Refold 1: 50 mM HEPES, 300 mM Urea, 150 mM NaCl, 1 mM DTT, 5 mM EDTA pH 7.5.
5. Refold 2: 50 mM HEPES, 150 mM NaCl, 1 mM DTT, 5 mM EDTA pH 7.5.
6. Refold 3: 50 mM HEPES, 150 mM NaCl, 1 mM DTT pH 7.5.

3. Methods

3.1. Amylose Affinity Chromatography at Small Scales Using Magnetic Beads

1. Pre-cool 2.5 mL of the wash and elution buffer on ice for 20 min.
2. Wash 100 μ L of amylose magnetic bead suspension by adding to 400 μ L of column preparation solution (*see Note 10*) in a microfuge tube and thoroughly vortex.
3. Pull beads to the side of the tube using a magnet (30–60 s) and decant the supernatant with a pipette.
4. Repeat step 2 by adding 500 μ L of ice-cold column buffer and repeat step 3.
5. Carefully add 500 μ L of clarified bacterial cell lysate (*see Note 11*) to the magnetic beads and gently mix to a suspension with the pipette slowly (*see Note 12*).
6. Incubate the suspension for 45 min at 4°C on a suitable shaker (e.g., rocking or platform) at a low speed setting (*see Note 13*).
7. Apply the magnet and decant supernatant as in step 3. Retain the supernatant as required in a separate microfuge tube for analysis of the unbound flow-through. Repeat this washing step three times.
8. The functionalized magnetic beads can now be used in a secondary capture system employing the passenger species as required.
9. If desired, elute the MBP-passenger protein complex from the magnetic beads by adding 50 μ L of elution buffer (*see Note 14*) and resuspend gently with a pipette and incubate for 10 min on ice.
10. Resuspend gently with a pipette and apply the magnet as in step 3 retaining the supernatant containing the MBP-passenger protein.

3.2. Calculation Experiment

1. Collect a 1 mL aliquot of cells at the time of harvesting the expression culture (*see Note 15*).
2. Pre-cool equilibration, wash and elution buffers on ice for 20 min.
3. Lyse bacterial cells (*see Note 11*) and prepare a cleared lysate in 500 μ L of wash buffer 1.
4. Place supernatant in a fresh microfuge tube.
5. Wash 0.1 mL of amylose agarose or amylose high flow suspension by adding to 0.9 mL of column preparation solution in a microfuge tube and thoroughly vortex.
6. Pellet the matrix by centrifugation at $2000 \times g$ for 1 min and decant the supernatant (*see Note 16*).
7. Wash once with 1 mL of pre-equilibration buffer, thoroughly vortex and repeat step 6.
8. Resuspend in 0.5 mL of equilibration buffer and transfer to a fresh microfuge tube (*see Note 17*).

9. Thoroughly vortex in the new microfuge tube, pellet the matrix and decant the supernatant as in step 6.
10. Resuspend in 1 mL of equilibration buffer, thoroughly vortex and incubate on ice for 10 min.
11. Pellet the matrix and decant the supernatant as in step 6.
12. Carefully add 500 μ L of clarified bacterial cell lysate (*see Note 11*) to the matrix and gently mix to a suspension by slow pipetting (*see Note 12*).
13. Incubate the suspension for 45 min at 4°C on a suitable shaker (e.g., rocking or platform) at a low speed setting (*see Note 13*).
14. Pellet the matrix and decant the supernatant as in step 6. Retain the supernatant as required in a separate microfuge tube for analysis of the unbound flow-through.
15. Carefully add 0.5 mL of wash buffer 1, gently mix and transfer to a fresh microfuge tube (*see Note 17*).
16. Pellet the matrix and decant the supernatant as in step 6.
17. Wash twice in the same microfuge tube by repeating an addition of 1 mL of wash buffer 2, pelleting the matrix and decanting the supernatant as in step 6. Check the decontamination of the matrix by measuring the $A_{280\text{ nm}}$ of the second wash from step 16 and repeat washes until the $A_{280\text{ nm}}$ is stable between 0.01 and 0.001 blanking with wash buffer 2.
18. Elute the MBP-passenger protein complex from the matrix by adding 200 μ L of elution buffer and resuspend gently with a pipette and incubate for 10 min on ice.
19. Centrifuge at $4000 \times g$ for 5 min, collecting the supernatant for protein estimation and ensure a relatively low contamination by SDS-PAGE analysis. Protein can also be tested for proteolytic separation of MBP-passenger complexes from these solutions.
20. Regenerate the matrix using the steps described on **Subheading 3.5**.

3.3. Amylose Affinity Chromatography in Batch and Semi-Batch Modes Using Agarose and High Flow Matrices

1. Measure the total protein concentration of the cleared lysate from the calculation experiment (*see Subheading 3.2*).
2. Calculate liquid handling requirements (amount of matrix) based on the expression level and handling capacity (*see Note 18*).
3. Pre-cool equilibration, wash and elution buffers on ice for 20 min.
4. Lyse bacterial cells (*see Note 11*) and prepare a cleared lysate in wash buffer 1.
5. Place supernatant in a fresh vessel.
6. Wash matrix suspension by adding to 5 resin volumes (RV) of resin preparation solution and thoroughly mix.
7. Pellet the matrix by centrifugation at $1000 \times g$ for 5 min and decant the supernatant (*see Note 16*).
8. Wash once with 10 RV of pre-equilibration buffer, thoroughly mixing and repeat step 7.

9. Resuspend in 10 RV of equilibration buffer, thoroughly vortex and incubate on ice for 30 min.
10. Pellet the matrix and decant the supernatant as in step 7.
11. Carefully add the clarified (bacterial cell lysate) to the matrix and gently mix to a suspension (*see Note 12*).
12. Incubate the suspension for 1 h at 4°C on a suitable shaker (e.g., rocking or platform) at a low speed setting (*see Note 13*).
13. Pellet the matrix and decant the supernatant as in step 7. Retain the supernatant as required in a separate vessel for analysis of the unbound flow-through.
14. Carefully add 5 RV of wash buffer 2, gently mix and transfer to a fresh vessel (*see Note 17*).
15. Pellet the matrix and decant the supernatant as in step 7.
16. Continue to washing using 5 RV wash buffer 2 until the $A_{280\text{ nm}}$ is stable between 0.01 and 0.001 blanking with wash buffer 2.
17. Pellet the matrix and decant the supernatant as in step 7.
18. Resuspend the matrix in 1 RV of wash buffer 2 and either transfer to a smaller vessel for elution (proceed to step 19), or load to a column (proceed to step 21).
19. Pellet the matrix and decant the supernatant as in step 7 and elute the MBP-passenger protein complex from the matrix by adding 1 RV of elution buffer (*see Note 14*) and resuspend gently with a pipette and incubate for 10 min on ice.
20. Centrifuge at $4000 \times g$ for 5 min, collecting the supernatant containing the MBP-passenger protein. Regenerate the matrix using the steps described in **Subheading 3.5**.
21. Wash the column with 1 RV of wash buffer 2 and elute the MBP-passenger protein by adding 0.5–1 mL aliquots of elution buffer (*see Note 14*) and allowing it to enter the matrix, collecting similar sized fractions.
22. Check the $A_{280\text{ nm}}$ to identify fractions containing the desired MBP-passenger protein.
23. Regenerate the matrix using the steps described in **Subheading 3.5**.

3.4. FPLC Purification: Amylose High Flow Affinity Chromatography

1. Measure the total protein concentration of the cleared lysate from the calculation experiment (*see Subheading 3.2*).
2. Calculate the amount of matrix for purification and pour column.
3. Attach the column to the FPLC, and wash with 5 column volumes (CV) of column preparation solution under operational conditions (*see Note 2*).
4. Wash once with 5 CV of pre-equilibration buffer.
5. Wash with 10 RV of column buffer. Confirm equilibration by measuring pH and conductivity. Continue equilibration until pH and conductivity from the column matches equilibration buffer.
6. Load the bacterial cell lysate (*see Note 11*) at 2.5 mg/mL protein concentration onto the column in accord with conditions given in **Note 2**.
7. Wash with 10 CV of column buffer.

8. Collect 1 mL fractions and elute the MBP-passenger protein with 15 CV of elution buffer.
9. Regenerate the column using the steps described in **Subheading 3.5**.

3.5. Regeneration Conditions for Amylose Agarose or Amylose High Flow Matrices

1. Wash the matrix with 5 RV/CV of final wash or column buffer (*see Note 19*).
2. Wash the matrix sequentially with 5 RV/CV of regeneration 1 and regeneration 2.
3. Wash the matrix with 10 RV/CV of ddH₂O (regeneration 3).
4. Wash the matrix with 5 RV/CV of regeneration 4 and store at 4°C.

3.6. Matrix-Assisted Dialysis Refolding Methods

1. Lyse bacteria in no more than 5 mL of denaturation buffer (*see Note 11*) and form a cleared lysate.
2. Place cleared lysate in a suitable dialysis cassette or membrane (10–30 kDa cut-off).
3. Dialyze at 4°C in 1 L (refold 1) for 8 h.
4. Transfer to 1 L (refold 2) and dialyze for 8–16 h.
5. Transfer to 1 L (refold 3) and dialyze for 8 h.
6. Remove from dialysis membrane (*see Note 20*) to a suitable vessel and proceed as for batch/column method (**Subheading 3.3., steps 7–23**).

4. Notes

1. MBP (New England BioLabs pMAL-C2 construct calculated as a Factor Xa cleaved product) has a molecular weight of 42,481.9 Da, an acidic theoretical pI of 5.08, a molar extinction coefficient of 1.541 M/cm ($A_{280\text{ nm}} 0.1\% = 1\text{ g/L}$) and favourable aliphatic index of 80.78 (**35**). The authors have found the molar extinction coefficient is not an accurate means to estimate MBP fusion protein concentration in non-denatured solutions and could be related to a change in spectral fluorescence noted at longer wavelengths (a tryptophan red shift) with conformational changes upon ligand binding (**36**). The aliphatic index indicates the relative volume occupied by aliphatic side chains is quite high in the protein and is a positive factor for increased thermostability (**35,37**), which may allow a protein to more easily refold by allowing the protein to undergo a rapid and stable hydrophobic collapse to conformations close to the native state (**38**). The thermostability and refolding ability of MBP has been noted in the literature and is maximal at pH 6 (T_m of 64.9°C, ΔH_m of 259.7 kcal/mol) (**19,20,39**). MBP is stable between pH 4 and 10.5 (25°C) and undergoes a reversible, two-state refolding mechanism at neutral pH in the presence of temperature variation and chemical denaturants (**22,39**). The association constant (K_a) for a range of maltodextrins to MBP is between approximately 2 and 4×10^{-7} M/s, and so

differences in equilibrium constants are reflected in different dissociation rates ($K_d \sim 3.5 \mu\text{M}$, maltose; $\sim 0.16 \mu\text{M}$, maltotriose) (36).

2. Three amylose affinity chromatography matrices are manufactured by New England BioLabs, being functionalized onto magnetic beads, agarose and a high flowing support matrix, though a custom matrix can be manufactured (40). Amylose magnetic beads have a binding capacity of up to $10 \mu\text{g}/\text{mg}$ (supplied as a $10 \text{ mg}/\text{mL}$ suspension). Amylose agarose has a binding capacity of $3 \text{ mg}/\text{mL}$ for MBP and $6 \text{ mg}/\text{mL}$ for an MBP- β -galactosidase protein. The typical flow velocity of the amylose resin is $1 \text{ mL}/\text{min}$ in a $2.5 \text{ cm} \times 10 \text{ cm}$ column, and the matrix can withstand small manifold vacuums (e.g., a “piglet”). The amylose matrix can suffer from flow restrictions, and so total protein loading should be $\leq 2.5 \text{ mg}/\text{mL}$. Amylose high flow has a binding capacity of approximately $7 \text{ mg}/\text{mL}$ for an MBP-paramyosin protein. The exact chemical nature of the matrix is not described but has a pressure limit of 0.5 MPa (75 psi), a maximum flow velocity of $300 \text{ cm}/\text{h}$, and recommended velocities are below $60 \text{ cm}/\text{h}$ being $10\text{--}25 \text{ mL}/\text{min}$ (for 1.6-cm and $\varnothing 2.5\text{-cm}$ columns respectively).
3. New England BioLabs provide simple lysis conditions to access MBP-passenger proteins located in the periplasm (2). The method involves lysis using sucrose, EDTA and MgSO_4 and low speed centrifugation. This is an effective means of purification as the periplasmic MBP-mediated transport system is susceptible to mild osmotic shock, causing the loss of transport activity and recovery of periplasmic-binding proteins in the osmotic medium (18).
4. When cloned using Eco RI, early systems would not be cleaved by Factor Xa and some constructs produced Factor Xa sites that were inefficiently cleaved – likely due to structural complications induced about the cleavage site. In general, the Factor Xa bioprocessing is also unfavoured by many users owing to the promiscuity of Factor Xa; it is well documented that Factor Xa cleaves non-canonical sites of the desired recombinant passenger protein in regions that contain arginine at the P1 site, possibly where regions are in proteolytically preferred extended conformations (41). Methods to reversibly acylate such sites have been described (42–44), but in the hands of these authors such methods are ineffective. Amylose was originally functionalized onto agarose and had earlier been reported to have a binding capacity of $>3 \text{ mg}/\text{mL}$, which has been revised (see Note 2). This seemed to be relatively low compared to other affinity purification systems and had a tendency to encounter viscosity problems at concentrated protein loadings, causing the columns to suffer flow restrictions and creating a need to work with dilute loadings (thereby imposing liquid handling and chromatographic scale limitations). Generally, the range of improved constructs, diversity in protease sites and development of the amylose high flow matrix overcome all these issues when a bioprocess is properly planned with the physico-chemical properties of the passenger protein or peptide carefully considered.
5. Where a detergent is required for the passenger protein to remain soluble, it is advisable to utilize the additive in buffers following amylose affinity chromatography or in the elution buffer. Where this is not possible, as a general rule, the

binding efficiency is reduced using 10% glycerol but appears to be tolerated, the binding efficiency is significantly reduced below 5% in the presence of 0.25% Triton X-100 or Tween 20, and is completely precluded in the presence of 0.1% SDS. It is advised not to use a detergent as an additive to assist lysis as varied results occur, however if used, the binding efficiency is often suitably restored by dilution prior to binding to the amylose matrix (2) (~0.05% Tween 20 and ~1:10 dilution for B-Per retains ~80% binding efficiency). The authors have found there can be batch-to-batch inconsistencies using B-Per extraction reagent that could be related to detergent effects dependent on MBP-passenger protein and total protein concentrations. We have specifically noted proteolysis inefficiencies following detergent extractions.

6. Under normal conditions defined as 15 mL of amylose agarose matrix processing, 1 L of Lauria Bertani media supplemented with 0.2% glucose (producing ~40 mg MBP fusion protein); the deterioration of the matrix is reported to be approximately 1–3% of the initial binding capacity each time it is used. It is stated that such a column may be used up to 5 times before a decrease in yield is detectable (5–15% lost binding capacity), and up to 10 times before the loss is significantly noticeable (10–30% lost binding capacity). However with different media producing heavier cell densities but a lower MBP fusion protein yield, the loss of amylose binding capacity will be more dramatic.
7. The inhibitor cocktail is a solution containing protease inhibitors to reduce the degradation of the recombinant protein due to the activity of proteases released from the bacterial cell upon lysis. They generally consist of broad specificity inhibitors of serine, cysteine, aspartic and aminopeptidases, with the activity of EDTA and EGTA influencing metalloenzymes and proteases (*see Note 8*). Inhibitor cocktails can be purchased from most chemical supply companies or made in-house using a combination of non-specific and/or specific protease inhibitors. The ExPASy peptide cutter tool (<http://au.expasy.org/tools/peptidecutter/>) (35) can be used to predict potential proteolysis issues or specific protease classes which may be an issue to a given MBP-passenger protein amino acid sequence. Using the peptide cutter tool, a particular set of potential proteolysis issues can be identified and addressed using protease inhibitors. In general, inhibitor cocktail comprise phenylmethylsulfonyl fluoride (1 mM), aprotinin (1 µg/mL), leupeptin (1 µg/mL) and pepstatin A (1 µg/mL) in the buffer. Specific care should be taken with washing steps following protease solutions if proteolysis is to follow purification.
8. EDTA and EGTA chelate metal ions that are important to metalloproteases and metalloenzymes. EDTA specifically chelates divalent and trivalent metal ions such as Mn(II), Cu(II), Fe(III) and Co(III). EGTA has a higher affinity for Ca(II) compared to EDTA, and calcium ions may be particularly relevant to MBP purification as a co-factor for some formulations of Factor Xa used to separate MBP from the passenger protein, but also as a known cofactor for potential contaminants of the maltose regulon (18).

9. When preparing all buffers add ingredients making the buffer to 90% of final volume and titrate the pH using HCl to the desired concentration, making up to the final volume. With urea-containing buffers, dilute dry ingredients to 50% of final volume and fully dissolve the solids. Urea dissolves in an endothermic reaction (turning the solution cold), therefore, allow the buffer solution to return to room temperature once fully dissolved before making to 90% of the final volume and adjusting the pH with HCl.
10. The different amylose matrices are supplied by New England BioLabs in a 20% ethanol solution that can negatively influence purification and requires removal. The amylose magnetic beads are supplied with 0.05% Tween-20 that can be significant at very small protein concentrations. In related applications, the authors have found that low levels of residual detergents (especially from regeneration solutions) can still remain and have found SDS and detergent mixed micelles particularly difficult to remove. The authors have analyzed removal of detergent and mixed micelles using surface plasmon resonance (BIAcore T100, BIAcore) and dual polarization interferometry (AnaLight 200, Farfield Instruments), finding dilute methanol-containing solutions are most efficient for removal of these agents.
11. The manner of lysis is dependent on available equipment, scale, bacterial strain (which may encode a lysozyme in the case of pLysS strains (45)) and whether periplasmic (*see Note 3*) or cytoplasmic expression is undertaken. For cytoplasmic expression, mechanical lysis is more effective for successful MBP purification than chemical lysis as chemical lysis employs agents that are frequently incompatible with MBP chemistry (*see Note 5*). Large scales may require a cell disruptor such as a French press to successfully achieve lysis and suitable viscosity, whereas moderate and small scales may employ sonication. Small-scale lysis can also be achieved using standard freeze-thaw cycling techniques but may have elevated viscosities due to intact nucleic acid being present. The standard method of the authors employs sonication on ice using a Branson B30 Sonifier at 70% duty cycle to 20 kHz (~5.5 output but varies with turbidity), for 3 min with 3 × 30-s bursts with rests between cycles. Typically, the authors form a cleared lysate by centrifugation at $45,700 \times g$ for 45 min at 4°C.
12. It is important to avoid vigorous mixing during all liquid handling steps as this causes loss of product by denaturation (foaming) of protein solutions. For this reason, liquid handling and mixing is conducted gently.
13. If suitable mixers or cold rooms are unavailable, place the suspension on ice and invert gently every 5 min. The binding reaction is enhanced by collisions between amylose and MBP species as opposed to passive diffusion and therefore gentle agitation of the suspension maximizes the capture to the amylose matrix.
14. Higher concentrations of maltose (50–100 mM) can influence storage of eluted protein samples as a cryoprotectant, and so, eluted samples sometimes require dilution below 20 mM for proper freezing.

15. Cells can be pelleted from the aliquot by centrifugation ($6000 \times g$, 15 min, 4°C). Ensure the supernatant is decanted completely. It is optional to store the cell pellet and process at a later date. Cell pellets should not be stored at 4°C , but may be stored for several weeks at -20°C before proceeding. Longer term storage should be at -80°C or under liquid nitrogen. If stored frozen, thaw the pellet in ice water when ready to proceed.
16. New England BioLabs state amylose resin can withstand centrifugation at up to $6000 \times g$ (2).
17. Transferring to a new vessel during washing steps decreases non-specific contaminants that adhere to vessel walls or remain in vessels and carry over to subsequent steps.
18. For amylose agarose, the total protein concentration should be ≤ 2.5 mg/mL for best binding to the amylose matrix with reduced viscosity (which causes severe flow restrictions). The dilution to 2.5 mg/mL restricts the batch mode by what volume can be handled in terms of the size of centrifuge tubes that can be used and column size. For batch and semi-batch modes, no more than 5 mL matrix is recommended.
19. The resin may be reused up to 10 times (2), but caution should be made if regenerating at 4°C as SDS can precipitate over time.
20. When using a dialysis cassette, it is necessary to cut away the membrane window using a scalpel to properly extract the amylose matrix to avoid foaming.

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Methods for Detection of Protein–Protein and Protein–DNA Interactions Using HaloTag™

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Summary

HaloTag™ is a protein fusion tag which was genetically engineered to covalently bind a series of specific synthetic ligands. All ligands carry two groups, the reactive group and the functional/reporter group. The reactive group, the chloroalkane, is the same in all the ligands and is involved in binding to the HaloTag™. The functional reporter group is variable and can carry many different moieties including fluorescent dyes, affinity handles like biotin or solid surfaces such as agarose beads. Thus, HaloTag™ can serve either as a labeling tag or as a protein immobilization tag depending on which ligand is bound to it. Here, we describe a procedure for immobilization of HaloTag™ fusion proteins and how immobilized proteins can be used to study protein–protein and protein–DNA interactions in vivo and in vitro.

Key Words: HaloTag™; immobilization; covalent; protein–protein interactions; protein–DNA interactions; in vivo; in vitro.

1. Introduction

One of the major limitations to understanding biological processes is our lack of knowledge of protein function and how they assemble into complex protein networks. In recent years, we have witnessed development of several powerful protein analysis technologies. Two of them in particular have profoundly effected how proteins are studied in vivo and in vitro: autofluorescent proteins

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and affinity purification tags. Autofluorescent proteins revolutionized the way protein function is studied in living cells (1–3). They are useful not only for protein localization studies but also for study of dynamic processes, conformational changes and protein–protein interactions. Similarly, affinity fusion tags transformed in vitro analysis of proteins. Affinity tags provide a selective, easy and efficient tool for protein isolation and immobilization (4–7). The number of new affinity tags and applications for their use continues to grow (8). However, there are limitations to both technologies. With autofluorescent proteins, we are limited with respect to fluorophores, and in addition, these proteins do not provide us with an easy option to isolate and immobilize proteins for in vitro studies. The use of an additional tag, for example, His tag, is required for protein immobilization when using fluorescent proteins. On the other hand, affinity tags provide a very efficient method for in vitro protein studies, but they do not enable specific labeling and imaging of proteins in live cells.

Our goal was to develop a new technology that will combine advantages of both of these technologies and overcome some of the limitations. Based on these criteria, we have developed the HaloTag™ technology that enables specific labeling, imaging and immobilization of proteins in vivo and in vitro. The technology is based on a new protein fusion tag, called HaloTag™, and a series of synthetic HaloTag™ ligands which specifically and covalently bind the HaloTag™ protein.

HaloTag™ is a monomeric protein of 33 kDa and can be genetically fused to the protein of interest either at the C or N terminus using a HaloTag™ expression vector. The HaloTag™ protein was derived from a hydrolase found in *Rhodococcus rhodochrous*, and therefore, it is not present in mammalian systems, insect cells, yeast and even *Escherichia coli*. Thus, HaloTag™ technology does not suffer from the interference of an endogenous protein or ligand, which enhances the specificity of this system. The first and most important modification of the wild-type enzyme was introduction of a mutation that leads to preservation of the covalent bond and a permanent association of the protein with the substrate. We used the natural substrate to develop a series of chemically modified HaloTag™ ligands (see Fig. 1).

In addition to the critical modification in the active site which leads to covalent binding of the ligand, other mutations were introduced into the binding pocket. These mutations dramatically increase the rate of binding between HaloTag™ protein and the HaloTag™ ligands. Fluorescence polarization analysis using fluorescent HaloTag™ ligand and purified GST-HaloTag™ fusion protein shows that the binding kinetics of the ligand to HaloTag™ protein is very rapid with an on-rate similar to that measured for the biotin–streptavidin interactions.

As mentioned above, the HaloTagTM system consists of chemically modified HaloTagTM ligands which bestow different functionalities onto HaloTagTM fusion proteins upon binding. To achieve efficient and specific binding with several different ligands, we designed the ligands so that they consist of two elements: the constant reactive group and a variable functional reporter group. The reactive group consists of the chloroalkane, which is the natural substrate for HaloTagTM protein. This part of the ligand is the same in all the ligands and is involved in the covalent and specific binding to the HaloTagTM polypeptide. The remaining part of the ligand, the functional group, encompasses many different entities including different fluorescent dyes, affinity handles (e.g., biotin) or the solid support (e.g. resin). Thus, binding of different HaloTagTM ligands to HaloTagTM fusion protein imparts different functionalities onto the fusion protein that allow imaging and/or immobilization. Consequently, one genetic construct can be used in various in vitro and in vivo (cell-based) assays (*see Fig. 1*).

Immobilization of proteins onto solid support surfaces is becoming increasingly important in characterization of protein function and protein interactions (9). We have developed a surface for immobilization of HaloTagTM fusion proteins, a nonmagnetic resin (HaloLinkTM), which enables covalent and oriented surface immobilization. HaloLinkTM resin consists of agarose

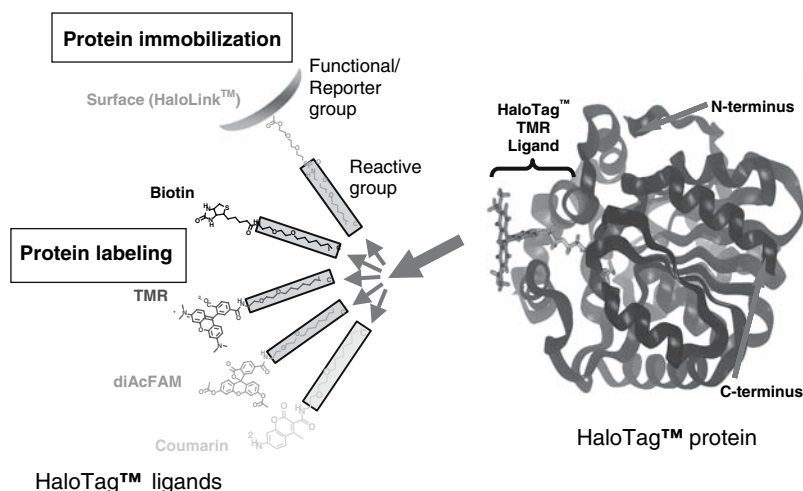


Fig. 1. Variable functionalities of the HaloTagTM Technology. The HaloTagTM Technology comprises the HaloTagTM protein and a system of interchangeable synthetic ligands that specifically and covalently bind to the HaloTagTM protein. These ligands bind to HaloTagTM impart multiple functions to a HaloTagTM fusion protein including imaging and immobilization. Thus, one genetic construct can be used in various in vitro and in vivo assays.

beads with HaloTag™ ligand covalently coupled to the surface. The resin shows very low nonspecific protein binding but high specific binding of HaloTag™ fusion proteins resulting in high binding capacity (7 mg of protein/ml resin). HaloLink™ resin can be used in a variety of applications including immobilization of enzymes, protein–protein interaction studies and analysis of protein–DNA interactions. Furthermore, purification of the fusion protein from the HaloLink™ resin can be achieved using protease cleavage (see **Fig. 2** and **Subheading 3.6.**).

The advantages of HaloLink™ technology over other methods used for immobilization are several. First, the covalent linkage between the HaloTag™ protein and HaloLink™ resin allows extensive washing to remove nonspecifically bound proteins without the danger of eluting the HaloTag™ fusion

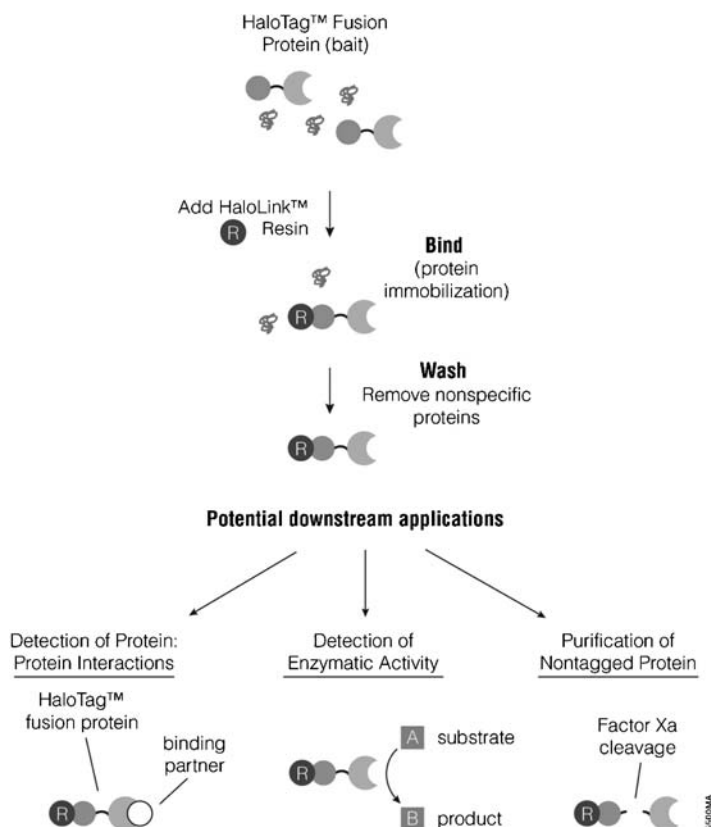


Fig. 2. Overview of HaloLink Resin immobilization protocol and potential downstream applications such as detection of protein–protein and protein–DNA interactions, detection of enzymatic activity and purification of nontagged proteins.

protein. Second, the rapid binding, high binding capacity and low nonspecific binding characteristics of HaloLink™ resin yield highly reproducible and reliable reagent with low background signal. Furthermore, rapid binding enables efficient immobilization of proteins at very low concentration without the need for long incubation times. Third, HaloTag™ binds directly onto HaloLink™ resin that eliminates the need for antibodies to precipitate protein complexes. This is especially important for isolation of protein–DNA complexes. Traditionally, protein–DNA complexes are isolated employing the chromatin immunoprecipitation method (10,11). This method requires use of specific antibodies which is the major obstacle due to lack of specific antibodies which efficiently recognize crosslinked protein–DNA complexes. With the HaloTag™ technology, formaldehyde crosslinked HaloTag™ protein–DNA complexes can be isolated directly from cells using HaloLink™ resin, therefore eliminating the need to use an antibody. In addition, the covalent nature of HaloTag™ binding to the HaloLink™ resin allows very stringent washing and removal of nonspecifically bound DNA and proteins, resulting in an increased signal-to-noise ratio, allowing for detection of small changes in protein–DNA interactions within the genome.

2. Materials

2.1. General Protocol for Immobilization of HaloTag™ Fusion Proteins onto the HaloLink™ Resin

1. HaloLink™ resin (cat. no. G1911 or G1912, Promega).
2. TNT® quick coupled transcription/translation system (cat. no. L1170, Promega).
3. Binding buffer: 100 mM Tris–HCl pH 7.6, 150 mM NaCl, 0.05% IGEPAL-CA630 (Sigma). Warning: Solutions containing IGEPAL-CA630 should be prepared fresh (*see Note 1*).
4. Wash buffer: 100 mM Tris–HCl pH 7.6, 150 mM NaCl, 1 mg/ml bovine serum albumin (BSA), 0.05% Igepal-CA630 (Sigma) (*see Notes 1 and 2*).

2.2. Detection of Protein–Protein Interactions by Pre-Binding of HaloTag™ Fusion Protein (Bait) to HaloLink™ Resin

1. HaloLink™ Resin (cat. no. G1911 or G1912, Promega).
2. TNT® quick coupled transcription/translation system (cat. no. L1170, Promega). [³⁵S] methionine 2 µl (1000 Ci/mmol at 10 mCi/ml) or FluoroTect Green in vitro Translation Labeling System (cat. no. L5001, Promega).
3. Binding buffer: Same as **Subheading 2.1., step 3**.
4. Wash buffer: Same as **Subheading 2.1., step 4**.
5. Elution buffer (4×): 0.24 M Tris–HCl (pH 6.8), 3 mM bromophenol blue, 50.4% glycerol, 0.4 M dithiothreitol, 8% sodium dodecyl sulfate (SDS).

2.3. Detection of Protein–Protein Interactions by Isolation of Pre-Formed Bait–Prey Complexes

Follow the steps as in **Subheading 2.2.**

2.4. Detection of Protein–Protein Interactions In Vivo

1. HaloLink™ Resin (cat. no. G1911 or G1912, Promega).
2. Binding buffer: Same as **Subheading 2.1., step 3**, except the concentration of IGEPAL-CA630 is reduced to 0.001%.
3. Wash buffer: Same as **Subheading 2.1., step 4**, except the concentration of BSA is reduced to 0.5%.
4. Elution buffer: Same as **Subheading 2.2., step 5**.

2.5. Detection of Protein–DNA Interactions

1. HaloLink™ resin (cat. no. G1911 or G1912, Promega).
2. HaloLink™ Equilibration Buffer: 1× Tris-EDTA buffer (TE) pH 7 (10 mM Tris–HCl pH 7.0, 1 mM EDTA) 0.05% IGEPAL or 0.5% Triton X-100.
3. Tris buffered saline (TBS) buffer, 1×: 100 mM Tris–HCl pH 7.6, 150 mM NaCl.
4. Phosphate-buffered saline (PBS) buffer, Dulbecco's PBS, 1× (cat. no. 14190, Invitrogen).
5. Lysis buffer: 50 mM Tris–HCl pH 7.5, 150 mM NaCl, 5 mM EDTA, 1% Triton X-100, 0.1% sodium deoxycholate (NaDOC).
6. High salt lysis buffer: 50 mM Tris–HCl pH 7.5, 700 mM NaCl, 5 mM EDTA, 1%, Triton X-100, 0.1% NaDOC.
7. Reversal buffer: 1× TE pH 7, 300 mM NaCl.

2.6. Enzyme Immobilization and Analysis of Enzymatic Activity on the Surface

1. HaloLink™ Resin (cat. no. G1911 or G1912, Promega).
2. Binding buffer: Same as **Subheading 2.1., step 3**.
3. Wash buffer: Same as **Subheading 2.1., step 4**.

2.7. One-Step Purification of Fusion Proteins

1. HaloLink™ Resin (cat. no. G1911 or G1912, Promega).
2. Binding buffer: Same as **Subheading 2.1., step 3**.
3. Wash buffer: Same as **Subheading 2.1., step 4**.

2.8. Cloning Vectors

The HaloTag™-containing Flexi® Vectors are available for the cloning of desired proteins. The protein of interest can be fused to HaloTag™ using Flexi®

Vectors designed for expression in mammalian cells or in the in vitro protein expression systems. Flexi® Vectors provide a rapid, highly reliable system for cloning and transfer of coding regions between vectors containing various tags and expression options.

3. Methods

This section provides guidelines on how to immobilize HaloTag™ fusion proteins onto HaloLink™ resin (*see Fig. 2*). Immobilized proteins can then be evaluated for in vitro protein–protein interactions (*see Subheadings 3.2.1. and 3.2.2.*), in vivo protein–protein interactions (*see Subheading 3.3.*), protein–DNA interactions (*see Subheading 3.4.*), enzymatic activity (*see Subheading 3.5.*) and for isolation of protein fused to HaloTag by proteolytic cleavage of the fusion protein bound to the resin (*see Subheading 3.6.*) (*see Fig. 2*).

3.1. General Protocol for Immobilization of HaloTag™ Fusion Proteins onto the HaloLink™ Resin

The protocol below is optimized for binding of proteins expressed in the in vitro expression systems (*see Fig. 2*). We used TNT® T7 Quick Coupled Transcription/Translation System (cat. no. L1170, Promega). Other in vitro expression systems can be used. These reactions are typically 50 µl, which may be sufficient for more than one immobilization reaction. This protocol can also be used for immobilization of proteins expressed in vivo in mammalian cells. If mammalian expression systems are used optimize amounts of resin and cells, follow the steps described in **Subheading 3.3** through phase 3 washing as a guideline. Different lysis conditions can be used, see also **Subheading 3.4**, step 10.

3.1.1. Phase 1

Synthesis of the HaloTag™ fusion protein in vitro using TNT® T7 Quick Coupled Transcription/Translation system following manufacturer protocol: During the incubation of the TNT® T7 Quick Coupled Transcription/Translation reaction equilibrate HaloLink™ resin (*see Subheading 3.1.2., steps 1–7*). Keep resin resuspended in the binding buffer until TNT® T7 Quick Coupled Transcription/Translation reaction is completed (if needed resin can be kept in this buffer overnight at 4°C).

3.1.2. Phase 2: Resin Equilibration

Mix resin by inverting the tube several times to obtain uniform suspension.

1. Dispense 50 µl of HaloLink™ resin into 1.5-ml Eppendorf tube and spin in centrifuge for 1 min at $800 \times g$ (*see Note 3*).

2. Carefully remove and discard the supernatant without disturbing the resin at the bottom of the tube.
3. Add 400 μ l of binding buffer, mix thoroughly by inverting the tube.
4. Centrifuge for 2 min at $800 \times g$ at room temperature.
5. Carefully remove and discard the supernatant without disturbing the resin at the bottom of the tube.
6. Repeat steps 3–5 two more times for a total of three washes.
7. After last wash, resuspend the resin in 50–100 μ l of binding buffer (*see Note 4*).

3.1.3. Phase 3: Binding the HaloTag™ Fusion Protein

1. To the equilibrated resin, add 20 μ l (or more if protein expression is low) of the in vitro Transcription/Translation reaction containing the HaloTag™ fusion protein (*see Note 5*).
2. Incubate by mixing on a tube rotator (*see Note 6*) for 30–60 min at room temperature (incubate at 4°C if proteins are unstable, longer incubation time may be required). Make sure resin does not settle to the bottom of the tube as that will reduce efficiency of binding.
3. Centrifuge for 2 min at $800 \times g$. Save supernatant for analysis if desired.

3.1.4. Phase 4: Washing

1. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
2. Centrifuge for 2 min at $800 \times g$. Discard the wash.
3. Repeat steps 1 and 2 two more times.
4. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
5. Incubate for 5 min with occasional mixing.
6. Centrifuge for 2 min at $800 \times g$. Discard the wash.
7. Repeat steps 4–6 one more time.
8. Resuspend resin carrying covalently attached HaloTag™ fusion protein in desired volume of buffer compatible with downstream applications, for example, detection of protein interactions (*see Subheadings 3.2.1, 3.2.2, 3.3. and 3.4.*), analysis of enzymatic activity (*see Subheading 3.5.*) of the fusion protein or cleavage of the fusion protein from the resin (*see Subheading 3.6.*).

3.2. Detection of Protein–Protein Interactions In Vitro Using Pull-Down Assay

There are two general approaches to study protein–protein interactions in vitro using “pull-down” method. In the first approach (described in **Subheading 3.2.1.**), a mixture of proteins containing the HaloTag™ fusion proteins (from here on referred to as bait) is added to the resin, and the bait is allowed

to bind to the resin during an incubation step. This step is also known as “pre-charging” of the resin. To this resin carrying the bait, a new protein mixture containing the binding partner (prey) is added. The bait–prey complexes are formed and then isolated from the protein mixture by resin precipitation (pull-down). During this procedure, several washes are performed to remove nonspecifically bound proteins. At the end, the prey protein is eluted and analyzed on SDS–polyacrylamide gel electrophoresis (PAGE) gel or by mass spectrometry.

In the second approach (*see Subheading 3.2.2.*), the bait and prey proteins are first mixed and allowed to form complexes. Resin is added to the pre-formed bait–prey complexes. Complexes bind to the resin during incubation and are then isolated from the rest of the proteins by resin precipitation (spinning). This approach may be closer to physiological conditions, but may be more challenging because the concentration of the complexes may be rather low. In the case of pre-charging of the resin, the local protein concentration (concentration of the bait on the resin) is increased which increases the likelihood of successful isolation of the prey.

It should be mentioned that in all the procedures described below (*see Subheadings 3.2. and 3.4.*), it is important to perform control reactions. Control reactions should contain all the components of the experimental sample, except for the bait protein, for example, control would consist of the resin, mixed with TNT® extract containing the prey. All the methods describe use of the control in detail.

3.2.1. Detection of Protein–Protein Interactions by Pre-Binding of HaloTag™ Fusion Protein (Bait) to HaloLink™ Resin

In the protocol below, we describe a “pull-down” method (*12*) for detection of protein–protein interactions in which the bait protein is first immobilized onto HaloLink™ resin. A protein mixture containing the binding partner (prey) is added to the immobilized bait and is allowed to bind. Bait–prey complexes are then isolated and prey protein is identified.

3.2.1.1. PHASE 1

Synthesis of the HaloTag™ fusion protein (bait) in vitro using TNT® T7 Quick Coupled Transcription/Translation system following manufacturer protocol: During the incubation of the TNT® T7 Quick Coupled Transcription/Translation reaction, equilibrate HaloLink™ resin (*see Subheading 3.2.1.2., steps 1–7*). Keep resin resuspended in the binding buffer until TNT® T7 Quick Coupled Transcription/Translation reaction is completed (if needed resin can be kept in this buffer overnight at 4°C) (*see Note 7*).

3.2.1.2. PHASE 2

Immobilization of HaloTagTM fusion protein onto HaloLinkTM resin: For each experimental sample, a negative control sample containing resin but no bait should be included. This control allows to separate the signal from the specific protein–protein interaction from the nonspecific background binding of prey to the resin.

3.2.1.3. RESIN EQUILIBRATION

Mix resin by inverting the tube several times to obtain uniform suspension.

1. Dispense 50 μ l of HaloLinkTM resin into two 1.5-ml Eppendorf tubes (experimental and control) and spin in centrifuge for 1 min at $800 \times g$ (*see Note 3*).
2. Carefully remove and discard the supernatant without disturbing the resin at the bottom of the tube.
3. Add 400 μ l of resin equilibration buffer, mix thoroughly by inverting the tube.
4. Centrifuge for 2 min at $800 \times g$ at room temperature.
5. Carefully remove and discard the supernatant without disturbing the resin at the bottom of the tube.
6. Repeat steps 3–5 two more times for a total of three washes.
7. After last wash, resuspend the resin in 50–100 μ l of binding buffer (*see Note 4*).

Add co-factors, detergents or other reagents needed for specific protein–protein interactions.

3.2.1.4. BINDING THE BAIT

1. To the experimental resin sample, add 20 μ l (or more if protein expression is low) of cell lysate containing the HaloTagTM fusion protein.
2. To the negative control sample (resin without the bait), add 20 μ l buffer or TNT® T7 Quick Coupled Transcription/Translation mix without the DNA template.
3. Incubate by mixing on a tube rotator (*see Note 6*) for 30–60 min at room temperature (incubate at 4°C if proteins are unstable, longer incubation time may be required). Make sure resin does not settle to the bottom of the tube as that will reduce efficiency of binding. During this incubation, you can set up TNT® T7 Quick Coupled Transcription/Translation for the prey, *see Subheading 3.2.1.6*.
4. Centrifuge for 2 min at $800 \times g$.

3.2.1.5. WASHING

1. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
2. Centrifuge for 2 min at $800 \times g$. Discard the wash.
3. Repeat steps 1 and 2 two more times.
4. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
5. Incubate 5 min with occasional mixing.

6. Centrifuge for 2 min at $800 \times g$. Discard the wash.
7. Repeat steps 4–6 one more time.
8. Resuspend resin in 100 μ l of wash buffer containing 1 mg/ml BSA (keep the resin with immobilized bait at 4°C until prey synthesis is finished).

3.2.1.6. PHASE 3

Synthesis of the prey in vitro using TNT® T7 Quick Coupled Transcription/Translation System following manufacturer protocol: Label the prey protein by adding [35 S] methionine (2 μ l) (1000 Ci/mmol at 10 mCi/ml) or FluoroTect Green in vitro Translation Labeling System (cat. no. L5001, Promega) into the in vitro TNT® T7 Quick Coupled Transcription/Translation reaction; follow instructions given by manufacturer (*see* **Notes 7** and **8**).

3.2.1.7. PHASE 4: CAPTURE AND ANALYSIS OF PREY PROTEIN CAPTURE

1. Add 20 μ l of the TNT® T7 Quick Coupled Transcription/Translation reaction from phase 3 to the resin carrying HaloTag™ fusion protein and to the negative control resin (no bait) prepared in phase 2.
2. Incubate by mixing on a tube rotator (*see* **Notes 6** and **9**) for 1 h at room temperature. Make sure resin does not settle to the bottom of the tube as that will reduce efficiency of binding.
3. Centrifuge for 3 min at $800 \times g$. Discard the supernatant.

3.2.1.8. WASHING

Stability of different protein–protein interactions is protein pair specific and depends on the affinity of interaction. If interaction is not very stable, the washing conditions used for these protein pairs may have to be optimized, for example, change the number and volume of washes. However, insufficient washing may result in detection of nonspecific interactions.

1. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
2. Centrifuge for 2 min at $800 \times g$. Discard the wash.
3. Repeat steps 1 and 2 two more times.
4. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
5. Incubate for 5 min with occasional mixing.
6. Centrifuge for 2 min at $800 \times g$. Discard the wash.
7. Repeat steps 4–6 one more time.

3.2.1.9. ELUTION

1. Add 20 μ l of 1 $\times$ SDS loading buffer (for composition *see* **Subheading 2**; 0.5 or 0.25 $\times$ elution buffer can also be used).
2. Incubate 2–5 min at 90°C (*see* **Note 10**).
3. Remove supernatant and load on a SDS–PAGE gel for analysis.

3.2.2. Detection of Protein–Protein Interactions by Isolation of Pre-Formed Bait–Prey Complexes (See **Subheading 3.2.**)

3.2.2.1. PHASE 1

1. Synthesis of the bait (HaloTag™ fusion protein) in vitro using TNT® T7 Quick Coupled Transcription/Translation system: Follow instructions given by manufacturer (see **Note 7**).
2. Synthesis of the prey in vitro using TNT® T7 Quick Coupled Transcription/Translation system following manufacturer protocol: Label the prey protein by adding [³⁵S] methionine (2 µl) (1000 Ci/mmol at 10 mCi/ml) or FluoroTect green in vitro translation labeling system (cat. no. L5001, Promega) into the in vitro TNT® T7 Quick Coupled Transcription/Translation reaction. Use instructions given by manufacturer.

3.2.2.2. PHASE 2—BAIT: PREY BINDING

For each experimental sample, a negative control sample containing resin but no bait should be included. This control allows to separate the signal from the specific protein–protein interaction from the nonspecific background binding of prey to the resin.

1. For the experimental sample, combine 20 µl of bait with 20 µl of prey of the TNT® T7 Quick Coupled Transcription/Translation reactions prepared in Phase 1.
2. For the negative control sample, combine 20 µl of prey with 20 µl TNT® Quick Master Mix or buffer.
3. Mix and incubate at room temperature for 1 h (see **Note 9**).

Add co-factors, detergents or other reagents needed for specific protein: protein interactions. During the incubation of bait and prey, equilibrate HaloLink™ resin (see **Subheading 3.2.2.3.**).

3.2.2.3. PHASE 3 ISOLATION OF THE BAIT–PROTEIN COMPLEXES

3.2.2.3.1. Resin Equilibration

For each experimental bait–prey complex sample, also set up a negative control sample (resin only, no bait). Mix resin by inverting to obtain uniform suspension.

1. Dispense 50 µl of HaloLink™ resin into two 1.5-ml Eppendorf tubes (experimental and control) and spin in centrifuge for 1 min at 800 × g (see **Note 3**).
2. Carefully remove and discard the supernatant without disturbing the resin at the bottom of the tube.
3. Add 400 µl of binding buffer, mix thoroughly by inverting the tube.
4. Centrifuge for 2 min at 800 × g at room temperature.
5. Carefully remove and discard the supernatant without disturbing the resin at the bottom of the tube.

6. Repeat steps 3–5 two more times for a total of three washes.
7. After last wash, resuspend the resin in 50–100 μl of binding buffer (*see* **Note 4**).

Add co-factors, detergents or other reagents needed for specific protein–protein interactions.

3.2.2.3.2. Bait–Prey Complex Capture and Analysis

1. To the resin samples, add 20 μl of the appropriate mix (experimental bait–prey or control) set up above (*see* **Subheading 3.2.2.2**).
2. Incubate by mixing on a tube rotator (*see* **Notes 6 and 9**) for 1–2 h at room temperature. Make sure resin does not settle to the bottom of the tube as that will reduce efficiency of binding.
3. Centrifuge for 2 min at $800 \times g$ and discard the supernatant.

3.2.2.3.3. Washing

Stability of different protein–protein interactions is protein pair specific and depends on the affinity of interaction. If interaction is not very stable, the washing conditions used for these protein pairs may have to be optimized, for example, change the number and volume of washes. However, insufficient washing may result in detection of nonspecific interactions.

1. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
2. Centrifuge for 2 min at $800 \times g$. Discard the wash.
3. Repeat steps 1 and 2 two more times.
4. Add 1 ml of wash buffer containing 1 mg/ml BSA and mix thoroughly by inverting the tube.
5. Incubate 5 min with occasional mixing.
6. Centrifuge for 2 min at $800 \times g$. Discard the wash.
7. Repeat steps 4–6 one more time.

3.2.2.3.4. Elution

1. Add 20 μl of $1\times$ SDS loading buffer (for composition *see* **Subheading 2**; 0.5 or $0.25\times$ elution buffer can also be used).
2. Incubate 2–5 min at 90°C (*see* **Note 10**).
3. Remove supernatant and load on a SDS–PAGE gel for analysis.

3.3. Detection of Protein–Protein Interactions In Vivo

This protocol is intended to serve as a guide. You should empirically optimize the cell culture protocol, transfection conditions, amount of HaloLink™ resin used and adjust buffers if necessary.

The following protocol was used with HeLa cells cultured in 10-cm Petri dish transfected with pFC8A(HT)-p65 (encoding human p65-HaloTag™ fusion protein). This protocol used a lipid-based transfection reagent and was performed according to the manufacturer's instructions.

3.3.1. Day 1: Plating Cells

HeLa cells ($1.5\text{--}2.5 \times 10^6$ cells) were plated in 10-cm plastic Petri dish and grown overnight in Dulbecco's Modified Eagle's Medium + 10% fetal bovine serum in atmosphere of 5% CO₂ at 37°C to 70–80% density.

3.3.2. Day 2: Transfecting Cells

Transfect cells following the manufacturer's instructions for the transfection reagent that you are using. In our case, cells were transfected using Lipofectamine 2000™ according to manufacturer's protocol using 1–2 µg DNA and 50 µl of Lipofectamine 2000™ per dish.

3.3.3. Day 3: Capturing and Analysis of the Protein Complexes

3.3.3.1. PHASE 1: PREPARATION OF CYTOSOLIC FRACTION

1. Twenty-four-hour post-transfection, aspirate off media and wash cells twice with 5 ml of ice-cold 10 mM N-(2 Hydroxyethyl piperazine-N'-(2-ethanesulfonic acid); 4-(2 Hydroxyethyl) piperazine-1-ethanesulfonic acid (HEPES) buffer, pH 7.5.
2. Resuspend cells in 1 ml of the HEPES buffer containing protease inhibitors and collect by scraping.
3. Lyse cells using mechanical disruption (e.g., use glass homogenizer 2 ml size; 25–30 strokes on ice or through a 27-gauge needle) followed by sonication on ice.
4. Centrifuge at $10,000 \times g$ for 7 min at 4°C.
5. Carefully remove supernatant and use immediately or store at –70°C for up to a month.

3.3.3.2. PHASE 2: RESIN EQUILIBRATION

Mix resin by inverting to obtain uniform suspension.

1. Dispense 100 µl of HaloLink™ resin into 1.5-ml Eppendorf tube and spin in microcentrifuge for 1 min at $800 \times g$.
2. Carefully remove and discard the supernatant leaving resin at the bottom of the tube.
3. Add 400 µl of binding buffer, mix thoroughly by inverting the tube.
4. Centrifuge for 2 min at $800 \times g$.
5. Carefully remove and discard the supernatant leaving resin at the bottom of the tube.
6. Repeat steps 3–5 two more times for a total of three washes.
7. After last wash, resuspend the resin in 40 µl of binding buffer.

3.3.3.3. PHASE 3: CAPTURE OF PROTEIN COMPLEXES

1. To the resin, add 100 μ l of the cytosol prepared as described above (preparation of cytosolic fraction; the volume of the cytosolic fraction may have to be adjusted, use 100 μ l only as a guideline).
2. Incubate with mixing using rotation for 1 h at room temperature or 4 h at 4°C (*see Note 9*). Make sure resin does not settle to the bottom of the tube as that will reduce efficiency of binding.
3. Centrifuge for 2 min at 800 $\times$ g.

3.3.3.4. WASHING

1. Add 1 ml of wash buffer containing 0.5 mg/ml BSA and mix thoroughly by inverting the tube.
2. Centrifuge for 2 min at 800 $\times$ g. Discard the wash.
3. Repeat steps 1 and 2 two more times.
4. Add 1 ml of wash buffer containing 0.5 mg/ml BSA and mix thoroughly by inverting the tube.
5. Incubate 5 min with occasional mixing.
6. Centrifuge for 2 min at 800 $\times$ g. Discard the wash.
7. Repeat steps 4–6 twice one more time.

3.3.3.5. ELUTION

1. Add 30 μ l of 1 $\times$ SDS loading buffer and heat to 95°C for 2–5 min.
2. Remove supernatant and analyze samples immediately or store at –20°C. Proteins can be resolved on a SDS–PAGE gel and analyzed by Western blotting.

3.4. Detection of Protein–DNA Interactions

This protocol is designed for the use of $1\text{--}5 \times 10^6$ cells at 70–80% confluency. Typically, this is 1–2 wells of a 6-well plate, each containing 2 ml of cells (*see Note 11*).

3.4.1. Resin Equilibration

1. Aliquot 100 μ l of HaloLink™ resin into a 1.5-ml microcentrifuge tube.
2. Centrifuge resin for 3 min at 800 $\times$ g and remove the supernatant.
3. Wash resin with 400 μ l HaloLink™ Equilibration Buffer.
4. Centrifuge for 3 min at 800 $\times$ g and remove the wash.
5. Repeat steps 3 and 4 two more times.
6. Remove the final wash and add 100 μ l of 1 $\times$ TBS (BSA at a final concentration of 1 mg/ml may be added if desired).

3.4.2. Crosslinking, Capture and Release of DNA

1. Grow approximately 1×10^6 cells to 70–80% confluency.
2. With constant swirling, slowly add formaldehyde (stock concentration of 37%) to a final concentration of 1% directly to cells.

3. Incubate for 10 min at room temperature.
4. Quench crosslinking by the addition of glycine, pH 7, to a final concentration of 125 mM directly to cells.
5. Incubate for 5 min at room temperature.
6. Aspirate off media and wash cells twice with 2 ml of ice-cold 1× PBS.
7. Add 1.5 ml of ice-cold PBS to cells and scrape cells into a 1.5-ml microcentrifuge tube.
8. Place cells immediately on ice.
9. Centrifuge and pellet cells at $2000 \times g$ for 5 min at 4°C.
10. Remove PBS and resuspend cells in 650 µl of lysis buffer.
11. Vortex and incubate on ice for 15 min.
12. Dounce cells or lyse them by passing them through 25–27-gauge needle several times using 1-ml syringe.
13. Sonicate on ice to obtain DNA fragments between 500–1000 bp (see recommendations below for a Misonix 3000 sonicator).
14. Clear lysates by centrifugation at $14000 \times g$ for 10 min at 4°C.
15. Add lysate (supernatant) directly to prepared HaloLink™ resin and incubate with rotation for 2 h at room temperature or 4–18 h at 4°C.
16. Spin lysates with HaloLink™ resin at $800 \times g$ for 3 min. Discard supernatant.
17. Wash resin twice with 1 ml of lysis buffer. Discard supernatant each time.
18. Wash resin twice with 1 ml with high salt lysis buffer. On the last wash, incubate resin with buffer for 5 min at room temperature with rotation. Discard supernatant each time.
19. Wash resin three times with 1 ml nuclease-free water. On the last wash, incubate resin with water for 5 min at room temperature with rotation. Discard supernatant each time.
20. Add 100 µl of reversal buffer to resin and place tubes at 65°C for 4–18 h to reverse crosslinks.
21. Centrifuge resin at $800 \times g$ for 3 min after reversal and save the supernatant containing released target DNA.
22. Purify DNA for PCR amplification using a PCR clean-up kit according to manufacturer's recommendations.

Misonix 3000 Sonication Recommendation (Microtip 418): Set the output to 2.5. For 1×10^6 cells in a volume of 500–700 µl, on ice, perform 6×10 -s pulses with 10 s of rest in between each pulse.

3.5. Enzyme Immobilization and Analysis of Enzymatic Activity on the Surface

Immobilization of enzymes and study of their enzymatic activities is very important. Covalent attachment of proteins to the HaloLink™ resin allows assaying of enzymatic activities over a long period of time in different buffer conditions without protein dissociation from the resin. Affinity purification

resins like His-Tag binding resins are often used to attach enzymes onto the surface. However, after incubation in the assay buffer, equilibrium will be established leading to dissociation of protein from the resin. Because HaloTag™ fusion proteins are bound covalently to HaloLink™, dissociation from the resin does not occur.

3.5.1. Phase 1

Immobilize HaloTag™ fusion protein according to the steps described in **Subheading 3.1.**

3.5.2. Phase 2

Optimize assay for detection of enzymatic activity according to the particular enzyme.

3.6. One-Step Purification of Fusion Proteins

The major application for HaloLink™ resin is permanent attachment of proteins onto the resin that does not allow purification of the HaloTag™ fusion proteins as they cannot be eluted off the resin. However, our plasmids pFC8A(HT) and pFC8K(HT) contain protease cleavage site (factor Xa) situated in the linker sequence between the HaloTag™ and the protein of interest. This allows the release of the pure, nontagged protein of interest from the HaloLink™ resin by factor Xa protease cleavage.

3.6.1. Phase 1

Immobilize HaloTag™ fusion protein according to the steps described in **Subheading 3.1.**

3.6.2. Phase 2

Add Factor Xa to the resin carrying HaloTag™ fusion protein. Optimize factor Xa cleavage reaction according to the manufacturer's recommendations.

4. Notes

1. IGEPAL-CA630 is added to prevent sticking of the resin to the sides of the tube. The range of effective of IGEPAL concentration is from 0.001 to 0.05%. Warning: Solutions containing IGEPAL-CA630 should be prepared fresh.
2. In case IGEPAL-CA630 interferes with the activity of the protein of interest, the concentration can be reduced to 0.001% or eliminated; however, this may result in higher nonspecific binding. We recommend that IGEPAL-CA630 be replaced

by 0.5% Triton X-100 or by 5% glycerol. BSA may also be eliminated if it interferes with the activity of the protein, but higher nonspecific binding may be detected. Other Tris-based buffers can be used in this protocol.

3. Appropriate speed in rpm can be calculated from the following formula, $RCF = (1.12)(r)(rpm/1000)^2$ where r = radius in mm measured from the center of spindle to bottom of rotor bucket; rpm = revolutions per minute. In a standard size microcentrifuge, $800 \times g$ corresponds to 3000 rpm.
4. Volume used for resuspending the resin can be adjusted for a specific experiment.
5. In case of proteins expressed in mammalian cells, we added 100 μ l of cytosolic fraction to 50 μ l of the HaloLink™ resin.
6. We used a tube rotator from Scientific Equipment Products; other mixing devices can be used (e.g., IKA-SCHÜTTLER MTS2).
7. In vitro Transcription/Translation (TNT®) reactions are typically 50 μ l, which may be sufficient for more than one pull-down reaction. Efficiency of the in vitro protein synthesis and the strength of protein–protein interaction may differ for different protein pairs, thus, the volume of the in vitro TNT® reaction added to the HaloLink™ resin may have to be adjusted for a specific pair. Smaller or larger volumes may be needed.
8. If immobilization of proteins onto HaloLink™ takes longer than incubation time required for TNT® T7 Quick Coupled Transcription/Translation, it is best to keep reactions at 30°C or on ice, if protein stability is in question. Prolonged incubation on ice may result in protein precipitation. An aliquot of 1–5 μ l of the reaction may be saved for analysis of the efficiency of the prey synthesis by SDS–PAGE gel.
9. Time of incubation may need optimization for different protein pairs.
10. Overheating may result in aberrant migration of proteins or even prevent protein migration into the gel. If this occurs, heat samples to 70°C for 3–5 min or 60°C for 10 min. When analyzing the efficiency of the prey synthesis, too much of the sample may cause coagulation of hemoglobin and cause aberrant migration in the gel. We suggest to reduce the volume of reaction loaded to 1–2 μ l.
11. When using $0.1\text{--}0.5 \times 10^6$ cells, reduce the amount of HaloLink™ resin to 50–75 μ l. When using $0.5\text{--}1 \times 10^7$ cells, increase the amount of HaloLink™ resin to 125 μ l.

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Site-Specific Cleavage of Fusion Proteins

Adam Charlton

Summary

Where an affinity tag has served its purpose, it may become desirable to remove it from the protein of interest. This chapter describes the removal of such fusion partners from the intended protein product by cleavage with site-specific endoproteases. Methods to achieve proteolytic cleavage of the fusion proteins are provided, along with techniques for optimizing the yield of authentic product.

Key Words: Fusion protein; affinity tag; site-specific proteolysis; protease; proteolytic cleavage.

1. Introduction

The use and benefits of affinity tags is the subject of this book; although when the tag has served its purpose, it is often desirable to remove it to obtain homogeneous protein product of native size and sequence. The use of site-specific endoproteases to facilitate this removal is an approach that has gained considerable favour in recent times. There are many reasons for this widespread adoption, but foremost amongst these is that site-specific proteases recognize long, uncommon amino acid sequences that are highly unlikely to be found within the protein of interest. Also, as proteases are themselves quite labile proteins, sensitive to extremes of temperature or chemical environment, proteolytic cleavage systems tend to function in mild conditions that may enhance protein product stability. Finally, many site-specific proteases act after their

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recognition sequence, rather than within it. This therefore provides the opportunity to generate the exact sequence for the target protein, as no contribution to catalysis needs to be made by any element of the target protein itself.

A very limited number of all proteases display suitable site specificity for a sufficiently long amino acid sequence to be useful for fusion protein cleavage. These proteases are frequently isolated as proprotein activation enzymes, where evolutionary pressure has led toward site specificity. This is the case with many of the proteases covered in this chapter, which represent those that are both readily commercially available and have a long history of application in fusion protein cleavage.

1.1. Fusion Proteins

The fusion protein strategy is a popular approach to the expression of recombinant proteins in bacteria. The fusion of the protein of interest to another unrelated protein, or fusion partner can improve yields of the target protein. The fusion partner can provide protection against proteolysis, enable *in vivo* folding of the target protein or facilitate recovery by acting as affinity tags (**1,2**).

The protease substrate numbering convention of Schechter and Berger (**3**) will be used for this chapter, where the amino acids of the substrate (the fusion protein) N terminal to the site of cleavage are designated P and those C terminal are P'. The residues are numbered with increasing distance from the scissile bond (*see* **Fig. 1**).

The fusion partner may be incorporated at the N- or C-terminal end of the target protein, but for the purposes of this chapter, N-terminal fusions will be specifically covered. As all the specific proteases detailed cleave on the carbonyl side of the P1 residue, less or no non-native sequence elements are retained from these fusions. The methods are valid for C-terminal fusions, but the recognition sequences will remain attached as a C-terminus extension of the protein product. **Figure 1** depicts an N-terminal fusion protein.

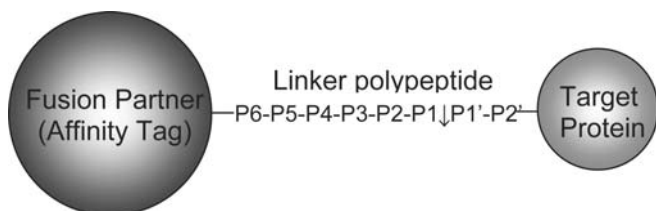


Fig. 1. A schematic representation of an N-terminal fusion protein.

In the design of a fusion protein strategy, the selection of the protease to affect the final cleavage may be as important as the selection of the fusion partner itself. Where available, sequence and structural information can guide this decision, as can the final application of the target protein. When a protease has been selected, the recognition sequence for that protease must be inserted between the fusion partner and the target protein as a linker peptide, as shown in **Fig. 1**.

1.2. Enterokinase

Enterokinase (EC 3.4.21.9) is a mammalian gastric serine protease. The *in vivo* function of this enzyme is the activation of trypsin by cleavage of the trypsinogen zymogen to its active form. The cleavage site for this enzyme with its natural substrate is C terminal to the recognition sequence pentapeptide (Aspartate)₄–Lysine (**4**). As Enterokinase cuts C terminal to its recognition sequence, without requiring the interaction of residues on the other side of the scissile bond, it is capable of generating a native N terminus for the protein product. The high charge density of the recognition sequence will increase the likelihood of solvent exposure at the site, maximizing protease accessibility and also serving to improve the overall solubility of the fusion protein (**5**).

The unique nature of the cleavage motif should preclude its occurrence within a protein product; however, Enterokinase largely recognizes the charge density of its recognition sequence rather than the precise amino acid sequence. Cleavage by Enterokinase is possible down to sequences as short as Asp-Asp-Lys (**4**), and activity is permitted with substitution of the motif residues with their charge equivalents (**6**). Therefore, similar apparent charge densities in the target protein may also be susceptible to Enterokinase cleavage.

Enterokinase is available as a recombinant enzyme, in many cases, as only the catalytic subunit of the holoenzyme. It must be noted that not all vendors offer the recombinant protein, so care must be taken in obtaining the enzyme if this is important.

1.3. Factor Xa

Factor Xa (EC 3.4.21.6) is an enzyme of the mammalian blood clotting cascade. Upon its own activation, this enzyme in turn activates the next enzyme in the cascade by cleavage of prothrombin, liberating active Thrombin. Factor Xa is highly specific for cleavage following the tetrapeptide sequence Isoleucine–(Glutamate/Aspartate)–Glycine–Arginine, allowing for the generation of an authentic N terminus for the protein product (**7**).

Factor Xa is not currently produced recombinantly and, therefore, must be isolated from mammalian plasma (usually bovine). This should be considered

when selecting Factor Xa for a fusion protein system, depending on the intended final use of the target protein product.

1.4. Thrombin

Thrombin (EC 3.4.21.5) is another enzyme of the mammalian blood clotting cascade, acting downstream of Factor Xa its function *in vivo* is the cleavage of fibrinogen to generate fibrin (8). Unlike the other specific proteases described in this chapter, thrombin does not have a long defined specificity sequence, with the only absolute requirement for cleavage being that it occurs after an Arginine, especially where the Arginine residue is preceded by a Glycine or a Proline at P2 and followed by a Glycine at P1' (9). Although lacking a long recognition sequence, thrombin cleavage can be further targeted by inclusion of hydrophobic residues in the P4 and P3 positions (9). Thrombin cleavage is also improved with non-acidic P1' and P2' residues, but these will be determined by the target protein's sequence and not usually available for substitution.

Thrombin distinctly prefers cleavage within a P-R↓G sequence, so much so that it should be considered to cleave within this recognition sequence, and as such a protein released from a fusion by this protease will have a residual N-terminal Glycine. Thrombin is therefore unlikely to produce the target protein with fully authentic sequence, except in cases where the first residue of the protein is Glycine. There are examples of thrombin cleavage prior to residues other than Glycine, but these are uncommon (10).

Thrombin possesses high intrinsic activity, so can function at relatively low enzyme concentrations and is tolerant of a wider range of buffer conditions than other mammalian proteases. Like Factor Xa, thrombin is not commercially available as a recombinant product, so consideration of the purpose for the target protein must be made before designing a fusion protein regime around this protease.

1.5. Genenase I

Genenase I is unique amongst the selected proteases, as it represents the only example of a bacterial enzyme and of a protease with engineered specificity. The parent enzyme for this rationally designed protease is subtilisin BPN' from the bacterium *Bacillus subtilis* (11). Genenase I was developed by mutation of a necessary active site Histidine residue to Alanine, resulting in a non-functional enzyme. The functionality of the protease can be restored if the side chain of the Histidine residue is supplied by the substrate at the P2 or P1' position; this mechanism is known as substrate-assisted catalysis (11,12).

Cleaving C terminal to its ideal recognition sequence, Genenase I is capable of producing the correct N terminus for the product. As this sequence is not

based around a charged amino acid, as is the case with many of the other proteases, Genenase I offers a quite different cleavage mechanism. It is tolerant of somewhat harsher conditions than its mammalian counterparts.

Owing to the requirement for substrate-assisted catalysis, the overall activity of this enzyme is considerably lower than other, fully self-functional proteases. This often translates to a requirement for higher enzyme : substrate ratios. As a licensed product, Genenase I is only available from one manufacturer and may impose a cost limitation to future scale-up of a cleavage system.

1.6. Viral Cysteine Proteases

To obtain novel site-specific proteases, attention has turned to the enzymes of RNA viruses. Upon infection, the genomes of these viruses are translated as one large polyprotein (**13**). The proteases act to specifically cleave the polyprotein into its individual structural and functional components. A major feature that distinguishes this group of proteases is that they employ a cysteine residue at the core of their catalytic mechanism, as opposed to the serine of the mammalian and bacterial proteases. The overall fold of these viral enzymes is very similar to that of the serine proteases; in some cases, the active site cysteine can be substituted with serine to achieve an active enzyme, albeit with significantly diminished activity (**14**).

Many viral proteases are highly specific for very long recognition sequences, but the two that have made the greatest impact in fusion protein cleavage are the proteases of Tobacco Etch Virus (TEV) and Human Rhinovirus (HRV). The recognition sequence for these enzymes spans at least seven and eight residues, respectively, with little divergence from the wild-type sequence of the natural polyprotein junctions possible. The minimum cleavage site for TEV protease is of the form E-X-X-Y-X-Q↓(G/S), with a consensus sequence of E-N-L-Y-F-Q↓(G/S) (**15,16**). The site for HRV follows a similar general theme, with a consensus sequence of L-E-V-L-F-Q↓G-P (**17**). As can be seen from these sequences, the viral proteases cleave within their recognition sequences and will hence leave a non-natural mono-peptide or di-peptide extension on the N terminus of the target protein. TEV protease is somewhat more flexible in its P1' requirements, with peptide studies suggesting that it may tolerate Glycine, Serine, Alanine or Methionine at P1' (**18**). Although for initial proof of concept cleavage trials, it would be advisable to maintain the wild-type Glycine or Serine.

High purity recombinant preparations of TEV and HRV proteases are available for fusion protein cleavage. Many manufacturers' implementations of these enzymes also bear an affinity tag to facilitate later removal of the protease from the protein preparation.

2. Materials

2.1. Reagents for Cleavage of Fusion Proteins with Serine Proteases

1. Cleavage buffer: 50 mM Tris-HCl (*see Note 1*), 50 mM NaCl, 2 mM CaCl₂, (*see Note 2*), pH 8.
2. Microfuge tubes.
3. Pipettes and tips for accurate liquid dispensation in the 10 µl, 100 µl and 1 ml ranges.
4. Ice.
5. Heating block.
6. Reducing sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) loading buffer, 2×: 50 mM Tris-HCl, 2% SDS, 10% glycerol. Bromophenol blue, 0.02%, pH 6.8, or as supplied for proprietary PAGE systems.
7. Dialysis equipment (if required) for example, tubing or centrifugal concentrator.
8. HCl, 1 M.

2.2. Reagents for Cleavage of Fusion Proteins with Cysteine Proteases

1. Cleavage buffer: 50 mM Tris-HCl (*see Note 1*), 150 mM NaCl, 1 mM ethylenediamine tetraacetic acid (EDTA), 1 mM dithiothreitol (DTT) (*see Note 3*), pH 7.5.
2. Microfuge tubes.
3. Pipettes and tips for accurate liquid dispensation in the 10 µl, 100 µl and 1 ml ranges.
4. Ice.
5. Reducing SDS-PAGE loading buffer, 2×: 50 mM Tris-HCl, 2% SDS, 10% glycerol. Bromophenol blue, 0.02%, pH 6.8, or as supplied for proprietary PAGE systems.
6. Dialysis equipment (if required) for example, tubing or centrifugal concentrator.
7. HCl, 1 M.

3. Method

3.1. Selecting the Appropriate Protease

1. Based on the background information and the data in **Table 1**, select a protease appropriate for the fusion protein of interest.
2. Examine the target protein amino acid sequence for complete or partial occurrences of the recognition sequence for the intended protease. Where that sequence, or the two or three residues around the cleavage site, exists in the target protein product, avoid the use of that protease.
3. Insert the cleavage sequence between the fusion partner and the protein product.
4. Sequence the construct to ensure the correct insertion of the protease recognition sequence.

Table 1
Properties of Specific Proteases for Fusion Protein Cleavage

Protease	Protease type	Cleavage site	Unlikely to cleave before	Suppliers	Notes
Enterokinase	Serine	D-D-D-D-K↓	P	I, R, M, N, S	
Factor Xa	Serine	I-E-G-R↓	P, R	S, M, P, R, Q, N, G	
Genenase I	Serine	P-G-A-A-H-Y↓	P, I, D*, E*	N	*4
Thrombin	Serine	(G/P)-R↓G	n/a	M, G, S, R	
Tobacco Etch Virus protease	Cysteine	E-N-L-Y-F-Q↓(G/S)	n/a	I, U	
Rhinovirus 3C proteinase	Cysteine	L-E-V-L-F-Q↓G-P	n/a	M,G	

1 G, GE Healthcare (Amersham Biosciences); 2 I, Invitrogen; 3 M, Merck Biosciences; 4 N, New England Biolabs; 5 P, Pierce; 6 Q, Qiagen; 7 R, Roche Diagnostics; 8 S, Sigma Aldrich; 9 U, U.S. Biological.

- Obtain the selected protease. Always use the highest purity, or restriction grade, protease preparations to avoid non-specific cleavage of the target protein by contaminating proteases.
- Refer **Subheading 3.2** for the protease type serine and **Subheading 3.3** for the protease type cysteine of the selected protease system.

3.2. Cleavage of Fusion Proteins with Serine Proteases

- If the fusion protein sample contains urea or guanidine (*see Note 5*), salts >250 mM (*see Note 6*), imidazole >50 mM, ionic detergents >0.01% (*see Note 7*), reducing agents or known protease inhibitors (*see Note 8*), dialyze into cleavage buffer.
- Concentrate or dilute the fusion protein preparation to approximately 0.5 mg/ml (*see Note 9*)
- Dilute the protease preparation to 0.05 units/μl (or 0.05 μg/μl) in cleavage buffer (*see Note 10*). Keep protease preparations and stock on ice until needed.
- Set up a pilot cleavage by mixing 100 μl of fusion protein (50 μg at 0.5 μg/μl, *see step 2*) with 10 μl of protease dilution (*see step 3*). Prepare a negative control reaction by adding 2 μl of cleavage buffer to 20 μl of fusion protein preparation. Incubate these reactions at approximately 21°C (*see Note 11*). If a positive cleavage control was supplied, prepare this reaction according to the manufacturer's directions.
- Remove 22-μl samples of the cleavage reaction at 1, 2, 4, 8 and 24 h. Terminate the reaction by adding 22 μl of 2× reducing SDS–PAGE loading buffer

- (see **Notes 12** and **13**). Terminate the negative control at 24 h. Store at -20°C until all of the samples are ready to run on SDS-PAGE (see **Note 14**).
6. Analyze the time point samples and the negative control on SDS-PAGE.
 7. If there is significant degradation of the target protein (see **Note 15**) go to step 8. If there is incomplete cleavage (see **Note 16**), or no cleavage apparent where a positive control was successful, go to step 10. If the cleavage was successful, go to step 12.
 8. Incubation with a lower amount of protease may help to minimize (see **Note 17**) internal cleavage of the target protein. Dilute the protease preparation to 0.005 and 0.0005 units/ μl (or 5 and 0.5 ng/ μl). To $2 \times 20 \mu\text{l}$ of fusion protein from **step 2**, add 2 μl each of these protease dilutions. Incubate at approximately 21°C (see **Note 11**) for 1 h (see **Note 18**). Terminate the reaction (see **Note 13**) and analyze. If these reactions yield sufficient correctly cleaved target protein, go to step 12.
 9. If overdegradation is still observed, reduce the concentration of protease further and repeat the reaction. Further improvement in the yield of correct protein product may be possible by altering the structural properties of the target protein (see **step 11**).
 10. Increasing the concentration of protease may enable cleavage. Dilute the protease stock to 0.25 and 0.5 units/ μl (or $\mu\text{g}/\mu\text{l}$). Add 4 μl of each protease dilution to 40 μl of fusion protein from step 2. To another 40 μl of fusion protein, add 4 μl of neat protease stock. Incubate at approximately 21°C (see **Note 11**). Remove 22 μl aliquots at 4 and 24 h. Terminate the reactions (see **Note 13**) and analyze by SDS-PAGE. If these reactions yield sufficient correctly cleaved target protein, go to step 12. If these protease concentrations remain unable to produce adequate levels of correctly cleaved material, or if significant degradation of the target protein is observed (see **Note 15**), go to step 11.
 11. Alter reaction conditions (see **Note 19**).
 - a. Select one factor at one concentration/level from **Table 2** to alter and prepare fusion protein at 0.5 mg/ml in this variant cleavage buffer by dialysis into the new system, or by adjustment of the original cleavage buffer to include the new factor.
 - b. Dilute the protease preparation to 0.05 units/ μl (or 0.05 $\mu\text{g}/\mu\text{l}$) in the variant cleavage buffer. Keep protease preparations and stock on ice until needed.
 - c. To 20 μl of the new fusion protein preparation (see **step 11a**), add 2 μl of the new protease dilution (see **step 11b**) and incubate at the desired temperature for either 1 h where reduction of internal cleavage is desired or 24 h where improvement of incomplete cleavage is the intended outcome. Terminate the reaction at the appropriate time and analyze.
 - d. If the degree of correct cleavage is increased, but not sufficiently, further improvement may be possible by altering the selected factor up or down, and repeating steps 11a–c. If further improvement within one factor class is not possible, hold this first factor constant at the level that gave the best result and introduce a second variant factor, repeating steps 11a–c with both factors.
 - e. See **Note 20** for other avenues to achieve successful cleavage.

Table 2
Conditions that can Alter Protease Specificity that are Compatible with Serine Proteases

pH	Non-ionic detergent (%, v/v)	Ionic detergent (%, w/v)	Chaotrope (M)	NaCl (mM)	Temperature (°C)
6.5	0.1	0.01	0.5	100	4
7.0	0.5	0.05	1	200	16
7.5	1	0.1	2	300	21–25
8.0	1.5	0.5	3	400	37
8.5	2	4	500		
9.0					
9.5					
Note 19a, b	Note 19c, d	Note 19c, e	Note 19c, f	Note 19 (g)	Note 19h

12. Scale-up the successful reaction conditions 10-fold to provide a working preparation of cleaved protein. Although individual reaction conditions and incubation times will vary depending on those determined in steps 4–11, a generic reaction protocol would be as follows: Mix 1 ml of fusion protein preparation (*see step 2*) with 100 μ l of protease dilution (*see steps 3, 8–10*); incubate at the required temperature (*see steps 4–11*) for 1 h (if step 8 was followed) or 4–24 h (if steps 10 and 11 were followed); terminate the reaction by addition of 50 μ l of 1 M HCl or addition of appropriate protease inhibitors (*see Table 3*).
13. For notes on product purification and reaction cleanup, *see Note 22*.

Table 3
Common Protease Inhibitors

Inhibitor	Protease class	Molecular weight	Effective concentration	Notes
Aprotinin	S	6500	10–250 μ g/ml	21
Leupeptin hemisulphate	S/C	475.6	1–100 μ M	
Phenylmethylsulfonyl fluoride (PMSF)	S	174.2	0.1–1 mM	
Iodoacetic acid	C	207.9	1–10 mM	
Pefabloc® SC (AEBSF)	S	239.7	0.1–2 mM	
Pepstatin A	A	685.9	0.5–1 μ g/ml	
Bestatin	M (E)	344.8	1–150 μ M	
EDTA	M	372.3	1–10 mM	
E-64	C	357.4	1–10 μ M	

A, Aspartic; C, Cysteine; (E), Exoprotease; M, Metalloprotease; S, Serine.

3.3. Cleavage of Fusion Proteins with Cysteine Proteases

1. If the fusion protein sample contains urea or guanidine (*see Note 5*), ionic detergents >0.01% (*see Note 6*), Zn^{++} >5 mM (*see Note 23*) or known protease inhibitors (*see Note 8*), dialyze into cleavage buffer.
2. Concentrate or dilute the fusion protein preparation to approximately 0.5 mg/ml (*see Note 9*).
3. Dilute the protease preparation to 0.05 units/ μl (or 0.05 $\mu\text{g}/\mu\text{l}$) in cleavage buffer (*see Note 10*). Keep protease preparations and stock on ice until needed.
4. Set up a pilot cleavage by mixing 100 μl of fusion protein (50 μg at 0.5 $\mu\text{g}/\mu\text{l}$, *see step 2*) with 10 μl of protease dilution (*see step 3*). Prepare a negative control reaction by adding 2 μl of cleavage buffer to 20 μl of fusion protein preparation. Incubate these reactions at 4°C (*see Note 24*). If a positive cleavage control was supplied, prepare this reaction according to the manufacturer's directions.
5. Terminate the reactions after 24 h by adding 22 μl of 2× reducing SDS-PAGE loading buffer (*see Notes 12 and 13*). Store at −20°C until the samples are ready to run on SDS-PAGE (*see Note 14*).
6. Analyze the time point samples and the control(s) on SDS-PAGE.
7. If there is significant degradation of the target protein (*see Note 25*) go to step 8. If there is incomplete cleavage (*see Note 16*), or no cleavage apparent where a positive control was successful, go to step 10. If the cleavage was successful, go to step 12.
8. Carefully analyze the negative (no protease) control (*see Note 26*); if degradation is observed in this reaction, consider expression in a host protease-deficient bacterial strain such as *Escherichia coli* BL21(DE3). The inclusion of protease inhibitors that do not affect cysteine proteases may also be beneficial, *see Table 3*. Return to step 4 with inhibitor inclusions or new host strain. Where the degradation is observed to be attributable to the viral protease, continue to step 9.
9. Incubation with a lower amount of protease may help to minimize (*see Note 17*) internal cleavage of the target protein. Dilute the protease preparation to 0.005 and 0.0005 units/ μl (or 5 and 0.5 ng/ μl). To 2 × 20 μl of fusion protein from step 2, add 2 μl each of these protease dilutions. Incubate at 4°C for 24 h. Terminate the reaction (*see Note 13*) and analyze by SDS-PAGE. If these reactions yield sufficient correctly cleaved target protein, go to step 12. Otherwise continue to step 11.
10. Increasing the concentration of protease may enable cleavage. Dilute the protease preparation to 0.25 and 0.5 units/ μl (or $\mu\text{g}/\mu\text{l}$). Add 4 μl of each protease dilution to 40 μl of fusion protein from step 2. To another 40 μl of fusion protein, add 4 μl of neat protease stock. Incubate the reactions at 4°C for 24 h. Terminate the reactions (*see Note 13*) and analyze by SDS-PAGE. If these reactions yield sufficient correctly cleaved target protein, go to step 12. If these protease concentrations remain unable to produce adequate levels of correctly cleaved material, go to step 11.

Table 4
Conditions that can Alter Protease Specificity that are Compatible with Cysteine Proteases

pH	Non-ionic detergent (% v/v)	NaCl (mM)	Temperature (°C)
6.5	0.1	200	4
7.0	0.5	300	16
7.5	1	400	21-25
8.0	1.5	500	34
8.5	2	800	
9.0	1000		
9.5			
Note 19a, b	Note 19c, d	Notes 19g and 27	Note 19 h

11. Alter reaction conditions (*see* **Note 19**):
 - a. Select one factor at one concentration/level from **Table 4** to alter and prepare fusion protein at 0.5 mg/ml in this variant cleavage buffer by dialysis into the new system, or by adjustment of the original cleavage buffer to include the new factor.
 - b. Dilute the protease preparation to 0.05 units/μl (or 0.05 μg/μl) in the variant cleavage buffer. Keep protease preparations and stock on ice until needed.
 - c. To 20 μl of the new fusion protein preparation (*see* **step 11a**), add 2 μl of the new protease dilution (*see* **step 11b**) and incubate at the desired temperature for 24 h. Terminate the reaction and analyze.
 - d. Increase or decrease the factor iteratively by repeating **steps 11a–c** until successful cleavage is obtained.
 - e. *See* **Note 20** for other avenues to achieve successful cleavage
12. Scale-up the successful reaction conditions 10-fold to provide a working preparation of cleaved protein. Although individual reaction conditions and incubation times will vary depending on those determined in steps 4-11, a generic reaction protocol would be as follows: Mix 1 ml of fusion protein preparation (*see* **step 2**) with 100 μl of protease dilution (*see* **steps 3, 9–11**); incubate at the required temperature (*see* **step 4** or **11**) for 24 h. Terminate the reaction by addition of 50 μl of 1 M HCl or addition of appropriate protease inhibitors (*see* **Table 3**).
13. For notes on product purification and reaction cleanup, *see* **Note 22**.

4. Notes

1. Other buffers for this pH are acceptable, such as N-(2-hydroxyethyl) piperazine-N'-(2-ethanesulfonic acid) (HEPES).
2. The action of these proteases is enhanced by inclusion of low levels of NaCl and trace CaCl₂ (**19**).

3. The catalytic mechanism of cysteine proteases relies on the active site cysteine thiol nucleophile. It is therefore vital to the activity of these enzymes that this thiol be preserved, with the state of this functional group ensured by maintaining a reducing environment. If this concentration of DTT causes reduction of labile disulphide bonds in the target protein (determined by incubation of the protein in cleavage buffer followed by analysis by a technique such as reversed phase High Performance Liquid Chromatography (rp-HPLC)), then a milder redox pair such as 3 mM reduced glutathione + 0.3 mM oxidized glutathione might be more appropriate.
4. Cleavage prior to aspartate and glutamate can be improved >10-fold by cleavage in 2 M KCl (manufacturer's recommendation).
5. Chaotropes such as urea or guanidine-HCl are known to severely inhibit cleavage by many proteases. The activity of these enzymes falls off sharply in the presence of any chaotrope, often with undetectable activity in concentrations above 2 M urea/1.5 M guanidine-HCl. Aside from decreased protease activity, the presence of chaotropes can alter the specificity profile of the enzyme, potentially giving rise to cleavage at unintended sites. It is therefore recommended that chaotropes be avoided in the pilot cleavage experiments to avoid their unexpected interference.
6. Enterokinase and Factor Xa are inhibited by concentrations of salts (such as NaCl) over 250 mM, as such it is recommended that the total concentration of all salts not exceed this level in initial experiments. Imidazole is known to inhibit these enzymes at concentrations over 50 mM. Although thrombin is generally more salt and imidazole tolerant, with successful cleavage reported in 500 mM NaCl and 500 mM imidazole (20), it is again advised that the total concentration be kept below the stated thresholds if possible.
7. SDS, an anionic detergent, can inhibit cleavage at concentrations as low as 0.001%, but in practice, the effect of less than 0.01% should be negligible. Although less information exists for enzyme inhibition by other charged detergents, it is likely that they too cause a very similar loss of activity, and as such, their presence in pilot cleavage experiments is not recommended.
8. Protease inhibitors may have been added at the cell lysis stage of protein purification.
9. Substrate concentration can have an effect on the rate of enzyme reactions. Keep fusion protein concentrations as consistent as possible in pilot cleavage experiments. Concentration of the fusion protein preparation can be performed simultaneously with step 1.
10. One percent concentration of protease (relative to fusion protein) is the goal. Use 1 unit of enzyme where the supplier defines a unit as having the ability to cleave >90% of 100 µg. Some manufacturers may use a different unit definition, in these circumstances; adjust the volume of protease added accordingly. For example, if a particular manufacturer's protease preparation defines one unit as having the ability to cleave 50 µg of control protein, then double the volume of protease added. Where both the mass (e.g., mg/ml) and the activity (units) of

the protease preparation are supplied, use the activity measure to determine the amount of protease to use.

11. Room temperature is acceptable if constant and within 20–25°C.
12. A reducing SDS-PAGE will show if the protease has cleaved protein product internally. An adventitiously cut protein product may appear intact on a non-reducing gel if held together by disulphide bonds.
13. The constituents of SDS-PAGE loading buffer, particularly the high concentration of SDS, will very effectively terminate all protease activity. If not using SDS-PAGE analysis, the reaction may be terminated by acidification, for example, add 3–5 μ l 1 M HCl, or by the addition of a protease inhibitors against the added enzyme, as listed in **Table 3**.
14. Select a SDS-PAGE system that will allow separation in the range between the size of the full-size fusion protein and the successfully cleaved target protein. Bear in mind that there may be smaller fragments present if the protein has been overdegraded.
15. Degradation of the protein product is indicated by a decreased abundance of material with the correct mass and the appearance of smaller products that were not present in the initial preparation or in the negative control sample. These effects will usually become more pronounced over the time course. In some cases, the fusion partner may be visible by SDS-PAGE. Ensure its presence is not mistaken for an internal cleavage fragment. If degradation is observed in the protease negative control, there may be contamination of the fusion protein sample by other proteases. Consider further purification.
16. In many cases, incomplete cleavage is preferable to overdegradation as intact fusion protein is more readily separated from the correct protein product than that protein will be from internal cleavage fragments, *see Note 22*.
17. Where internal cleavage of the protein has occurred, it is unlikely to be completely avoided. If the presence of these breakdown fragments or the associated yield losses cannot be tolerated, consider using another protease system.
18. It is assumed that the protein was overdegraded at the 1-h point in the initial time course. In most cases, a lower protease concentration will not change the cleavage profile (the products that are generated) substantially, but will instead increase the time taken to achieve the same profile. Performing the reaction at a lower protease concentration can be thought of as somewhat analogous to expanding the time taken to create the reaction products. Thus, it is possible to collect the reaction products at time points that would have been impractical to capture at the initial reaction ratio, such as those that formed in the first few minutes or seconds of the reaction.
19. Alteration of reaction buffer conditions may promote correct specificity. **Table 2** (*see Subheading 3.2.*, serine proteases) or **Table 4** (*see Subheading 3.3.*, cysteine proteases) suggest a range of potential reaction condition variations in which the specificity of the protease may be sufficiently altered to enable hydrolysis at the intended site. The concentration/level value ranges provided are intended as a guide only, with any amount within those ranges acceptable as circumstances

may dictate. However, deviation outside the upper and lower limits specified is unlikely to meet with a successful cleavage reaction outcome. The tertiary structure of the protein can either inhibit protease action at the intended site by sterically hindering accessibility, or promote incorrect internal cleavage by exposing labile surface motifs. The non-exhaustive list in **Table 2** or **Table 4** suggests conditions that will mildly alter the protein structure without denaturing the protease or protein product. Modification of multiple factors in concert may be required for optimal outcomes. If the degree of correct cleavage is increased by a factor, but not sufficiently so at any concentration/level, further cleavage improvements may be made by holding this first factor constant at the level that gave the best result and introduce a second variant factor and repeating the optimization experiments.

- a. Whilst not significantly altering the structure of the protein, the pH at which the reaction is performed may be particularly useful for reducing non-specific cleavage within the protein. As can be seen in **Table 1**, many of the proteases recognize charge amino acid groupings; therefore, altering the pH of the buffer can move toward or away from the pKa of the side chains of ionizable amino acids. This can alter local charge environments and can be sufficient to mask the secondary sites and prevent cleavage. Similarly, varying the pH can cause localized charge modifications in the protease active site that can shift the specificity of the enzyme enough to discourage secondary cleavage.
- b. **Table 5** lists some common buffers that will be effective at the stated pH points. Fifty millimolar solutions of each will provide sufficient buffering.
- c. Inclusion of chaotropes or detergents will relax the structure of the protein. These agents allow the normally buried hydrophobic residues of the protein to become more solvent exposed by disrupting hydrogen bonding and hydrophobic interactions. This can perturb the original structure of the protein, providing greater exposure of the expected target cleavage site,

Table 5
Suitable Buffers at Given pH Ranges

6.5	7.0	7.5	8.0	8.5	9.0	9.5
citrate MES	MOPS	MOPS	Tris-HCl	Tris-HCl		
		Tris-HCl		Tricine		
		HEPES	HEPES			
		Tricine	Tricine	Tricine		
				borate		
					borate	borate
					CHES	CHES

potentially shifting the equilibrium of the cleavage reaction away from the secondary site and toward the primary. Although **Note 5** cautions against the use of chaotropes, successful cleavage is indeed possible under these conditions, with successful cleavage reported by Enterokinase in 2 M urea (21), and Genenase I in 2.5 M urea (22,23). However, the activity of the proteases will most likely be significantly decreased, requiring a higher concentration of enzyme. The inclusion of chaotropes will most likely require a concurrent re-examination of the amount of enzyme used, as in step 10.

- d. Examples of common non-ionic detergents are Tween-20 and Triton X-100.
 - e. An example of a common ionic detergent is SDS. Ionic detergents should be used sparingly, as they are powerful protein denaturants.
 - f. The most commonly used chaotropes are urea and guanidine-HCl. The concentrations given in **Tables 2** and **4** are based on urea; if guanidine-HCl is used instead, decrease these values by 25%.
 - g. The inclusion of NaCl can relax protein structure by reducing the stabilizing effect of salt bridges. The inclusion of NaCl alone is unlikely to alter the initial cleavage profile, but can synergistically act with the other suggested factors to improve the overall specificity of the protease.
 - h. Aside from directly contributing to the rate of the protease reaction in a manner much similar to alteration in the enzyme : substrate ratio, the temperature of the incubation can also have an effect on protein structure. As decreased temperatures weaken hydrophobic interactions and strengthen hydrogen bonds and vice versa, there exists the potential to alter the cleavage profile of the system by simply altering the incubation temperature (author's personal observations).
20. If successful cleavage is still not obtained but the use of the selected protease is still desired, consider the insertion of a tetra- to hexa-peptide spacer sequence N terminal to the protease recognition sequence. The inclusion of a flexible spacer peptide sequence can allow greater access to the intended cleavage site by minimizing steric inhibition by the fusion partner. The steric inhibition effect can be particularly prevalent when dealing with small, largely unstructured peptide fusions that are able to fold back onto the protein structure, occluding the cleavage site (author's personal observations). For serine proteases, sequences such as S₃G (24), SG₄A (25) and SG₅ (26) have been used successfully for this purpose. As viral proteases tolerate little deviation from the wild-type recognition sequence, an upstream spacer derived from their wild-type polypeptide sequences may be more useful than an artificial polypeptide at reducing steric interference. In the case of TEV, such a sequence is DYDIPTT (27), and for HRV, a similar candidate is KMQITDS (28). Return to **Subheading 3.2., step 1** or **Subheading 3.3., step 1** with the new fusion construct
21. Leupeptin may also inhibit viral cysteine proteases at concentrations over 100 μM (29).

22. The full-length fusion protein and the separated affinity tag will bind to the affinity column under the same conditions employed to generate the fusion protein initially. The correctly cleaved protein, now lacking an affinity tag, will not be bound by the column and will thus flow-through. It should be noted that internal cleavage fragments of the product (if generated) will not be separated by this technique. If an internally cut protein is held together by disulphide bonds (*see Note 12*), it may be successfully separated from intact protein by ion-exchange chromatography due to the extra surface charges provided by the hydrolysis sites. Where the internal cleavage fragments are not held together, size-exclusion chromatography may provide separation.
23. Zinc ions are quite potent inhibitors of cysteine protease activity, with concentrations as low as 5 mM resulting in significant loss of activity. This inactivation is thought to occur due to the formation of a complex between the zinc ion and three amino acids in the active site pocket, including the catalytic cysteine (**21**).
24. Although not the optimal temperature for these enzymes, it has been shown, at least in the case of TEV protease, that incubation at 4°C results in only a 3-fold reduction in overall activity compared to room temperature (20°C) (**30**). The benefit to product stability at low temperature is, in most cases, well worth a slightly longer incubation time.
25. Internal cleavage by viral cysteine proteases is highly unlikely, with no reported cleavage at sites other than the minimum penta- or hexa-peptide recognition sequences in fusion proteins.
26. Degradation may be due to the action of bacterial host proteases that have co-purified with the fusion protein.
27. Viral proteases are far more salt tolerant than the serine proteases with activity reported in 800 mM NaCl (**18**).

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The Use of TAGZyme for the Efficient Removal of N-Terminal His-Tags

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Summary

The use of affinity tags and especially histidine tags (His-tags) has become widespread in molecular biology for the efficient purification of recombinant proteins. In some cases, the presence of the affinity tag in the recombinant protein is unwanted or may represent a disadvantage for the projected use of the protein, like in clinical, functional or structural studies. For N-terminal tags, the TAGZyme system represents an ideal approach for fast and accurate tag removal. TAGZyme is based on engineered aminopeptidases. Using human tumor necrosis factor α as a model protein, we describe here the steps involved in the removal of a His-tag using TAGZyme. The tag used (UZ-HT15) has been optimized for expression in *Escherichia coli* and for TAGZyme efficiency. The UZ-HT15 tag and the method can be applied to virtually any protein. A description of the cloning strategy for the design of the genetic construction, two alternative approaches and a simple test to assess the performance of the tag removal process are also included.

Key Words: Histidine tags; N-terminal tag; affinity tag removal; aminopeptidases; TAGZyme; downstream processing; recombinant protein.

1. Introduction

Affinity chromatography has become the method of choice to simplify and improve recovery in the purification of recombinant proteins. Affinity chromatography currently represents the most powerful tool available to

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downstream processing, in terms of both selectivity and recovery. Using an affinity tag and a single purification step, it is possible to achieve a yield of over 95% compared to the yields typically obtained using three or more standard chromatographic steps (40–50%).

Histidine tags (His-tags) are the most widely used affinity tags in research and protein structural studies (**1**). Compared to similar approaches, tagging with a His-tag offers several advantages: low levels of toxicity and immunogenicity, a smaller size and no net charge at neutral pH. The incorporation of a His-tag allows for single-step purification using Immobilized Metal Affinity Chromatography (IMAC) resins.

Purification using His-tag proteins relies on the high affinity displayed by short histidine tracks for chelated nickel, cobalt or zinc at neutral or weak basic pH. Metal ions are immobilized to a chromatographic support such as nitriloacetate, and metal binding occurs via the imidazole side chain of histidine. IMAC matrices display high protein-binding capacity and recovery (typically more than 90%). Importantly, IMAC is chemically stable to the extensive cleaning-in-place procedures widely used in pharmaceutical production.

For pharmaceutical applications, the affinity tag may need to be removed before the protein can be used for clinical or structural studies. A common approach is to include an unusual cleavage site between the His-tag and the native protein sequence. This tag removal step is then performed by the addition of the specific endoprotease to the purified tagged protein. In spite of the specificity, unspecific cleavage can often occur at cryptic sites or during long treatments (**2,3**), representing a challenge for the purification process and the intactness of the protein.

By engineering the specific endoprotease to include the same affinity tag as the target protein, an efficient removal of the process enzyme(s), the unprocessed fusion protein and the released tag can be designed. An affinity-tagged endoprotease can also be used for on-column cleavage. Furthermore, simultaneous affinity purification and on-column processing can be achieved. Immobilization of process enzymes is especially important for large-scale applications, as it may result in cost reductions, for example, with the use of lower amounts of enzyme.

TAGZyme is an enzymatic system based on engineered aminopeptidases designed for the efficient and accurate removal of N-terminal affinity tags such as His-tags. Because TAGZyme is designed for the removal of N-terminal tags by exopeptidases and not endoproteases, the native protein sequence is not affected during tag removal. The major enzyme in the TAGZyme system is DAPase, a recombinant dipeptidyl peptidase I. DAPase cleaves sequentially dipeptides from the N terminus of virtually any protein, provided the amino acid sequence does not contain (i) an arginine or lysine at the N terminus or at an uneven position in the sequence; (ii) a proline anywhere in the tag.

Upon cleavage, DAPase will stall if any of the above residues is found in the sequence (4). Additionally, different cleavage rates have been observed for certain dipeptide sequences ((5), see **Note 1**).

Removal of tags using TAGZyme is effective (typically >95 %) and can be performed with short treatments (<1 h) using low amounts of enzyme even at low temperature. These aspects are important when compared to endoprotease cleavage that requires high amounts of enzyme and prolonged treatments at high temperature increasing the risk of protein degradation or denaturation (3).

It is also of paramount importance to understand the properties of the TAGZyme components to enable the appropriate design of the affinity tag. Alternatively, it is possible to use commercially available cloning vectors that already include a suitable His-tag (e.g., Qiagen's TAGZyme pQE vectors). These vectors use a 6× His sequence and a suitable multiple cloning site. The use of this tag sequence for commercial applications requires a license from Qiagen.

We have developed an alternative his-tag sequence (UZ-HT15) that can be used for research and commercial purposes without a licensing requirement or patent infringement (see **Fig. 1**). It contains alternating histidine and glutamine (HQ), and it has been optimized for use with TAGZyme, for optimal DAPase cleavage and for high expression levels of virtually any protein in *E. coli* (4). Importantly, the second residue is K to minimize the effect of N-terminal methionine excision in *E. coli* (6) but also to prevent DAPase to cleave molecules where the first amino acid (M) has been removed, which may lead to further dipeptidase cleavage of the protein (see **Fig. 1A**). In this way, the small fraction of protein molecules affected by methionine excision will be eliminated during subtractive IMAC because they maintain a functional His-tag.

After tag removal, the exoproteolytic enzymes, which contain a His-tag at the C terminus, are removed by subtractive IMAC (see **Fig. 1B**). As the His-tag of TAGZyme enzymes is at the C terminus, they are not subject to cleavage by DAPase as they all contain suitable stop position for DAPase at the N terminus (see below). PCR fragments and cDNAs can be cloned into a TAGZyme pQE vector in such a way that a glutamine (Gln, Q) residue is introduced into the position between the last cleavable dipeptide of the tag and the first authentic amino acid of the target protein. DAPase cleavage is performed in the presence of a second enzyme, glutamine cyclotransferase (Qcyclase), which catalyzes the cyclization of an N-terminal glutamine residue to pyroglutamate. The presence of excess Qcyclase ensures immediate cyclization of the inserted glutamine residue when the His-tag has been digested and glutamine appears at the N terminus. A protein possessing an N-terminal pyroglutamate residue is thereby protected against further DAPase digestion (see **Fig. 1A**).

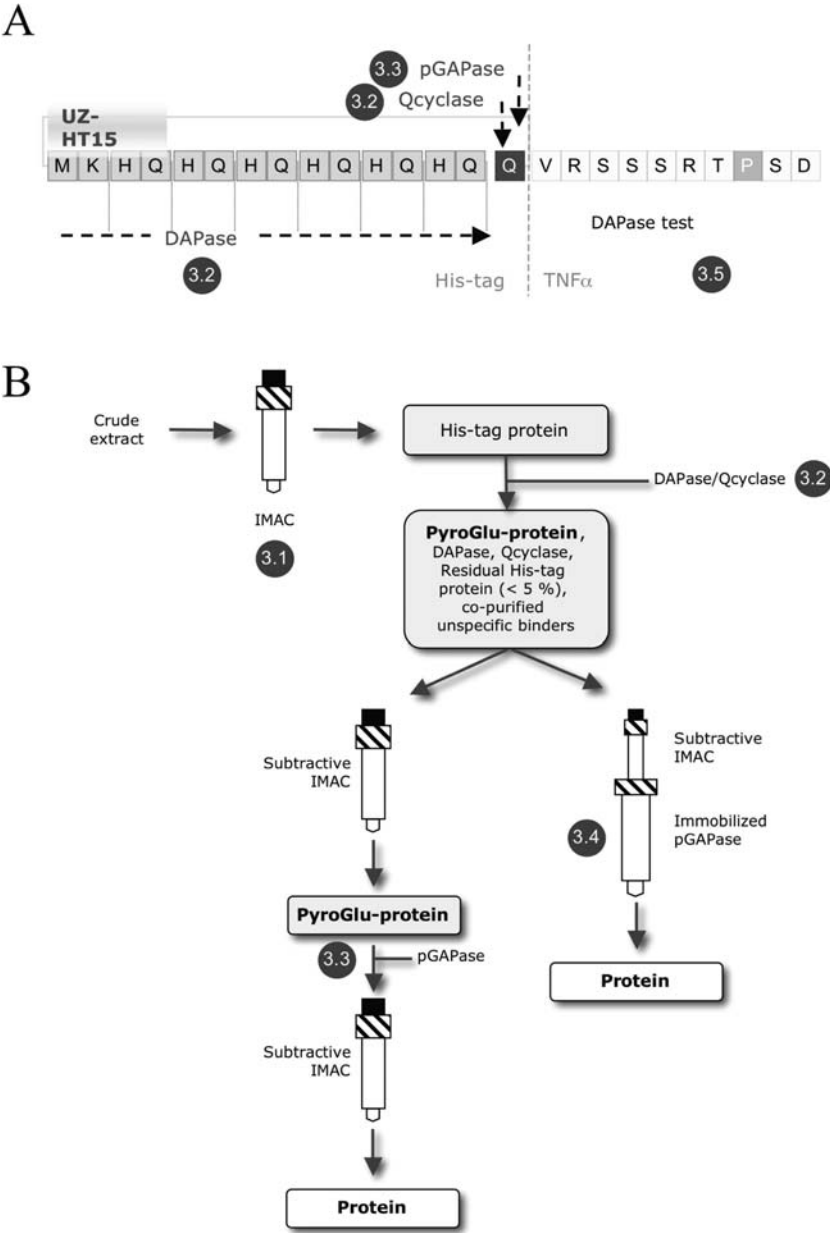


Fig. 1. Overview of histidine tag (His-tag) tumor necrosis factor α (TNF α) (A) and TAGZyme process for His-tag removal (B). The N-terminal sequence of the His-tag TNF α protein contains an even number of residues (the UZ-HT15 His-tag: MK HQ HQ HQ HQ HQ HQ HQ) that are cleaved by the DAPase before a Q residue adjacent to the native start of TNF α . DAPase cleavage is performed in the presence of excess Qcyclase

After removal of DAPase and Qcyclase, the processed protein is treated with pGAPase, a pyroglutamyl aminopeptidase that removes the N-terminal pyroglutamyl residue rendering a purified tag-free protein with the native N terminus. This step can be performed in batch mode or on-column where pGAPase is immobilized. The yield for the complete tag removal process using TAGZyme is typically over 90%.

A number of potentially therapeutic proteins contain a natural stop position for DAPase at their N terminus (e.g., R or K at position 1) in the mature or active form found in vivo. To produce and purify these proteins using recombinant DNA technology, a His-tag without the additional Gln can be added to the N terminus, and a process that only requires DAPase for tag removal can be developed. DAPase cleavage will proceed until the stop position is reached. Removal of DAPase and elution of the tag-free, purified protein can be achieved in a single step. This type of process is not explained further in this chapter but information can be found elsewhere (5).

The precise amino acid sequence of a His-tag and the nucleotide sequence selected to encode it are of great importance for the overall performance of the resulting construct during expression, post-translational processing, purification and tag removal. For these reasons, vectors have been optimized for use with TAGZyme ((5), see **Fig. 2**). Other vectors may be used following the guidelines for TAGZyme tag design and gene construction strategy (see **Subheading 2.1**).

Additionally, custom-optimized His-tag sequences for expression in *E. coli* or in other hosts and for tag removal can also be generated via mutagenesis of UZ-HT15. For expression in eukaryotic hosts, a signal peptide may be placed upstream of the His-tag to facilitate secretion. It is important to use a well-characterized signal peptide with known a cleavage site to ensure that the correct number of amino acid residues in the secreted protein will be suitable for TAGZyme removal of the tag (7).



Fig. 1. (see **Subheading 3.2.**) that acts when an N-terminal Q is found resulting in the formation of a pyroglutamyl. DAPase cleavage is blocked when a pyroglutamyl is present at the N terminus. After DAPase/Qcyclase treatment and subtractive Immobilized Metal Affinity Chromatography (IMAC) for enzyme removal, the protein is treated with pGAPase to remove the pyroglutamyl residue (see **Subheading 3.3.**). This step can also be performed using pGAPase bound to an IMAC to simplify the process (see **Subheading 3.4.**). Finally, a DAPase test (see **Subheading 3.5.**) can be performed on the purified tag-free protein to ensure that the final product does not contain tag residues. A DAPase stop position is found at the N terminus (P) that results in a truncated TNF α where the first six amino acids (VR SS SR) are cleaved. This can be confirmed by sodium dodecyl sulfate–polyacrylamide gel electrophoresis (see **Fig. 4**).

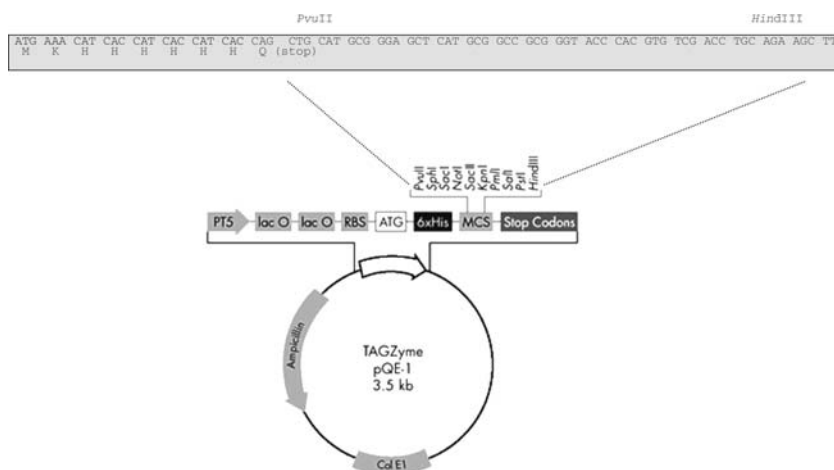


Fig. 2. The pQE-1 vector for N-terminal histidine tag (His-tag) constructions to facilitate removal of N-terminal residues using TAGZyme. Restriction sites within the multiple cloning sites, DNA sequences and the corresponding N-terminal amino acid sequences are shown. DAPase cleaves off dipeptides from the N terminus, and DAPase digestion stops at the glutamine residue (Q) in the presence of excess Qcyclase. See <http://www.qiagen.com> for more information about pQE vectors.

2. Materials

2.1. Molecular Biology: Cloning Strategy to Incorporate the UZ-HT15 His-Tag Sequence as a Universal Tag

The coding sequence of interest can be amplified using a primer that includes the His-tag adjacent to the target gene sequence and provides a restriction site for cloning in any vector that contains a, for example, *Nco*I site (C_CATGG) or a *Bsp*HI site (T_CATGA) or a *Pci*I site (A_CATGT) at the start codon. This design provides a good translation start in *E. coli*. It is also important that the amino acid sequence starts with MK to ensure a high expression level and an effective cleavage with DAPase ((5), see also **Subheading 1**).

General primer design for UZ-HT15 His-tag sequence with added Gln stop (67 nucleotides) (see **Note 2**).

MetLysHisGlnHisGlnHisGlnHisGlnHisGlnHisGlnGln

NNNNTCATGAACACCAACACCAACATCAACATCAACATCAACATCAACAG...18 bp

overlap (target gene)

Cloning vectors for use with TAGZyme are also available. The Gln DAPase stop point for DAPase can be introduced by cloning the protein coding sequence into TAGZyme vector pQE-2 (5). Here, any uneven amino acid position can

be chosen for the Gln residue, and the first amino acid of the mature target protein must immediately follow. Alternatively, TAGZyme pQE-1 is the vector of choice whenever the sequence of the protein allows cloning into the blunt-ended *PvuII* restriction site, which links the first amino acid of the target protein to the Gln stop point (see **Fig. 2**).

2.2. The Model Protein: Human TNF α

Tumor necrosis factor α (TNF α) is a multifunctional pro-inflammatory cytokine with effects on lipid metabolism, coagulation, insulin resistance and endothelial function (8). We have used TNF α as a model to illustrate the properties of TAGZyme. Similar approaches can be adapted for other proteins. The N-terminal sequence of mature TNF α is VRSSSRTPSD. The sequence of the His-tag of the recombinant TNF α used here is shown in **Fig. 1A**. Basically, the sequence includes the UZ-HT15 His-tag (4) with the additional Gln residue adjacent to the first residue (V) of TNF α .

2.3. Initial IMAC Purification of His-Tag protein from *E. coli*

An *E. coli* strain containing a plasmid that carries the sequence of human TNF α as a fusion with the UZ-HT15 His-tag sequence (see **Fig. 1**) was used. This strain was cultured in shake flasks (600 mL) essentially as described in **ref. 4**. Briefly, the strain was grown to an OD_{600 nm} between 0.4 and 0.6. Gene expression was induced by addition of 0.5 mM Isopropyl 1-thio- β -D-galactopyranoside (IPTG). Cells were harvested after 4–5 h of induction.

2.3.1. Buffers

1. Lysis buffer: 25 mM Tris-HCl, 300 mM NaCl, pH 8.
2. Buffer A: 20 mM NaH₂PO₄, 300 mM NaCl, 20 mM imidazole, pH 7.5.
3. Buffer B: 20 mM NaH₂PO₄, 300 mM NaCl, 1 M imidazole, pH 7.5 (see **Note 3**).
4. Buffer C: 20 mM sodium phosphate, 150 mM NaCl, pH 7.0.
5. Buffer D: 20 mM sodium phosphate, 150 mM NaCl, 2 mM cysteamine, pH 7.0.

2.3.2. Step A

IMAC Stationary Phase: Ni-Chelating Sepharose 6 FF column (2 cm² $\times$ 6 cm).

1. Preparation of Ni-chelating Sepharose 6 FF is performed according to the method described by the manufacturer.
2. Lysis treatment: Lysozyme (30 mg/ml; Sigma) and Benzonase (250 units/ μ L; Merck) in lysis buffer.
3. Buffer A for wash and buffer B for elution.

2.3.3. Step B

Buffer exchange on Sephadex G25 F (see **Note 4**) stationary phase: Sephadex G25 column (5.3 cm² × 30 cm) equilibrated with buffer C.

Cysteamine-HCl and Imidazole were obtained from Sigma. Sephadex G-25 F and Ni-chelating Sepharose 6 FF.

2.4. DAPase and Qcyclase Treatment

DAPase (10 units/ml) and Qcyclase (50 units/ml).

2.5. Removal of DAPase and Qcyclase Followed by Removal of Pyroglutamyl Using pGAPase and Subtractive IMAC

Stationary support: Freshly prepared HisTrap column (1 ml) and equilibrated with pGAPase (25 units/ml, Qiagen) prepared in buffer C.

2.6. Subtractive IMAC Using On-Column-Bound pGAPase

Stationary supports:

1. Column 1: 5 ml freshly prepared HisTrap equilibrated with 25 ml buffer C.
2. Column 2: 5 ml HiTrap equilibrated with buffer C.
3. Column 3: 20 ml pGAPase-chelating Sepharose FF (50 units/ml) is prepared by the following method at room temperature:
 - a. 20 ml chelating Sepharose FF packed in a 2 cm² × 20-cm column is loaded with 200 ml 10 mM ZnSO₄ pH 7 at a flow rate of 2 ml/min.
 - b. Wash the column (2 ml/min) with 40 ml H₂O.
 - c. Equilibrate (2 ml/min) with 30 ml buffer C.
 - d. Load the Zn-chelating Sepharose FF column (2 ml/min) with 1000 units pGAPase in 200 ml buffer C.
 - e. Mix the contents in the column to ensure a homogeneous material and pack the column again.
 - f. Equilibrate (2 ml/min) with 30 ml buffer D (see **Note 5**).
 - g. Equilibrating (2 ml/min) with 60 ml buffer C.
4. Chelating Sepharose, HisTrap and HiTrap were from GE Healthcare.

2.7. DAPase Test for Pyroglutamyl Removal in TNFα by pGAPase

After removal of pyroglutamyl by pGAPase and production of a tag-free protein, the first dipeptides of TNFα (ValArg SerSer SerArg) can be further processed before a stop position is encountered (ThrPro) if DAPase is added (see **Fig. 1A**). Thus, treatment of tag-free TNFα using DAPase for 2 h (as described in **Subheading 3.5.**) would result in a truncated TNFα only if

pGAPase removal of the N-terminal pyroglutamyl residue has been effectively performed. The truncated TNF α displays a different migration that is detectable by sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE). If pyroglutamyl has not been removed from the N terminus during DAPase/Qcyclase treatment, then DAPase will not cleave and no size alteration will be observed on this test. Thus, DAPase treatment can be used as a diagnostic method to test the efficiency of pyroglutamyl removal by pGAPase.

3. Methods

3.1. Initial IMAC Purification of His-Tag Protein from *E. coli*

1. Harvest cells from 2 $\times$ 600 ml culture by centrifugation (15 min, 4°C, 5000 $\times$ g) and resuspend in 80 ml pre-cooled lysis buffer.
2. Freeze/thaw the cell pellets to aid cell lysis and add 60 mg lysozyme (2 ml; 30 mg/ml) and 1250 units benzonase (5 μ l; 250 units/ μ l). Incubate for 1 h at 4°C (no mixing required) and centrifuge for 30–45 min (4°C, 13,000 $\times$ g).
3. Apply the sample ($\sim$ 80 ml) at a flow rate of 2 ml/min onto a Ni-chelating Sepharose 6 FF column (2 cm² $\times$ 6 cm) pre-equilibrated with lysis buffer at 4°C.
4. Wash the column with 20 ml lysis buffer at a flow rate of 2 ml/min.
5. Wash the column with 50 ml buffer A at a flow rate of 2 ml/min.
6. Elute the His-tag protein using a linear gradient (80 ml) from buffer A to buffer B at a flow rate of 1 ml/min, collecting 2 ml fractions (*see Note 3*).
7. Run diagnostic SDS–PAGE/activity assay with the obtained fractions to identify fractions containing the His-tag protein (*see Fig. 3A*).
8. Pool fractions containing the purified His-tag protein.
9. Apply the pooled fractions of the purified His-tag protein (typically 30–40 ml) at a flow rate of 4–5 ml/min onto a Sephadex G25 F column (5.3 cm² $\times$ 30 cm) equilibrated with buffer C.
10. Wash the column with buffer B at a flow rate of 4–5 ml/min and collect the desalted His-tag protein (measured by absorbance at 280 nm) in one pool.
11. Measure the protein concentration and proceed to removal of the tag.

3.2. Removal of the Tag, Step 1: DAPase and Qcyclase Treatment (for $\sim$ 80 mg protein)

1. Prepare a DAPase/Qcyclase mixture:
 - a. Mix 2 units DAPase (200 μ l) and 10 μ l 20 mM cysteamine and incubate 5–10 min at room temperature (*see Notes 5 and 6*).
 - b. Add 240 units Qcyclase (4.8 ml). For information on the required buffer pH, *see Note 7*. For information on reducing enzyme needs *see Note 8*.

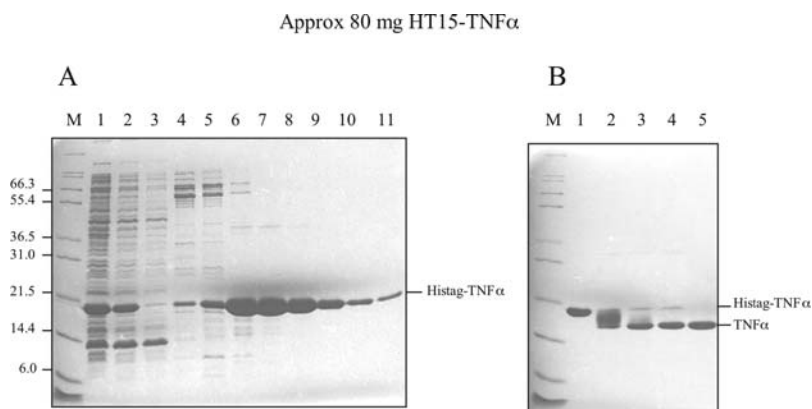


Fig. 3. Immobilized Metal Affinity Chromatography (IMAC) purification of histidine tag (His-tag) tumor necrosis factor α (TNF α) in *Escherichia coli* (A, see **Subheading 3.1.**) and subsequent tag removal using TAGZyme (B, see **Subheadings 3.2.** and **3.3.**). (A) Lane M: molecular weight markers (sizes in kDa); lane 1: crude extract; lane 2: crude extract after centrifugation; lane 3: flow through from the IMAC column (see **Subheading 3.1.**); lanes 4–11: eluted fractions 24, 26, 28, 30, 32, 34, 36 and 38, respectively. (B) Cleavage of His-tag TNF α obtained from the initial IMAC. Lane 1: His-tag TNF α ; lane 2: DAPase/Qcyclase 10-min treatment; lane 3: DAPase/Qcyclase 20-min treatment; lane 4: DAPase/Qcyclase 30-min treatment; lane 5: eluted tag-free TNF α after pGAPase treatment and subsequent elution from IMAC.

2. Add the 5 ml DAPase/Qcyclase mixture to the desalted sample of His-tag protein (typically 35–50 ml, in the example shown containing $\sim$ 80 mg for TNF α).
3. Incubate (no mixing required) at 37°C for 30 min. At time 10, 20 and 30 min, take 25 μ l aliquots to follow the cleavage of the His-tag and mix with 2 $\times$ SDS–PAGE sample buffer containing Dithiothreitol (DTT) (see **Fig. 3B**). See **Note 8** for the use of lower DAPase amounts.

3.3. Removal of the Tag, Step 2: Removal of DAPase and Qcyclase Using Subtractive IMAC Followed by Removal of Pyroglutamyl Using pGAPase ($\sim$ 80 mg protein)

1. After the DAPase/Qcyclase reaction, the enzyme reaction mixture is passed through a 5-ml HisTrap column to remove DAPase, Qcyclase, unprocessed His-tagged TNF α and other unspecific IMAC binders using a flow rate of 2 ml/min. This step is called subtractive IMAC because the primary role is the removal (“subtraction”) of His-tag proteins (DAPase and Qcyclase together with poorly processed protein molecules resulting from, e.g., removal of initial Met during expression) and to elute the tag-free protein.

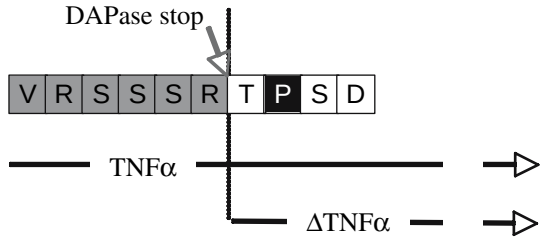
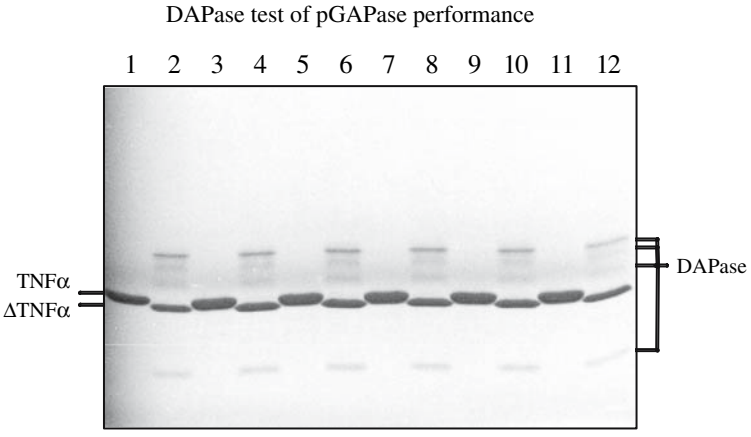


Fig. 4. DAPase test for pyroglutamyl removal by pGAPase (*see Subheading 3.5.*). DAPase treatment of tumor necrosis factor α (TNF α) results in cleavage only if pyroglutamyl has been removed by pGAPase. Thus, addition of DAPase results in the cleavage of the first six amino acids resulting in truncated TNF α (Δ TNF α). DAPase cleavage stalls at TP (**3,4**).

2. Collect the flow-through (measured by absorbance at 280 nm) and pool fractions containing pyroglutamyl-TNF α protein.
3. Prepare a pGAPase mixture: Mix 75 units pGAPase (3 ml) and 300 μ l 20 mM cysteamine and incubate 5–10 min at room temperature. *See Note 9* for ratios between pGAPase and target proteins.
4. Add pGAPase to the pooled sample.
5. Incubate at 37°C for 1 h (no mixing required).
 - a. After the pGAPase reaction, the mixture is passed through a 5-ml HisTrap column as above to remove pGAPase.

6. Collect the flow-through (measured by absorbance at 280 nm) and pool fractions containing the tag-free protein (TNF α).

3.4. Removal of the Tag Alternative, Step 2: On-Column-Bound pGAPase Treatment for Pyroglutamyl Removal

1. Place column 1 on top of column 2 and the set on top of column 3 (see **Fig. 1B** and Subheading 2.6.).
2. Apply sample (~40 ml) and set flow to 1 ml/min (see **Note 10**).
3. Wash the column with buffer C at a flow rate of 1 ml/min. Collect the flow-through (measured by absorbance at 280 nm) and pool fractions containing the tag-free protein (TNF α).

3.5. DAPase Test for Pyroglutamyl Removal by pGAPase

Treatment of purified, tag-free TNF α with DAPase can be monitored as it yields a truncated form where the first six residues are removed when pyroglutamyl has been removed from the N terminus. This results in a change in size that can be monitored by SDS–PAGE (see **Fig. 4**). If removal of pyroglutamyl is not effective, the DAPase test will result in a fraction of the protein not been cleaved.

1. Prepare a mixture containing 13.5 μ l DAPase (10 units/ml) and 13.5 μ l cysteamine (200 mM).
2. Mix 25 μ l purified protein (or 40–50 μ g processed protein eluted from the subtractive IMAC) and 27 μ l DAPase mix.
3. Incubate at 37°C for 2 h (no mixing required).
4. Use 25 μ l sample to run an SDS–PAGE comparing to the untreated processed protein (see **Fig. 4**).

4. Notes

1. Sequence-dependent cleavage efficiency by DAPase (see **Table 1**).
2. NNNN depicts a stretch of four bases to allow for effective digestion with restriction enzymes. In an alternative strategy, the cloning vector can be modified to incorporate the UZ-HT15 sequence. Then, it is possible to engineer restriction sites overlapping the last codon of the His-tag sequence. One example of this is shown with PvuII (for blunt-end cloning). vector... ATGAAACACCAACACCAACATCAACATCAACATCAACAT (CAA)CAGCTG... vector or NdeI, where the last Gln is substituted with a Met (it can be removed with DAPase). Here again, the stop Gln residue has to be added at the 5' end of the cloned fragment to ensure precise cleavage of the tag. vector... ATGAAACACCAACACCAACATCAACATCAACATCAACATATG.....vector
3. The high imidazole concentration in buffer B is required for the elution of TNF α as the protein is a trimer with high affinity for IMAC. For other proteins, 0.5 M imidazole should be a reasonable concentration. Optimization can be performed to further reduce the concentration of imidazole.

Table 1
Sequence-Dependent Cleavage Efficiency of DAPase

Rapid	Medium	Slow	No cleavage
Xaa-Arg	Asp-Asp ^{a,b}	Gly-Ser	Lys-Xaa
His-Gln	Glu-Glu ^{a,b}	Ser-Met	Arg-Xaa
His-Gly	Glu-His ^b	Gly-Met	Xaa-Xbb-Pro
Xaa-Lys	Gly-Phe ^c	Xaa-Phe ^c	Xaa-Pro
Gly-His	Ser-Tyr		Gln-Xaa (in the presence of Qcyclase)
His-Ala	Ala-Ala		
His-His	Phe-Xaa ^c		
His-Met	Xaa-Asp ^b		
Ala-His	Xaa-Glu ^b		
Met-His			

^a Medium-to-slow cleavage rate.

^b Positively or negatively charged side chains inhibit DAPase cleavage.

^c With a few exceptions, slow cleavage rate apply to all dipeptides containing Phe, Ile, Leu, Tyr and Trp in either of the two positions.

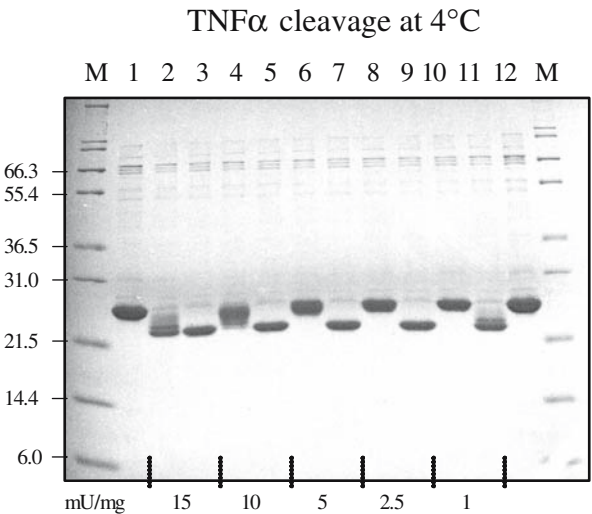


Fig. 5. DAPase/Qcyclase treatment of histidine tag (his-tag) tumor necrosis factor α (TNF α) using lower enzyme amounts and incubation at 4°C. Lane M: molecular weight marker (in kDa); lanes 1 and 12: untreated His-tag TNF α ; lanes 2, 4, 6, 8 and 10: 1-h treatment; lanes 3, 5, 7, 9 and 11: overnight treatment. The amount of DAPase per mg protein is shown.

4. Desalting is necessary to remove imidazole that would otherwise inhibit DAPase activity during tag removal.
5. DAPase requires the presence of a reducing thiol group for activity. It is thought that at physiological pH, cysteamine and its oxidized form cystamine act as a hydrogen donor for the reduction of disulfides in the enzymes. Therefore, it is recommended to use freshly prepared enzyme cocktails with cysteamine. Similarly, pGAPase bound to IMAC requires activation using cysteamine prior to running the sample and after binding of the enzyme to IMAC.
6. Enzyme activation by cysteamine is performed in small volumes to reduce the amount needed.
7. Tag sequences containing Asp or Glu can only be digested at acidic pH, while sequences containing His require pH above 6. Therefore, His-tag sequences containing Glu or Asp can only be processed at pH 6–6.5.
8. Especially for upscaling purposes, it is important to reduce the amount of enzyme used for tag removal. One approach is to run the cleavage at 4°C overnight (see Fig. 5).
9. If the target protein concentration is 1–2 mg/ml, then 1 unit pGAPase per mg of protein is recommended. For lower concentrations of target protein, higher amounts of pGAPase are required, for example, at 0.75 mg/ml, 2 units/mg pGAPase should be used.
10. The flow rate has great impact in the degree of completion of pyroglutamyl removal. It is therefore not recommended to use higher flow rates.

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III

VARIOUS APPLICATIONS OF AFFINITY CHROMATOGRAPHY

Affinity Processing of Cell-Containing Feeds Using Monolithic Macroporous Hydrogels, Cryogels

Igor Yu. Galaev and Bo Mattiasson

Summary

Monolithic macroporous hydrogels, “cryogels,” are produced by polymerization in a partially frozen state when the ice crystals perform as a porogen. Cryogels have a unique combination of properties: (i) large (10–100 μm) pores; (ii) minimal non-specific interactions due to the hydrophilic nature of the polymers; (iii) porosities exceeding 80–90%; (iv) good mechanical stability. These properties of cryogels allow for their application for direct capture of extracellularly expressed histidine-tagged protein from the fermentation broth and separation of different cell types.

Key Words: Monolithic macroporous hydrogel; cryogel; cell separation; lymphocyte fractionation; protein A; Immobilized Metal Affinity Chromatography; cell labeling.

1. Introduction

A variety of polymeric gels are used at present in different areas of biotechnology as chromatographic materials, carriers for the immobilization of molecules and cells, matrices for electrophoresis and immunoanalysis, as a gel basis for solid cultural media. Polymer gels enable us to solve numerous technical problems in biotechnology and biomedicine; however, new, often contradictory requirements for the gels are permanently emerging and stimulate the development and the commercialization of new gel materials for biological applications. One of the new types of polymer gels with considerable potential

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in biotechnology is cryogels (from the Greek κρυος (kryos) meaning frost or ice). Cryogels are produced by polymerization in a partially frozen state when the ice crystals perform as a porogen. After completing the polymerization and melting ice crystals, a system of interconnected pores is formed (1).

Cryogels have a unique combination of properties:

- Pores of 10–100 μm in size allow even large (at molecular scale) objects like microbial or mammalian cells to pass easily through the cryogel without being trapped.
- Hydrophilic nature of the polymers, which form pore walls, minimizes the non-specific interactions with the pore walls.
- High polymer concentration in the pore walls and hence a good mechanical stability of the cryogels.

The large pore size and the interconnected morphology of pores allow unhindered mass transport of solutes of practically any size. The cryogel columns have porosities exceeding 80–90%. High porosity and the interconnected morphology of the pores result in a very small flow resistance of cryogel columns. The columns can be operated at flow rates of about 750–2000 cm/h, at hydrostatic pressure approximately 0.01 MPa (2). Due to the convective flow of the mobile phase through the interconnected pores, the mass transfer resistance is practically negligible, and the height equivalent to a theoretical plate (HETP) is practically independent either of flow rate or of the size of the marker (from acetone to *Escherichia coli* cells) (3).

Mechanically, the cryogel adsorbent is very stable. The continuous matrix could easily be removed from the column, dried at 60°C and kept in a dry state. The dry matrix has a slightly smaller diameter than the swollen one and could be easily inserted inside the empty chromatographic column. After re-hydration in the running buffer which takes usually less than a minute, the cryogel column is ready for operation. The elasticity of the cryogel ensures the tight connection of cryogel monoliths with the column walls and the absence of by-pass of liquid in between the cryogel monolith and the column walls (4).

Commercially available pre-activated cryogel matrices are produced by Protista Biotechnology AB as 0.25-, 2- or 5-ml monolithic columns. The monolithic columns are made of cross-linked polyacrylamide or polydimethylacrylamide (polyDMAAm) and contain 20–30 μmole epoxy groups/ml column volume (CV).

The presence of epoxy groups allows easy coupling of a variety of ligands to monolithic cryogel columns, for example, ion-exchange ligands (5), Immobilized Metal Affinity Chromatography (IMAC) ligands (2,6), protein A and antibodies (7,8). The produced monolithic chromatographic columns have been used for the direct capture of histidine-tagged proteins from crude cell homogenate (2) and from cell fermentation broth (6), for specific isolation

of microbial (9) and mammalian (7,8) cells, inclusion bodies (10) and mitochondria (11). The use of monolithic cryogel columns will be illustrated using the example of direct capture of extracellularly expressed histidine-tagged protein from the fermentation broth (see **Subheading 3.3.**) and separation of two different cell types, namely T and B lymphocytes (see **Subheading 3.4.**).

2. Materials

2.1. Coupling IMAC Ligand, Iminodiacetic Acid

1. Na_2CO_3 , 0.5 M.
2. Na_2CO_3 , 1 M.
3. Iminodiacetic acid (IDA) solution, 0.5 M, in 1 M Na_2CO_3 , pH 10, 0.5 M CuSO_4 .
4. Ethylenediaminetetraacetic acid (EDTA), 0.1 M, pH 7.6.

2.2. Coupling Affinity Ligand, Protein A

1. Na_2CO_3 , 0.2 M.
2. Ethylenediamine solution, 0.5 M, in 0.2 M Na_2CO_3 .
3. Sodium phosphate buffer, 0.1 M, pH 7.2.
4. Glutaraldehyde solution, 5% v/v, in 0.1 M sodium phosphate buffer, pH 7.2.
5. Protein A solution, 1.6 mg/ml, in 0.1 M sodium phosphate buffer, pH 7.2.
6. NaBH_4 solution, 0.1 M, in sodium carbonate buffer, pH 9.2.
7. Bicinchoninic acid (BCA) solution for protein assay (Sigma).

2.3. Direct Capture of (His)₆-Tagged Single-Chain Fv Antibody Fragments (See Note 1)

1. Running buffer: 20 mM 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), 200 mM NaCl, 2 mM imidazole, pH 7.
2. Elution buffer: 0.2 M imidazole in 20 mM HEPES, 200 mM NaCl, pH 7.
3. Regeneration buffer: 20 mM EDTA in 20 mM HEPES, 200 mM NaCl, pH 7.
4. Charge solution: 0.25 M CuSO_4 in distilled water.
5. Feed: The starter culture of the recombinant strain of *E. coli* producing extracellular (His)₆-tagged single-chain Fv antibody fragments was grown in Luria-Bertani (LB) medium (tryptone 10 g; yeast extract 5 g and sodium chloride 5 g in 1 l distilled water, pH 7.2) containing 0.1 mg/ml ampicillin. Expression of the target protein was carried out in Terrific Broth (TB) medium (pancreatic digest of casein 12 g; yeast extract 24 g; dipotassium phosphate 9.4 g and monopotassium phosphate 2.2 g in 1 l distilled water, pH 7.2) supplemented with glycerol 4 ml/l and ampicillin 0.1 mg/ml and induced by 0.1 mM isopropyl β -D-thiogalactopyranoside at $\text{OD}_{600 \text{ nm}} = 0.5$. The batch was cultivated at 37°C for 24 h with shaking at 175 rpm. The obtained fermentation broth (turbidity 18–23 units $\text{OD}_{450 \text{ nm}}$; protein = 8–10 mg/ml) was used directly with no pretreatment.
6. Stationary support: 5 ml monolithic pre-activated cryogel column produced from polyDMAAm (Protista Biotechnology AB).

2.4. Separation of T and B Lymphocytes Using Protein A-Cryogel Monolithic Column (See Note 9)

1. Running buffer: 20 mM HEPES buffer, pH 7.4, containing 0.2 M NaCl.
2. Elution buffer: Dog IgG (30 mg/ml) in 20 mM HEPES buffer, pH 7.4, containing 0.2 M NaCl.
3. Regeneration buffer: 0.1 M glycine-HCl buffer, pH 2.5, containing 0.1 M NaCl.
4. Feed: Lymphocytes are isolated from freshly collected human buffy coat using Ficoll-Paque. The buffy coat (20 ml) is diluted with an equal volume of balanced salt solution (0.145 M Tris-HCl, pH 7.6, containing 0.1% glucose, 0.05 mM CaCl_2 , 0.98 mM MgCl_2 , 5.4 mM KCl and 14 mM NaCl). Six milliliters of the diluted buffy coat is overlaid on 5-ml Ficoll-Paque in 15-ml tissue culture plastic tube and then centrifuged at $400 \times g$ for 40 min at room temperature. The lymphocytes are collected at the interface. To minimize the contamination of red blood cells, the lymphocytes collected in the above procedure are re-centrifuged on Ficoll-Paque as above. The cells are washed twice with 10 ml of balanced salt solution and centrifuged at $200 \times g$ for 10 min. The washed lymphocytes are then suspended in balanced salt solution and used within 24 h for further experiments. Stationary support: 2 ml monolithic pre-activated cryogel produced from polyDMAAm (Protista Biotechnology AB).

3. Methods

3.1. Coupling IMAC Ligand: Iminodiacetic Acid

1. Pass 50 ml 0.5 M Na_2CO_3 followed by 50 ml 1 M Na_2CO_3 solutions through the column at a flow rate of 75 cm/h.
2. Recycle 0.5 M IDA solution in 1 M Na_2CO_3 , pH 10, for 24 h at room temperature through the column at a flow rate of 75 cm/h.
3. Wash the modified cryogel in the column with 0.5 M Na_2CO_3 (100 ml) and then with water until pH is around neutrality.
4. Load the IDA-cryogel with Cu(II) ions by passing 50 ml 0.5 M CuSO_4 (dissolved in distilled water) through the column at flow rate of 75 cm/h.
5. Determine the amount of immobilized IDA for IDA-cryogel by assaying the amount of bound copper ions at saturation assuming a stoichiometric ratio after the adsorbent is saturated with Cu(II) ions. Elute the Cu(II) ions from the column with 0.1 M EDTA, pH 7.6, and determine spectrophotometrically as absorbance of Cu(II) complex formed in 0.1 M EDTA solution, pH 7.6 at $\lambda_{\text{max}} = 730$ with $\epsilon_{730} = 46.8 \text{ M/cm}$.
6. After elution, wash the IDA-cryogel column with 100 ml water at a flow rate of 75 cm/h and then dry at 60°C overnight.
7. Insert a dry IDA-cryogel column in Pharmacia chromatographic column (i.d. of 1 cm) supplied with adapters (or any other suitable column with i.d. of 1 cm).
8. Re-swell the IDA-cryogel column in the running buffer and adjust the ends of the IDA-cryogel monolith.

3.2. Coupling Affinity Ligand: Protein A

1. Connect 2-ml cryogel column to a pump and wash with 20 ml of water at a flow rate of 1 ml/min and then with 0.2 M Na₂CO₃ (20 ml).
2. Apply ethylenediamine solution (0.5 M in 0.2 M Na₂CO₃; 30 ml) to the column at a flow rate of 75 cm/h in recycle mode for 4 h.
3. Wash with water until pH is close to neutral.
4. Wash with 20 ml, 0.1 M sodium phosphate buffer, pH 7.2.
5. Apply glutaraldehyde solution (5% v/v; in 0.1 M sodium phosphate buffer, pH 7.2, 30 ml) to the column at a flow rate of 75 cm/h in recycle mode for 5 h.
6. Recycle protein A solution (1.6 mg/ml; in 0.1 M sodium phosphate buffer, pH 7.2, 12 ml) through the column at a flow rate of 75 cm/h at 4°C for 24 h.
7. Apply the freshly prepared NaBH₄ solution (0.1 M in sodium carbonate buffer, pH 9.2, 30 ml) to the column at a flow rate of 75 cm/h for 3 h in recycle mode to reduce Schiff base formed between the protein and the aldehyde-containing matrix.
8. The amount of protein A immobilized on polyDMAAm monolithic cryogel matrix is determined by the BCA method according to a modified method given by Smith et al. (12). A suitable amount of dried protein A cryogel pieces are well suspended in water by finely grinding and ultrasonication. To different amounts of the protein A gel suspension (20–100 µl) is added 2 ml of the BCA solution, and the mixture is incubated at 37°C with thorough shaking for 30 min. The absorbance is measured at 562 nm. Appropriate controls are taken using native poly DMAAm cryogel. The standard curve is made by quantitative additions of the protein A to the native polyDMAAm cryogel and absorbance measured under the same conditions.

3.3. Direct Capture of (His)₆-Tagged Single-Chain Fv Antibody Fragments (See Note 1)

1. Wash IDA-cryogel column with 4 CV of distilled water, followed by 4 CV of 0.25 M CuSO₄ in distilled water and finally by 4 CV of distilled water (see Note 2).
2. Equilibrate column with 5 CV of 20 mM HEPES, 200 mM NaCl, 2 mM imidazole, pH 7 (see Note 3).
3. Load 1-ml sample containing non-diluted cell culture fluid containing 24–32 µg/ml His single-chain Fv at a flow rate of 300 cm/h (see Note 4).
4. Wash with 5 CV of 20 mM HEPES, 200 mM NaCl, 2 mM imidazole, pH 7, at a flow rate of 300 cm/h (see Note 5).
5. Elute with 0.2 M imidazole in 20 mM HEPES, 200 mM NaCl, pH 7, at a flow rate of 300 cm/h (see Note 6).
6. Regenerate the column with 10 CV of 20 mM EDTA in 20 mM HEPES, 200 mM NaCl, pH 7. Store column at 4°C, preferably in the presence of antimicrobial agent (see Note 7).
7. Analyze eluted fractions for protein content, for example, using BCA assay according to ref. 12, and for the content of target protein (see Note 8).

3.4. Separation of T and B Lymphocytes Using Protein A-Cryogel Monolithic Column (See Note 9)

1. Equilibrate protein A-cryogel column with 10 CV of 20 mM HEPES buffer, pH 7.4, containing 0.2 M NaCl (*see Note 10*).
2. Treat lymphocytes (1 ml, $2-4 \times 10^7$ cells/ml) with 50 μ l (0.1 μ g/ μ l) of goat anti-human IgG(H+L) by incubating at 4°C for 15 min. Centrifuge the cells at $200 \times g$ for 10 min and re-suspend in 1 ml of balanced salt solution (*see Note 11*).
3. Apply the antibody-treated lymphocytes to the top of the column and let 1.5 ml of liquid to flow through before allowing cells to run completely into the monolithic column bed. Close the column outlet and allow cells to bind efficiently to the matrix by incubating the column at room temperature for 10 min without any buffer flow (*see Note 12*).
4. Apply 20 ml of HEPES buffer, pH 7.4, containing 0.2 M NaCl through the column at a flow rate of 110 cm/h. Collect first 4 ml (*see Note 13*).
5. Apply 2 ml of dog IgG (30 mg/ml) to the column and incubate at 37°C for 1 h. Apply 4 ml more of dog IgG (30 mg/ml) and collect the eluted fraction (*see Note 14*).
6. Regenerate the column with 10 CV of 0.1 M glycine-HCl buffer, pH 2.5, containing 0.1 M NaCl at a flow rate of 110 cm/h. Store column at 4°C (*see Note 15*).
7. Analyze the breakthrough and eluted fractions for the content of particular cell lines (*see Note 16*).

4. Notes

1. General comments on protein purification using traditional IMAC adsorbents (*see Chapters 2 and 10*) are applicable to the IMAC purification of histidine-tagged proteins directly from crude extracts or fermentation broth using IMAC cryogels.
2. IDA-cryogel column washing and all the following steps are operated at a flow rate of 12 ml/min (600 cm/h). This step is carried out to charge the column with Cu(II) ions and washout all non-bound Cu(II) ions.
3. A small concentration of imidazole, 2 mM, in the running buffer favors washing loosely bound Cu(II) ions and prevents non-specific binding of impurities to Cu(II)-IDA ligands.
4. Do not exceed total load of 30 μ g of His-tagged protein per 5 ml monolithic IMAC-cryogel column. The feed loaded on the column could contain cell debris or even the whole cells as the pores in the monolithic column are big enough to allow for the free passage of particulate material through the column without blocking the flow. Moreover, due to large pores, the flow resistance of the column is very low allowing the use of flow rates as high as 600 cm/h without deteriorating the column performance.
5. This step is carried out to wash cells and unbound soluble impurities. The cell content is monitored by measuring absorbance at 450 nm.

6. The bound His-tagged proteins are usually eluted from IMAC-cryogel columns within 2 CV. The collection of 1–2 ml fractions is recommended.
7. The regeneration with EDTA strips the column from Cu (II) ions. The regenerated column if needed could be cleaned in place with 3–5 CV of 0.2 M NaOH followed by washing with distilled water till neutrality. The regenerated column is ready for re-charging with Cu (II) ions. Make sure that antimicrobial agent is washed out properly before re-charging column with Cu (II) especially when sodium azide is used as antimicrobial agent as sodium azide forms strong complexes with Cu (II) ions.
8. The content of target protein in the eluted fractions could be analyzed either by assaying the biological activity of the target protein (e.g., enzymatic activity) or using immunoanalysis such as ELISA.
9. The separation of individual cell types is based on the specific interaction of one cell type with the affinity ligands covalently coupled to the cryogel monolithic column and hence adsorption of this cell type to the cryogel column. The other type(s) of cell incapable of specific interaction with the coupled ligand pass non-retained, through the column due to the large interconnected pores. Bound cells are recovered from the column. The success of cell separation is mainly determined by the selection of an affinity ligand capable of selective recognition of the given cell type. Antibodies could be developed against numerous specific targets present on the surface of cells of a particular cell type. The cells specifically labeled with antibody are discriminated from non-labeled cells via binding to protein A ligands. Protein A presents a ligand capable of selective binding to Fc fragments of many types of IgG antibodies. Fc fragments are not involved in specific recognition of the target by antibodies, hence when the cells are specifically labeled with antibodies, the Fc fragments of the antibodies remain free for the interaction with protein A.
10. The buffer composition is selected in order to favor the interactions of protein A ligands with Fc fragments of antibodies used for specific cell labeling.
11. At this step, B lymphocytes are specifically labeled with antibodies, whereas T lymphocytes remain non-labeled. Non-bound antibodies are removed by centrifugation and re-suspension of lymphocytes.
12. Due to the large size of cells as compared to protein molecules, the kinetics of cell binding is relatively slow, and some time is required to achieve efficient binding of antibody-labeled cells to protein A ligands. T and B lymphocytes are very fragile, so low flow rates should be used to maintain the viability of cells. Two-milliliter protein A-cryogel column retains about 5×10^7 B lymphocytes.
13. This fraction contains non-bound cells, predominantly T lymphocytes.
14. When high concentration of dog IgG is added, IgG molecules start to compete for binding to protein A ligands with already bound antibody-labeled cells. Slowly the desorption of bound B lymphocytes takes place. Due to the slow kinetics of the desorption process, long incubation time of about 1 h is needed.
15. Protein A is a stable ligand and harsh regeneration conditions like pH 2.5 are not detrimental for its performance. On the other hand, harsh regeneration conditions

ensure elution of bound dog IgG and killing the residual cells which were not washed or eluted from the column.

16. As an initial step in monitoring the binding and recovery of the cells on the column, the absorbance measured at 470 nm (turbidity) of the fractions obtained from the column, could be determined. The viability of the initial cell load, the cells collected in the breakthrough fractions and cells in the eluted fractions were checked using the Trypan blue dye exclusion method (13). The dead cells stained dark blue and could be differentiated from the live cells, which remained unstained. Alternatively, the cells could be labeled with fluorescent conjugated antibodies followed by sorting and counting in a flow cytometer like FACSscan (Becton-Dickinson).

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Monolithic Bioreactors for Macromolecules

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Summary

Enzymes immobilized on solid-phase matrices have found various applications in biotechnology, molecular biology and molecular diagnostics and can serve as industrial catalysts and as specific reagents for analytical procedures. A wide range of supports have been utilized for immobilization among which particle-based supports are the most commonly implemented. Type of support used for immobilization is one of the key considerations in practical application due to different immobilization efficiency, ligand utilization and the mass transfer regime. The mass transfer between the mobile and the particulate stationary phase is often a bottleneck for the entire process due to slow pore diffusion of large molecules. In contrast, monoliths due to their structure enable almost flow-independent properties. Consequently, the overall behavior of the immobilized ligand reflects its intrinsic reaction kinetics. Therefore, such an immobilized system is expected to allow higher throughput because of higher enzyme efficiency, especially pronounced for macromolecular substrates having low mobility. In this work, different methods for immobilization of enzymes on Convective Interaction Media monolithic supports are presented. In particular, enzymes acting on macromolecular substrates, such as trypsin, deoxyribonuclease and ribonuclease, are described in detail. Immobilized efficiency is evaluated for different immobilization procedures in terms of biologic activity and long-term stability. Finally, their performance on real samples is demonstrated.

Key Words: Immobilization; monoliths; CIM; deoxyribonuclease; ribonuclease; trypsin.

1. Introduction

Enzymes immobilized on solid-phase matrices have found various applications in biotechnology, molecular biology and molecular diagnostics and can serve as industrial catalysts and as specific reagents for analytical procedures. The advantages of using immobilized enzymes instead of an enzyme solution include increased stability and an opportunity to work with a continuous system over long periods of time. A wide range of supports have been utilized for immobilization among which particle-based supports are the most commonly implemented. The type of support used for immobilization is one of the key considerations in practical application due to different immobilization efficiency, ligand utilization and mass transfer regime. The mass transfer between the mobile phase and the stationary phase has a pronounced effect on the performance. In the case of particulate porous supports, the substrate has to diffuse from the mobile phase into the pores in order to reach the catalytic sites of the immobilized enzyme. Because the diffusion, especially for large molecules, is commonly slower than the reaction process at the active site, the overall kinetic behavior of the immobilized enzyme is governed by mass transfer, causing a decrease in efficiency.

To overcome this drawback, a new group of supports called monoliths was introduced (**1**). Contrary to conventional stationary phases that are in the form of a few micrometer particles, monoliths are made of a single piece of porous material. Pores are highly interconnected forming a channel network through which the mobile phase flows. As the main transport mechanism is convection, mass transfer resistance can be neglected under operating conditions. Consequently, the overall behavior of the immobilized ligand reflects its intrinsic reaction kinetics. Therefore, such an immobilized system is expected to allow higher throughput because of higher enzyme efficiency, especially pronounced for macromolecular substrates having low mobility.

Among different types of monoliths, methacrylate-based monoliths were most frequently used for immobilization of various ligands. As such, they were used either as an affinity support for purification of target compounds (**2,3**) or as bioreactors (**2,4,5**). In this work, some examples of macromolecular bioreactors based on Convective Interaction Media (CIM)[®] (CIM is a registered trademark of BIA Separations, Ljubljana, Slovenia) supports (methacrylate-based monoliths) are presented.

2. Materials

1. CIM Convective Interaction Media[®] epoxy groups containing poly (glycidyl methacrylate-co-ethylene dimethacrylate) monolithic columns (BIA Separations) with a diameter of 12 mm and thickness of 3 mm (monolith volume 0.34 ml) having median pore size of approximately 1.5 or 6 μm (CIM epoxy disk).

2. CIM Convective Interaction Media[®] imidazole carbamate-activated groups containing poly (glycidyl methacrylate-co-ethylene dimethacrylate) monolithic columns (BIA Separations) with a diameter of 12 mm and thickness of 3 mm (monolith volume 0.34 ml) having median pore size of approximately 1.5 or 6 μm (Carbonyl diimidazole (CDI) CIM disk).
3. Trypsin from bovine pancreas, type XI, lyophilized powder, ≥ 6000 , N α -benzoyl-L-arginine ethyl ester hydrochloride (BAEE) units/mg protein (Sigma, Taufkirchen, Germany).
4. Deoxyribonuclease I (DNase I) from bovine pancreas (Sigma).
5. Highly polymerized calf thymus DNA (0.006–0.08 g/l) (Sigma).
6. Ribonuclease A (RNase A) (Sigma).
7. BAEE, 3×10^{-4} M (Sigma).
8. Cytidin-2,3-cyclic monophosphate (Sigma).
9. BCA protein assay (Sigma).
10. Benzamidine hydrochloride, 50 mM.
11. Tris-HCl buffer, 50 mM, pH 9.
12. Borate buffer, 0.1 M, pH 8.
13. Tris-HCl buffer, 20 mM, pH 8.
14. Acetate buffer, 50 mM, pH 5, containing 1 mM CaCl_2 .
15. Tris-HCl buffer, 50 mM, pH 7 and 9, containing 1 mM CaCl_2 .
16. Tris-HCl buffer, 10 mM, pH 7.5, 2 mM EDTA, 0.1 M NaCl.
17. Tris-HCl buffer, 40 mM, pH 8, 1 mM MgCl_2 , 1 mM CaCl_2 , 0.1 M NaCl.
18. Tris-HCl buffer, 40 mM, pH 8, 1 mM MgCl_2 , 1 mM CaCl_2 .

2.1. Equipment

1. HPLC system KNAUER (Berlin, Germany).

3. Methods

Methods described below outline (i) static and dynamic immobilization method on CDI and epoxy-activated monolith (*see Subheading 3.1.*); (ii) determining biological activity of immobilized enzymes (*see Subheading 3.2.*); (iii) immobilization of DNase (*see Subheading 3.3.*), RNase (*see Subheading 3.4.*) and trypsin (*see Subheading 3.5.*); and (iv) application of DNase or RNase reactor for removal of DNA or RNA from sample (*see Subheading 3.6.*).

3.1. Immobilization Methods

Two methods, static (*see Subheading 3.1.1.*) and dynamic (*see Subheading 3.1.2.*), could be used for enzyme immobilization (*see Note 1*). The monolith has a disk shape (CIM disk) and can be immobilized as such or when placed in a CIM housing forming CIM disk monolithic column (**6,7**).

3.1.1. Static Immobilization Method

1. Place CIM disk into CIM housing to obtain CIM disk monolithic column.
2. Connect CIM disk monolithic column to HPLC system and equilibrate it by washing with at least 5 column volumes of water and at least 5 column volumes of immobilization buffer (*see Note 1*).
3. Prepare 3 ml of a protein solution (2 g/l) by dissolving the protein in appropriate immobilization buffer.
4. Remove CIM disk from the CIM housing and immerse it into 3 ml of the immobilization solution (*see Fig. 1A*).
5. Incubate CIM disk in the immobilization solution at given temperature for a defined period of time (*see Note 1*).
6. After immobilization is completed, place CIM disk again into CIM housing, connect CIM disk monolithic column (named further “enzyme reactor”) to HPLC system, remove the residual non-bound protein by washing the enzyme reactor with 10 column volumes of immobilization buffer containing 0.1 M NaCl and finally with deionized water.
7. Disconnect enzyme reactor from HPLC system, remove CIM disk from CIM housing and store immobilized CIM disk at 4°C either in water, 20% ethanol, or suitable buffer.

3.1.2. Dynamic Immobilization Method

1. Place CIM disk into CIM housing to obtain CIM disk monolithic column.
2. Connect a syringe filled with water to one side of the CIM disk monolithic column.
3. Equilibrate CIM disk monolithic column by pushing with a syringe at least 5 column volumes of water (~2 ml), fill the syringe with immobilization buffer and wash the column with at least 5 column volumes.
4. Prepare 3 ml of a protein solution (2 g/l) by dissolving the protein in appropriate buffer and fill the syringe with it.

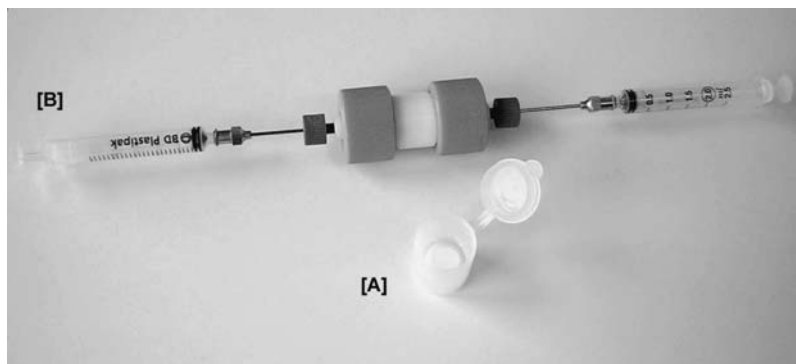


Fig. 1. Static (A) and dynamic (B) immobilization method.

5. Connect the filled syringe to CIM disk monolithic column from one side and an empty syringe to the other side (*see Fig. 1B*).
6. Push the immobilization solution through the CIM disk monolithic column by pressing filled syringe and leave empty syringe free so the solution passing through the CIM disk is collected into it. Repeat this procedure in regular time intervals of 15 min.
7. After immobilization is completed, disconnect one syringe, exchange the solution in a syringe and remove the residual protein by washing the CIM monolithic column (named further “enzyme reactor”) with 10 column volumes of immobilization buffer containing 0.1 M NaCl and finally with deionized water.
8. Remove immobilized CIM disk from the CIM housing and store it at 4°C either in water, 20% ethanol, or suitable buffer.

3.2. Biological Activity

Immobilization efficiency was determined via measurement of biological activity and amount of immobilized enzyme. If the biological activity is manifested as a change in absorbance at designated wavelength, on-line frontal analysis could be used to determine biological activity of immobilized enzyme.

3.2.1. On-line Frontal Analysis

1. Place immobilized CIM disk into CIM housing to obtain enzyme reactor.
2. Connect the enzyme reactor to the HPLC system.
3. Pump the reagent solution stream carrying substrate at constant temperature through enzyme reactor at different flow rates. When the substrate solution at a certain concentration is pumped through the enzyme reactor, substrate is hydrolyzed which results in an increase of absorbance at the column outlet that becomes constant when the system is in equilibrium (*see Fig. 2A*).
4. Plot absorbance values at the outlet against residence time (*see Fig. 2B*). The residence time of substrate inside the enzyme reactor is calculated from the flow rate of substrate and pore volume of monolith, (which is approximately 60% of monolith volume being 0.197 ml (6–8)) by the following equation:

$$t = \frac{V}{\Phi}, \quad (1)$$

where t = residence time; V = pore volume of the monolith; Φ = flow rate.

5. Express biological activity as a change of absorbance per minute (dA/dt) at low residence time or as substrate consumption per minute knowing the calibration curve absorbance versus substrate concentration (*see Fig. 2B*).
6. Calculate specific biological activity from biological activity divided by amount of immobilized enzyme (*see Subheading 3.2.2*).
7. Kinetic parameters can be calculated as described in **Note 2**.

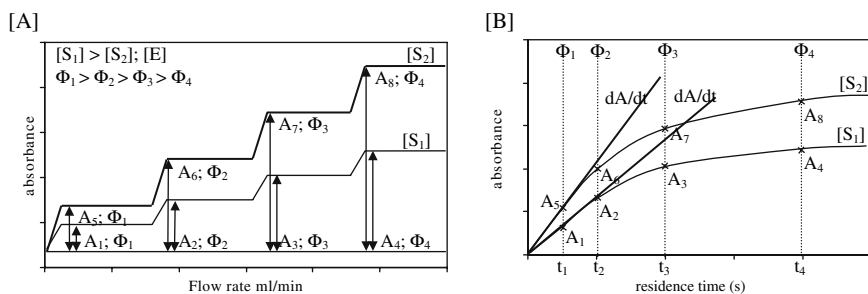


Fig. 2. Schematic presentation of results obtained by on-line frontal analysis of biological activity. (A) Absorbance of substrate passed through an enzyme reactor at different flow rates. (B) Absorbance of substrate at calculated residence time. $[S_1]$ and $[S_2]$, concentration of substrate; $[E]$, amount of enzyme immobilized to CIM disk; Φ_{1-4} , flow rates; A_{1-8} , absorbance calculated as difference between absorbance of substrate passed through enzyme reactor and of substrate passed through CIM disk monolithic column of the same chemistry (epoxy or CDI) but without enzyme.

3.2.2. Amount of Immobilized Enzyme Using BCA Kit

Quantity of enzyme immobilized on the CIM disk was determined from a difference in enzyme concentration in the immobilization solution before and after immobilization using BCA protein determination kit according to the manufacturer's instructions.

3.2.3. Stability of Enzyme Reactor

Stability of enzyme reactor was determined by monitoring biological activity regularly for prolonged periods of time. For all measurements, experimental conditions had to be identical.

3.3. DNase Immobilization

The static and dynamic immobilization methods were used to immobilize DNase on CIM disk via epoxy groups (9). The efficiency of DNase immobilization was determined by hydrolysis of DNA as substrate, as described in **Subheading 3.3.2**. Different immobilization conditions like temperature, pH and immobilization time were tested (conditions are described in **Table 1**).

Immobilized DNase activity is presented in **Table 1** and long-term stability of enzyme reactor in **Table 2**. The highest specific biological activity of immobilized enzyme was detected for immobilization on epoxy groups at pH 7 and at 22°C (see **Table 1**). Immobilized CIM disk stored in buffer had better long-term stability compared to the one stored in water (see **Table 2**). The apparent (see **Note 3**) Michaelis–Menten constant, k_m^{app} , and turnover number, k_3^{app} , were,

Table 1
Effect of Temperature, Immobilization Time and pH on Deoxyribonuclease (DNase) Immobilization.

Temperature (°C)	Time (h)	pH value	Immobilization method	Biological activity (dA _{260 nm} /min)	Amount of enzyme (mg DNase/g support)	Specific biological activity (dA _{260 nm} /min/mg)
37	3	5	Static	0.1	4.3	0.15
37	24	5	Static	1.9	9.4	1.26
37	3	7	Static	0.1	1.9	0.33
22	24	7	Static	1.2	3.4	2.21
22	0.5	7	Dynamic	0.9	2.9	1.94
22	2	7	Dynamic	1.2	3.5	1.96
37	24	7	Static	1.32	5.0	1.65
37	24	9	Static	0	5.6	0

DNase (2 g/l) in 50 mM Tris, pH 7 or 9, or 50 mM acetate buffer, pH 5, containing 1 mM CaCl₂ was immobilized on a CIM epoxy disk, median pore size 6 µm. The amount of immobilized enzyme was determined as described in **Subheading 3.2.2**. Biological activity was determined as described in **Subheading 3.3.2**. DNA concentration was 0.02 g/l in 40 mM Tris buffer, pH 8, 1 mM MgCl₂, 1 mM CaCl₂ and detection wavelength 260 nm. Specific biological activity is expressed as DNase activity per mg of DNase. Adapted from ref (9).

respectively, 0.28 g of DNA/l and 16 dA_{260 nm}/min/mg of immobilized DNase (see **Fig. 3**) (see **Note 4**). The long-term stability of enzyme reactor was determined by measuring biological activity (see **Subheading 3.2.3**) immediately after immobilization and repeatedly for up to 1 month.

The immobilization procedure to obtain the highest biological activity of the immobilized enzyme and measurement of biological activity are presented below.

3.3.1. Immobilization Procedure

1. Prepare DNase solution by dissolving enzyme (2 g/l) in 50 mM acetate buffer, pH 5, containing 1 mM CaCl₂.
2. Apply static immobilization procedure described in **Subheading 3.1.1** for 24 h at 37°C.
3. After immobilization is completed, wash the enzyme reactor first with 40 mM Tris-HCl buffer, pH 8, containing 1 mM MgCl₂, 1 mM CaCl₂, 0.1 M NaCl, followed by 40 mM Tris-HCl buffer, pH 8, containing 1 mM MgCl₂, 1 mM CaCl₂ buffer.
4. Immobilized CIM disk should be stored in immobilization buffer to better preserve biological activity (see **Table 2**).
5. Determine quantity of immobilized enzyme as described in **Subheading 3.2.2** if specific biological activity is of interest.

Table 2
Long-Term Stability of Deoxyribonuclease Enzyme Reactor Stored Either in Water or in Buffer at 4°C.

Days	Biological activity (dA _{260nm} /min)	% initial activity
Water		
0	0.041	100
2	0.023	57
4	0.020	49
8	0.011	26
Buffer		
0	0.118	100
1	0.105	89
2	0.103	87
3	0.098	83
6	0.089	75
8	0.089	75
13	0.034	29
21	0.021	18
31	0.012	10

Biological activity was determined as described in **Subheading 3.3.2**. DNA concentration was 0.02 g/l in 40 mM Tris buffer, pH 8, 1 mM MgCl₂, 1 mM CaCl₂ and detection wavelength 260 nm.

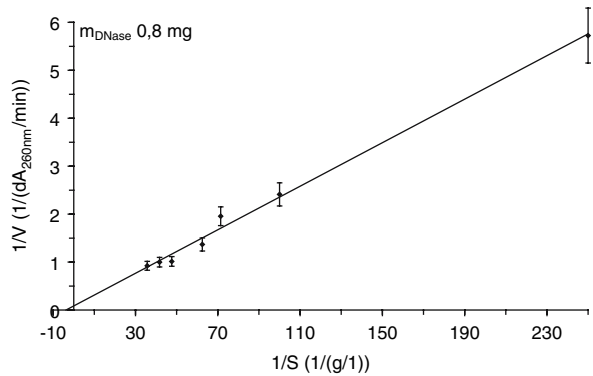


Fig. 3. Double reciprocal ($1/v$ versus $1/[S]$) Lineweaver–Burk plot of immobilized deoxyribonuclease (DNase) (9). The intercept with x-axis represents $-1/K_m$ and intercept with y-axis represents $1/v_{max}$. The biological activity of immobilized DNase is determined by on-line frontal analysis as described in **Subheading 3.3.2**. Enzyme reactor: CIM disk, median pore size 6 μ m. Chromatographic conditions: flow rates 0.1–10 ml/min, calf thymus DNA 0.006–0.08 g/l in 40 mM Tris-HCl buffer, pH 8, 1 mM MgCl₂, 1 mM CaCl₂, detection wavelength 260 nm.

3.3.2. Biological Activity: DNase

The modified Kunitz hyperchromicity assay (**10**) was used to determine DNase biological activity. The DNase activity is manifested as an increase in absorbance at 260 nm.

1. Prepare 40 mM Tris–HCl buffer, pH 8, containing 1 mM MgCl₂, 1 mM CaCl₂ (buffer A).
2. Prepare substrate of calf thymus DNA at concentration of 0.006–0.08 g/l in buffer A.
3. Connect DNase enzyme reactor to the HPLC system.
4. Set the wavelength on HPLC detector at 260 nm for monitoring the substrate conversion.
5. Equilibrate enzyme reactor by washing it with at least 10 column volumes of buffer A.
6. Set to zero HPLC detector to compensate background absorbance of buffer A.
7. Pump different substrate solutions of calf thymus DNA at 25°C through the enzyme reactor and change the residence time by altering the flow rate in the range of 0.1–10 ml/min. When the substrate solution at a certain concentration is pumped through the enzyme reactor at fixed flow rate, immobilized DNase has been hydrolyzing DNA which results in an increase of the absorbance at the column outlet.
8. Read the steady-state absorbance for each flow rate (*see Subheading 3.2.1*).
9. Draw a graph showing absorbance at 260 nm versus the residence time (*see Fig. 2A*).
10. The DNase biological activity (v) is determined as the slope of the linear increase in absorbance at low residence time, and specific biological activity is calculated from biological activity divided by the amount of immobilized enzyme.
11. Changing the substrate concentration enables calculation of kinetics parameters $v_{\max}$ and K_m using Michaelis–Menten equation (*see Note 2*) (*see Fig. 3*).

3.4. RNase Immobilization

The dynamic immobilization procedure (*see Subheading 3.1.2.*) was used to immobilize RNase on CIM disk via epoxy and imidazole carbamate groups. The efficiency of RNase immobilization was determined by hydrolysis of the low molecular weight substrate cytidine-2,3-cyclic monophosphate as described in **Subheading 3.4.2**. Immobilization was performed at different pH values of immobilization buffer as indicated in **Table 3** and described for optimal case in **Subheading 3.4.1**.

Biological activity of immobilized RNase is presented in **Table 3**. RNase immobilized on CDI-activated monolith at pH 9 was six-fold more active than the one immobilized on epoxy-activated monolith (*see Table 3*). Furthermore, there was almost no change in activity over 42 days (*see Table 4*). The

Table 3

Effect of Buffer pH on Ribonuclease (RNase) Immobilization via Epoxy or Carboxydiimidazole Groups

pH of immobilization buffer	RNase biological activity (dA _{288 nm} /min)	Amount of enzyme (mg RNase/ disk)	Specific biological activity (dA _{288 nm} /min/mg)
Epoxy			
5	14	0.6	24
7	25	1.1	23
9	44	0.7	65
11	75	0.9	83
13	8	0.7	11
CDI			
7	341	1.3	262
9	349	0.7	499
11	16	0.9	17

RNase (2g/l) in 50 mM Tris buffer, pH 7 and 9, or 50 mM acetate buffer, pH 5, or 50 mM sodium carbonate buffer, pH 11, or 50 mM KCl NaOH buffer, pH 13, was immobilized on a Connective Interaction Media (CIM) epoxy or CIM CDI disk, median pore size 6 μ m. The amount of immobilized enzyme was determined as described in **Subheading 3.2.2**. Biological activity was determined as described in **Subheading 3.4.2**. Cytidine-2,3-cyclic monophosphate concentration was at 0.57 mM in 10 mM Tris-HCl pH 7.5, 2 mM EDTA, 0.1 M NaCl buffer and detection wavelength 288 nm. Specific activity was expressed as RNase activity per mg of RNase.

Michaelis–Menten constant, K_m , and turnover number, k_3 , for immobilized RNase are 0.52 mM and 4.6 per second (see **Fig. 4**). The stability of RNase enzyme reactor was determined as described in **Subheading 3.2.3**.

The immobilization procedure to obtain the highest biological activity and measurement of biological activity are presented below.

3.4.1. Immobilization Procedure

1. Prepare RNase solution by dissolving enzyme (2 g/l) in 50 mM Tris buffer pH 9 (immobilization buffer).
2. Apply dynamic immobilization procedure described in **Subheading 3.1.2** for 2 h at room temperature.
3. After immobilization is completed, wash the enzyme reactor first with immobilization buffer containing 0.1 M NaCl and with immobilization buffer afterwards.
4. Immobilized CIM disk should be stored in 20% ethanol solution to preserve biological activity (see **Table 4**).
5. Determine quantity of immobilized enzyme as described in **Subheading 3.2.3** if specific biological activity is of interest.

Table 4**Long-Term Stability of Ribonuclease Enzyme Reactor Stored in 20% Ethanol at 4°C.**

Days	Biological activity (dA _{288 nm} /min)	% initial activity
epoxy-CIM		
0	75	100
7	57	76
28	58	77
CDI-CIM		
0	349	100
14	342	98
42	343	98

Biological activity was determined as described in Subheading 3.4.2. Cytidine-2,3-cyclic monophosphate concentration was at 0.57 mM in 10 mM Tris, pH 7.5, 2 mM EDTA, 0.1 M NaCl buffer and detection wavelength 288 nm.

3.4.2. Biological Activity-RNase

The biological activity of immobilized RNase (**II**) was determined by on-line frontal analysis using cytidine-2,3-cyclic monophosphate as substrate. The initial velocity was calculated as the slope of linear increase in absorbance at 288 nm ($\epsilon_{288 \text{ nm}} = 1308/\text{M}/\text{cm}$) of cytidine-2,3-cyclic monophosphate at low residence time.

1. Prepare 10 mM Tris-HCl, pH 7.5, 2 mM EDTA, 0.1 M NaCl buffer (buffer A).
2. Prepare substrate cytidine-2,3-cyclic monophosphate at concentration of 0.39–0.68 mM in buffer A.
3. Connect RNase enzyme reactor in to the HPLC system.
4. Set the wavelength on HPLC detector at 288 nm for monitoring the substrate conversion.
5. Equilibrate enzyme reactor by washing it with at least 10 column volumes of buffer A.
6. Set to zero HPLC detector to compensate background absorbance of buffer A.
7. Pump different substrate solutions through the enzyme reactor and change the residence time by altering the flow rate in the range of 0.1–10 ml/min at 25°C. When the substrate solution at a certain concentration is pumped through the enzyme reactor at fixed flow rate, immobilized RNase has been hydrolyzing cytidine-2,3-cyclic monophosphate which results in an increase of the absorbance at the column outlet.

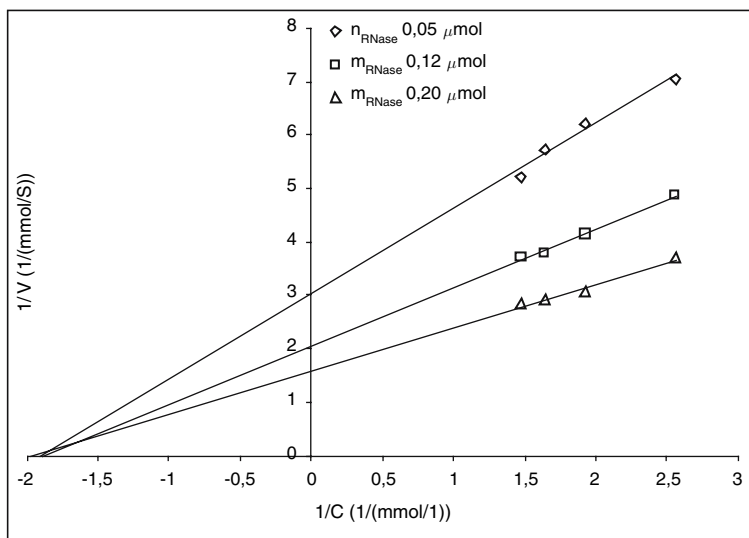


Fig. 4. Double reciprocal ($1/v$ versus $1/[S]$) Lineweaver–Burk plot of immobilized ribonuclease (RNase). The intercept with x-axis represents $-1/K_m$ and intercept with y-axis represents $1/v_{max}$. The biological activity of immobilized RNase is determined by on-line frontal analysis as described in **Subheading 3.4.2**. Enzyme reactor: CIM disk, median pore size 6 μm . Chromatographic conditions: flow rates 0.2–1 ml/min, cytidine-2,3-cyclic monophosphate (0.39–0.68 mM) in 10 mM Tris-HCl, pH 7.5, 2 mM EDTA, 0.1 M NaCl buffer, detection wavelength 288 nm ($\epsilon_{288 \text{ nm}} = 1308 \text{ M/cm}$).

8. Read the steady-state absorbance for each flow rate (see **Subheading 3.2.1**).
9. Draw a graph showing absorbance at 288 nm versus the residence time (see **Fig. 2A**).
10. The RNase biological activity is determined as a slope of the linear increase in absorbance at low residence time. Specific biological activity is calculated from biological activity divided by amount of immobilized enzyme.
11. Changing the substrate concentration enables calculation of kinetics parameters v_{max} and K_m using Michaelis–Menten equation (see **Note 2**) (see **Fig. 4**).

3.5. Trypsin Immobilization

Immobilization of trypsin was performed on CIM disk via epoxy or imidazole carbamate groups by the dynamic or static immobilization procedure with and without benzamidine hydrochloride (see **Note 5**) as indicated in **Table 5 (7)**.

Table 5
Effect of Immobilization Procedure on Biological Activity of Trypsin Enzyme Reactor

Monolith	Static method		Immobilization time ^c (min)				
	Trypsin ^a (mAU/min)	Trypsin ^b (mAU/min)	5	30	60 (mAU/min)	120	1440
Epoxy	1005	7066	115	436	1937	1981	5883
CDI	11530	11222	8430	5312	6329	7987	8369

^a Immobilization performed without benzamidine hydrochloride.

^b Immobilization performed with benzamidine hydrochloride.

^c Immobilization of trypsin under dynamic conditions.

Trypsin (2 g/l) in 0.1 M borate buffer pH 8 with or without 50 mM benzamidine hydrochloride was immobilized on CIM epoxy or CDI CIM disk, median pore size 1.5 µm. Biological activity was determined as described in **Subheading 3.5.2**. Hydrolysis of Nα-benzoyl-L-arginine ethyl ester at concentration of 3×10^{-4} M in 20 mM Tris-HCl buffer pH 8 at wavelength of 254 nm was monitored. Adapted from ref (7).

The efficiency of trypsin immobilization could be determined by hydrolysis of high molecular weight substrates (**12**) or low molecular weight substrates, for example, BAEE (**7,8**), as described in **Subheading 3.5.2**.

Biological activity of immobilized trypsin at different conditions is presented in **Table 5**. Immobilization efficiency via imidazole carbamate groups is 10 times higher than those obtained for epoxy groups if the immobilization procedure was performed without addition of benzamidine hydrochloride (see **Table 5**). Dynamic immobilization method was completed in 120 min while static immobilization method lasted 24 h (see **Table 5**). Furthermore, biological activity of trypsin enzyme reactor was not changed over 2 years (see **Table 6**).

The immobilization procedure to obtain the highest biological activity and measurement of biological activity are presented below.

3.5.1. Immobilization Procedure

1. Prepare 0.1 M borate buffer, pH 8, containing 50 mM benzamidine hydrochloride (immobilization buffer).
2. Prepare trypsin solution 2 g/l by dissolving the trypsin in immobilization buffer.
3. Apply dynamic immobilization procedure described in **Subheading 3.1.2** at room temperature for 5 min.
4. After immobilization is completed, the residual protein is removed by washing the enzyme reactor with 10 column volumes of immobilization buffer and finally with deionized water.

Table 6
Long-Term Stability of Trypsin Enzyme Reactor Stored in Water at 4°C

Days	Biological activity (mAU/min)	% initial activity
epoxy-CIM		
1	7111	100
99	7101	100
190	6972	98
220	8114	114
687	6615	93
CDI-CIM		
1	9994	100
99	10960	100
190	9388	94
220	10676	107
687	10540	105

Biological activity was determined as described in Subheading 3.5.2. Digestion of N α -benzoyl-arginine ethyl ester at concentration of 3×10^{-4} M in 20 mM Tris-HCl buffer pH 8 at wavelength of 254 nm is monitored.

5. Immobilized CIM disk should be stored at 4°C in distilled water to preserve biological activity.
6. Determine quantity of immobilized enzyme as described in **Subheading 3.2.2** if specific biological activity is of interest.

3.5.2. Biological Activity: Trypsin

The biological activity of the immobilized trypsin is determined by on-line frontal analysis using low molecular substrates BAEE (7,8).

1. Prepare 20 mM Tris-HCl buffer, pH 8 (buffer A).
2. Prepare the substrate solution of BAEE at concentration of 3×10^{-4} M in buffer A.
3. Connect trypsin enzyme reactor to the HPLC system.
4. Set the wavelength on HPLC detector at 254 nm for monitoring substrate conversion.
5. Equilibrate enzyme reactor by washing with at least 10 column volumes of buffer A.
6. Set to zero HPLC detector to compensate background absorbance of buffer A.

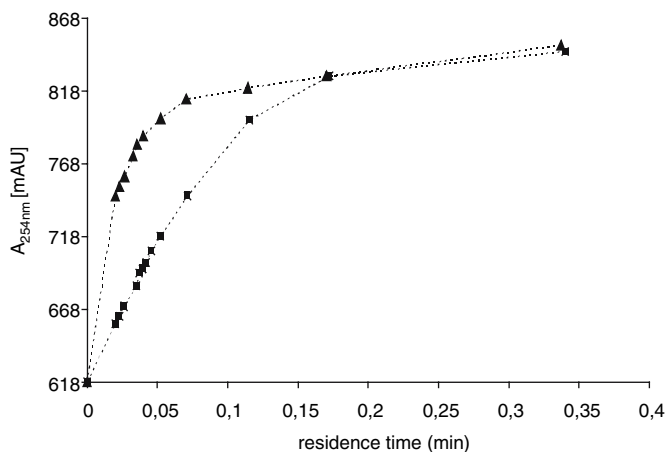


Fig. 5. The biological activity of trypsin immobilized via epoxy (■) or imidazole carbamate (▲) monolith (7). Enzyme reactor: CIM disk, median pore size 1.5 μm . Chromatographic conditions: flow rate 0.1–20 ml/min, N α -benzoyl-L-arginine ethyl ester concentration 3×10^{-4} M in 20 mM Tris-HCl buffer, pH 8, detection wavelength 254 nm

7. Pump the BAEE solution through the enzyme reactor and change residence time by altering the flow rate in the range of 0.2–18 ml/min.
8. Read the steady-state absorbance for each flow rate (*see Subheading 3.2.1.*).
9. Draw a graph showing absorbance at 254 nm versus residence time (*see Fig. 2*).
10. The slope of the linear increase in absorbance at low residence time is a measure for biological activity (*see Fig. 5*).

3.6. Use of Enzyme Reactor

To remove either DNA or RNA contaminants from RNA or DNA isolates, respectively, samples are often treated with specific nucleases that are removed from the reaction mixture by phenol extraction when reaction is completed. To omit unnecessary purification steps that represent a danger for contamination of samples, immobilized nucleases (RNase or DNase) could be used to remove DNA and RNA in different samples.

Use DNA or RNA sample (200 μl) obtained according to the procedure described elsewhere (13) dissolved in TE buffer containing 10 mM Tris, 1mM EDTA, pH 7.6 and passed through the DNase or RNase enzyme reactors at flow rate of 0.1 ml/min. For a control experiment, an enzyme reactor was replaced with a CIM disk monolithic column of the same chemistry but without enzyme (epoxy or CDI). The process was monitored with UV detection at wavelength

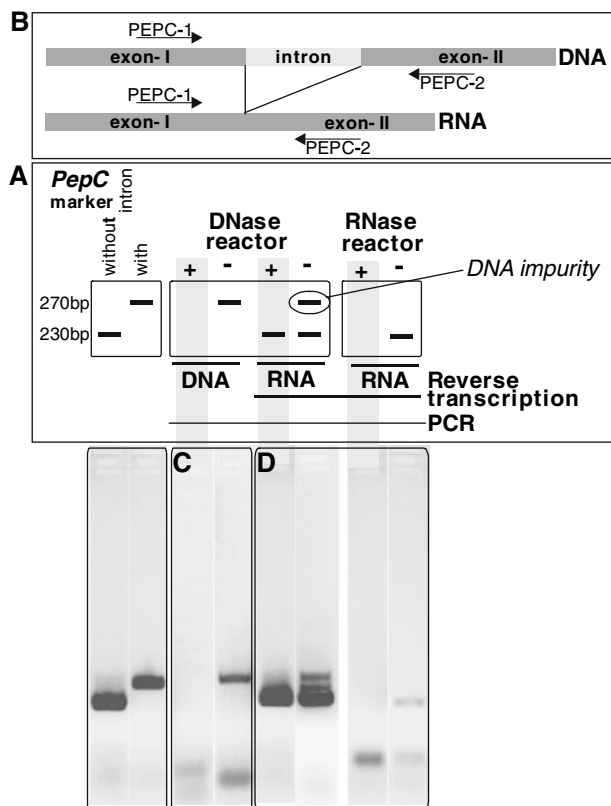


Fig. 6. (A) In order to distinguish between RNA and DNA, primers used in PCR and RT-PCR were chosen to anneal at different exons of DNA, which causes that PCR fragment of DNA is 40 base pairs larger than RT-PCR product of RNA. (B) Schematic presentation of either PCR products of DNA or RT-PCR products of RNA and DNA passed enzyme reactor [deoxyribonuclease (DNase) or ribonuclease (RNase)] or Convective Interaction Media disk monolithic column of the same chemistry (epoxy or CDI) but without enzyme. (C) The PCR products of genomic DNA isolated from *Aspergillus niger* passed through DNase reactor and control reactor. (D) The RT-PCR products of total RNA isolated from *A. niger* passed through DNase or RNase reactor and control. Simultaneously with samples, PCR or RT-PCR was performed on plasmids containing *PepC* gene with and without intron, and presence of PCR products was determined together with the PCR or RT-PCR products of the sample. Products (10 μ l) were loaded on 1.6% agarose gel stained with ethidium bromide. Flow rate was 0.1 ml/min.

260 nm. Hydrolyzed DNA/RNA was collected at outlet of CIM disk monolithic column, and RT-PCR or PCR was performed as described in the literature (14,15). Products of RT-PCR and PCR were analyzed with gel electrophoresis (16).

The results of RT-PCR and PCR on DNA or RNA hydrolyzed by enzyme reactors are presented in Fig. 6.

4. Notes

1. Choice of immobilization method highly depends on (i) activated groups of monolith, (ii) immobilization conditions and (iii) enzyme to be immobilized.
2. Changing the substrate concentration enables calculation of kinetics parameters $v_{\max}$ and K_m using Michaelis–Menten equation (see Fig. 2).

$$v = v_{\max} \cdot \frac{[S]}{K_m + [S]}, \quad (2)$$

where v = initial velocity (the instantaneous velocity, $d[P]/dt$ at given substrate concentration); $v_{\max}$ = maximum biological activity; $[S]$ = fixed substrate concentration; K_m = Michaelis–Menten constant.

$$v_{\max} = k_3 \times [E], \quad (3)$$

where $[E]$ = amount of immobilized enzyme; k_3 = rate constant for the breakdown of enzyme substrate complex, turnover constant, catalytic rate constant.

3. Apparent values are used as absolute values cannot be determined due to the nature of the polymeric substrate, the unknown number and type of different substrate binding sites, and the unclear relationship between the absorbance signal and the actual catalytic events.
4. K_m and k_3 values of free DNase were 0.07 g/l and 76 dA_{260 nm}/min/mg, respectively (9).
5. Addition of a benzamidine hydrochloride in immobilization buffer prevents undesired autodigestion of trypsin.

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Plasmid DNA Purification Via the Use of a Dual Affinity Protein

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Summary

Methods are presented for the production, affinity purification and analysis of plasmid DNA (pDNA). Batch fermentation is used for the production of the pDNA, and expanded bed chromatography, via the use of a dual affinity glutathione S-transferase (GST) fusion protein, is used for the capture and purification of the pDNA. The protein is composed of GST, which displays affinity for glutathione immobilized to a solid-phase adsorbent, fused to a zinc finger transcription factor, which displays affinity for a target 9-base pair sequence contained within the target pDNA. A Picogreen™ fluorescence assay and/or an ethidium bromide agarose gel electrophoresis assay can be used to analyze the eluted pDNA.

Key Words: Plasmid DNA; affinity purification; fermentation; chromatography; expanded bed adsorption.

1. Introduction

One of the central challenges in delivering vaccines and gene therapy products is to find a vector that is able to safely introduce the product to the target cells (1). The use of viral vectors has been questioned due to safety and regulatory concerns over their toxicity and immunogenicity (2). This led to the study of plasmid deoxyribonucleic acid (plasmid DNA (pDNA)) as a non-viral gene therapy expression vector, which has the dual advantages of being free from specific safety concerns associated with viruses and generally simpler to develop (3). In medical therapy, pDNA may be used to treat monogenic

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diseases, cancer and infectious diseases. The potential use of pDNA in vaccines has also been shown (**4–8**) through the expression of specific antigens on cell membranes that help to stimulate the immune system's response and memory. As a result of these findings, there has been an increasing demand on the biotechnology industry to supply purified pDNA for gene therapy, vaccine and research applications.

The contaminants that pose a particular problem in the production of purified pDNA are anionic polymers of a similar structure, charge and physical behavior to pDNA. These contaminating anionic polymers include genomic DNA (gDNA), RNA and lipopolysaccharides or endotoxins (**9**). Current commercial pDNA purification techniques typically require at least three chromatographic stages to remove all of these afore-mentioned contaminating species and to meet the evermore demanding purity levels required by customers (e.g., <0.1 endotoxin unit/ μ g pDNA, <1% g DNA w/w pDNA, <0.5% RNA w/w pDNA, >95% w/w of the pDNA in supercoiled form). Affinity chromatography offers a method to obtain highly pure, supercoiled pDNA in a single chromatographic unit operation. Chase (1998) (**10**) identified an overwhelming need for novel affinity ligands for use in nucleic acid purification.

The proposed affinity purification mechanism utilizes a dual affinity GST fusion protein that displays affinity for both immobilized glutathione ligands and a very specific binding region contained with the target pDNA. This would enable the use of an identical purification strategy that is not dependent upon the physico-chemical characteristics of the target pDNA, but rather relies upon the presence of the binding region.

Presented are methods for the production and purification of pDNA without the co-purification of contaminating anionic polymers (gDNA, RNA and endotoxins) in a single unit operation of affinity chromatography. The optical density at 280 nm ($OD_{280\text{ nm}}$) shows the presence of material that adsorbs light at that wavelength but cannot be used to determine the concentrations of particular biomolecules in the column exit stream (e.g., pDNA concentration in elution fractions that also contain protein) and does not give information about the form of the pDNA (i.e., supercoiled versus open circular versus linear). Off-line analysis of the pDNA can be performed using a PicogreenTM fluorescence assay or ethidium bromide agarose gel electrophoresis. The Picogreen fluorescence assay is used to accurately and quickly determine the concentration of double-stranded DNA (dsDNA) in the presence of contaminants due to selective intercalation and fluorescence of Picogreen with dsDNA (**11**). Ethidium bromide agarose gel electrophoresis enables analysis of the form of the eluted pDNA and a method to determine if contaminating gDNA and RNA

are present in the elution fractions. With an appropriate gel analysis system and via comparison to DNA markers, the concentration of pDNA in the elution fractions can also be calculated via densitometry studies.

2. Materials

2.1. Biomolecules

The target pDNA, named pTS, is a pUC19 plasmid that has had a zinc finger binding domain inserted into the *Sma*I site. Hence, the pTS plasmid is a molecule of dsDNA 2715 base pairs in size. The pUC19 plasmid (accession number L09137) has historically been used for general cloning (*12*) and has ampicillin resistance as its method of selection. DNA sequencing of pTS confirmed that the zinc finger binding domain sequence was present. The plasmid molecule has a molecular weight of approximately 1800 kDa. The pTS plasmid was produced by Dr. David Palfrey at the Department of Pharmaceutical Sciences, Aston University (UK), and was kindly supplied by Dr. Anna Hine. The other biomolecules used in this work (pM6, GST-ZnF and glutathione) are described in Chapter 9.

2.2. Buffers and Reagents

Where required, use 1 M HCl or 1 M NaOH to adjust the buffer pH.

1. Growth media: Terrific broth containing 12 g tryptone, 24 g yeast extract, K_2HPO_4 12.5 g, 2.3 g KH_2PO_4 in 1 L of deionized (DI) water, pH 7.
2. Phosphate-buffered saline (PBS): PBS is used as the equilibration and running buffer. The buffer can be prepared by dissolving a PBS tablet in 200 ml of DI water to yield a buffer containing 10 mM phosphate buffer, 2.7 mM potassium chloride and 137 mM sodium chloride, pH 7.4.
3. Cell resuspension solution: 50 mM Tris-HCl, 10 mM ethylenediamine tetraacetic acid (EDTA), pH 7.5.
4. Cell lysis solution: 0.2 M NaOH, 1% sodium dodecyl sulfate.
5. Neutralization solution: 1.32 M potassium acetate, pH 4.8.
6. Elution buffer: 20 mM reduced glutathione ($\geq 99\%$, MW 307), 100 mM Tris-HCl, pH 9 (prepare elution buffer on day to be used as glutathione should be stored at 4°C).
7. High pH adsorbent regeneration buffer: 0.1 M Tris-HCl, 0.5 M NaCl, pH 8.5.
8. Low pH adsorbent regeneration buffer: 0.1 M sodium acetate, 0.5 M sodium chloride, pH 4.5.
9. Adsorbent storage buffer: 20% v/v ethanol (Sigma-Aldrich), 80% PBS.
10. Tris-EDTA (TE) buffer: 10 mM Tris-HCl, 1 mM EDTA, pH 7.
11. Tris-acetate-EDTA (TAE) buffer 50 $\times$ stock: 242 g Tris, 57.1 ml glacial acetic acid, 9.3 g EDTA, total volume adjusted to 1 l with DI water. Dilute the 50 $\times$ stock to 1 $\times$ buffer on the day of use.

12. Sample loading buffer: 50% v/v glycerol, 0.25% w/v Bromophenol Blue in 1× TAE.

3. Methods

3.1. Bacterial Fermentation for Plasmid DNA Production

1. Prepare an inoculum of growth media that is 10% v/v that of the final fermentation volume. Pick a freshly transformed colony of cells and grow the inoculum overnight (~16 h) at 37°C and 200 rpm on a shaker incubator in an unbaffled shake flask. To ensure good aeration, use a shake flask that is at least 2.5 times the volume of the cell broth (*see Note 1*).
2. Fill the fermenter vessel with growth medium and autoclave for 30 min at 121°C (*see Note 2*).
3. Attach the vessel to a fermenter control unit to maintain the required process parameters (*see Note 3*).
4. Once the fermentation culture has cooled to less than 60°C, add 50 µg/ml of antibiotic (where an antibiotic resistance marker exists), 0.1% v/v polypropylene glycol (organic antifoam) and 1% w/v glucose aseptically (*see Note 4*).
5. Set the dissolved oxygen (DO) level (*see Note 5*).
6. Before adding the inoculum, check that the OD_{600 nm} reading of the inoculum is above 1.5 and preferably above 4 before adding to the fermentation vessel (*see Note 6*). Add the inoculum when the medium temperature and pH readings obtain the levels set at step 3.
7. After fermentation is complete (*see Note 7*), remove the cell broth from the vessel and harvest the cells by centrifuging at 5000 × g (5300 rpm in a JA-10 centrifuge) for 10 min at room temperature.
8. A clarified cell lysate can be prepared immediately or the cell pellet can be used stored at -80°C until further use. Where required, cell pellets can be resuspended in PBS buffer before storage (i.e., if pellets need to be removed from centrifuge tubes).

3.2. Preparation of Clarified Cell Lysis

1. Add 3 ml of cell resuspension solution for every 100 ml of pelleted cell culture. If cells were stored in PBS buffer, centrifuge at 5000 × g (5300 rpm in a JA-10 centrifuge) for 10 min in a room temperature rotor and then pour off the supernatant before resuspending in the cell resuspension solution.
2. Add 3 ml of cell lysis solution for every 100 ml of pelleted cell culture and gently mix by inverting several times. Cell lysis is complete when the solution becomes clear and viscous (*see Note 8*).
3. Add 3 ml of neutralization solution for every 100 ml of pelleted cell culture and gently mix by inverting several times.
4. Centrifuge the solution at 14 000 × g (8900 rpm in a JA-10 centrifuge) for 15 min in a room temperature rotor (*see Note 9*).

5. Decant the clarified cell lysate (pDNA containing supernatant) into a clean container. Prepare cell lysate on the day it is to be used.

3.3. Expanded Bed Adsorption Plasmid DNA Purification

1. Load a chromatography column via gravity settling with the adsorbent prepared as described in Chapter 9 (*see Subheading 3.1.*).
2. Equilibrate the column with at least 10 settled bed column volumes of PBS buffer using upward flow to expand the column. Expand the bed to twice its settled bed height. In a 1-cm diameter column, a flow rate of approximately 150 cm/h is required to expand the bed to twice its settled bed height.
3. Using upward flow, load GST–ZnF-containing lysate into the column. Some column expansion should be expected due to the higher density and viscosity of the feed. To prevent loss of adsorbent through the top of the column, the flow may need to be reduced or the position of the top column frit adjusted.
4. Wash the column with at least 5 settled bed column volumes of PBS buffer. Ensure that the OD_{280 nm} of the column outlet stream returns to base-line levels.
5. Still in expanded mode, load the pDNA containing clarified cell lysate into the column, followed by 5 settled bed column volumes of PBS buffer to wash the column.
6. Reverse the flow to downward flow and lower the top adaptor. Continue washing with PBS until the OD_{280 nm} of the column outlet stream returns to base-line levels.
7. Elute the GST–ZnF–pTS complex in packed bed mode with elution buffer and collect the elution fractions for off-line analysis via ethidium bromide agarose gel electrophoresis and Picogreen assays (*see Note 10*).

3.4. Affinity Adsorbent Regeneration

1. After elution is complete, signified by a stable OD_{280 nm}, reverse the flow to the upward flow direction and expand the column to twice its settled bed height using high pH adsorbent regeneration buffer. Pump 5 settled bed column volumes of high pH adsorbent regeneration buffer through the column.
2. Still in expanded bed mode, pump 5 settled bed column volumes of low pH adsorbent regeneration buffer through the column.
3. Repeat steps 1 and 2 a further two times or until no more material is eluted from the affinity adsorbent, which is shown by a stable OD_{280 nm} for the column outlet stream (*see Note 11*).
4. Wash the column with 5 bed volumes of PBS.
5. For long-term storage (i.e., several weeks or more), wash the column with 5 bed volumes of adsorbent storage buffer and store at 4°C.

3.5. Picogreen Fluorescence Assay

1. Mix one part of Picogreen as supplied with 199 parts of TE buffer to produce a Picogreen working solution (*see Note 12*).

2. Mix 100 μ l of sample, 100 μ l of the Picogreen working solution from step 1 above and 1800 μ l of TE buffer.
3. Allow the Picogreen to intercalate with the dsDNA for 30 s at room temperature.
4. Take a fluorescence reading at an excitation wavelength of 480 nm and an emission wavelength of 520 nm.
5. Compare fluorescence readings for samples with those of a calibration curve constructed using known concentrations of pDNA (supercoiled form) to determine the concentration of dsDNA in the sample. Some dilution of the sample may be required in order for the fluorescence reading to be within the linear range as determined by the calibration curve.

3.6. Ethidium Bromide Agarose Gel Electrophoresis

1. Prepare the agarose gel by mixing 1 $\times$ TAE buffer with 1.0% w/v agarose followed by boiling to dissolve the agarose and homogenize the solution (*see Note 13*).
2. Allow the agarose solution to cool to below 60°C, then add ethidium bromide to a concentration of 0.5 μ g/ml of gel (*see Note 14*).
3. Pour the gel into a cast with a toothed comb to create the wells and allow to set.
4. Carefully remove the toothed comb from gel, then remove the gel from the cast and place into the electrophoresis apparatus.
5. Pour 1 $\times$ TAE buffer into the electrophoresis apparatus tank until the gel is just covered.
6. Add 20% by volume sample loading buffer to each sample before loading into the wells of the gel.
7. Run gels at 60 V for a minimum of 1 h or until the required resolution between the bands had been obtained (*see Note 15*).
8. Photograph the gel using an appropriate gel documentation system (*see Note 16*).

4. Notes

1. A colony of DH5a *Escherichia coli* cells transformed with pTS were grown overnight in 200 ml of inoculum media in a 2 l unbaffled shake flask. This cell line was selected for the production of pDNA as it is relatively easy to transform with pDNA, is well characterized and displays a high copy number (**12**). A high copy number means that compared to other strains of bacteria, the number of pDNA molecules that it produces per cell is high.
2. A 2 l working volume Applicon fermentation vessel linked to an Applicon ADI 1010 Bio Controller was used.
3. A cell culture temperature of 37°C was maintained via a water jacket and a pH of 7 by use of 3 M NaOH and 3 M HCl additions.
4. The pTS plasmid confers ampicillin resistance to transformed *E. coli*.
5. DO was controlled to 30% of the maximum DO level by altering the agitation speed (rpm) and compressed air or O₂ addition. The DO probe was calibrated by running 4 l/min of pure O₂ through the system at an agitation speed of 800 rpm.

Initially, compressed air at 4 l/min was supplied to the vessel. The gas was changed from compressed air to pure O₂ when the system failed to maintain a DO level of 30% at the maximum agitation speed of 800 rpm.

6. It is good practice to check the OD_{600 nm} reading of the inoculum to ensure that the transformed colony has indeed grown. A low cell concentration in the inoculum results in a long lag phase of cell growth and is an inefficient use of time and resources. The preparation of more than one inoculum will assist in the success of the bacterial fermentation protocol.
7. The length of a fermentation run is dependent upon the system used: cell line, pDNA, inoculum used, temperature, medium, pH, DO and so on. Generally, for the system described above, a lag time of approximately 2 h was witnessed before the system entered the exponential growth phase. Smaller inoculum volumes extend the initial lag time. OD_{600 nm} readings of up to 23.9 were obtained after 24 h of fermentation; however, fermentation runs of this length are not necessarily required as the maximum volumetric yield of supercoiled pDNA can be obtained as soon as 10 h after inoculum addition.
8. The length of time for this step is dependent upon the cell concentration. Cell lysis is normally complete after 5–10 min. Leaving the solution too long may result in the yield, and/or supercoiled nature of the pDNA being compromised as the pDNA is then unable to renature upon neutralization.
9. This step removes the precipitated floc formed after neutralization. The majority of host cell-derived contaminants (gDNA, proteins and cell debris) precipitate to form fragile salt aggregated flocs after neutralization. The advantages of alkaline lysis are that it has a high capacity for cell-derived contaminant removal and is fully scalable. Care must be taken to prevent high shear during lysis as this results in a lower yield of supercoiled pDNA and fragmentation of gDNA.
10. Protein and reduced glutathione will be present in this eluted product. The pDNA and free fusion protein elute at different rates, so this enables some removal of free fusion protein from the fractions that contain the highest concentration of pDNA. EDTA is a very potent zinc-chelating agent. EDTA treatment (2 mM) leads to irreversible denaturation and aggregation of the zinc-binding domain that cannot be restored by addition of an excess of zinc (**13**). If further removal of protein and reduced glutathione from the pDNA is required, incubate the elution fractions in a solution containing 2 mM EDTA, then separate the denatured protein and reduced glutathione from the pDNA using size exclusion chromatography or a buffer exchange method.
11. The binding capacity of the affinity adsorbent can be affected by the accumulation of precipitate, denatured or nonspecifically bound proteins that are not removed by the relatively mild high and low pH adsorbent regeneration buffers. Precipitated and/or denatured substances can be removed by washing with 2 column volumes of 6 M guanidine hydrochloride followed by washing with 5 column volumes of PBS. Hydrophobically bound substances can be removed by washing with 4 column volumes of 70% v/v ethanol followed by washing with 5 column volumes of PBS (**14**).

12. Safety Warning: Picogreen must be treated as a potential mutagen as it binds with nucleic acid, so must be handled with appropriate care. It is recommended to use double gloves when handling the stock solution. Picogreen reagent should be poured through activated charcoal before disposal. The charcoal must then be incinerated to destroy the dye.
13. Safety Warning: Be careful when opening bottles containing heated agarose gel as the solution can become superheated and inflict burns. The solution will initially be cloudy when the agarose is suspended in the buffer, and then becomes clear once the agarose has dissolved.
14. Safety Warning: Ethidium bromide is a potential carcinogen and mutagen. Always wear gloves when handling ethidium bromide and equipment that may have been in contact with ethidium bromide.
15. The gel can be run at a higher voltage (i.e., 100 V), however, the resolution of the gel may be compromised and the DNA may be degraded under high temperatures.
16. Ensure that your eyes and skin are adequately protected from sources of UV light.

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Affinity Chromatography of Phosphorylated Proteins

Grigoriy S. Tchaga

Summary

This chapter covers the use of immobilized metal ion affinity chromatography (IMAC) for enrichment of phosphorylated proteins. Some requirements for successful enrichment of these types of proteins are discussed. An experimental protocol and a set of application data are included to enable the scientist to obtain high-yield results in a very short time with pre-packed phospho-specific metal ion affinity resin (PMAC).

Key Words: Phosphorylated proteins; immobilized metal ion affinity chromatography; ferric protein purification.

1. Introduction

Protein phosphorylation is a highly important mechanism for signal transduction in eukaryotic cells, and there are examples of phosphorylation events occurring in prokaryotic organisms as well (1–5).

Signal transduction, transcriptional regulation, and cell division are just three examples of the many metabolic processes regulated by the phosphorylation and dephosphorylation of proteins by kinases and phosphatases. Despite the broad use of phosphorylation to regulate cellular processes, only a small percentage of all cellular proteins are phosphorylated at any given time (6–7).

The target proteins are prevalently phosphorylated on side chains that contain a hydroxyl group, such as serine, threonine, and tyrosine residues. However, an increasing number of examples of histidine phosphorylation have also been described (4). Abundance of the four different phosphorylated side chains in

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proteins is variable. However, phosphohistidine is estimated to be 10- to 100-fold more abundant than phosphotyrosine, but less abundant than phosphoserine and phosphothreonine (8).

The only currently viable method for enrichment of the complete phosphoprotein complement is immobilized metal ion affinity chromatography (IMAC) with hard metal ions.

IMAC was introduced by Porath and coworkers (9) in 1975 under the name of metal chelate affinity chromatography. This short publication reported for the first time the use of immobilized zinc and copper metal ions for the fractionation of proteins from human serum.

The classical system cited by most scientists in the IMAC field is that of Pearson (10), who postulated that metal ions can be divided into three categories according to their preferential reactivity with nucleophiles: hard, intermediate, and soft. To the group of hard metal ions belong Fe^{3+} , Ca^{2+} , and Al^{3+} all of which have a preference for oxygen.

Hundreds of papers have been published since, describing the use of immobilized hard metal ions in group separations of phosphorylated proteins, and the future of this particular application field looks very bright indeed (11–24). These adsorbents are also finding broad application for enrichment of phosphorylated peptides (25–29).

In this chapter, an outline is presented of a typical experimental protocol that ensures reproducible and quantitative enrichment of all phosphorylated proteins with exposed phosphorylated side chains.

When attempting to enrich the phosphorylated proteins from any given biological sample, one needs to take into consideration the following issues:

1. Phosphorylation–dephosphorylation processes are generally quick processes. Important consideration must therefore be given to the time for extraction, loading of the sample and the initial washes (30). Speedy removal of phosphatases is important as the presence of phosphatase inhibitors such as sodium ortho-vanadate, might be undesirable during the chromatography.
2. In general, gaining an as complete as possible enrichment is more important than obtaining a higher purification factor that results in losses of phosphorylated proteins in the non-adsorbed fraction (*see Table 1 and Fig. 1* for typical yields of phosphorylated proteins from different sources). It is clear, therefore, that further reduction of complexity has to occur after this first step (before one would be able to identify and quantify the individual phosphorylated proteins from the total proteome).
3. Selective and complete enrichment of the total phosphorylated proteome is impossible under native conditions. A simple example is the formation of homodimeric and heterodimeric Stat protein complexes upon their phosphorylation and transport to the nucleus (31,32). In this case, a phosphorylated side chain of tyrosine is involved in the formation of the Stat protein dimers. Accordingly, this

Table 1
Typical Yields of Enriched Phosphoprotein From Various Cell Lines

Cell line	Loaded (mg)	Non-adsorbed (mg)	Washes (mg)	Eluate (mg)	Eluate, % of loaded
HEK 293	2.5	1.9	0.23	0.41	16
Jurkat	3.3	2.4	0.30	0.52	16
Cos-7	3.1	2.4	0.26	0.47	15
NIH 3T3	2.7	1.9	0.21	0.45	17
HeLa	3.4	2.5	0.24	0.46	14

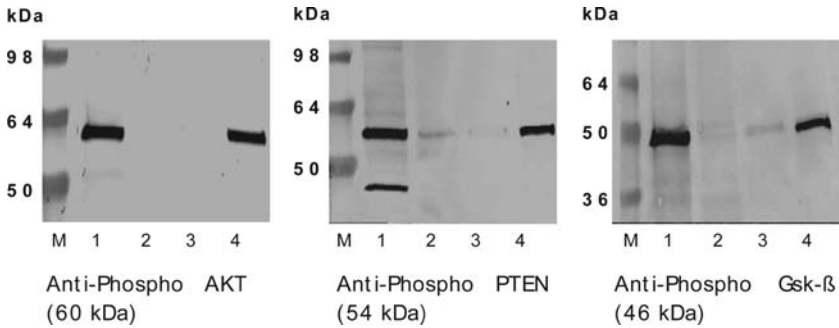


Fig. 1. Western blot data for three phosphoproteins from HEK 293 enriched using PMAC Phosphoproteins Enrichment Kit (Cat. no. 635624) according to the protocol on page 288. Fractions from PMAC chromatography were run on SDS gel, transferred to PVDF membrane, and stained with phospho-peptide specific antibodies for the three proteins.

MW-Marker

Lane 1: Original Sample (total protein loaded on the column)

Lane 2: Flow through

Lane 3: Washes

Lane 4: Eluate

Western blot data for phosphoproteins from HEK 293 enriched using phosphoprotein resin and buffers given in protocol for running sample on phosphoprotein column. Samples from the column were analyzed by Western blotting using phosphoprotein-specific antibodies. Phosphorylated proteins were clearly detected in the eluate fraction.

residue is buried and is not exposed for binding to the adsorbent. In our group, we have observed that native phosphorylated Stat1 cannot bind to a number of phosphotyrosine-specific antibodies (unpublished data).

2. Materials

1. Phosphoprotein Enrichment Kit (Clontech Cat. no. 635624). The kit comes with the following reagents and materials suitable for six purifications.
 - Six Phosphoprotein Affinity Columns (1 ml, disposable).
 - 220 ml Buffer A (Extraction/Loading Buffer)—Clontech proprietary buffer.
 - 45 ml Buffer B (Elution Buffer)—20 mM sodium phosphate, 0.5 M sodium chloride, pH 7.2.
2. 2-ml microcentrifuge tubes.
3. 5-ml screw-cap centrifuge tubes.
4. pH meter or pH paper.
5. Micropipettor.
6. BCA Protein Assay Reagent Kit (Pierce Biotechnology, Rockford, IL, USA)—provides a detergent-compatible BCA reagent for quantifying total protein (*see Note 1*). Required for tissue extraction:
7. Mortar and Pestle.
8. Alumina (Sigma, St. Louis, MO, USA). The following materials may be required depending on your purification:
9. Sterile Syringes and syringe filters (0.45 μ m) for filtering lysates.
10. Phosphatase Inhibitors (if phosphatase inhibitors are desired).
11. Sodium orthovanadate (1–2 mM).
12. Sodium fluoride (10–50 mM).
13. Gel Filtration Column (for phosphatase inhibitor removal or buffer exchange). PD-10, (GE Healthcare, Piscataway, NJ, USA).
14. Microconcentrators for sample concentration (optional).
15. Millipore 4-ml centrifugal filter and tube (Millipore) and
16. Millipore 0.5-ml centrifugal filter and tube (Millipore).

3. Methods

The protocol outlined below covers the experimental setup when using Clontech's phospho-specific metal ion affinity resin (PMAC) Phosphoproteins Enrichment Kit (Clontech, Palo Alto, USA).

This kit has been developed with the goal to enrich as great an amount of phosphorylated proteins in as quick a time as possible, reducing unwanted de-phosphorylation and/or proteolysis by running the purification at 4°C (**Fig. 2**).

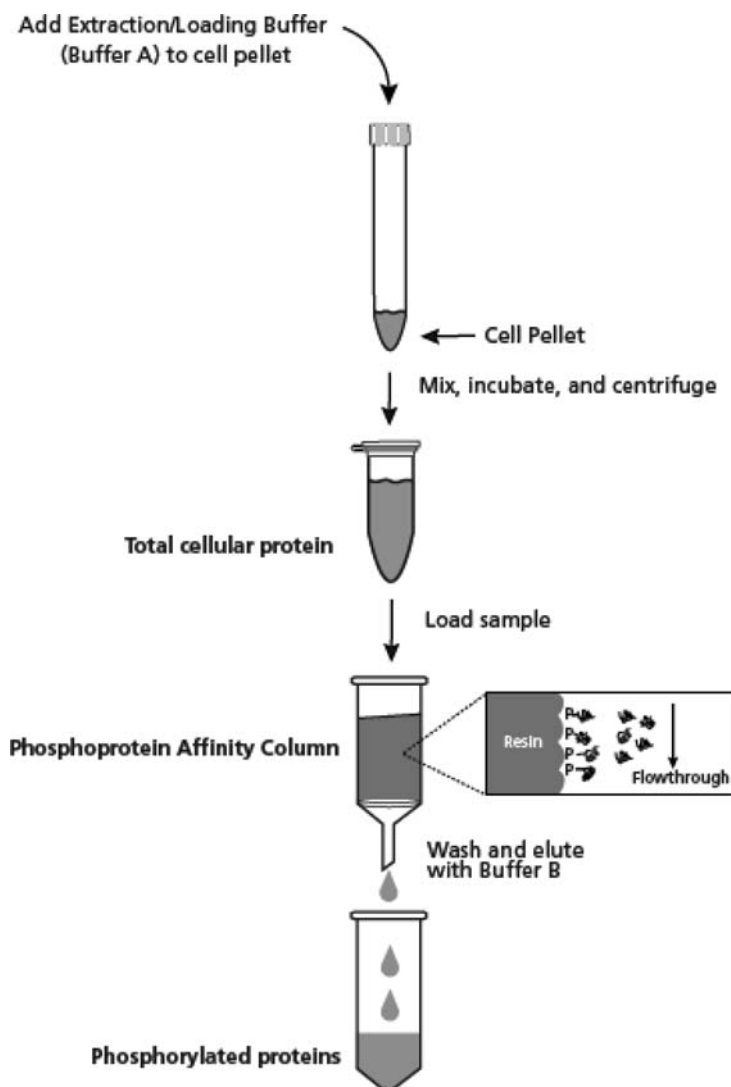


Fig. 2. Overview of the purification procedure with Clontech's PMAC Phosphoproteins Enrichment Kit.

3.1. Extracting Proteins from Cells

1. Wash 50–150 mg of cells three times with 20 vol of phosphate-buffered saline (PBS) by centrifuging at $500 \times g$ in a pre-weighed centrifuge tube (*see Note 2*).
2. After washing, centrifuge cells as above and then decant the supernatant and aspirate the residual liquid.

3. Centrifuge the tube again (for ~2 min) and aspirate any residual traces of liquid. Reweigh the tube to determine the weight of the cell pellet.
4. Freeze your samples by placing them in liquid nitrogen or in a -80°C freezer.
5. Re-suspend the cell pellet (~100 mg) in 30 μl of Buffer A for each mg of cells (see **Note 3**).
6. Disperse the pellet by gently pipetting up and down approximately 20 times.
7. Incubate at 4°C for 10 min, inverting the tube every minute during incubation. Transfer the cell lysate to a microcentrifuge tube.
8. Centrifuge the cell extract at $10,000 \times g$ for 20 min at 4°C to remove insoluble material (see **Note 4**).
9. Transfer the supernatant to a clean tube without disturbing the pellet. This is the starting clarified sample used in the PMAC chromatography.
10. Reserve a small portion of the clarified sample at 4°C for phosphate, protein, and other analysis (see **Note 5**). Proceed to **Subheading 3.3**.

3.2. Extracting Protein from Crude Tissue

1. Before starting, chill the following items on ice or at 4°C .
 - 5 ml Buffer A.
 - one mortar & pestle.
 - two 2-ml microcentrifuge tubes.
 - one 5-ml tube.
2. Transfer 100–200 mg of frozen tissue to a pre-chilled mortar.
3. Add 0.25–0.5 g of Alumina to the mortar.
4. Use the pestle to grind the tissue until a paste is formed.
5. Add 2 ml of pre-chilled Buffer A.
6. Mix the buffer into the paste using the pestle. When complete, use a micropipette tip or sterile instrument to scrape any paste that adheres to the pestle back into the mortar.
7. Transfer the extract to a pre-chilled 2-ml microcentrifuge tube.
8. While holding the pestle over the mortar, rinse the pestle with 2 ml of Buffer A pre-chilled at 4°C .
9. Combine the rinse with the original extract in a 2-ml tube. (Use a second 2-ml tube if the volume exceeds the tube's capacity.)
10. Centrifuge the suspension at $10,000 \times g$ and 4°C for 20 min (see **Note 6**).
11. While taking care not to disturb the pellet, transfer the supernatant to a pre-chilled 5-ml tube.
12. Gently invert the tube to mix the lysate (see **Note 7**).
13. Reserve a small portion of the clarified sample at 4°C for phosphate, protein, and other analysis. Proceed to **Subheading 3.3**: Column Enrichment (see **Note 8**).

3.3. Column Enrichment

1. Allow the column to stand at room temperature in an upright position until the resin settles out of suspension.

2. Remove the column top cap and then the end cap, and allow the storage buffer to drain out until it is flush with the top of the Resin bed.
3. Wash the column with 5 ml of distilled water or 5 column volumes (5 CVs).
4. Add 5 ml (5 CVs) of pre-chilled at 4°C Buffer A to equilibrate the column and allow the buffer to flow through.
5. Repeat **step 4** once.
6. Collect and measure the pH of the last 2 ml of flow through. If the pH is not less than or equal to 6.0, then continue washing with Buffer A.
7. Close the column with the end cap.
8. Add your clarified sample to the column (*see Note 9*).
9. Close the column with the top cap.
10. Gently agitate column with sample at 4°C for 20 min on a platform shaker to allow the phosphorylated proteins to bind to the column (*see Note 10*).
11. Let the column stand for 5 min in the upright position to allow the resin to settle out of suspension (*see Note 11*).
12. Remove the column top cap and then the end cap and allow non-adsorbed material to flow through. Collect the non-adsorbed material, if analysis of non-phosphorylated proteins is necessary.
13. Wash the column by adding 5 ml (5 CVs) of Buffer A and allowing it to flow through under gravity.
14. Repeat this wash three more times for a total of 4×5 ml washes.
15. Add 1 ml of Buffer B (elution buffer) and collect the fraction on ice.
16. Repeat **step 15** four times with 1 ml of Buffer B each time (collect fractions every time). Store all fractions on ice immediately. Note: The enriched phosphorylated proteins are generally present in the second and third fractions—approximately 2 ml of elution volume.
17. Run a BCA analysis to determine protein concentration in the cell extract as well as the eluted fractions (**5**). Eluted fractions 2 and 3 will most likely have the highest concentration of phosphorylated protein.

Multiple downstream steps can be applied for additional complexity reduction such as 2D-Gel Electrophoresis or Multidimensional LC/MS-MS (MuD LC/MS-MS). One possible intermediate step is group-specific separation of phospho-tyrosine proteins from the rest of the phosphorylated proteins (unpublished observations).

4. Notes

1. Pierce's BCA Protein Assay Reagent Kit should be used for all Phosphoprotein Enrichment Kit analyses. Using other protein assays or BCA reagents (or kits) could lead to errors in protein estimation, as PMAC buffers contain substances known to interfere with protein assays.
2. We find that two 150-mm culture plates of 80–90% confluent cells yield approximately 150 mg of cells.
3. If your sample comprises 100 mg of cells, add 3 ml of Buffer A.
4. Start preparing the column (*see Subheading 3.3.*) while centrifuging the samples.

5. Use the BCA Protein Assay (Pierce; Cat. no. 23235) for protein quantitation.
6. Start preparing the columns while centrifuging the samples.
7. If extract or lysate is not translucent, you can clarify the sample by passing it through a 0.45- μ m filter or filter paper.
8. Use the BCA Protein Assay (Pierce; Cat. no. 23235) for protein quantitation.
9. We recommend a maximum sample load of 8 mg of total protein over a single column. If loading higher amounts, additional washing steps should be performed. Up to 5 ml of extract can be added to the column at a time. If your sample volume is larger than 5 ml, then add the extract in steps.
10. This type of purification is referred to as mixed batch/gravity flow chromatography in which the adsorption of the target proteins is carried under batch mixing of the sample with the resin, followed by gravity-based adsorption, washing, and elution.
11. Optional: If a cold room environment is not available perform this and the following steps at room temperature, otherwise continue at 4°C.

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Protein Separation Using Immobilized Phospholipid Chromatography

Tzong-Hsien Lee and Marie-Isabel Aguilar

Summary

The chromatographic support containing monolayers of phospholipids offers novel modes in analyzing and separating proteins. The polar choline head groups on immobilized phosphatidylcholine were used for the affinity purification of phospholipase A (PLA). The purification process involves removing the contaminating proteins with detergent additives to the elution buffer such as short-chain alkylsulfonates. The lipid-bound PLA was eluted with acetonitrile or octyllysophosphatidylcholine. The purity of PLA was approximately 70% based on densitometric scans of gel electrophoresis. These results suggest that the lipid-immobilized chromatography may be applied to develop purification methods for PLA, enzymes, and membrane proteins obtained from diverse cells.

Key Words: Immobilized lipid chromatography; membrane proteins; detergent; organic solvent.

1. Introduction

Analysis of genomic sequence data estimated that 30% of the proteins derived from *Homo sapiens*, *Escherichia coli*, and *Saccharomyces cerevisiae* will be integral membrane proteins (1–3). However, while the number of predicted gene sequences for integral membrane proteins has increased over the last few years, there is considerably less information about their structure and the nature of their function within the membrane.

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The primary difficulty encountered in the study of membrane proteins is that of obtaining the protein of interest. The difficulties in the investigation and separation of membrane proteins originate from their nature as membrane proteins (1). Membrane proteins are usually present at very low levels in biological membranes (2). They are very hydrophobic and have single or several transmembrane parts, or closely associate with the membrane (3). In the functional form, many of them comprise (homologous or heterologous) multi-subunit complexes (4). Such membrane protein complexes contain many cofactors and, inevitably, lipids (5). Some membrane protein complexes have several peripheral proteins, which are functionally important but easily detached during the isolation process. Despite the inherent difficulties of working with membrane proteins, they remain an important area for study because of their role in the control of fundamental biochemical process and their importance as pharmaceutical targets (3).

In general, the methods available for the purification of membrane proteins are basically the same as those employed to purify water-soluble, non-membrane-associated proteins (4–6). These methods include precipitation, gel filtration, ion exchange, reversed phase, and affinity chromatography. Several unique characteristics of membrane proteins, however, often make it difficult to apply these methods successfully. It is important to stress that, just as with soluble proteins, there is no way to present a single, precise set of methods for the purification of all membrane proteins. Each membrane protein possesses a unique set of physical characteristics, and conditions that are suitable for the purification of one protein may not be suitable for others. As a single chromatographic separation is not always successful in analyzing and isolating the protein of interest, the combination of various modes of chromatography is being developed for the study and separation of complex membrane proteomes (7).

Owing to the hydrophobic nature and the complexity of proteins that reside in biomembranes, immobilization of various modified phospholipids onto the surface of chromatographic supports which potentially mimics the physico-chemical properties of biomembrane surfaces provides an additional dimension in analyzing and separating membrane proteins (8–14). The chromatographic supports modified with various phospholipid molecules, such as phosphatidylcholine, phosphatidylglycerol, phosphatidylethanolamine, phosphatidylserine, and phosphatidic acids, have been applied mainly for the analysis of drug–membrane partition (15,16) and peptide–membrane interactions (10). However, only columns packed with phosphatidylcholine-immobilized spherical particles are commercially available, the structure of which is shown in **Fig. 1**.

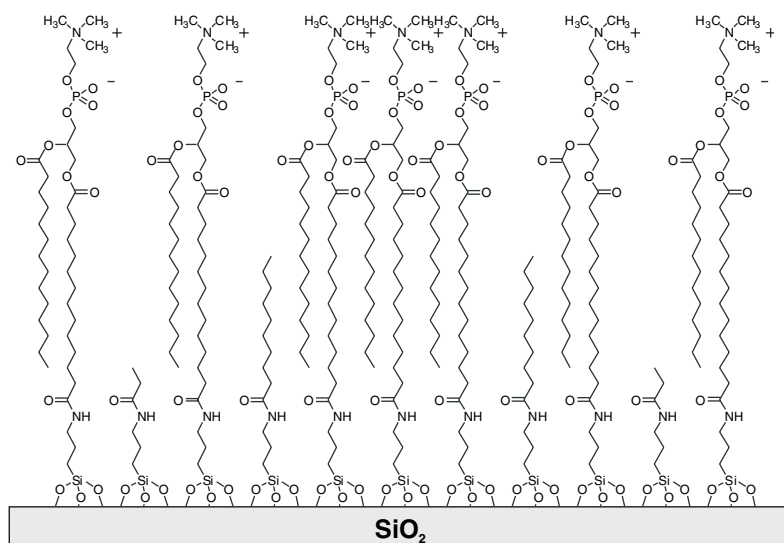


Fig. 1. General structure of phosphatidylcholine immobilized silica substrate. The phosphatidylcholine is covalently bound to propylamine groups and the residual amines are blocked with decanoic anhydride (C10 groups) followed by propionic anhydride (C3 groups).

In this chapter, a protocol for the isolation of proteins with affinity to membrane lipids is described using the immobilized phosphatidylcholine column.

2. Materials

2.1. Chemicals and Reagents

1. Milli-Q water.
2. Tris base.
3. Sodium octanesulfonate.
4. Acetonitrile (CH_3CN), HPLC grade.
5. Ethylene glycol.
6. 0.1 M phenylmethylsulfonylfluoride (PMSF) dissolved in isopropanol.
7. Trypsin solution: 10 $\mu\text{g}/\text{mL}$ trypsin in sample buffer.
8. 0.1 M NaCl.
9. $\text{CaCl}_2 \cdot \text{hexahydrate}$.
10. $(\text{NH}_4)_2\text{SO}_4$.
11. Sample buffer: 25 mM CaCl_2 , 50 mM Tris-HCl, pH 7.6.
12. Tissue-homogenizing solution: 0.1 M NaCl in Milli-Q water.
13. 0.1 M solution of PMSF in isopropanol.

2.2. Equipment and Supplies

1. HPLC solvent delivery system equipped with quaternary gradient capability and a variable wavelength UV detector. Typically, the detector is set to a range of a 0.08 bandwidth and a response time of 1.0 s.
2. IAM.PC.DD2 guard column 12- μ m particle size, 300Å pore size, 3.0 mm i.d. $\times$ 1 cm length (Regis Technologies Inc., Morton Grove, IL, USA).
3. IAM.PC.DD2 12- μ m particle size, 300Å pore size, 4.6 mm i.d. $\times$ 15 cm length (Regis Technologies Inc.).
4. Solvent filtration apparatus equipped with a 0.22- μ m Durapore filter (Millipore, Billerica, MA, USA).
5. Sample filter, 0.22 μ m cellulose acetate membrane.
6. Buffer A: 0.1 M Tris-HCl (pH 7.2), 0.2 M KCl, 20% ethylene glycol, 0.05% NaN₃ (see **Note 1**).
7. Buffer B: 1% sodium octanesulfonate in Buffer A.
8. Buffer C: 4% acetonitrile in Buffer B.
9. A programmable fraction collector.

3. Methods

3.1. Sample Preparation

1. Solubilize 100 mg lyophilized *Crotalus artox* venom powder from Sigma (St. Louis, MO, USA) containing phospholipase A₂ in 20 mL sample buffer (25 mM CaCl₂, 50 mM Tris-HCl, pH 7.6) and filtered through a 0.22- μ m cellulose acetate membrane syringe filter. The protein concentration of this solution is approximately 4 mg/mL.
2. For the preparation of phospholipase A₂ directly from pancreatic tissue, homogenize 300 g tissue in 300 mL of tissue homogenizing solution 0.1 M NaCl using a blender for 30 s at 4°C. After homogenization, adjust the tissue homogenate solution to pH 4.0 with concentrated HCl and heat the solution at 70°C for 2–3 min. Cool the homogenate solution in an ice water bath for 30 min and readjust the pH to 7 with concentrated NH₄OH. Centrifuge the sample at 3500 $\times$ g for 5 min at 4°C and then gradually add solid (NH₄)₂SO₄ to the supernatant with constant stirring until the concentration of (NH₄)₂SO₄ reaches 60% saturation at room temperature. Precipitate the proteins in an ice bath for 1 h and collect the precipitate by centrifugation at 5000 $\times$ g for 10 min at 4°C. Dissolve the pellet in 2.5 mL Milli-Q water followed by 50 μ L of a 0.1 M solution of PMSF in isopropanol. Incubate the sample further on ice for 1 h and lyophilize. To lyophilize the proteins, the solution is kept in a -75°C deep-freezer or placed in a dried ice/acetone bath till the solution completely frozen. The sample is then lyophilized overnight at -75°C in a vacuum lyophilizer. Dissolve the lyophilized proteins in sample buffer with volume which gives the protein concentration approximately 4 mg/ml (see **Note 2**). Before use, activate the lyophilized sample by adding Trypsin relative to the total protein. Trypsin converts the inactive phospholipase A₂ to its active form by selectively cleaving an N-terminal octapeptide.

3.2. HPLC Buffer Preparation

1. Filter all solvents through a 0.22- μ m Durapore filter membrane in a filtration apparatus fitted with vacuum. This removes particulates that could block the column and the solvent tubing.
2. For HPLC systems without an on-line degassing capability, subject the solvent to degassing before use in the HPLC instrument.

3.3. Column Equilibration and Blank Run

1. Connect the guard and the separation column to the tubing according to the HPLC system requirements and equilibrate the column with 100% Buffer A at a flow rate of 0.5 mL/min until the baseline is stable monitored at 280 nm for 30 min (*see Note 3*).
2. Maintain the column temperature at $25 \pm 1^\circ\text{C}$ during the equilibration and the separation. If the HPLC system is not equipped with a column thermostat, ambient room temperature is also appropriate for the equilibration and separation. Monitor the baseline and protein separation at 280 nm.
3. Once the stable baseline is obtained, inject 10 μ L of Milli-Q water or Buffer A either manually or through an autosampler to the column (*see Note 4*).

3.4. Chromatography

1. Injection volume: 50 μ L.
2. Inject the sample at a flow rate of 0.2 mL/min and run for 8 min which facilitates affinity adsorption between the injected proteins and the immobilized lipid surface. After protein loading, increase the flow rate from 0.2 to 0.5 mL/min over 2 min. Maintain this flow rate throughout the whole separation process.
3. After the protein loading, elute the proteins with Buffer A for 10 min. Program a change in the elution solvent from 100% Buffer A to 100% Buffer B over 10 min and then maintain 100% buffer B for 25 min. Finally, change the solvent from 100% Buffer B to 100% Buffer C over 1 min and maintain these conditions at 100% Buffer C for 30 min (*see Notes 5 and 6*).
4. After each chromatographic separation, it is strongly recommended that columns are washed with 50 mL isopropanol followed by about 50 mL of Milli-Q water before re-equilibrating the column with aqueous mobile phase column. Owing to the high viscosity of isopropanol, it is also necessary to avoid the high back pressure. Adjust the flow rate for column washing with isopropanol to 0.2 mL/min and wash for 250 min. For Milli Q wash, set the flow rate initially at 0.2 mL/min for 100 min and then raise it to 0.5 mL/min for 150 min (*see Note 7*).
5. Store the column at 4°C in either 100% methanol or 100% acetonitrile.
6. A typical chromatographic result is shown in **Fig. 2** for the separation of PLA2. The UV chromatogram at 280 nm shows two early eluting peaks that do not have any enzymatic activity. PLA2 elutes at approximately 60 min and is well separated from contaminating proteins.

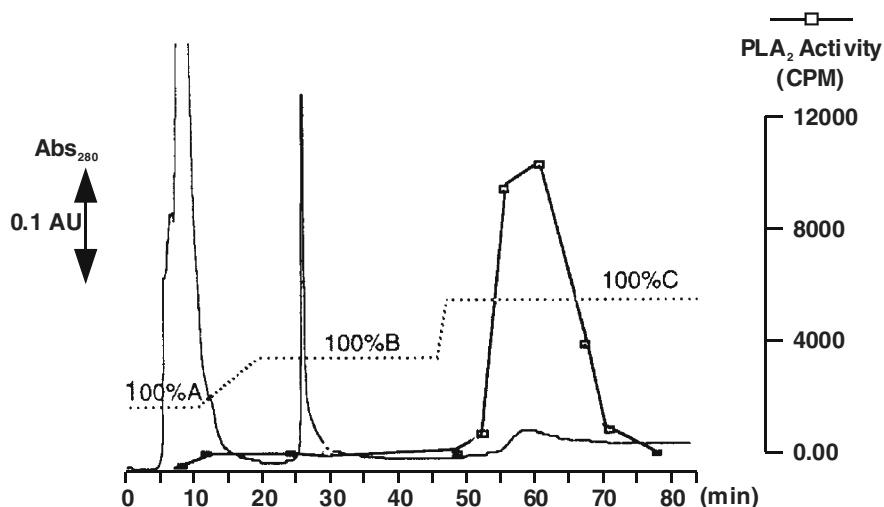


Fig. 2. Elution of proteins in the Sigma PLA_2 using sodium octanesulfonate and acetonitrile gradients. Two hundred micrograms of protein in approximately 200 μl is injected to the phosphatidylcholine immobilized column (4.6 i.d. $\times$ 100 mm). Mobile phase A contained 0.1 M Tris (pH 7.2), 0.2 M KCl, 20% ethyleneglycol, and 0.05% NaN_3 . Mobile phase B contained 1% sodium octanesulfonate in mobile phase A. Mobile phase C contained 4% acetonitrile in mobile phase B. The dotted line represents the chromatography gradient. (●) PLA_2 activity. Each square represents a 1 ml chromatographic fraction assayed for PLA_2 activity. Reproduced with permission from **ref. 13**.

7. 1 mL fractions are collected from the column. The protein content in each fraction is determined using the bicinchoninic acid (BCA) protein assay kit (Pierce), and the purity is further analyzed using SDS-PAGE. A 12% polyacrylamide gel is routinely used for analyzing the protein species in each collected fractions. Silver stain is then used to visualize the protein bands.

4. Notes

1. The immobilized phospholipids are labile under acid and base conditions. The addition of organic acid modifiers such as trifluoroacetic acid and acetic acid into the separation buffer has to be avoided.
2. The addition of low levels of detergents or lysophospholipids with a high critical micellar concentration (cmc) of detergent additives is often required to maintain the activity of the protein of interest. An additive with a low cmc is preferable to facilitate their subsequent removal by, for example, dialysis.
3. Some proteins and non-protein materials can be strongly retained on the column and failure to flush out these materials may affect the separation result. It is

therefore recommended to wash the column before commencing the separation, with 100% Buffer C until a stable baseline is reached followed by re-equilibration in Buffer A conditions.

4. The blank run may need to be repeated two to three times to ensure proper equilibration, particularly for a newly purchased column or if the column has been stored for a long time.
5. The addition of detergent to the mobile phase may be varied depending on the separation efficiency. The detergent, chaotrope additives, and phospholipids present in the collected fractions may affect typical methods such as the BCA method in determining the protein content. The detergent, chaotropes, and lipid-compatible methods (such as DC/RC protein kit from Bio-Rad or 2D protein Quant kit from Amersham Bioscience, Piscataway, NJ, USA) are required to accurately determine the amount of proteins. Compatibility of detergent to further 1D or 2D gel electrophoresis also needs to be considered for testing the purity of protein.
6. Removal of detergent from the collected fraction may be required to recover the activity of membrane proteins, which can be achieved by the selective adsorption of the detergent to hydrophobic substrates of Bio-Beads.
7. Column regeneration is typically achieved by continued washing with the starting or running buffer. However, because of the hydrophobic nature of membrane proteins, the binding of membrane proteins to the lipid ligands may be very strong. Hence, high stringency wash buffers are necessary to completely remove the residual bound membrane proteins.

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Analysis of Proteins in Solution Using Affinity Capillary Electrophoresis

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Summary

Analysis of protein interactions by means of capillary electrophoresis (CE) has unique challenges and rewards. The choice of analysis conditions, especially involving electrophoresis buffers, are crucial and not universal for protein analysis. If conditions for analysis can be worked out, it is possible to utilize CE quantitatively and qualitatively to characterize protein-ligand binding involving unmodified molecules in solution and taking place under physiological conditions. This chapter deals with the most important practical considerations in capillary electrophoretic affinity approaches, affinity CE (ACE). The text emphasizes the most critical factors for successful analyses and has application examples illustrating various types of information offered by ACE-based studies. Also included are step-by-step accounts of the two main classes of experimental design: the pre-equilibration ACE (in the form of CE-frontal analysis (CE-FA)) and mobility shift ACE together with examples of their use. The ACE approaches for binding assays of proteins should be considered when the biological material is scarce, when any kind of labeling is not possible or desired, when the interacting molecules are the same size and when rapid and simple method development is a priority.

Key Words: Affinity capillary electrophoresis; binding assay; analytical conditions; pre-equilibration ACE; mobility shift ACE.

1. Introduction

Very highly efficient separations that are reminiscent of the capabilities of high-performance liquid chromatography (HPLC) but do not require reversed phase conditions can be achieved by capillary electrophoresis (CE). High-voltage electrophoresis in solution in sub-millimetre diameter quartz tubes was introduced in the end of the 1980s (1–7), and CE has been used much since, especially for characterizing small molecules. The technique has unique capabilities, e.g. for separating impurities and enantiomers using simple, short procedures. Also, the potential for automation has been extremely successfully combined with parallel processing and laser-induced fluorescence detection in DNA-sequencing where CE has become the most important separation method. As for other biological macromolecules, the chemically more complex polypeptides and proteins have proved to be challenging to analyse using CE. Despite this, CE offers unique possibilities for functional characterization of proteins (8) by exploiting and characterizing binding interactions in which the protein takes part. Protein CE involving reversible molecular interactions, known as protein affinity CE (ACE), is the topic of this chapter.

Proteinaceous biomolecules are complicated analytes because they are not always structurally or conformationally homogeneous, since they may be composed of distinct subdomains with widely different properties and because they are often only available in limiting amounts which hampers method development. The goal of functional characterization of proteins is typically to understand their role at their point of origin, i.e., under physiological conditions or conditions as near physiological as possible. This chapter is intended to give a discussion of CE methods for this purpose from a practical perspective with an emphasis on the most critical factors for successful analyses and with application examples illustrating various types of information garnered from CE-based affinity studies. The literature is not reviewed comprehensively. A number of recent publications may be consulted for more systematic reviews of interaction applications and the theory of CE (9–20).

2. Objectives and Limitations

There are no simple and universal rules as to the size, isoelectric point, amino acid composition, conformational characteristics, solubility or other molecular features that predict whether CE investigations of proteins are going to be applicable and how they should be carried out. Some generalizations, however, can be made: If a protein is small, structurally homogeneous, conformationally stable, globular, well-soluble and negatively charged, then chances are good that characterizing this particular protein by CE separations in buffers near physiological pH and ionic strength values will be feasible. The objectives of using molecular interactions in CE can be different: discovery and mapping of

and screening for ligand-binding sites, optimization of CE resolution, conformation structure-function studies, detection of functional heterogeneity and estimation of quantitative binding parameters such as binding (stability) and rate constants. The reasons for using CE for such studies will typically be the scarcity of biological material, the unique ability of CE to be applied to unlabelled interacting compounds of similar size and the convenient and fast analyses.

An attractive feature of using CE for affinity studies is the versatility, i.e., the variety of approaches available making it possible to accommodate CE to a wide range of interactions. The ACE methods may be divided into two major groups: (1) the mobility shift and (2) the pre-equilibrium (pre-eq) assays. Changes in analyte mobility as a function of ligand concentration are used for determination of binding constants in the mobility shift assays. The pre-eq assays are characterized by introduction of a pre-incubated sample containing both of the interacting species into a capillary containing only neat electrophoresis buffer. Upon separation, peak heights or areas are used in the subsequent data analysis. The main features of the various ACE methods are summarized in **Table 1**. Unfortunately, a wealth of different names, acronyms and abbreviations have been assigned to ACE methods by different groups (*see Note 1*).

3. Experimental Variables

In any CE experiment, the most important decisions deal with the choice of capillary column, the washing solutions, the electrophoresis and sample buffers and the choice of running conditions. In affinity electrophoresis, these choices are all very dependent on the type of analyte and ligand. For a thorough evaluation of the relative importance of the parameters affecting the separation performance of CE experiments, it is worthwhile to consult (21). Factors and practical considerations that affect molecular interactions, recovery and reproducibility of peak shape, area and appearance time in the CE analyses of proteins will be focused on here.

3.1. Capillaries

The standard approach is to use columns of uncoated fused synthetic silica of 50 μm in internal diameter (i.d.) and of 40–70 cm in length. This i.d. in most instruments gives an appropriate detection path length at the same time as the induced Joule heating is efficiently removed. In this regard, instruments equipped with liquid cooling may be advantageous over forced air-cooled instruments and definitely over instruments with no active cooling (22). Even though different approaches exist to estimate the distribution of temperature inside a capillary buffer during a run (21–23), the cooling may not so much be used to secure a given temperature during an analysis but more to make

Table 1
CE Methods and Their Corresponding Acronyms Used for Characterization of Molecular Interactions^a

Name	Sample	Electrophoresis buffer	Quantitation parameter
<i>Mobility shift assays</i>			
Mobility shift ACE (100)	Analyte	Ligand added	Shift in mobility of analyte
Partial-filling ACE (84,101,101)	Analyte	Ligand added	Shift in mobility of analyte
Vacancy affinity capillary electrophoresis (VACE) (102)	Neat buffer	Analyte + ligand added	Shift in mobility of analyte (vacancy peak)
Capillary affinity gel electrophoresis (CAGE) (103)	Analyte	Ligand immobilized in gel	Shift in mobility of analyte
Hummel–Dreyer principle (104,105)	Analyte + ligand	Ligand added	Peak area corresponding to complex concentration (vacancy peak)
Vacancy peak analysis (VP) (104)	Neat buffer	Analyte + ligand added	Peak area of analyte vacancy peak
<i>Pre-equilibrium (pre-eq) assays</i>			
Pre-eq CZE (106)	Analyte + ligand	Neat buffer	Peak area
Capillary electrophoresis frontal analysis (CE-FA) (9,104)	Analyte + ligand	Neat buffer	Analyte plateau height
Frontal analysis continuous capillary electrophoresis (FACCCE) (107)	Analyte + ligand	Neat buffer	Analyte plateau height
Affinity probe capillary electrophoresis (APCE) (108,109)	Analyte + ligand 1	Neat buffer or ligand 2 added	Peak area

CE, capillary electrophoresis; ACE, affinity CE.

^a See **Note 1** for alternative names of the methods.

sure that temperature conditions are constant in all the experiments in a series. Owing to the decrease in viscosity with temperature and the increase in current, the observed peak migration is extremely dependent on the temperature conditions. Therefore, the uniformity and consistency of the cooling are of great importance for ACE experiments. Also, the UV-absorbance of most buffers is dependent on the temperature of the buffer. The capillary length is chosen to provide enough separation with minimal diffusion (peak broadening) and enough resistance to minimize current. It should also be considered that there is a relation between the stability of a molecular complex and the separation time. Short-lived, low-affinity pre-incubated complexes require short separation times (capillaries with short effective lengths i.e., distance to the detector point and/or high field strengths) to be detected before they dissociate. As a rule, to ensure 10% or less-complex dissociation during separation, the dissociation rate constant of the complex should be less than $0.105/t$, where t is the time it takes to separate the peaks (24). Sometimes, it will therefore be advantageous to inject and separate pre-equilibrated samples from the short end of the capillary or use custom-made apparatus on microchips and/or flow-gated capillaries to achieve separations as short as 1 s or lower. This also enables on-line immunochemical monitoring of biofluids (25–28). There are, however, practical limits to how short a capillary can be fitted into commercial instruments, and decreasing resolution also determines how short a capillary can be. In many cases, when studying low-affinity interactions, the mobility shift ACE approaches (c.f. **Subheading 6**) may instead be considered.

Very narrow capillaries will allow very high field strengths to be applied and will thus increase separation efficiency – however, at the expense of detection limits. Regarding the handling of capillaries, it is sensible to pay attention to the cut edges of the capillary ends; the less frayed and irregular these can be made, the less is the risk of carry-over, irreproducible pressure injection volumes and capillary blockage (*see Fig. 1*).

Having taken into account stable temperature and current and sufficient detection path length, the by far most prevalent problem is the recovery of protein analytes in uncoated fused silica capillaries at the neutral pH conditions that normally will be favoured for binding experiments.

Coated capillaries may overcome some protein adsorption problems and come in many different versions, but overall such capillaries have not been used much, probably because any kind of coating whether being dynamic or static (29) will be associated with its own set of problems. Also, the great feature of electroendosmosis (EEO)-assisted electroseparations is that it makes all analytes analyzable in one operation without changing polarity although various coatings may eliminate or reverse the EEO flow. Coated capillaries also generally have shorter life-spans than plain capillaries.

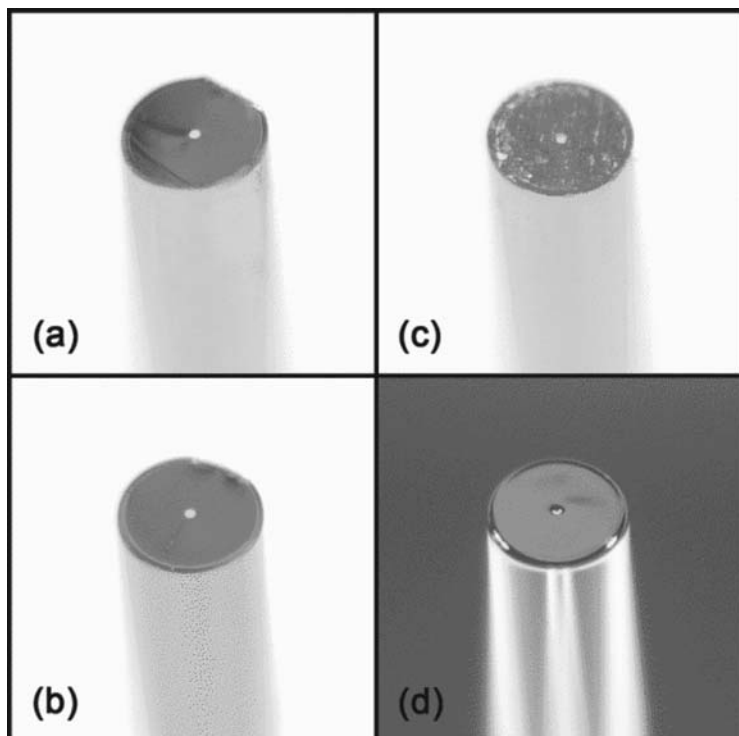


Fig. 1. Images of capillaries cut by different methods. The capillary is a 375- μm o.d., 25- μm i.d. polyimide-coated fused-silica capillary. The following cutting methods were used: (1), standard cleave using a ceramic cleaving stone; (2) precision cleave using a cleaving device (Polymicro); (3) saw cut and (4) laser cut using a programmable CO_2 laser station. Reproduced by permission from Polymicro Technologies, LCC AZ, USA.

3.2. Washing, Conditioning, Electrophoresis and Sample Buffers

At neutral pH in an uncoated capillary the wall charge is negative and creates an electroendosmotic (EEO) flow towards the cathode, which in the conventional set-up is situated at the detector end of the capillary. Actually, full protonation of the siloxide groups (zero charge) first occurs at a pH as low as 2.0 (30). In addition, the magnitude of the EEO flow decreases with increasing buffer ionic strength. All protein analytes/ligands that display positive charge will be prone to attach to the fixed capillary wall charges by electrostatic interactions. Therefore, proteins with low isoelectric points, i.e., negatively charged at neutral pH, will be more likely to be recoverable than basic proteins. However, even acidic proteins may – despite a low pI – contain patches of positively charged side chains and display the hallmarks of disruptive wall interactions: variable peak areas, tailing or other asymmetry or disappearance.

The starting point in electrophoresis buffer selection is the condition that best mimics the environment in which it is interesting to characterize the interaction in question. Thus, for serum proteins, an isotonic buffer (corresponding to 154 mM NaCl), pH 7.4 will be appropriate, while for proteins functioning in specialized sites, e.g. in kidney compartments, at infectious sites or intracellularly, very different conditions may be appropriate. If this first choice of buffer turns out to be incompatible with analysis one may try to modify it, (e.g. if protein adsorption is the problem, by modifying pH in small steps to determine the smallest pH-shift from the ideal value that allows for a reproducible analysis with full recovery of the analyte (31)) or by adding various non-ionic detergents to disaggregate interacting hydrophobic patches (32). High ionic strength may by itself be sufficient to counteract wall interactions and increase resolution (33). Also, ion-pairing agents (34,35), as known from reversed phase high pressure liquid chromatography (RP-HPLC), may be used to the same effect as long as it is ensured that these agents do not themselves interact with analytes or ligand additives, and that the current increases that are bound to occur with increased charged ions in the buffer, are not detrimental for the temperature inside the capillary. This may be an issue for easily denatured proteins. Some strategies to counteract wall interactions rely on utilizing the pH hysteresis effect of fused silica (36). This is conveniently achieved by an acid pre-rinse solution (for example, 0.1 M HCl instead of 0.1 M NaOH), which will diminish capillary wall deprotonation and negative charge at the ensuing neutral pH analysis (37,38).

The single most important analyte parameter influencing electrophoretic mobility is charge, i.e. electrophoresis buffer pH (5,6). The buffer choice is also specifically influenced in binding experiments with ligand addition to the electrophoresis buffer by solubility and other ligand characteristics in particular buffers. When deciding on the pH of a separation, all the usual buffer considerations such as buffer capacity and buffering range apply. In addition, some CE-specific features such as the UV-transparency and heat capacity influence the choice of electrophoresis buffer. Also, it is important to remember that buffer components such as ions added may adhere in a charge-dependent fashion to the inner capillary surface. It is always instructive to watch the EEO flow for changes as an indicator of immobilized wall-charge changes. In special cases, for instance, when performing low-temperature electrophoresis, the viscosity characteristics of the electrophoresis buffer also become important (39).

The UV-transparency of buffers is extremely important for low-wavelength (200 nm) detection, which is most often employed in work with proteins. Even under the best of conditions, the polypeptide limit-of-detection (LOD) rarely exceeds 1 μ M. A high UV-absorbance by the buffer decreases the linear dynamic range of the detector and thus peak heights. Specific buffer

characteristics may be desired, e.g. when studying metal-ion-binding proteins (40,41). Calcium ions will, for example, precipitate in phosphate buffers. Very reliable results may instead be obtained with HEPES buffers that have minimal cation-binding (42). In work involving, for example Ca^{2+} , it may be necessary to use chelating agents such as ethylenediaminetetraacetic acid (EDTA) in the washing solutions to remove all divalent cations between runs. The magnitude of the EEO flow will be a sensitive indicator of the amount of immobilized cations in such experiments.

The interplay between sample solution and electrophoresis buffer also requires attention. Conductivity differences may be detrimental but may also be exploited to increase detection limits by taking advantage of stacking phenomena. It is important to realize that even though considerable increases in detection limits may be achieved by dissolving the analyte in a sample buffer with lower (typically 1/10 diluted electrophoresis buffer) conductivity than the electrophoresis buffer (43,44), the resulting temperature increase in the sample zone may be very high leading to, for example, heat-induced partial or complete denaturation or the induction of other artifacts, such as, aggregation of the protein analyte (22) (see Fig. 2). Conversely, when the conductivity of the sample is higher (e.g. because of a high salt content), analyte peak broadening is to be expected. Also, in any CE experiment where buffer and sample conductivity is not the same, the precise concentration of the analyte in the sample zone is bound to be different from the concentration in the sample and will change during the initial electrophoresis steps. This complicates affinity experiments where the exact concentration of analyte during the run is required for the subsequent calculations.

In addition to conductivity differences, the correspondence of pH in the sample solution and in the electrophoresis buffer also warrants attention because the crossing of the pH boundary created upon initiation of electrophoresis may lead to analyte aggregation and precipitation.

Finally, the vial strategy should be considered for two main reasons: one is that repeated electrophoresis from the same buffer vial will lead to so-called buffer depletion, (a change, caused by electrolysis, in the ionic composition of the anodic and cathodic buffer solutions) leading to changes in mobility when the electrolyzed buffer is used as a running buffer. Thus, fresh buffer should always be used to replenish the electrophoresis buffer, for example, by using different reservoirs for running and for rinsing. This will ensure a reproducible composition of the buffer inside the capillary. Another detail regarding vial and washing strategies is that in affinity electrophoresis with ligand addition to the electrophoresis buffer, it is normally not the intention to introduce ligand into the sample solution. Carry over of ligand into the sample solution when sample injection follows immediately after washing the capillary with the ligand-

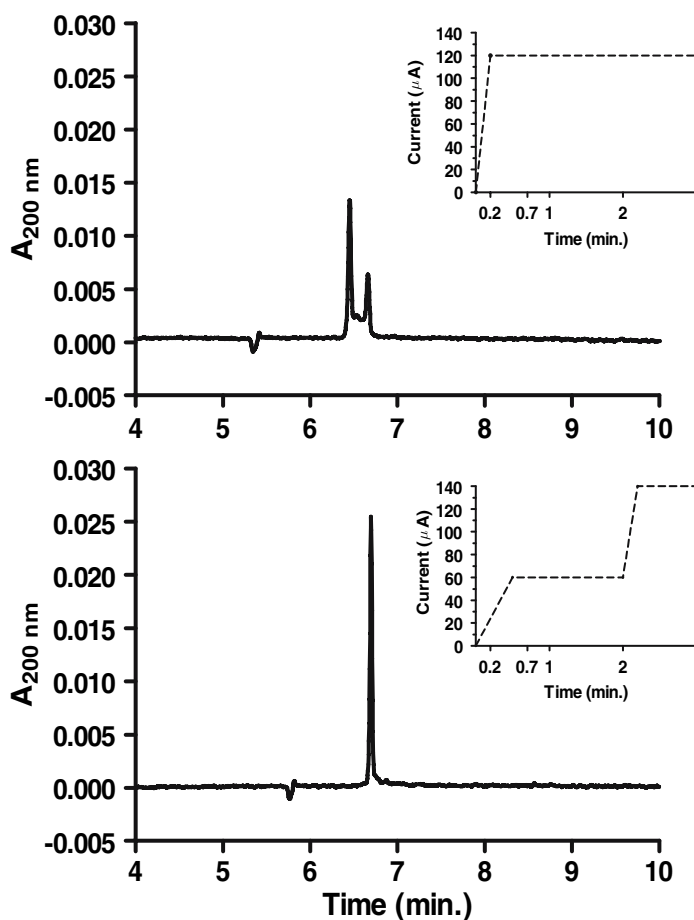


Fig. 2. Sample zone temperature influences the peak profile of β_2 -microglobulin (β_2 m). This protein displays conformational heterogeneity at elevated temperatures (45). β_2 m diluted from 9.4 mg/ml in phosphate-buffered saline (PBS) to 0.5 mg/ml by water was injected for 4 s. Capillary electrophoresis (CE) was performed in 0.1 M phosphate, pH 7.4, using stepwise constant current profiles as indicated by the inserted graphs. The capillary was liquid thermostated at 18°C. Under CE conditions with a rapid current ramping after sample injection (upper graph), β_2 m separates into two peaks representing different conformations, while a slow ramping, even with a higher final current, (lower graph) results in a single peak with no signs of conformational heterogeneity.

containing electrophoresis buffer is in practice easily prevented by introducing a 1 s injection step, of water. This step is programmed to occur before injection of sample and after rinsing the capillary with ligand-containing electrophoresis buffer. It is then possible to perform tests with multiple ligands using the same

sample. A final note regarding the buffer vials is that a hydrodynamic force (siphoning) will be added to electrophoresis and EEO during a separation if the capillary ends are at different fluid heights, and this may be detrimental to efficiencies (21). Thus, it is important to ensure that inlet and outlet vial buffer levels are equal.

3.3. Running Conditions

After appropriate conditioning/washing, and pre-rinsing, the affinity electrophoresis experiment is initiated by injecting the sample and applying a current. The controllable parameters here include sample injection mode and settings for time/current/field strength, and in some cases sample temperature. In addition, there are choices to be made regarding constant current/voltage/power, rise times, detection mode, run time and capillary temperature control.

With regard to the sample solution temperature, this is controllable in some instruments by an external circulating water bath, and this may be very helpful in instances when studying protein folding–unfolding processes (45) and when the sample stability or pre-incubated binding interaction is temperature dependent. For sensitive experiments, it is advisable to control the actual temperature with a temperature probe into a sample vial. When working with different sample temperatures, it is also worth considering that solution viscosity, and thus injected volume in pressure injection modes, is changing with temperature. The viscosity of aqueous solutions increases with decreasing temperature. The peak area of a marker (e.g. a non-interacting peptide) may be used to normalize such injection volume fluctuations. Because sample volumes usually are in the 5- to 50 μL range (with injected volumes in zone electrophoresis usually being in the 1- to 15 nL range), another issue that merits attention is sample evaporation. Again, an internal calibrant may be used to correct for changes in analyte concentration caused by evaporation, but for larger time series where maybe many hundred injections are going to be performed from the same solution, the use of a protective layer of light mineral oil on top of the sample (as known from PCR experiments) will prevent evaporation (46).

In zone electrophoretic applications, the sample volume injected is normally not much more than 1–5% of the total capillary volume which usually is 1–2 μL . Injection may be performed by positive or negative pressure (hydrodynamic injection) or by current. The latter mode has the disadvantages of being selective (relatively more of high mobility components will be sampled), of altering the electrolyte composition in the sample and of being less reproducible than hydrodynamic injections (21). There are few reasons to use this sampling method in free solution electrophoresis except maybe to enrich for a specific high mobility analyte component.

If temperature in a sample-stacking zone is a concern, one may program a step-wise increase to ensure electrophoretic transport of the analyte into the electrophoresis buffer before the full field strength is applied (*see Fig. 2*). The choice of electrical parameters is otherwise an interplay between efficiency, time and induced temperature characteristics of the electrophoresis buffer (and thus on the efficiency of the cooling system). If as high a field strength as possible is desirable, one may use an Ohm's law plot to estimate the breakthrough-current (where the linearity of current as a function of applied potential is lost because the resistance drops with uncontrollable increase in temperature caused by inadequate Joule heat dissipation) (*21,47*). Performing separations under constant current settings has the advantage that the amount of induced Joule heat is constant. With constant field strength, more constant migration times will be obtained. However, there will be current and thus temperature fluctuations. These are usually of minor importance if the temperature is kept constant and the conductivity in sample solution and electrophoresis buffer is not too different.

Detector choices depend on the nature of the compounds involved in the affinity interaction and the scope of the analysis. In UV-absorbance detection, the concentration LOD is only in the low micromolar range for polypeptides (*see Note 2*). This confers a problem when measuring binding of low-concentration analytes and molecules involved in strong binding interactions. In these situations, much lower detector sensitivity is required. Labelling of the interacting molecules with fluorescent probes will increase the sensitivity of the system, sometimes down to sub attoM concentration LOD (*48*), but also modifies the structure of the analyte covalently possibly changing analyte electrophoretic mobility and binding behaviour. Laser-induced fluorescence detection principles are reviewed in (*49*). The types of fluorescent probes available are diverse, and thus in many cases, it is possible to avoid the interfering effect caused by the labelling. One example is to carbohydrate-tag an analyte with fluorescein-thiosemicarbazide as example in studies of the binding interactions of rHLA-DR4 complex with influenza virus hemagglutinin peptide ligand. The fluorescein-thiosemicarbazide probe is attached at the carbohydrate moiety of the protein complex which is not involved in the interaction (*32*).

Alternatives to the commonly used UV-absorbance and laser induced fluorescence (LIF) detectors are electrochemical detectors which have proven advantageous when analyzing for metal ions and small inorganic molecules in biological fluids (*50*), but which are difficult to use in conjunction with physiological buffers. Radioactivity based detectors may be very sensitive (*51*) but entail the use of non-standard detector equipment and require labelling of analytes.

Especially useful for applications involving binding interactions should be information-rich detector systems such as mass spectrometry (MS) and nuclear magnetic resonance (NMR) spectroscopy, but the experience with and practice of CE-NMR (52) is still limited. CE-MS in the form of CE coupled with electrospray ionization (ESI) mass analysers (*see Note 3*) (53,54) has been of utility in affinity studies of proteins (55–59). Also, ionization on surfaces using laser-desorption (MALDI) has been CE-interfaced (60), but ESI is suitable for on-line work and is more commonly used. The major issue is the junction between the separation capillary and the spray capillary/needle and the CE-buffer compatibility with the ionization process (61–63). Three general types of CE-ESI-MS interfaces have been developed: the sheathless interface, the liquid junction or split-flow interface and the more commonly used coaxial sheath-flow interface. Buffers for CE-MS applications are typically 10–30 mM aqueous high vapour pressure (volatile) acids such as formic and acetic acid or aqueous ammonium acetate or ammonia for positive and negative ionization modes, respectively (53). These types of buffers display minimal ionization suppression and adduct formation, but are not very well suited for working with separations in the pH 4–8 range. Although sheath-flow interfaces are relatively simple, the sheathless interfaces give higher detection sensitivity (*see Note 4*). However, they may be technically demanding (53). Split-flow interfaces (54), however, overcome the problems with analyte dilution, decrease in resolution, intricate fabrication and bubble formation inside capillaries associated with the other types of interfaces.

Evolving CE-detector combinations of potential utility for ACE of proteins in addition to NMR (64,65) include Fourier transform infrared spectroscopy (66), Raman spectroscopy (67,68), flame-heated furnace atomic absorption spectrometry (69), electrothermal atomic absorption spectroscopy (70), X-ray (71) and surface plasmon resonance (72,73).

4. Discovery and Mapping of Ligand-Binding Sites

If a given protein has a well-defined ligand-binding function, CE may be used as an adjunct technique to map binding site(s) in that protein. For linear binding sites, the standard approach will be to cleave the protein into tryptic fragments and then perform CE peak profiling in the presence and the absence of ligand in the electrophoresis buffer. In **Fig. 3**, the approach is shown with serum amyloid P component and its ligand heparin (*see Note 5*). Changes in the tryptic digest peptide peak profile are indicative of ligand interactions, and after identification of ligand-binding peptides – e.g. by CE-MS or by purification by HPLC followed by MS and spiking analysis by CE – the identified peptide may be purified or synthesized and quantitative binding

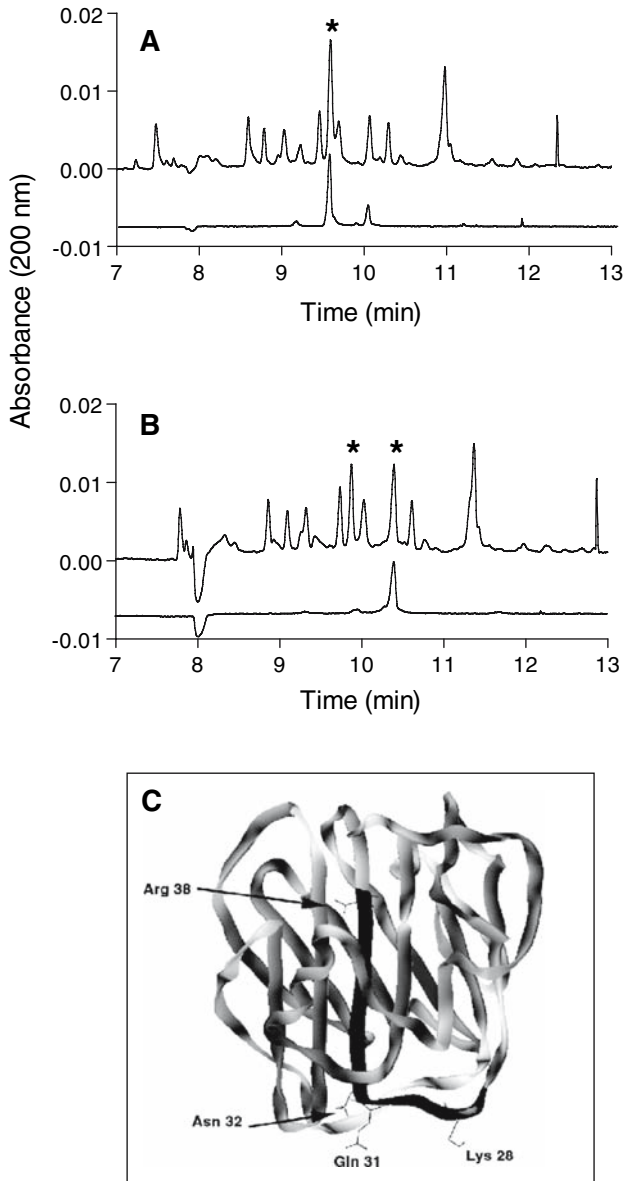


Fig. 3. Mapping of a heparin-binding site in human serum amyloid P component (SAP) using affinity capillary electrophoresis (CE) (76,110). (A, B) tryptic peptide map of SAP analysed by CE at 15 kV in a 50- μ m i.d. $\times$ 50/57 cm capillary in 0.1 M phosphate, pH 7.4, obtained in the absence (A) or presence (B) of 5 mg/ml heparin in the electrophoresis buffer. SAP was S-pyridylated and trypsinized (90) and 200 μ L digest was dried down and resolubilized with 50 μ L water + 20 μ L acetonitrile.

parameters extracted. In principle, the quantitative characterization may be performed using the complex tryptic digest mixture directly. This is the case if there is a suitable resolution and if the interaction kinetics enable migration shift experiments because then the exact concentration of receptor molecules need not be known (c.f. **Subheading 6**, below).

4.1. Method

Most laboratories will have experience in methods for trypsin digestion of proteins, compared with (74). A very convenient reagent for S-pyridylethylation of cysteine residues is 4-vinylpyridine (75). A short outline of trypsin digestion and test for heparin binding is given here:

1. Protein at >1mg/ml is reduced and S-alkylated/amidated/pyridylated, dialyzed against water and trypsinized in 0.1 M NH_4HCO_3 using 1–5% (w/w) sequencing grade trypsin at 37°C with gentle stirring.
2. The trypsin cleavage is followed by HPLC or by CE to ensure complete digestion (typically overnight).
3. The digest is dried down (in a Speed-vac centrifuge) in polypropylene tubes.
4. Re-dissolve in 10–20 μL water and subject to CE in 0.1 M phosphate, pH 7.4 (see **Note 6**) in the absence or presence of heparin.
5. A concentration-dependent anodic displacement/disappearance of tryptic peaks in the profile indicates heparin-binding activity (see **Fig. 3**).
6. Reactive peptides are purified for identification by preparative CE (see **Note 7**), or peaks are mapped by HPLC-MS and collected purified material is used for spiking analysis to identify the peaks in the CE-profile.
7. Based on the findings, synthetic peptides can now be made and used to characterize binding quantitatively (76) (see **Fig. 4**).

5. Conformation Structure-Function Studies

Few possibilities exist for the simultaneous separation of protein conformers and performance of binding studies. CE is unique in sometimes being able to achieve such a separation, and thus, folding parameters such as interconversion



Fig. 3. (Continued) Asterisks mark an interacting component and the lower trace in each figure shows the behaviour of an RP-HPLC-purified tryptic peptide (T3) corresponding to amino acid residues 14–38 of the parent SAP. The T3 peptide was identified by mass spectrometry/amino acid composition analysis (III), and its placement in the structure of an SAP monomer is indicated in (C) by the dark colour (Adapted with permission from (17)).

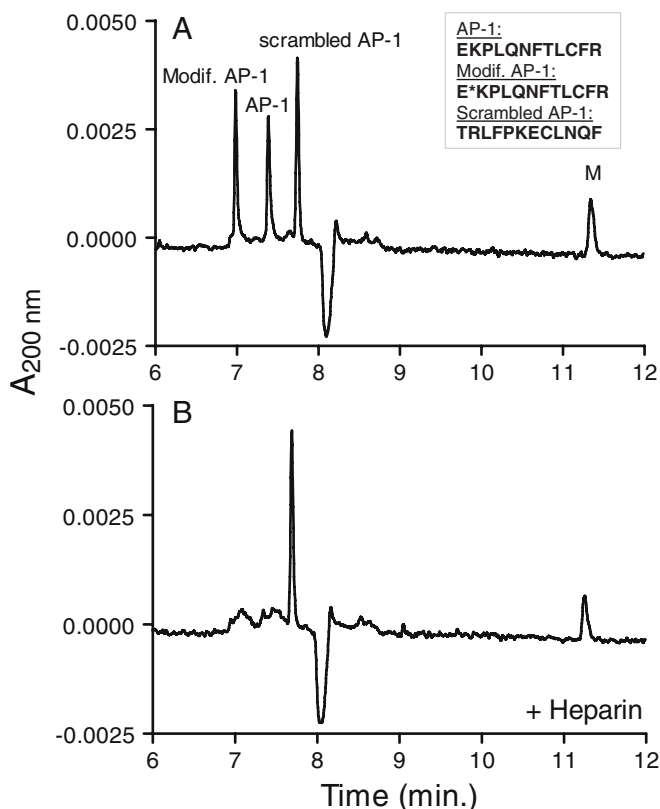


Fig. 4. Capillary electrophoresis (CE)-based binding study using synthetic SAP-T3-derived peptides (c.f. **Fig. 3**) elucidate structure-function relationships of heparin-binding peptides. Electrophoresis buffer was 0.1 M sodium phosphate, pH 7.4. The separation took place in a 50- μ m inner diameter uncoated fused silica capillary with 50 cm to the detector window and of 57 cm total length. Separations were carried out at 18 kV at liquid cooling at 20 C. Samples were pressure injected for 8 s after a 2-s pre-injection of water and were subjected to electrophoresis from a separate set of buffer vials than those used for pre-rinse. Peptide structures are indicated using single-letter amino acid abbreviations. The AP-1 peptide preparation used for the CE experiments was found to contain a mixture of regular AP-1 and modified (dehydrated) AP-1 (modif. AP-1), while scrambled AP-1 was homogeneous. A 1:1 mixture of AP-1 peptide and the scrambled AP-1 peptide (both 0.5 mg/ml (334 β M) in water) were analysed using CE in the absence (**A**) or presence (**B**) of 1 mg/ml (200 β M) LMW heparin in the electrophoresis buffer (Adapted with permission from (110)).

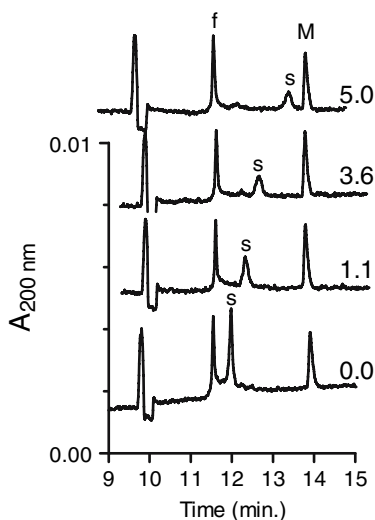


Fig. 5. Mobility shift ACE for studying binding activity of CE-separated β_2 -microglobulin (β_2 m) conformers. Congo red dye was added to the electrophoresis buffer in separations of conformationally heterogeneous β_2 m. The sample was 0.17 mg/ml β_2 m and 0.05 mg/ml peptide marker (M) in 33% (v/v) trifluoroethanol and injections took place for 4 s. CE was performed at 17 kV. Under these conditions the β_2 m analyte separates into two conformer peaks (*see f and s*), corresponding to a native (*see f*) and a more unfolded (*s*) conformation. Congo red was added to the running buffer from a 0.144 mM stock solution in electrophoresis buffer to the final concentrations (μ M) given in the figure. The s-peak is much more sensitive to the presence of Congo red than the f-peak (Adapted with permission from (112)).

rates and activation enthalpy and energy are accessible by CE (77–79) at the same time as the binding activities of the individual conformers may be estimated (45). Binding assays such as these are executed exactly as any other CE-based binding assays but exploit the high-resolution capabilities of the technique (*see Fig. 5*).

6. Quantitative Protein-Binding Parameters

6.1. Theory and Strategy

The binding strength is an important parameter in the functional evaluation of a protein and its interactions with established and putative ligands. While ACE may be used for identification of ligands (c.f. 4 above), it may also be used quantitatively, i.e. for the determination of binding stoichiometries (80) and binding constants. In special cases, also determination of the association rate and dissociation rate constants relating to the equilibrium constant is possible

(81–83). As indicated in **Table 1**, a number of approaches exist. The most widely used modes for the determination of binding constants are mobility shift ACE, pre-eq CZE and CE-FA. The principles of these methods will be outlined here. Partial-filling ACE and FACCE may be considered as specialized variants of mobility shift ACE and CE-FA, respectively. The workflow for conducting affinity experiments using these two methods has previously been described in the *Methods in Molecular Biology* series (84,85). After a brief description of mobility shift ACE, pre-eq CZE and CE-FA, a few general comments on how to approach interaction studies using ACE are provided. Practical examples on how to conduct mobility shift ACE and CE-FA are presented in **Subheading 6.2**.

The fundamental parameter of all CE experiments is the electrophoretic mobility, μ . The value of μ is determined by

$$\mu = \frac{q_{\text{eff}}}{6\pi\eta r} \quad (1)$$

where η is the viscosity of the background electrolyte; q_{eff} and r are the effective charge and radius of the analyte, respectively (86). After introduction of a molecule into an electrical field, a steady state is attained in which the ionic attraction is balanced by the frictional drag acting on the molecule. The charge-to-size ratio of **Eq. 1** represents this balance between forces, which makes a charged molecule (analyte or ligand) migrate with constant velocity in an electrical field of constant magnitude. The interaction of an analyte with another molecule present in the electrophoresis medium is likely to alter the charge-to-size ratio of the analyte. This will make the analyte migrate with a different velocity in the presence of interacting species. In other words, the analyte–ligand complex formed most often has an electrophoretic mobility different from that of the free analyte. This complexation-induced change in mobility is the basis of ACE. The high efficiency of CE makes it possible to detect even subtle differences in μ and consequently makes CE a strong tool for interaction analysis.

Mobility shift ACE is especially well suited for low-to-medium affinity interactions. A prerequisite for mobility shift ACE is that the dynamics of the binding equilibrium is fast, i.e. that the association and dissociation processes are rapid. If a 1:1 binding stoichiometry for the interaction between the analyte A and the ligand L is assumed, the corresponding binding equilibrium and stability constant for the interaction will be given by **Eqs. 2** and **3**, respectively.



$$K = \frac{[AL]}{[A][L]} \quad (3)$$

where AL is the formed complex, [A], [L] and [AL] the equilibrium concentrations of the analyte, ligand and complex, respectively, and K the stability (association) constant. Mobility shift ACE is conducted by performing a series of CE experiments in which a small volume of the analyte and a non-interacting marker are introduced into the capillary while the electrophoresis buffer contains various known concentrations of the ligand. Provided that free and complexed analyte have different electrophoretic mobilities, the effective electrophoretic mobility of the analyte, μ_{eff} , will depend on the concentration of the ligand added to the electrophoresis buffer according to

$$\mu_{\text{eff}} = \frac{[A]}{[A] + [AL]} \mu_A + \frac{[AL]}{[A] + [AL]} \mu_{AL} \quad (4)$$

where μ_A and μ_{AL} are the electrophoretic mobilities of the free analyte and the AL complex, respectively. **Equation 4** may be combined with **Eq. 3** and rearranged to give

$$\mu_{\text{eff}} = \frac{\mu_A + \mu_{AL} K[L]}{1 + K[L]} \quad (5)$$

A plot of μ_{eff} as a function of the free ligand concentration, [L], will give the binding isotherm, and the stability constant may be obtained by non-linear regression analysis using a suitable software package. Given the use of an internal marker and use of the same buffer, temperature and field strength conditions in a series of mobility shift ACE experiments, the peak appearance time t can be used directly in plots to estimate binding constants (c.f. **Subheading 6.2.1.**, below). The free ligand concentration in **Eq. 5** is assumed to be equal to the total ligand concentration in the electrophoresis buffer. For this to be approximately true, the analyte concentration in the sample needs to be more than 10–100 times lower than the ligand concentration (**87,88**). Note, however, that in contrast to the ligand concentration, the concentration of the analyte does not need to be accurately known. If the binding kinetics is not fast relative to the separation time, it will be evident in the mobility shift experiments as disappearance, broadening, tailing or splitting of the analyte peak (**87**). As a rule, averaged, weighted peaks reflecting the association–dissociation time distribution will only occur if the dissociation half-time $\ln 2/k_{\text{off}}$ is equal to or less than 1% of the peak appearance time (**89**). If the $1/k_{\text{off}}$ -value is getting close to the analyte peak appearance time, the complexes are too stable for the mobility shift approach to be useful (**87**) (see **Fig. 6** see **Subheading 6.2.1**). The figure illustrates a mobility shift experiment (of a monoclonal antibody interacting with its antigen) where the analyte peak is displaced by the anionic ligand (synthetic oligonucleotide) but otherwise unperturbed. Thus, the experiments can be used to estimate the binding constant for this interaction. In addition, in

the same series of experiments, a considerable portion of the antibody solution that is not binding is uncovered when the active antibody fraction is displaced.

Pre-eq capillary zone electrophoresis (pre-eq CZE) is complementary to mobility shift ACE in the sense that it is suitable for characterization of interactions with slow complex dissociation kinetics only. It is conducted by introducing a small volume of equilibrated sample into the capillary containing neat buffer. Owing to the sample introduction step, which is usually accomplished by hydrodynamic injection, the binding equilibrium between the interacting molecules has to be established. In addition to the free and complexed interacting species, the sample may contain an inert marker molecule that allows for correction of changes in peak areas because of variation in the EEO flow and injection volume (90). In contrast to the mobility shift assay, the electrophoresis buffer in pre-eq CZE does not contain the interacting species; thus, equilibrium is not maintained during electrophoresis. Separation of three peaks corresponding to the analyte, ligand and complex may be achieved when the complex dissociation kinetics is slow relative to the time scale of separation. The approach is feasible as long as it is possible to detect and separate one of the interacting molecules from the complex. A calibration curve is constructed for one of the interacting species (the analyte). The concentration of free analyte is usually determined from peak areas. A series of pre-incubated samples containing a constant concentration of the ligand and various concentrations of the analyte is analysed. To extract quantitative information, the total concentrations of both the analyte and the ligand have to be accurately known. Binding isotherms can be constructed by depicting $[AL]/[L]_{\text{total}}$ as a function of the free analyte concentration, $[A]$ from which the stoichiometry and the stability constant(s) can be obtained. In contrast to mobility shift ACE methods, pre-eq CZE is readily amendable to higher order stoichiometries.

CE-frontal analysis (CE-FA) is experimentally very similar to pre-eq CZE. The difference lies in the volume of sample introduced into the capillary. This volume is much larger in CE-FA than in pre-eq CZE. The large sample volume leads to the formation of plateaus or plateau peaks (*see Fig. 7*) instead of the narrow peaks characteristic of CZE. Owing to the increased sample volume, CE-FA is also feasible for studying interactions with rapid on-and-off kinetics (9,91). The FA principle is illustrated in **Fig. 7A** using the warfarin–human serum albumin (HAS) interaction as an example. The warfarin migration profile of the warfarin-HSA sample is characterized by three regions – a plateau corresponding to free warfarin (a), a plateau corresponding to the total warfarin concentration (free + bound) in the sample (b) in the region where equilibrium is sustained and a zone with a decreasing concentration of warfarin (c) caused by the depletion of warfarin and the ensuing disturbance of the equilibrium (9). **Figure 7A** also depicts migration profiles acquired by separate injections of

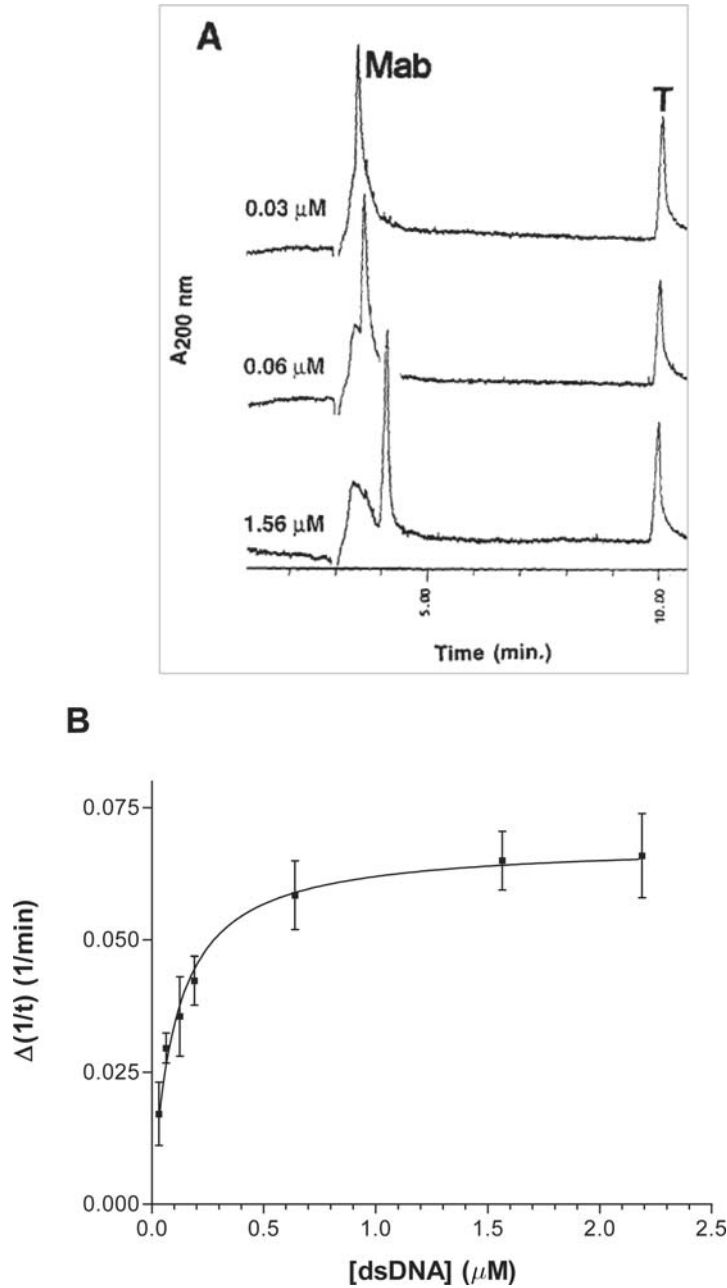


Fig. 6. Mobility shift affinity CE (ACE) for quantitative assessment of a binding interaction. (A) Monoclonal anti-DNA antibody (Mab, 0.7 mg/ml in 0.01 M phosphate, pH 8.13 with 0.03 mg/ml tyrosine phosphate (T) added as an internal marker) was injected for 2 s into a 27-cm, 50 μm i.d., untreated fused silica capillary with 20 cm

HSA (d) and warfarin (e). The concentration of free warfarin is proportional to the height of the free ligand plateau (region (a) in **Fig. 7A**). In zone (b) where both warfarin and HSA is present the equilibrium is preserved. Thus, the amount of warfarin leaving this zone and entering zone (a) is equal to the free warfarin concentration in the original pre-incubated sample. CE-FA is well established for studying interactions between low-molecular weight ligands and macromolecules where the mobility of the macromolecule is equal to that of the complex (9). However, theory indicates that the free concentrations are overestimated when these mobility requirements are not fulfilled, and this may be the case for some low-affinity protein–protein interactions (92). For systems characterized by slow binding kinetics where re-equilibration does not occur during the separation CE-FA performs as pre-eq CZE, and the mobilities of the complex relative to the free species is not an issue. Quantitation and data analysis are usually conducted as described for pre-eq CZE except that plateau heights are used rather than peak areas.

The first step in the characterization of an interaction system is to demonstrate binding. This is most easily accomplished using the mobility shift ACE format by conducting electrophoresis with and without the putative ligand added to the electrophoresis buffer. The sample should contain the analyte and a non-interacting marker molecule to correct for changes in the EEO flow. The existence of interactions will be revealed as a change in analyte mobility. The selection of the interacting species to be added to the electrophoresis buffer should be based on properties such as size, charge, UV-absorption properties and availability. Provided that the interaction kinetics is rapid and a 1:1 stoichiometry can be expected, mobility shift ACE may be used for further characterization of the system. If higher order stoichiometries are likely, one of the pre-incubation approaches should be considered if quantitative data are desired. In that case, the FA approach should be used initially as it is conducive to the study of interactions characterized by both fast- and slow-dissociation kinetics. In case of slow kinetics, however, the use of pre-eq CZE may be advantageous as compared with CE-FA because resolution is much



Fig. 6. to the detector. Electrophoresis took place at 8.5 kV in 0.1 M phosphate, pH 8.13, with additions of double-stranded 32mer biotin-DNA (dsDNA) at the concentrations given in the figure. Detection at 200 nm. **(B)** Data from binding experiments such as those presented in **(A)** plotted as outlined in **Subheading 6.2.1**. Data points represent the mean and the standard deviation of triplicate experiments. The curve represents a non-linear curve fit using a one-site binding hyperbola (GraphPadPrism). $R^2 = 0.99$. The equation for the curve yields a K_d for the Mab–dsDNA interaction of 0.10 μ M (Adapted with permission from (113)).

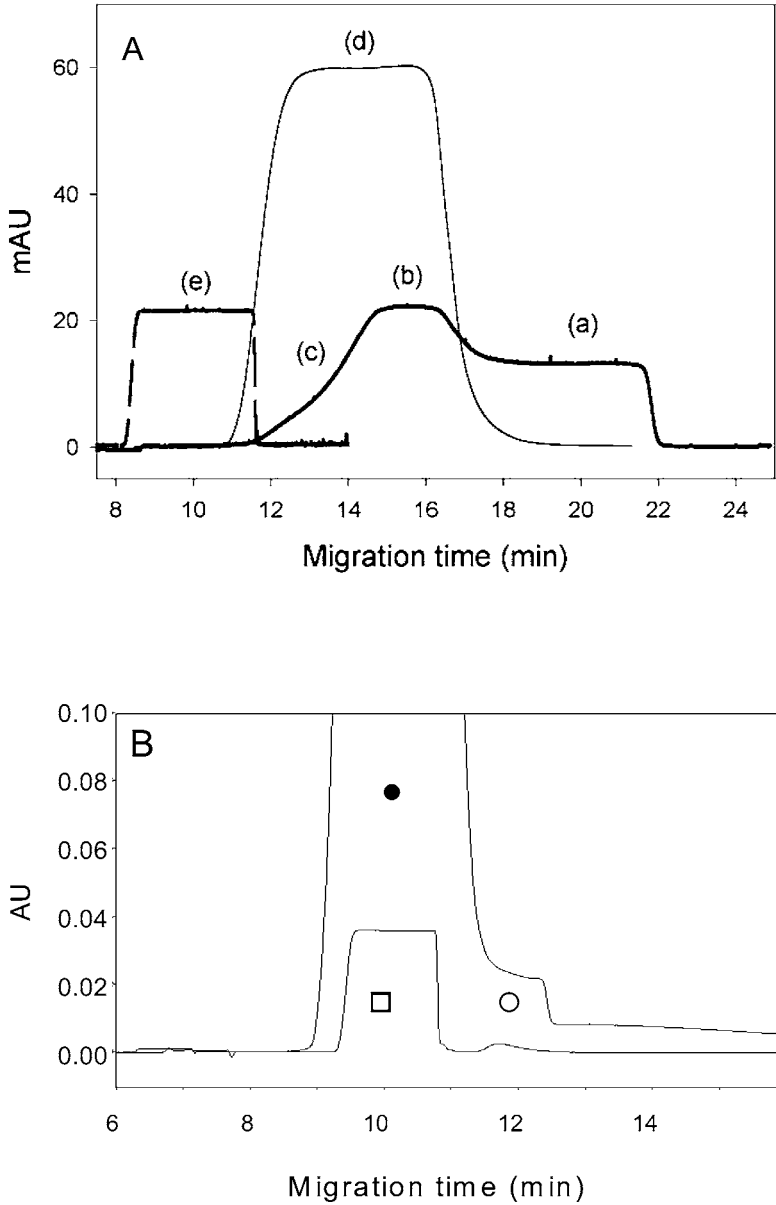


Fig. 7. Drug-protein binding studied by capillary electrophoresis-frontal analysis (CE-FA) in 0.067 M phosphate buffer, pH 7.4. (A) Electropherograms of 391 μ M warfarin with or without 65 μ M human serum albumin (HSA) (—, warfarin with HSA; — — —, (e) warfarin without HSA; —, (d) HSA blank). Experiments were performed on a Hewlett-Packard ^{3D}CE-instrument. Conditions: Uncoated fused silica capillary (48.5 cm $\times$ 50 μ m i.d., 40 cm effective length); applied voltage +10 kV ($\sim$ 46 μ A);

improved and less interference from impurities is anticipated. Selection of the analyte and ligand concentration ranges allowing a complete binding isotherm to be constructed is to a large extent dependent on the affinity of the system. However, due attention should be paid to the sensitivity of the detection system.

6.2. Binding Constants

6.2.1. Procedures for Mobility Shift ACE (87)

6.2.1.1. MATERIALS AND INSTRUMENTATION

1. Analyte solution containing non-interacting internal marker, e.g. a small synthetic peptide, dimethylsulphoxide or other molecule that is not binding to the ligand.
2. Protect sample against evaporation by carefully overlaying 10–20 μL light mineral oil (Sigma M-3516).
3. Mobility shift ACE is best performed in instruments with good thermostating capabilities to ensure reproducible temperature conditions in each analysis.

6.2.1.2. ELECTROPHORESIS BUFFER

For many ACE experiments, a phosphate electrophoresis buffer will provide sufficient neutral pH-buffering capabilities and high enough ionic strength to suppress unwanted electrostatic interactions (*see Note 8*), e.g. 0.1 M phosphate, pH 7.4:

40.5 ml 0.2 M Na_2HPO_4 (35.61 g/l of $\text{Na}_2\text{HPO}_4 \cdot 2\text{H}_2\text{O}$).

9.5 ml 0.2 M NaH_2PO_4 (27.6 g/l of $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$).

50 ml H_2O .

Fig. 7. detection 311 nm (200 nm for HSA blank); hydrodynamic injection for 100 s (50 mbar). *See Subheading 6.1* for explanation of (a)–(e). In contrast to warfarin, HSA absorbs very little at 311 nm. It is observed that the migration time of warfarin alone (e) is shorter than for free warfarin (a) in the HSA-containing sample. This is most probably caused by adsorption of HSA to the capillary wall leading to decreased electroosmotic flow and thus longer migration times for warfarin in the sample mixtures. (B) Electropherograms of 200 μM warfarin with or without 54 μM HSA; free warfarin concentration 72 μM . Experiments were performed on a Beckman P/ACE 5010 instrument. Conditions: Uncoated fused silica (57 cm $\times$ 75 μm i.d., 50 cm effective length); applied voltage +15 kV; detection 200 nm; hydrodynamic injection for 60 s (0.5 psi) (●, HSA; ○, warfarin sample; □, warfarin standard). Modified and reproduced from (9,114).

6.2.1.3. EQUATIONS

Data analysis may be conducted in several ways (see **Subheading 6.1**, e.g. **Eq. 5**). Here, two equivalent approaches based on differences in effective electrophoretic mobility and in peak appearance times are described:

1. Mobility change in experiment with ligand concentration [L] added to the electrophoresis buffer in comparison with no ligand added:

$$\Delta\mu = \Delta\mu_{\max} - K_d \times \Delta\mu/[L]$$

μ = effective, corrected electrophoretic mobility. $\mu = (lc \times ld)/[V \times (t - t_m)]$ where $t - t_m$ is the difference between the peak appearance time and the appearance time of a non-interacting marker. lc is the total capillary length and ld is the length of the capillary to the detection window. $\Delta\mu$, the mobility shift, i.e. the difference in μ between experiments with and without added ligand. $\Delta\mu_{\max}$, the maximum mobility shift (in a fully saturated system).

2. Mobility change and corrected peak appearance time (t) using internal (added to the sample) reference marker:

$$\Delta\mu = lc/E \times [(1/t - 1/t_r) - (1/t_0 - 1/t_{r0})] = lc/E \times \Delta(1/t)$$

lc is total capillary length, E is field strength, subscript 0 denotes reference experiment without ligand addition.

$\Delta(1/t) = (1/t - 1/t_r) - (1/t_0 - 1/t_{r0})$, i.e. difference in corrected inverse peak appearance time in experiment with and without added ligand.

3. Mobility change expressed using corrected peak appearance times:

$$\Delta(1/t) = \Delta(1/t)_{\max} - K_d \times \Delta(1/t)/[L]$$
4. Plots of $\Delta\mu$ as a function of [L] or $\Delta(1/t)$ as a function of [L] should show a definite curvature (saturation).
5. Non-linear curve fitting to the plot using a one binding site-hyperbola function yields the K_d if binding behaves according to a 1:1 molecular association binding isotherm of the equation: $\Delta(1/t) = (\Delta(1/t)_{\max} \times [L]) / (K_d + [L])$ (see **Fig. 6B**)

6.2.1.4. MOBILITY SHIFT ACE PROCEDURE

1. Preconditioning and inter-run washing procedures correspond to those given below for the pre-eq/FA-CE experiments.
2. Establish reproducible and suitable (e.g. physiological) analysis conditions for analyte, marker molecule and ligand separately, and ensure that they migrate differently.
3. Perform electrophoresis in the presence of various known concentrations of ligand added to the electrophoresis buffer. Depending on the availability, it will be advantageous to use the most charged molecule as the ligand (the buffer additive). Mix analyte in a suitable proportion with the marker molecule and perform the CE analysis. Look for migration shifts not affecting peak shape and size.
4. Perform affinity electrophoresis in the presence of ligand molar concentrations from 10 to 500 times the expected dissociation constant value while keeping the

approximate analyte concentration at least 10 times lower than the lowest ligand concentration.

5. Process peak appearance shift data according to the relations given above. A direct binding curve of $(1/t)$ as a function of $[L]$ is plotted to estimate the saturability of the system and to fit the binding isotherm to the experimental data using non-linear curve fitting methods. This yields the K_d from the formula for a one site-binding hyperbola.

6.2.2. Procedures for CE-FA as Applied to Drug–Plasma Protein Interactions

The procedures used for studying low-molecular weight drug binding to human serum albumin (HSA) (**93**) by CE-FA are given below. The approach was found to be applicable to a range of ligands with different physicochemical properties and should be useful for investigating the interactions of other ligands and proteins with minor modifications.

6.2.2.1. MATERIALS AND INSTRUMENTATION

1. HSA and drug samples of adequate purity.
2. Sample and electrophoresis buffer solution: 0.067 M sodium phosphate buffer (pH 7.4).
3. Deionized water, 1 M NaOH and 0.1 M NaOH for capillary conditioning and rinse procedures.
4. Uncoated fused silica capillary, suitable dimensions may be 57 cm $\times$ 50 μ m ID, 50 cm effective length. Condition capillaries by flushing with 1 M NaOH, water and electrophoresis buffer for 30 min each.
5. Commercially available CE-instrument with programmable autosampler.

6.2.2.2. SAMPLE SOLUTIONS

1. Prepare HAS-stock solution in electrophoresis buffer and drug-stock solutions in a suitable solvent. Filter HSA and buffer solutions through 0.45 or 0.22 μ m pore size filter before use.
2. Prepare a series of samples containing a constant and known concentration of HSA, e.g. 55 μ M and varying drug concentrations. Include a sample without drug added to check for impurities in the protein sample (*see Note 9*).
3. Prepare a series of standard solutions containing only the drug for construction of a calibration curve.

6.2.2.3. FA EXPERIMENTS

The standards and pre-incubated samples are all subjected to the procedure listed below:

1. Rinse capillary between measurements by flushing for 2 min each with 0.1 M NaOH and running buffer.

2. Introduce pre-incubated samples into the capillary by applying pressure (0.5 psi) for 99 s (injection volume ~ 121 nL) (*see Note 10*).
3. Perform electrophoretic separation of drug standard solutions and HAS-containing samples using a voltage of +15 kV in the normal polarity mode and a detection wavelength of 200 nm (*see Note 11*).
4. Construct a calibration curve by plotting plateau peak heights as a function of drug concentration of the standard solutions and determine the free drug concentration from the plateau heights by aid of the calibration curve.
5. Determine binding parameters by suitable data analysis (*see Note 12*).

Electropherograms obtained by CE-FA using experimental setups very similar to the one outlined above for warfarin-HSA binding are depicted in **Fig. 7**. The rectangular plateau peaks of the standard solution and the consecutive plateaus of the ligand-protein solution are characteristic of CE-FA. HSA and warfarin are both negatively charged with the apparent mobility of warfarin being slightly smaller than that of HAS, which leads to incomplete separation and the free warfarin plateau passing the detector after HSA. A positively charged ligand would be detected as a plateau before HAS, and complete separation would be obtained because of the large difference in mobility. Note that **Fig. 7A** was prepared for illustration of the CE-FA principle. With the long analysis time and very broad plateaus, the method would be of little practical interest. **Figure 7B** represents a more normal CE-FA experiment.

7. Conclusions

To the extent that proteins are recovered during conditions that are relevant for their native or in vivo function, there is a great deal to be learnt about their function from ACE experiments. Close attention to peak shapes and analyte recovery, reproducible temperature conditions, inclusion of non-interacting markers and proper coverage of binding isotherms will make useful characterization of protein interactions possible also in cases where only few other methods succeed.

8. Notes

1. The term ACE is normally used to cover both the mobility shift and the pre-eq formats. A number of alternative names for mobility shift ACE methodology have appeared; ACE, classical ACE (*17*), dynamic complexation CE (DCCE) (*94*) and mobility change analysis (*95*). Pre-eq CZE has been termed CZE (*96*), equilibrium-mixture analysis (*95*), CE mobility shift assay (CEMSA), pre-incubation ACE (PI-CE) (*10*) and a variant hereof non-equilibrium CE of equilibrium mixtures (NECEEM) (*82,83*). The recommended acronym for CE in the FA mode is CE-FA as the abbreviation FACE (*97*) has been used for fluorescence anisotropy CE.

2. The linear dynamic range of the detector is decreased when using buffers with high UV-absorbance. The result is a decrease in peak height, resolution and increase in noise in the electropherogram.
3. ESI is a mild ionization method compared with fast atom bombardment. ESI facilitates characterization of non-covalent molecular complexes in the gas phase.
4. To maintain the electrical circuit at the MS electrospray source, addition of surplus electrolyte is crucial for the liquid junction and coaxial sheath-flow interface. The analyte thus is diluted, and this gives a decrease in detection sensitivity as well as an interference with the resolution of the CE separation.
5. Heparins are strongly anionic, highly sulphated glycosaminoglycans. Their charges make them ideal for use as ligands in ACE. Heparin preparations contain mixtures of polymers of different chain length and of different sulphation and carboxylation (98). Heparin from bovine lung represents one of the most highly sulphated types (Dorothe Spillmann, personal communication).
6. CE buffers should routinely be prepared using deionized water (of Milli Q quality) and should be filtered through 0.22-pore size filters (e.g. cellulose acetate filter system (Corning 430767)) before use. The buffers usually can then be kept at 4°C for months.
8. Other useful binding buffers are
 - (a) Isotonic borate, pH 7.4
 - (A) 10 ml 0.05 M $\text{Na}_2\text{B}_4\text{O}_7$ (19.11 g/l of $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10 \text{H}_2\text{O}$)
 - (B) 90 ml 0.2 M HBO_4 (12.40 g/l)
 - (C) 270 mg NaCl
 - (b) To scan for analyte recovery at a range of electrophoresis buffer pH values (31), borate buffers of the following compositions may be helpful:

pH 6.8: 3 ml (A), 97 ml (B), 270 mg (C)
pH 7.8: 20ml (A), 80 ml (B), 260 mg (C)
pH 8.1: 30 ml (A), 70 ml (B), 240 mg (C)
pH 8.4: 45 ml (A), 55 ml (B), 210 mg (C)
pH 8.6: 55 ml (A), 45 ml (B), 190 mg (C)
pH 8.8: 70 ml (A), 30 ml (B), 140 mg (C)
pH 9.1: 90 ml (A), 10 ml (B), 70 mg (C)
- (c) HEPES, pH 7.4: 10 mM *N*-2-hydroxyethylpiperazine-*N'*-ethanesulphonic acid (HEPES) (2.38 g/l), adjusted with NaOH to pH 7.4, 150 mM NaCl (8.77 g/l).
- (d) Tricine, pH 8.15: This buffer will absorb strongly at 200 nm, 20 mM *N*-Tris(hydroxymethyl)methylglycine (Tricine) (3.58 g/l) adjusted with NaOH to pH 8.15, 150 mM NaCl (8.77 g/l).
- (e) Tris-buffered saline, pH 7.4: This buffer will absorb strongly at 200 nm. 5 mM Tris(hydroxymethyl)aminomethane (Tris) (0.61 g/l) adjusted with HCl to pH 7.4, 150 mM NaCl (8.77 g/l)

9. The sample solution should match the electrophoresis buffer with respect to ionic strength and pH to avoid stacking phenomena, which will perturb the binding equilibrium in the sample zone and thus invalidate results. If organic solvents have been used in the drug stock solution, the content must be diluted to $\leq 1\%$. In addition, UV-absorbing solvents may be detected as extra plateau peaks and thus hamper interpretation of the plateau patterns.
10. As a part of the method development, the effect of sample volume on the determined degree of binding should be examined (**91**). The injection time (volume) must be of a sufficient duration to provide plateau peak conditions that will ensure that the degree of binding is constant, independent of sample volume, and reflect the true equilibrium within the original sample. Equilibrium is usually attained very rapidly in drug-plasma protein solutions and only short time is needed for pre-equilibration. This, however, may be very different for other binding systems. The time required for attaining equilibrium may be established using CE by introducing the ligand-protein sample repeatedly over a period of time (**87**). Equilibrium has been reached when the plateau height of the analyte becomes invariant with time.
11. The applied voltage and capillary cassette temperature may be selected to avoid excessive Joule heating. For most drug substances, a detection wavelength of 200 nm appears to be optimal.
12. For drug-HSA interactions, binding parameters are often determined from

$$r = \frac{[L]_{\text{bound}}}{[P]_{\text{total}}} = \sum_{i=1}^m \frac{n_i K_i [L]_{\text{free}}}{1 + K_i [L]_{\text{free}}}$$

where r is the number of bound ligand molecules per molecule of protein; $[L]_{\text{free}}$, $[L]_{\text{bound}}$ and $[P]_{\text{total}}$ are the free ligand, bound ligand and total protein concentrations, respectively; m is the number of identical independent binding classes; n_i is the number of sites of class i and K_i is the corresponding association constant. The parameters are determined using non-linear regression analysis assuming one or two classes of independent binding sites ($m = 1$ or $m = 2$).

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